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GENERAL INFORMATION

DIAGNOSTIC TROUBLE CODE INDEX - INGENIUM I4 2.0L PETROL, DTC: POWERTRAIN CONTROL MODULE B10A2-31 TO P060D-00 [G2781177]

DESCRIPTION AND OPERATION

POWERTRAIN CONTROL MODULE [PCM] - INGENIUM I4 2.0L PETROL - B10A2-31 TO P060D-00

CAUTION:

Diagnosis by substitution from a donor vehicle is **NOT** acceptable. Substitution of control modules does not guarantee confirmation of a fault, and may also cause additional faults in the vehicle being tested and/or the donor vehicle.

NOTES:

- Guided Diagnostics must be followed and the repair advised by the process completed. Failure to do so may result in the rejection of any warranty claim made.
- If a control module or a component failure is suspected and the vehicle remains under manufacturer warranty, refer to the Warranty Policy and Procedures manual, or determine if any prior approval program is in operation, prior to the installation of a new module/component.
- Generic scan tools may not read the codes listed, or may read only 5-digit codes. Match the 5 digits from the scan tool to the first 5 digits of the 7-digit code listed to identify the fault (the last 2 digits give extra information read by the Jaguar Land Rover approved diagnostic equipment).
- When performing voltage or resistance tests, always use a digital multimeter that has the resolution ability to view 3 decimal places. For example, on the 2 volts range can measure 1 mV or 2 kΩ range can measure 1 Ω. When testing resistance always take the resistance of the digital multimeter leads into account.
- Check and rectify basic faults before beginning diagnostic routines involving pinpoint tests.
- Inspect connectors for signs of water ingress, and pins for damage and/or corrosion.
- If Diagnostic Trouble Code(s) are recorded and, after performing the pinpoint tests, a fault is not present, an intermittent concern may be the cause. Always check for loose connections and corroded terminals.
- Check the JLR claims submission system for open campaigns. Refer to the corresponding bulletins and SSMs which may be valid for the specific customer complaint and perform the recommendations as required.

The table below lists all Diagnostic Trouble Code(s) that could be set in the Powertrain Control Module (PCM). For additional diagnosis and testing information, refer to the relevant Diagnosis and Testing section in the workshop manualFor additional information, refer to: Electronic Engine Controls (303-14C Electronic Engine Controls - INGENIUM I4 2.0L Petrol, Diagnosis and Testing).

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION

B10A2-31	Crash Input - No signal	- Electrical Cause
		  - Yes
		- Mechanical Cause
		  - Yes
		- Control Module Cavity
		  - Circuit reference - I_F_INS -
		- Prioritized List of Possible Causes
		  - Crash signal circuit short circuit to ground, short circuit to power, open circuit, high resistance
		  - Restraints control module power or ground circuit open circuit, high resistance
		  - Supplementary restraint system fault
			- Prioritized Checks to Perform
			  - Refer to the electrical circuit diagrams and check the crash signal circuit for short circuit to ground, short circuit to power, open circuit, high resistance. Repair the wiring harness as necessary
			  - Refer to the electrical circuit diagrams and check the restraints control module power and ground circuits for open circuit, high resistance. Repair the wiring harness as necessary
			  - Using the Jaguar Land Rover approved diagnostic equipment, check the restraints control module for related Diagnostic Trouble Code(s) and refer to the relevant DTC index

B10A2-32	Crash Input - Signal low time < minimum	- Electrical Cause
		  - Yes
		- Mechanical Cause
		  - Yes
		- Control Module Cavity
		  - Circuit reference - I_F_INS -
		- Prioritized List of Possible Causes
		  - Crash signal circuit short circuit to ground, short circuit to power, open circuit, high resistance
		  - Restraints control module power or ground circuit open circuit, high resistance
		  - Supplementary restraint system fault
			- Prioritized Checks to Perform
			  - Refer to the electrical circuit diagrams and check the crash signal circuit for short circuit to ground, short circuit to power, open circuit, high resistance. Repair the wiring harness as necessary
			  - Refer to the electrical circuit diagrams and check the restraints control module power and ground circuits for open circuit, high resistance. Repair the wiring harness as necessary
			  - Using the Jaguar Land Rover approved diagnostic equipment, check the restraints control module for related Diagnostic Trouble Code(s) and refer to the relevant DTC index

B10A2-35	Crash Input - Signal high time > maximum	- Electrical Cause
		  - Yes
		- Mechanical Cause
		  - Yes
		- Control Module Cavity
		  - Circuit reference - I_F_INS -
		- Prioritized List of Possible Causes
		  - Crash signal circuit short circuit to ground, short circuit to power, open circuit, high resistance
		  - Restraints control module power or ground circuit open circuit, high resistance
		  - Supplementary restraint system fault
			- Prioritized Checks to Perform
			  - Refer to the electrical circuit diagrams and check the crash signal circuit for short circuit to ground, short circuit to power, open circuit, high resistance. Repair the wiring harness as necessary
			  - Refer to the electrical circuit diagrams and check the restraints control module power and ground circuits for open circuit, high resistance. Repair the wiring harness as necessary
			  - Using the Jaguar Land Rover approved diagnostic equipment, check the restraints control module for related Diagnostic Trouble Code(s) and refer to the relevant DTC index

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION
B10A2-36	Crash Input - Signal frequency too low	▪ Electrical Cause ▪ Yes ▪ Mechanical Cause ▪ Yes ▪ Control Module Cavity ▪ Circuit reference - I_F_INS - ▪ Prioritized List of Possible Causes ▪ Crash signal circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Restraints control module power or ground circuit open circuit, high resistance ▪ Supplementary restraint system fault	▪ Prioritized Checks to Perform ▪ Refer to the electrical circuit diagrams and check the crash signal circuit for short circuit to ground, short circuit to power, open circuit, high resistance. Repair the wiring harness as necessary ▪ Refer to the electrical circuit diagrams and check the restraints control module power and ground circuits for open circuit, high resistance. Repair the wiring harness as necessary ▪ Using the Jaguar Land Rover approved diagnostic equipment, check the restraints control module for related Diagnostic Trouble Code(s) and refer to the relevant DTC index
B10A2-37	Crash Input - Signal frequency too high	▪ Electrical Cause ▪ Yes ▪ Mechanical Cause ▪ Yes ▪ Control Module Cavity ▪ Circuit reference - I_F_INS - ▪ Prioritized List of Possible Causes ▪ Crash signal circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Restraints control module power or ground circuit open circuit, high resistance ▪ Supplementary restraint system fault	▪ Prioritized Checks to Perform ▪ Refer to the electrical circuit diagrams and check the crash signal circuit for short circuit to ground, short circuit to power, open circuit, high resistance. Repair the wiring harness as necessary ▪ Refer to the electrical circuit diagrams and check the restraints control module power and ground circuits for open circuit, high resistance. Repair the wiring harness as necessary ▪ Using the Jaguar Land Rover approved diagnostic equipment, check the restraints control module for related Diagnostic Trouble Code(s) and refer to the relevant DTC index
B1206-68	Crash Occurred - Event information	▪ Electrical Cause ▪ Yes ▪ Mechanical Cause ▪ Yes ▪ Control Module Cavity ▪ Circuit reference - I_F_INS - ▪ Prioritized List of Possible Causes ▪ Crash signal circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Restraints control module power or ground circuit open circuit, high resistance ▪ Supplementary restraint system fault	▪ Prioritized Checks to Perform ▪ Refer to the electrical circuit diagrams and check the crash signal circuit for short circuit to ground, short circuit to power, open circuit, high resistance. Repair the wiring harness as necessary ▪ Refer to the electrical circuit diagrams and check the restraints control module power and ground circuits for open circuit, high resistance. Repair the wiring harness as necessary ▪ Using the Jaguar Land Rover approved diagnostic equipment, check the restraints control module for related Diagnostic Trouble Code(s) and refer to the relevant DTC index

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION

B15C1-64	Integrated Power Brake Signal Not Plausible - No Loss in Functionality - Signal plausibility failure	▪ Electrical Cause
		  ▪ Yes
		▪ Mechanical Cause
		  ▪ Yes
		▪ Prioritized List Of Possible Causes:
		  ▪ Unexpected signals from the anti-lock brake system control module
		  ▪ Anti-lock brake system failure
			▪ Using the Jaguar Land Rover approved diagnostic equipment, check and update the anti-lock brake system control module with the latest level software. Clear the Diagnostic Trouble Code(s) and retest
			▪ Check anti-lock brake system control module for related Diagnostic Trouble Code(s) and refer to the relevant DTC index

B15C4-68	Integrated Power Brake Signal Not Plausible - Speed Limiter Activated - Event information	▪ Electrical Cause
		  ▪ Yes
		▪ Mechanical Cause
		  ▪ No
		▪ Prioritized List of Possible Causes
		  ▪ Diagnostic Trouble Code(s) have been set in the Anti-Lock Brake System (ABS) Control Module
		  ▪ Diagnostic Trouble Code(s) have been set in the Driver Assistance Domain Controller (DADC)
		  ▪ Anti-Lock Brake System (ABS) Control Module requires software update
			▪ Using the Jaguar Land Rover approved diagnostic equipment, check for Diagnostic Trouble Code(s) in the Anti-Lock Brake System (ABS) Control Module
			  ▪ If a DTC is set in the ABS module, diagnose, and rectify the issue. Refer to the relevant DTC index
			  ▪ When the DTC is cleared in the ABS module, Use the Jaguar Land Rover approved diagnostic equipment to check and clear Diagnostic Trouble Code(s) set in the PCM
			▪ Using the Jaguar Land Rover approved diagnostic equipment, check for Diagnostic Trouble Code(s) in the Driver Assistance Domain Controller (DADC)
			  ▪ If a DTC is set in the DADC module, diagnose, and rectify the issue. Refer to the relevant DTC index
			  ▪ When the DTC is cleared in the DADC module, Use the Jaguar Land Rover approved diagnostic equipment to check and clear Diagnostic Trouble Code(s) set in the PCM

C0031-81	Left Front Wheel Speed Sensor - Invalid serial data received	▪ Electrical Cause
		  ▪ Yes
		▪ Mechanical Cause
		  ▪ Yes
		▪ Prioritized List of Possible Causes
		  ▪ Anti-lock brake system fault
		  ▪ Front left wheel speed sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance
		  ▪ Magnetic reluctor ring damaged
			▪ Vehicle Conditions to Enable DTC Logging Strategy
			  ▪ Engine running
			  ▪ Vehicle moving
			▪ Prioritized Checks to Perform
			  ▪ Using the Jaguar Land Rover approved diagnostic equipment, check the anti-lock brake system control module for related Diagnostic Trouble Code(s) and refer to the relevant DTC index
			  ▪ Refer to the electrical circuit diagrams and check the front left wheel speed sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance. Repair the wiring harness or install a new front left wheel speed sensor as necessary
			  ▪ Check the magnetic reluctor ring for damage. Rectify as necessary

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION
C0034-81	Right Front Wheel Speed Sensor - Invalid serial data received	▪ Electrical Cause ▪ Yes ▪ Mechanical Cause ▪ Yes ▪ Prioritized List of Possible Causes ▪ Anti-lock brake system fault ▪ Front right wheel speed sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Magnetic reluctor ring damaged	▪ Vehicle Conditions to Enable DTC Logging Strategy ▪ Engine running ▪ Vehicle moving ▪ Prioritized Checks to Perform ▪ Using the Jaguar Land Rover approved diagnostic equipment, check the anti-lock brake system control module for related Diagnostic Trouble Code(s) and refer to the relevant DTC index ▪ Refer to the electrical circuit diagrams and check the front right wheel speed sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance. Repair the wiring harness or install a new front right wheel speed sensor as necessary ▪ Check the magnetic reluctor ring for damage. Rectify as necessary
C0037-81	Left Rear Wheel Speed Sensor - Invalid serial data received	▪ Electrical Cause ▪ Yes ▪ Mechanical Cause ▪ Yes ▪ Prioritized List of Possible Causes ▪ Anti-lock brake system fault ▪ Rear left wheel speed sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Magnetic reluctor ring damaged	▪ Vehicle Conditions to Enable DTC Logging Strategy ▪ Engine running ▪ Vehicle moving ▪ Prioritized Checks to Perform ▪ Using the Jaguar Land Rover approved diagnostic equipment, check the anti-lock brake system control module for related Diagnostic Trouble Code(s) and refer to the relevant DTC index ▪ Refer to the electrical circuit diagrams and check the rear left wheel speed sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance. Repair the wiring harness or install a new rear left wheel speed sensor as necessary ▪ Check the magnetic reluctor ring for damage. Rectify as necessary
C003A-81	Right Rear Wheel Speed Sensor - Invalid serial data received	▪ Electrical Cause ▪ Yes ▪ Mechanical Cause ▪ Yes ▪ Prioritized List of Possible Causes ▪ Anti-lock brake system fault ▪ Rear right wheel speed sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Magnetic reluctor ring damaged	▪ Vehicle Conditions to Enable DTC Logging Strategy ▪ Engine running ▪ Vehicle moving ▪ Prioritized Checks to Perform ▪ Using the Jaguar Land Rover approved diagnostic equipment, check the anti-lock brake system control module for related Diagnostic Trouble Code(s) and refer to the relevant DTC index ▪ Refer to the electrical circuit diagrams and check the rear right wheel speed sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance. Repair the wiring harness or install a new rear right wheel speed sensor as necessary ▪ Check the magnetic reluctor ring for damage. Rectify as necessary

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION

C0081-00	ABS Malfunction Indicator - No sub type information	• Electrical Cause
		  ◦ Yes
		• Mechanical Cause
		  ◦ No
		• Prioritized List of Possible Causes
		  ◦ Connector is disconnected, connector pin is backed out, connector pin corrosion
		  ◦ High speed CAN bus failure
		  ◦ Anti-lock brake system control module fuse failure
		  ◦ Anti-lock brake system control module circuit short circuit to ground, short circuit to power, open circuit, high resistance
			• Vehicle Conditions to Enable DTC Logging Strategy
			  ◦ Ignition on / engine running
			• Prioritized Checks to Perform
			  ◦ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion
			  ◦ Refer to the electrical circuit diagrams and check high speed CAN network for short circuit to ground, short circuit to power, open circuit
			  ◦ Check the integrity of the anti-lock brake system control module fuse. Rectify as necessary
			  ◦ Refer to the electrical circuit diagrams and check the anti-lock brake system control module circuit for short circuit to ground, short circuit to power, open circuit, high resistance. Repair the wiring harness as necessary

C1001-87	Vision System Camera - Missing message	• Electrical Cause
		  ◦ Yes
		• Mechanical Cause
		  ◦ Yes
		• Prioritized List of Possible Causes
		  ◦ Image processing module power or ground circuit open circuit, high resistance
		  ◦ High speed CAN bus (chassis) circuit short circuit to ground, short circuit to power, open circuit, high resistance
		  ◦ FlexRay bus circuit short circuit to ground, short circuit to power, open circuit, high resistance
		  ◦ Intelligent emergency braking system fault
			• Prioritized Checks to Perform
			  ◦ Refer to the electrical circuit diagrams and check the image processing module power and ground circuits for open circuit, high resistance. Repair the wiring harness as necessary
			  ◦ Refer to the electrical circuit diagrams and check the high speed CAN bus (chassis) circuit for short circuit to ground, short circuit to power, open circuit, high resistance. Repair the wiring harness as necessary
			  ◦ Refer to the electrical circuit diagrams and check the FlexRay bus circuit for short circuit to ground, short circuit to power, open circuit, high resistance. Replace the wiring harness as necessary
			  ◦ Using the Jaguar Land Rover approved diagnostic equipment, check the image processing module for related Diagnostic Trouble Code(s) and refer to the relevant DTC index

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION

P000A-00	"A" Camshaft Position Slow Response Bank 1 - No sub type information	▪ Electrical Cause
		  ▪ No
		▪ Mechanical Cause
		  ▪ Yes
		▪ Monitor Description
		  ▪ Inlet camshaft unable to achieve desired setpoint angle within a timely manner (slow to respond)
		▪ Prioritized List of Possible Causes
		  ▪ Camshaft position sensor signal faults
		  ▪ Camshaft solenoid wiring incorrect (Exhaust controlling Inlet and Inlet controlling Exhaust)
		  ▪ Lack of oil pressure to VVT actuator
		  ▪ Oil level below minimum fill
		  ▪ Camshaft actuator internal mechanical fault
			▪ Vehicle Conditions to Enable Diagnostic Trouble Code Setting Strategy
			  ▪ Engine oil temperature warmer than -10°C
			  ▪ Make sure no other camshaft or oil temperature related faults are present
			  ▪ Drive vehicle in part load condition such that the Inlet Variable Camshaft Timing would normally be active
			  ▪ Hold various speed and load conditions for 6-10 seconds (this is to make sure Variable Camshaft Timing is moving sufficiently), conduct at least 5 load changes
			▪ Prioritized Checks to Perform
			▪ Using the Jaguar Land Rover approved diagnostic equipment, check for related Diagnostic Trouble Code(s):
			  ▪ If P000A-00 is reported with other camshaft actuator performance DTCs, check the engine oil level and oil pressure is correct as per the JLR specification. Rectify as required
			  ▪ If P000A-00 is reported with camshaft solenoid DTCs, follow the repair procedures for the solenoid DTCs before continuing with the diagnostic steps below
			  ▪ If P000A-00 is reported with camshaft position sensor DTCs, follow the repair procedures for the sensor DTCs before continuing with the diagnostic steps below
			▪ Refer to the electrical circuit diagrams and check the VVT actuator wiring is correct (Inlet and Exhaust wiring/connectors are not switched). Rectify as required
			▪ If P000A-00 is reported as a single DTC, replace the camshaft actuator assembly and oil control valve bolt

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION
P000B-00	"B" Camshaft Position Slow Response Bank 1 - No sub type information	▪ Electrical Cause ▪ No ▪ Mechanical Cause ▪ Yes ▪ Monitor Description ▪ Exhaust camshaft unable to achieve desired setpoint angle within a timely manner (slow to respond) ▪ Prioritized List of Possible Causes ▪ Camshaft position sensor signal faults ▪ Camshaft solenoid wiring incorrect (Exhaust controlling Inlet and Inlet controlling Exhaust) ▪ Lack of oil pressure to VVT actuator ▪ Oil level below minimum fill ▪ Camshaft actuator internal mechanical fault	▪ Vehicle Conditions to Enable Diagnostic Trouble Code Setting Strategy ▪ Engine oil temperature greater than -10°C ▪ No other camshaft related Diagnostic Trouble Code(s) set ▪ No oil temperature related Diagnostic Trouble Code(s) set ▪ Drive vehicle in part load condition such that the inlet Variable Camshaft Timing would normally be active ▪ Hold various speed and load conditions for 6 to 10 seconds (this is to make sure Variable Camshaft Timing is moving sufficiently). Conduct at least 5 load changes ▪ Prioritized Checks to Perform ▪ Using the Jaguar Land Rover approved diagnostic equipment, check for related Diagnostic Trouble Code(s): ▪ If P000B-00 is reported with other camshaft actuator performance DTCs, check the engine oil level and oil pressure is correct as per the JLR specification. Rectify as required ▪ If P000B-00 is reported with camshaft solenoid DTCs, follow the repair procedures for the solenoid DTCs before continuing with the diagnostic steps below ▪ If P000B-00 is reported with camshaft position sensor DTCs, follow the repair procedures for the sensor DTCs before continuing with the diagnostic steps below ▪ Refer to the electrical circuit diagrams and check the VVT actuator wiring is correct (Inlet and Exhaust wiring/connectors are not switched). Rectify as required ▪ If P000B-00 is reported as a single DTC, replace the camshaft actuator assembly and oil control valve bolt
P0010-00	"A" Camshaft Position Actuator Control Circuit/Open Bank 1 - No sub type information	▪ Electrical Cause ▪ Yes ▪ Mechanical Cause ▪ No ▪ Control Module Cavity ▪ Circuit reference - O_T_VVTIN - ▪ Monitor Description ▪ Inlet camshaft solenoid open circuit ▪ Prioritized List of Possible Causes ▪ Inlet camshaft solenoid circuit fault ▪ Inlet camshaft solenoid internal failure	▪ Vehicle Conditions to Enable Diagnostic Trouble Code Setting Strategy ▪ Start and idle engine for at least 2 seconds ▪ Prioritized Checks to Perform ▪ Clear the DTC. Refer to the electrical circuit diagrams and swap the camshaft solenoid and the exhaust solenoid circuit connectors then retest ▪ If the same DTC persists (P0010-00), then check the connector and wiring between the PCM and the inlet camshaft solenoid for short circuit to power. Repair wiring as required ▪ If an exhaust camshaft solenoid DTC is reported (P0013-00), then replace the inlet camshaft solenoid. Reconnect the correct corresponding circuit connectors and retest to confirm the repair

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION

P0011-71	"A" Camshaft Position - Timing Over-Advanced or System Performance Bank 1 - Actuator stuck	▪ Electrical Cause
		  ▪ No
		▪ Mechanical Cause
		  ▪ Yes
		▪ Monitor Description
		  ▪ Inlet camshaft unable to achieve desired setpoint angle (camshaft stuck)
		▪ Prioritized List of Possible Causes
		  ▪ Camshaft position sensor signal faults
		  ▪ Camshaft solenoid wiring incorrect (Exhaust controlling Inlet and Inlet controlling Exhaust)
		  ▪ Lack of oil pressure to VVT actuator
		  ▪ Oil level below minimum fill
		  ▪ Camshaft actuator internal mechanical fault
			▪ Vehicle Conditions to Enable Diagnostic Trouble Code Setting Strategy
			  ▪ Engine oil temperature warmer than -10°C
			  ▪ Make sure no other camshaft or oil temperature related faults are present
			  ▪ Drive vehicle in part load condition such that the Inlet Variable Camshaft Timing would normally be active
			  ▪ Hold various speed and load conditions for 6-10 seconds (this is to make sure Variable Camshaft Timing is moving sufficiently), conduct at least 5 load changes
			▪ Prioritized Checks to Perform
			▪ Using the Jaguar Land Rover approved diagnostic equipment, check for related Diagnostic Trouble Code(s):
			  ▪ If P0011-71 is reported with other camshaft actuator performance DTCs, check the engine oil level and oil pressure is correct as per the JLR specification. Rectify as required
			  ▪ If P0011-71 is reported with camshaft solenoid DTCs, follow the repair procedures for the solenoid DTCs before continuing with the diagnostic steps below
			  ▪ If P0011-71 is reported with camshaft position sensor DTCs, follow the repair procedures for the sensor DTCs before continuing with the diagnostic steps below
			▪ Refer to the electrical circuit diagrams and check the VVT actuator wiring is correct (Inlet and Exhaust wiring/connectors are not switched). Rectify as required
			▪ If P0011-71 is reported as a single DTC, replace the camshaft actuator assembly and oil control valve bolt

P0013-00	"B" Camshaft Position Actuator Control Circuit/Open Bank 1 - No sub type information	▪ Electrical Cause
		  ▪ Yes
		▪ Mechanical Cause
		  ▪ No
		▪ Control Module Cavity
		  ▪ Circuit reference - O_T_VVTEX -
		▪ Monitor Description
		  ▪ Exhaust camshaft solenoid open circuit
		▪ Prioritized List of Possible Causes
		  ▪ Exhaust camshaft solenoid circuit fault
		  ▪ Exhaust camshaft solenoid internal failure
			▪ Vehicle Conditions to Enable Diagnostic Trouble Code Setting Strategy
			  ▪ Start and idle engine for at least 2 seconds
			▪ Prioritized Checks to Perform
			▪ Clear the DTC. Refer to the electrical circuit diagrams and swap the camshaft solenoid and the exhaust solenoid circuit connectors then retest
			  ▪ If the same DTC persists (P0013-00), then check the connector and wiring between the PCM and the exhaust camshaft solenoid for short circuit to power. Repair wiring as required
			  ▪ If an inlet camshaft solenoid DTC is reported (P0010-00), then replace the exhaust camshaft solenoid. Reconnect the correct corresponding circuit connectors and retest to confirm the repair

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION

P0014-71	"B" Camshaft Position - Timing Over-Advanced or System Performance Bank 1 - Actuator stuck	- Electrical Cause
		  - No
		- Mechanical Cause
		  - Yes
		- Monitor Description
		  - Exhaust camshaft unable to achieve desired setpoint angle (Camshaft stuck)
		- Prioritized List of Possible Causes
		  - Camshaft position sensor signal faults
		  - Camshaft solenoid wiring incorrect (Exhaust controlling Inlet and Inlet controlling Exhaust)
		  - Lack of oil pressure to VVT actuator
		  - Oil level below minimum fill
		  - Camshaft actuator internal mechanical fault
			- Vehicle Conditions to Enable Diagnostic Trouble Code Setting Strategy
			  - Engine oil temperature greater than -10°C
			  - No other camshaft related Diagnostic Trouble Code(s) set
			  - No oil temperature related Diagnostic Trouble Code(s) set
			  - Drive vehicle in part load condition such that the Inlet Variable Camshaft Timing would normally be active
			  - Hold various speed and load conditions for 6 to 10 seconds (this is to make sure Variable Camshaft Timing is moving sufficiently). Conduct at least 5 load changes
			- Prioritized Checks to Perform
			- Using the Jaguar Land Rover approved diagnostic equipment, check for related Diagnostic Trouble Code(s):
			  - If P0014-71 is reported with other camshaft actuator performance DTCs, check the engine oil level and oil pressure is correct as per the JLR specification. Rectify as required
			  - If P0014-71 is reported with camshaft solenoid DTCs, follow the repair procedures for the solenoid DTCs before continuing with the diagnostic steps below
			  - If P0014-71 is reported with camshaft position sensor DTCs, follow the repair procedures for the sensor DTCs before continuing with the diagnostic steps below
			- Refer to the electrical circuit diagrams and check the VVT actuator wiring is correct (Inlet and Exhaust wiring/connectors are not switched). Rectify as required
			- If P0014-71 is reported as a single DTC, replace the camshaft actuator assembly and oil control valve bolt

P0016-00	Crankshaft Position - Camshaft Position Correlation Bank 1 Sensor A - No sub type information	- Electrical Cause
		  - Yes
		- Mechanical Cause
		  - Yes
		- Monitor Description
		  - The powertrain control module has detected incorrectly installed components
		- Prioritized List of Possible Causes
		  - Engine timing chain at least one tooth misalignment between camshaft and crankshaft
		  - Camshaft position sensor fault
		  - Crankshaft position sensor fault
		  - Camshaft actuator failure
			- Vehicle Conditions to Enable Diagnostic Trouble Code Setting Strategy
			  - Engine cranking and running
			- Prioritized Checks to Perform
			- Using the Jaguar Land Rover approved diagnostic equipment, check for related Diagnostic Trouble Code(s):
			  - If this DTC is reported with additional camshaft or crankshaft position sensor DTCs, follow the repair procedures for these DTCs before continuing with the diagnostic steps below
			  - If this DTC is reported with camshaft slow response, target error or mechanical linkage DTCs, follow the repair procedures for these DTCs before continuing with the diagnostic steps below
			  - If this DTC is reported with P0017-00 or as a single DTC, check the base engine timing for at least a one tooth out misalignment between the camshaft and crankshaft and rectify as required

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION

P0016-76	Crankshaft Position - Camshaft Position Correlation Bank 1 Sensor A - Wrong mounting position	▪ Electrical Cause
		  ▪ Yes
		▪ Mechanical Cause
		  ▪ Yes
		▪ Monitor Description
		  ▪ The powertrain control module has detected incorrectly installed components
		▪ Prioritized List of Possible Causes
		  ▪ Camshaft position sensor installation or circuit fault
		  ▪ Crankshaft position sensor installation or circuit fault
		  ▪ Engine timing chain at least one tooth misalignment between camshaft and crankshaft
			▪ Vehicle Conditions to Enable Diagnostic Trouble Code Setting Strategy
			  ▪ Engine cranking and running
			▪ Prioritized Checks to Perform
			▪ Using the Jaguar Land Rover approved diagnostic equipment, check for related Diagnostic Trouble Code(s):
			  ▪ If this DTC is reported with additional camshaft or crankshaft position sensor DTCs, follow the repair procedures for these DTCs before continuing with the diagnostic steps below
			  ▪ If this DTC is reported with P0017-76 or as a single DTC, check the base engine timing for at least a one tooth out misalignment between the camshaft and crankshaft and rectify as required

P0016-79	Crankshaft Position - Camshaft Position Correlation Bank 1 Sensor A - Mechanical linkage failure	▪ Electrical Cause
		  ▪ No
		▪ Mechanical Cause
		  ▪ Yes
		▪ Monitor Description
		  ▪ Average inlet camshaft position outside of allowable during engine start
		▪ Prioritized List of Possible Causes
		  ▪ Camshaft locking pin is not engaged during start
		  ▪ Poor or inconsistent signal from camshaft position sensor
		  ▪ Camshaft chain tensioner fault
			▪ Vehicle Conditions to Enable Diagnostic Trouble Code Setting Strategy
			  ▪ Start and idle engine for at least 2 seconds
			▪ Prioritized Checks to Perform
			▪ Using the Jaguar Land Rover approved diagnostic equipment, check for related Diagnostic Trouble Code(s):
			  ▪ If P0016-79 is reported with other camshaft actuator performance DTCs, check the engine oil level and oil pressure is correct as per the JLR specification. Rectify as required
			  ▪ If P0016-79 is reported with camshaft position sensor DTCs, follow the repair procedures for the sensor DTCs before continuing with the diagnostic steps below
			▪ If P0016-79 is reported as a single DTC, replace the camshaft actuator assembly and oil control valve bolt

P0017-00	Crankshaft Position - Camshaft Position Correlation Bank 1 Sensor B - No sub type information	▪ Electrical Cause
		  ▪ Yes
		▪ Mechanical Cause
		  ▪ Yes
		▪ Monitor Description
		  ▪ The powertrain control module has detected incorrectly installed components
		▪ Prioritized List of Possible Causes
		  ▪ Engine timing chain at least one tooth misalignment between camshaft and crankshaft
		  ▪ Camshaft position sensor fault
		  ▪ Crankshaft position sensor fault
		  ▪ Camshaft actuator failure
			▪ Vehicle Conditions to Enable Diagnostic Trouble Code Setting Strategy
			  ▪ Engine cranking and running
			▪ Prioritized Checks to Perform
			▪ Using the Jaguar Land Rover approved diagnostic equipment, check for related Diagnostic Trouble Code(s):
			  ▪ If this DTC is reported with additional camshaft or crankshaft position sensor DTCs, follow the repair procedures for these DTCs before continuing with the diagnostic steps below
			  ▪ If this DTC is reported with camshaft slow response, target error or mechanical linkage DTCs, follow the repair procedures for these DTCs before continuing with the diagnostic steps below
			  ▪ If this DTC is reported with P0016-00 or as a single DTC, check the base engine timing for at least a one tooth out misalignment between the camshaft and crankshaft and rectify as required

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION

P0017-76	Crankshaft Position - Camshaft Position Correlation Bank 1 Sensor B - Wrong mounting position	▪ Electrical Cause
		  ▪ Yes
		▪ Mechanical Cause
		  ▪ Yes
		▪ Monitor Description
		  ▪ The powertrain control module has detected incorrectly installed components
		▪ Prioritized List of Possible Causes
		  ▪ Camshaft position sensor installation or circuit fault
		  ▪ Crankshaft position sensor installation or circuit fault
		  ▪ Engine timing chain at least one tooth misalignment between camshaft and crankshaft
			▪ Vehicle Conditions to Enable Diagnostic Trouble Code Setting Strategy
			  ▪ Engine cranking and running
			▪ Prioritized Checks to Perform
			▪ Using the Jaguar Land Rover approved diagnostic equipment, check for related Diagnostic Trouble Code(s):
			  ▪ If this DTC is reported with additional camshaft or crankshaft position sensor DTCs, follow the repair procedures for these DTCs before continuing with the diagnostic steps below
			  ▪ If this DTC is reported with P0016-76 or as a single DTC, check the base engine timing for at least a one tooth out misalignment between the camshaft and crankshaft and rectify as required

P0017-79	Crankshaft Position - Camshaft Position Correlation Bank 1 Sensor B - Mechanical linkage failure	▪ Electrical Cause
		  ▪ No
		▪ Mechanical Cause
		  ▪ Yes
		▪ Monitor Description
		  ▪ Average exhaust camshaft position outside of allowable during engine start
		▪ Prioritized List of Possible Causes
		  ▪ Camshaft locking pin is not engaged during start
		  ▪ Poor or inconsistent signal from camshaft position sensor
		  ▪ Camshaft chain tensioner fault
			▪ Vehicle Conditions to Enable Diagnostic Trouble Code Setting Strategy
			  ▪ Start and idle engine for at least 2 seconds
			▪ Prioritized Checks to Perform
			▪ Using the Jaguar Land Rover approved diagnostic equipment, check for related Diagnostic Trouble Code(s):
			  ▪ If P0017-79 is reported with other camshaft actuator performance DTCs, check the engine oil level and oil pressure is correct as per the JLR specification. Rectify as required
			  ▪ If P0017-79 is reported with camshaft position sensor DTCs, follow the repair procedures for the sensor DTCs before continuing with the diagnostic steps below
			▪ If P0017-79 is reported as a single DTC, replace the camshaft actuator assembly and oil control valve bolt

P0030-00	HO2S Heater Control Circuit Bank 1 Sensor 1 - No sub type information	▪ Electrical Cause
		  ▪ Yes
		▪ Mechanical Cause
		  ▪ Yes
		▪ Control Module Cavity
		  ▪ Circuit reference - O_T_LSH Htr -
		▪ Monitor Description
		  ▪ Sensor heater duty cycle holds a constant value
		▪ Prioritized List of Possible Causes
		  ▪ Heated oxygen sensor heater control circuit short circuit to ground, short circuit to power, open circuit, high resistance
		  ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion
		  ▪ Heated oxygen sensor failure
			▪ Vehicle Conditions to Enable DTC Logging Strategy
			  ▪ Start engine and allow to idle for 5 minutes
			▪ Prioritized Checks to Perform
			  ▪ Refer to the electrical circuit diagrams and check heated oxygen sensor heater control circuit for short circuit to ground, short circuit to power, open circuit, high resistance
			  ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion
			  ▪ Using the Jaguar Land Rover approved diagnostic equipment, run application - ECU Diagnostics / Powertrain Control Module (PCM) / Clear Diagnostic Trouble Code(s). Retest
			  ▪ Install a new heated oxygen sensor as required

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION

P0031-00	HO2S Heater Control Circuit Low Bank 1 Sensor 1 - No sub type information	- Electrical Cause
		  - Yes
		- Mechanical Cause
		  - Yes
		- Control Module Cavity
		  - Circuit reference - O_T_LSH Htr -
		- Monitor Description
		  - Sensor heater duty cycle below a threshold
		- Prioritized List of Possible Causes
		  - Heated oxygen sensor heater control circuit short circuit to ground, short circuit to power, open circuit, high resistance
		  - Connector is disconnected, connector pin is backed out, connector pin corrosion
		  - Heated oxygen sensor failure
			- Vehicle Conditions to Enable DTC Logging Strategy
			  - Start engine and allow to idle for 5 minutes
			- Prioritized Checks to Perform
			  - Refer to the electrical circuit diagrams and check heated oxygen sensor heater control circuit for short circuit to ground, short circuit to power, open circuit, high resistance
			  - Inspect connectors for signs of water ingress, and pins for damage and/or corrosion
			  - Using the Jaguar Land Rover approved diagnostic equipment, run application - ECU Diagnostics / Powertrain Control Module (PCM) / Clear Diagnostic Trouble Code(s). Retest
			  - Install a new heated oxygen sensor as required

P0032-00	HO2S Heater Control Circuit High Bank 1 Sensor 1 - No sub type information	- Electrical Cause
		  - Yes
		- Mechanical Cause
		  - Yes
		- Control Module Cavity
		  - Circuit reference - O_T_LSH Htr -
		- Monitor Description
		  - Sensor heater duty cycle above a threshold
		- Prioritized List of Possible Causes
		  - Heated oxygen sensor heater control circuit short circuit to ground, short circuit to power, open circuit, high resistance
		  - Connector is disconnected, connector pin is backed out, connector pin corrosion
		  - Heated oxygen sensor failure
			- Vehicle Conditions to Enable DTC Logging Strategy
			  - Start engine and allow to idle for 5 minutes
			- Prioritized Checks to Perform
			  - Refer to the electrical circuit diagrams and check the compressor recirculation valve circuit for short circuit to ground, short circuit to power, open circuit, high resistance
			  - Inspect connectors for signs of water ingress, and pins for damage and/or corrosion
			  - Using the Jaguar Land Rover approved diagnostic equipment, run application - ECU Diagnostics / Powertrain Control Module (PCM) / Clear Diagnostic Trouble Code(s). Retest
			  - Install a new compressor recirculation valve

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION

P0034-11	Turbocharger/Supercharger Bypass Valve "A" Control Circuit Low - Circuit short to ground	▪ Electrical Cause
		  ▪ Yes
		▪ Mechanical Cause
		  ▪ Yes
		▪ Monitor Description
		  ▪ Short circuit to ground of control circuit
		▪ Prioritized List of Possible Causes
		  ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion
		  ▪ Compressor bypass control actuator fuse failure
		  ▪ Compressor bypass control actuator circuit short circuit to ground, short circuit to power, open circuit, high resistance
		  ▪ Compressor bypass control actuator failure
			▪ Vehicle Conditions to Enable DTC Logging Strategy
			  ▪ Drive vehicle in boosted regions and complete numerous negative transients (Acceleration followed by abrupt back outs (throttle closure))
			▪ Prioritized Checks to Perform
			  ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion
			  ▪ Check the integrity of the compressor bypass control actuator fuse. Rectify as necessary
			  ▪ Refer to the electrical circuit diagrams and check the compressor bypass control actuator circuit for short circuit to ground, short circuit to power, open circuit, high resistance. Repair the wiring harness as necessary
			  ▪ Install a new compressor bypass control actuator only when diagnosed as failed
			  ▪ Using the Jaguar Land Rover approved diagnostic equipment, run application - ECU Diagnostics / Powertrain Control Module (PCM) / Clear Diagnostic Trouble Code(s). Retest

P0035-12	Turbocharger/Supercharger Bypass Valve "A" Control Circuit High - Circuit short to battery	▪ Electrical Cause
		  ▪ Yes
		▪ Mechanical Cause
		  ▪ Yes
		▪ Monitor Description
		  ▪ Short circuit to battery of control circuit
		▪ Prioritized List of Possible Causes
		  ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion
		  ▪ Compressor bypass control actuator fuse failure
		  ▪ Compressor bypass control actuator circuit short circuit to ground, short circuit to power, open circuit, high resistance
		  ▪ Compressor bypass control actuator failure
			▪ Vehicle Conditions to Enable DTC Logging Strategy
			  ▪ Drive vehicle in boosted regions and complete numerous negative transients (Acceleration followed by abrupt back outs (throttle closure))
			▪ Prioritized Checks to Perform
			  ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion
			  ▪ Check the integrity of the compressor bypass control actuator fuse. Rectify as necessary
			  ▪ Refer to the electrical circuit diagrams and check the compressor bypass control actuator circuit for short circuit to ground, short circuit to power, open circuit, high resistance. Repair the wiring harness as necessary
			  ▪ Install a new compressor bypass control actuator only when diagnosed as failed
			  ▪ Using the Jaguar Land Rover approved diagnostic equipment, run application - ECU Diagnostics / Powertrain Control Module (PCM) / Clear Diagnostic Trouble Code(s). Retest

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION
P0036-00	HO2S Heater Control Circuit Bank 1 Sensor 2 - No sub type information	▪ Electrical Cause ▪ Yes ▪ Mechanical Cause ▪ Yes ▪ Control Module Cavity ▪ Circuit reference - O_T_LSFH1 - ▪ Monitor Description ▪ Heater driver behavior not consistent with a correctly connected heater ▪ Prioritized List of Possible Causes ▪ Heated oxygen sensor heater circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Heated oxygen sensor failure	▪ Vehicle Conditions to Enable DTC Logging Strategy ▪ Fully warmed-up engine, drive off-idle for a minimum of 5 minutes ▪ Prioritized Checks to Perform ▪ Refer to the electrical circuit diagrams and check heated oxygen sensor heater circuit for short circuit to ground, short circuit to power, open circuit, high resistance ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion ▪ Using the Jaguar Land Rover approved diagnostic equipment, run application - ECU Diagnostics / Powertrain Control Module (PCM) / Clear Diagnostic Trouble Code(s). Retest ▪ Install a new heated oxygen sensor as required
P0037-00	HO2S Heater Control Circuit Low Bank 1 Sensor 2 - No sub type information	▪ Electrical Cause ▪ Yes ▪ Mechanical Cause ▪ Yes ▪ Control Module Cavity ▪ Circuit reference - O_T_LSFH1 - ▪ Monitor Description ▪ Heater driver behavior not consistent with a correctly connected heater ▪ Prioritized List of Possible Causes ▪ Heated oxygen sensor heater circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Heated oxygen sensor failure	▪ Vehicle Conditions to Enable DTC Logging Strategy ▪ Fully warmed-up engine, drive off-idle for a minimum of 5 minutes ▪ Prioritized Checks to Perform ▪ Refer to the electrical circuit diagrams and check heated oxygen sensor heater circuit for short circuit to ground, short circuit to power, open circuit, high resistance ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion ▪ Using the Jaguar Land Rover approved diagnostic equipment, run application - ECU Diagnostics / Powertrain Control Module (PCM) / Clear Diagnostic Trouble Code(s). Retest ▪ Install a new heated oxygen sensor as required

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION
P0038-00	HO2S Heater Control Circuit High Bank 1 Sensor 2 - No sub type information	▪ Electrical Cause ▪ Yes ▪ Mechanical Cause ▪ Yes ▪ Control Module Cavity ▪ Circuit reference - O_T_LSFH1 - ▪ Monitor Description ▪ Heater driver behavior not consistent with a correctly connected heater ▪ Prioritized List of Possible Causes ▪ Heated oxygen sensor heater circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Heated oxygen sensor failure	▪ Vehicle Conditions to Enable DTC Logging Strategy ▪ Fully warmed-up engine, drive off-idle for a minimum of 5 minutes ▪ Prioritized Checks to Perform ▪ Refer to the electrical circuit diagrams and check heated oxygen sensor heater circuit for short circuit to ground, short circuit to power, open circuit, high resistance ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion ▪ Using the Jaguar Land Rover approved diagnostic equipment, run application - ECU Diagnostics / Powertrain Control Module (PCM) / Clear Diagnostic Trouble Code(s). Retest ▪ Install a new heated oxygen sensor as required
P0039-13	Turbocharger/Supercharger Bypass Valve "A" Control Circuit Range/Performance - Circuit open	▪ Electrical Cause ▪ Yes ▪ Mechanical Cause ▪ Yes ▪ Monitor Description ▪ Open circuit of the dump valve detected ▪ Prioritized List of Possible Causes ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Compressor bypass control actuator fuse failure ▪ Compressor bypass control actuator circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Compressor bypass control actuator failure	▪ Vehicle Conditions to Enable DTC Logging Strategy ▪ Drive vehicle in boosted regions and complete numerous negative transients (Acceleration followed by abrupt back outs (throttle closure)) ▪ Prioritized Checks to Perform ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion ▪ Check the integrity of the compressor bypass control actuator fuse. Rectify as necessary ▪ Refer to the electrical circuit diagrams and check the compressor bypass control actuator circuit for short circuit to ground, short circuit to power, open circuit, high resistance. Repair the wiring harness as necessary ▪ Install a new compressor bypass control actuator only when diagnosed as failed ▪ Using the Jaguar Land Rover approved diagnostic equipment, run application - ECU Diagnostics / Powertrain Control Module (PCM) / Clear Diagnostic Trouble Code(s). Retest

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION

P0042-00	HO2S Heater Control Circuit Bank 1 Sensor 3 - No sub type information	▪ Electrical Cause
		  ▪ Yes
		▪ Mechanical Cause
		  ▪ Yes
		▪ Control Module Cavity
		  ▪ Circuit reference - O_T_LSFH2 -
		▪ Monitor Description
		  ▪ Heater driver behavior not consistent with a correctly connected heater
		▪ Prioritized List of Possible Causes
		  ▪ Heated oxygen sensor heater circuit short circuit to ground, short circuit to power, open circuit, high resistance
		  ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion
		  ▪ Heated oxygen sensor failure
			▪ Vehicle Conditions to Enable DTC Logging Strategy
			  ▪ Fully warmed-up engine, drive off-idle for a minimum of 5 minutes
			▪ Prioritized Checks to Perform
			  ▪ Refer to the electrical circuit diagrams and check heated oxygen sensor heater circuit for short circuit to ground, short circuit to power, open circuit, high resistance
			  ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion
			  ▪ Using the Jaguar Land Rover approved diagnostic equipment, run application - ECU Diagnostics / Powertrain Control Module (PCM) / Clear Diagnostic Trouble Code(s). Retest
			  ▪ Install a new heated oxygen sensor as required

P0043-00	HO2S Heater Control Circuit Low Bank 1 Sensor 3 - No sub type information	▪ Electrical Cause
		  ▪ Yes
		▪ Mechanical Cause
		  ▪ Yes
		▪ Control Module Cavity
		  ▪ Circuit reference - O_T_LSFH2 -
		▪ Monitor Description
		  ▪ Heater driver behavior not consistent with a correctly connected heater
		▪ Prioritized List of Possible Causes
		  ▪ Heated oxygen sensor heater circuit short circuit to ground, short circuit to power, open circuit, high resistance
		  ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion
		  ▪ Heated oxygen sensor failure
			▪ Vehicle Conditions to Enable DTC Logging Strategy
			  ▪ Fully warmed-up engine, drive off-idle for a minimum of 5 minutes
			▪ Prioritized Checks to Perform
			  ▪ Refer to the electrical circuit diagrams and check heated oxygen sensor heater circuit for short circuit to ground, short circuit to power, open circuit, high resistance
			  ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion
			  ▪ Using the Jaguar Land Rover approved diagnostic equipment, run application - ECU Diagnostics / Powertrain Control Module (PCM) / Clear Diagnostic Trouble Code(s). Retest
			  ▪ Install a new heated oxygen sensor as required

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION

P0044-00	HO2S Heater Control Circuit High Bank 1 Sensor 3 - No sub type information	▪ Electrical Cause
		  ▪ Yes
		▪ Mechanical Cause
		  ▪ Yes
		▪ Control Module Cavity
		  ▪ Circuit reference - O_T_LSFH2 -
		▪ Monitor Description
		  ▪ Heater driver behavior not consistent with a correctly connected heater
		▪ Prioritized List of Possible Causes
		  ▪ Heated oxygen sensor heater circuit short circuit to ground, short circuit to power, open circuit, high resistance
		  ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion
		  ▪ Heated oxygen sensor failure
			▪ Vehicle Conditions to Enable DTC Logging Strategy
			  ▪ Fully warmed-up engine, drive off-idle for a minimum of 5 minutes
			▪ Prioritized Checks to Perform
			  ▪ Refer to the electrical circuit diagrams and check heated oxygen sensor heater circuit for short circuit to ground, short circuit to power, open circuit, high resistance
			  ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion
			  ▪ Using the Jaguar Land Rover approved diagnostic equipment, run application - ECU Diagnostics / Powertrain Control Module (PCM) / Clear Diagnostic Trouble Code(s). Retest
			  ▪ Install a new heated oxygen sensor as required

P0054-00	HO2S Heater Resistance Bank 1 Sensor 2 - No sub type information	▪ Electrical Cause
		  ▪ Yes
		▪ Mechanical Cause
		  ▪ Yes
		▪ Control Module Cavity
		  ▪ Circuit reference - O_T_LSFH1 -
		▪ Monitor Description
		  ▪ Sensor heater not heating the sensor element enough
		▪ Prioritized List of Possible Causes
		  ▪ Heated oxygen sensor heater circuit short circuit to ground, short circuit to power, open circuit, high resistance
		  ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion
		  ▪ Heated oxygen sensor failure
			▪ Vehicle Conditions to Enable DTC Logging Strategy
			  ▪ Fully warmed-up engine, drive off-idle for a minimum of 5 minutes
			▪ Prioritized Checks to Perform
			  ▪ Refer to the electrical circuit diagrams and check heated oxygen sensor heater circuit for short circuit to ground, short circuit to power, open circuit, high resistance
			  ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion
			  ▪ Using the Jaguar Land Rover approved diagnostic equipment, run application - ECU Diagnostics / Powertrain Control Module (PCM) / Clear Diagnostic Trouble Code(s). Retest
			  ▪ Install a new heated oxygen sensor as required

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION

P0055-00	HO2S Heater Resistance Bank 1 Sensor 3 - No sub type information	▪ Electrical Cause
		  ▪ Yes
		▪ Mechanical Cause
		  ▪ Yes
		▪ Control Module Cavity
		  ▪ Circuit reference - O_T_LSFH2 -
		▪ Monitor Description
		  ▪ Sensor heater not heating the sensor element enough
		▪ Prioritized List of Possible Causes
		  ▪ Heated oxygen sensor heater circuit short circuit to ground, short circuit to power, open circuit, high resistance
		  ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion
		  ▪ Heated oxygen sensor failure
			▪ Vehicle Conditions to Enable DTC Logging Strategy
			  ▪ Fully warmed-up engine, drive off-idle for a minimum of 5 minutes
			▪ Prioritized Checks to Perform
			  ▪ Refer to the electrical circuit diagrams and check heated oxygen sensor heater circuit for short circuit to ground, short circuit to power, open circuit, high resistance
			  ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion
			  ▪ Using the Jaguar Land Rover approved diagnostic equipment, run application - ECU Diagnostics / Powertrain Control Module (PCM) / Clear Diagnostic Trouble Code(s). Retest
			  ▪ Install a new heated oxygen sensor as required

P0070-84	Ambient Air Temperature Sensor Circuit "A" - Signal below allowable range	▪ Electrical Cause
		  ▪ Yes
		▪ Mechanical Cause
		  ▪ Yes
		▪ Control Module Cavity
		  ▪ Circuit reference - I_A_ATS -
		▪ Monitor Description
		  ▪ Check of ambient temperature plausibility
		▪ Prioritized List of Possible Causes
		  ▪ Ambient air temperature sensor offset low
		  ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion
		  ▪ Ambient air temperature sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance
		  ▪ Ambient air temperature sensor failure
			▪ Vehicle Conditions to Enable DTC Logging Strategy
			  ▪ Start engine and drive vehicle at steady state condition for 20 minutes
			▪ Prioritized Checks to Perform
			  ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion
			  ▪ Refer to the electrical circuit diagrams and check the ambient air temperature sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance (greater than 20 ohms). Repair the wiring harness as necessary
			  ▪ Install a new ambient air temperature sensor only when diagnosed as failed
			  ▪ Using the Jaguar Land Rover approved diagnostic equipment, run application - ECU Diagnostics / Powertrain Control Module (PCM) / Clear Diagnostic Trouble Code(s). Retest

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION

P0070-85	Ambient Air Temperature Sensor Circuit "A" - Signal above allowable range	▪ Electrical Cause
		  ▪ Yes
		▪ Mechanical Cause
		  ▪ Yes
		▪ Control Module Cavity
		  ▪ Circuit reference - I_A_ATS -
		▪ Monitor Description
		  ▪ Check of ambient temperature plausibility
		▪ Prioritized List of Possible Causes
		  ▪ Ambient air temperature sensor offset high
		  ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion
		  ▪ Ambient air temperature sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance
		  ▪ Ambient air temperature sensor failure
			▪ Vehicle Conditions to Enable DTC Logging Strategy
			  ▪ Start engine and drive vehicle at steady state condition for 20 minutes
			▪ Prioritized Checks to Perform
			  ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion
			  ▪ Refer to the electrical circuit diagrams and check the ambient air temperature sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance (greater than 20 ohms). Repair the wiring harness as necessary
			  ▪ Install a new ambient air temperature sensor only when diagnosed as failed
			  ▪ Using the Jaguar Land Rover approved diagnostic equipment, run application - ECU Diagnostics / Powertrain Control Module (PCM) / Clear Diagnostic Trouble Code(s). Retest

P0071-23	Ambient Air Temperature Sensor Circuit "A" Range/Performance - Signal stuck low	▪ Electrical Cause
		  ▪ Yes
		▪ Mechanical Cause
		  ▪ Yes
		▪ Control Module Cavity
		  ▪ Circuit reference - I_A_ATS -
		▪ Monitor Description
		  ▪ Check of ambient air temperature at cold start
		▪ Prioritized List of Possible Causes
		  ▪ Ambient air temperature sensor offset low
		  ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion
		  ▪ Ambient air temperature sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance
		  ▪ Ambient air temperature sensor failure
			▪ Vehicle Conditions to Enable DTC Logging Strategy
			  ▪ Start engine and allow to idle for 10 seconds following an engine off period greater than 8 hours
			▪ Prioritized Checks to Perform
			  ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion
			  ▪ Refer to the electrical circuit diagrams and check the ambient air temperature sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance (greater than 20 ohms). Repair the wiring harness as necessary
			  ▪ Install a new ambient air temperature sensor only when diagnosed as failed
			  ▪ Using the Jaguar Land Rover approved diagnostic equipment, run application - ECU Diagnostics / Powertrain Control Module (PCM) / Clear Diagnostic Trouble Code(s). Retest

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION
P0071-24	Ambient Air Temperature Sensor Circuit "A" Range/Performance - Signal stuck high	▪ Electrical Cause ▪ Yes ▪ Mechanical Cause ▪ Yes ▪ Control Module Cavity ▪ Circuit reference - I_A_ATS - ▪ Monitor Description ▪ Check of ambient air temperature at cold start ▪ Prioritized List of Possible Causes ▪ Ambient air temperature sensor offset high ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Ambient air temperature sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Ambient air temperature sensor failure	▪ Vehicle Conditions to Enable DTC Logging Strategy ▪ Start engine and allow to idle for 10 seconds following an engine off period greater than 8 hours ▪ Prioritized Checks to Perform ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion ▪ Refer to the electrical circuit diagrams and check the ambient air temperature sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance (greater than 20 ohms). Repair the wiring harness as necessary ▪ Install a new ambient air temperature sensor only when diagnosed as failed ▪ Using the Jaguar Land Rover approved diagnostic equipment, run application - ECU Diagnostics / Powertrain Control Module (PCM) / Clear Diagnostic Trouble Code(s). Retest
P0071-85	Ambient Air Temperature Sensor Circuit "A" Range/Performance - Signal above allowable range	▪ Electrical Cause ▪ Yes ▪ Mechanical Cause ▪ Yes ▪ Control Module Cavity ▪ Circuit reference - I_A_ATS - ▪ Monitor Description ▪ Check of ambient air temperature upper range ▪ Prioritized List of Possible Causes ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Ambient air temperature sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Ambient air temperature sensor failure	▪ Vehicle Conditions to Enable DTC Logging Strategy ▪ Ignition on ▪ Prioritized Checks to Perform ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion ▪ Refer to the electrical circuit diagrams and check the ambient air temperature sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance (greater than 20 ohms). Repair the wiring harness as necessary ▪ Install a new ambient air temperature sensor only when diagnosed as failed ▪ Using the Jaguar Land Rover approved diagnostic equipment, run application - ECU Diagnostics / Powertrain Control Module (PCM) / Clear Diagnostic Trouble Code(s). Retest

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION

P0072-00	Ambient Air Temperature Sensor Circuit "A" Low - No sub type information	▪ Electrical Cause
		  ▪ Yes
		▪ Mechanical Cause
		  ▪ Yes
		▪ Control Module Cavity
		  ▪ Circuit reference - I_A_ATS -
		  ▪ Circuit reference - G_R_ATS / G_R_WGPT -
		▪ Monitor Description
		  ▪ Ambient air temperature signal, reading high, short circuit above threshold
		▪ Prioritized List of Possible Causes
		  ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion
		  ▪ Ambient air temperature sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance
		  ▪ Ambient air temperature sensor failure
			▪ Vehicle Conditions to Enable DTC Logging Strategy
			  ▪ Ignition on, engine running
			▪ Prioritized Checks to Perform
			  ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion
			  ▪ Refer to the electrical circuit diagrams and check the ambient air temperature sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance (greater than 20 ohms). Repair the wiring harness as necessary
			  ▪ Install a new ambient air temperature sensor only when diagnosed as failed
			  ▪ Using the Jaguar Land Rover approved diagnostic equipment, run application - ECU Diagnostics / Powertrain Control Module (PCM) / Clear Diagnostic Trouble Code(s). Retest

P0073-00	Ambient Air Temperature Sensor Circuit "A" High - No sub type information	▪ Electrical Cause
		  ▪ Yes
		▪ Mechanical Cause
		  ▪ Yes
		▪ Control Module Cavity
		  ▪ Circuit reference - I_A_ATS -
		  ▪ Circuit reference - G_R_ATS / G_R_WGPT -
		▪ Monitor Description
		  ▪ Ambient air temperature signal reading low, short circuit below threshold
		▪ Prioritized List of Possible Causes
		  ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion
		  ▪ Ambient air temperature sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance
		  ▪ Ambient air temperature sensor failure
			▪ Vehicle Conditions to Enable DTC Logging Strategy
			  ▪ Ignition on, engine running
			▪ Prioritized Checks to Perform
			  ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion
			  ▪ Refer to the electrical circuit diagrams and check the ambient air temperature sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance (greater than 20 ohms). Repair the wiring harness as necessary
			  ▪ Install a new ambient air temperature sensor only when diagnosed as failed
			  ▪ Using the Jaguar Land Rover approved diagnostic equipment, run application - ECU Diagnostics / Powertrain Control Module (PCM) / Clear Diagnostic Trouble Code(s). Retest

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION

P0087-77	Fuel Rail/System Pressure - Too Low Bank 1 - Commanded position not reachable	▪ Electrical Cause
		  ▪ No
		▪ Mechanical Cause
		  ▪ Yes
		▪ Prioritized List of Possible Causes
		  ▪ Fuel leaking outside of the fuel system
		  ▪ Fuel leaking into the low pressure system
			▪ Vehicle Conditions to Enable DTC Logging Strategy
			  ▪ Engine running greater than 1 minute
			▪ Prioritized Checks to Perform
			  ▪ Check for fuel leaks external of the fuel rail
			  ▪ Check high pressure fuel pumps are not leaking from the high pressure system into the low pressure system
			  ▪ Using the Jaguar Land Rover approved diagnostic equipment, run application - ECU Diagnostics / Powertrain Control Module (PCM) / Clear Diagnostic Trouble Code(s). Retest

P0087-84	Fuel Rail/System Pressure - Too Low Bank 1 - Signal below allowable range	▪ Electrical Cause
		  ▪ Yes
		▪ Mechanical Cause
		  ▪ Yes
		▪ Control Module Cavity
		  ▪ Circuit reference - I_A_RAILPS -
		▪ Monitor Description
		  ▪ Fuel rail pressure below threshold
		  ▪ To detect if the fuel rail pressure is below the expected minimum
		▪ Prioritized List of Possible Causes
		  ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion
		  ▪ Fuel rail pressure sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance
		  ▪ Fuel rail pressure sensor failure
			▪ Vehicle Conditions to Enable DTC Logging Strategy
			  ▪ Engine running greater than 1 minute
			▪ Prioritized Checks to Perform
			  ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion
			  ▪ Refer to the electrical circuit diagrams and check the fuel rail pressure sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance. Repair the wiring harness as necessary
			  ▪ Install a new fuel rail pressure sensor only when diagnosed as failed
			  ▪ Using the Jaguar Land Rover approved diagnostic equipment, run application - ECU Diagnostics / Powertrain Control Module (PCM) / Clear Diagnostic Trouble Code(s). Retest

P0088-77	Fuel Rail/System Pressure - Too High Bank 1 - Commanded position not reachable	▪ Electrical Cause
		  ▪ No
		▪ Mechanical Cause
		  ▪ Yes
		▪ Prioritized List of Possible Causes
		  ▪ Fuel leaking outside of the fuel system
		  ▪ Fuel leaking into the low pressure system
			▪ Vehicle Conditions to Enable DTC Logging Strategy
			  ▪ Engine running greater than 1 minute
			▪ Prioritized Checks to Perform
			  ▪ Check for fuel leaks external of the fuel rail
			  ▪ Check high pressure fuel pumps are not leaking from the high pressure system into the low pressure system
			  ▪ Using the Jaguar Land Rover approved diagnostic equipment, run application - ECU Diagnostics / Powertrain Control Module (PCM) / Clear Diagnostic Trouble Code(s). Retest

<table>
<thead><tr><th>DTC</th><th>DESCRIPTION</th><th>POSSIBLE CAUSES</th><th>ACTION</th></tr></thead>
<tbody>
<tr><td>P0088-85</td><td>Fuel Rail/System Pressure - Too High Bank 1 - Signal above allowable range</td><td>Electrical Cause
Yes
Mechanical Cause
Yes
Control Module Cavity
Circuit reference - I_A_RAILPS -
Monitor Description
Fuel rail pressure above threshold
To detect if the fuel rail pressure is over the expected maximum
Prioritized List of Possible Causes
Connector is disconnected, connector pin is backed out, connector pin corrosion
Fuel rail pressure sensor circuit short circuit to power
Stuck pressure relief valve
Fuel rail pressure sensor failure</td><td>Vehicle Conditions to Enable DTC Logging Strategy
Engine running greater than 1 minute
Prioritized Checks to Perform
Inspect connectors for signs of water ingress, and pins for damage and/or corrosion
Refer to the electrical circuit diagrams and check the fuel rail pressure sensor circuit for short circuit to power. Repair the wiring harness as necessary
Check pressure relief valve for mechanical integrity
Install a new fuel rail pressure sensor only when diagnosed as failed
Using the Jaguar Land Rover approved diagnostic equipment, run application - ECU Diagnostics / Powertrain Control Module (PCM) / Clear Diagnostic Trouble Code(s). Retest</td></tr>
<tr><td>P0089-22</td><td>Fuel Pressure Regulator 1 Performance - Signal amplitude > maximum</td><td>Electrical Cause
Yes
Mechanical Cause
Yes
Control Module Cavity
Not applicable
Monitor Description
The command signal to the fuel pump driver module is below a threshold
Prioritized List of Possible Causes
Fuel system leakage from the fuel tank and lines
Low pressure fuel pump failure</td><td>Vehicle Conditions to Enable DTC Logging Strategy
Engine running greater than 2 minutes
Prioritized Checks to Perform
Check for fuel system leakage from the fuel tank and lines both internal and external of the fuel tank
Install a new low pressure fuel pump only when diagnosed as failed
Using the Jaguar Land Rover approved diagnostic equipment, run application - ECU Diagnostics / Powertrain Control Module (PCM) / Clear Diagnostic Trouble Code(s). Retest</td></tr>
<tr><td>P0089-64</td><td>Fuel Pressure Regulator 1 Performance - Signal plausibility failure</td><td>Electrical Cause
Yes
Mechanical Cause
Yes
Control Module Cavity
Not applicable
Monitor Description
The command signal to the fuel pump driver module is above a threshold
Prioritized List of Possible Causes
Fuel system leakage from the fuel tank and lines
Low pressure fuel pump failure</td><td>Vehicle Conditions to Enable DTC Logging Strategy
Engine running greater than 2 minutes
Prioritized Checks to Perform
Check for fuel system leakage from the fuel tank and lines both internal and external of the fuel tank
Install a new low pressure fuel pump only when diagnosed as failed
Using the Jaguar Land Rover approved diagnostic equipment, run application - ECU Diagnostics / Powertrain Control Module (PCM) / Clear Diagnostic Trouble Code(s). Retest</td></tr>
</tbody>
</table>

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION

P0089-84	Fuel Pressure Regulator 1 Performance - Signal below allowable range	▪ Electrical Cause
		  ▪ Yes
		▪ Mechanical Cause
		  ▪ Yes
		▪ Control Module Cavity
		  ▪ Not applicable
		▪ Monitor Description
		  ▪ The command signal to the fuel pump driver module is below a threshold
		▪ Prioritized List of Possible Causes
		  ▪ Fuel leak from the low pressure fuel system
		  ▪ Low pressure fuel pump failure
			▪ Vehicle Conditions to Enable DTC Logging Strategy
			  ▪ Engine running greater than 2 minutes
			▪ Prioritized Checks to Perform
			  ▪ Check for fuel leaks from the low pressure fuel system, including inside the fuel tank. Rectify as necessary
			  ▪ Install a new low pressure fuel pump
			  ▪ Using the Jaguar Land Rover approved diagnostic equipment, run application - ECU Diagnostics / Powertrain Control Module (PCM) / Clear Diagnostic Trouble Code(s). Retest

P0089-85	Fuel Pressure Regulator 1 Performance - Signal above allowable range	▪ Electrical Cause
		  ▪ Yes
		▪ Mechanical Cause
		  ▪ Yes
		▪ Control Module Cavity
		  ▪ Not applicable
		▪ Monitor Description
		  ▪ The command signal to the fuel pump driver module is below a threshold
		▪ Prioritized List of Possible Causes
		  ▪ Fuel leak from the low pressure fuel system
		  ▪ Low pressure fuel pump failure
			▪ Vehicle Conditions to Enable DTC Logging Strategy
			  ▪ Engine running greater than 2 minutes
			▪ Prioritized Checks to Perform
			  ▪ Check for fuel leaks from the low pressure fuel system, including inside the fuel tank. Rectify as necessary
			  ▪ Install a new low pressure fuel pump
			  ▪ Using the Jaguar Land Rover approved diagnostic equipment, run application - ECU Diagnostics / Powertrain Control Module (PCM) / Clear Diagnostic Trouble Code(s). Retest

P008A-07	Low Pressure Fuel System Pressure - Too Low - Mechanical failures	▪ Electrical Cause
		  ▪ Yes
		▪ Mechanical Cause
		  ▪ Yes
		▪ Monitor Description
		  ▪ Monitoring of the low fuel pressure control deviation
		▪ Prioritized List of Possible Causes
		  ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion
		  ▪ Low pressure fuel system fuel pressure sensor circuit short circuit to ground, open circuit, high resistance
		  ▪ Low pressure fuel system fuel pressure sensor failure
			▪ Vehicle Conditions to Enable DTC Logging Strategy
			  ▪ Engine running
			▪ Prioritized Checks to Perform
			  ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion
			  ▪ Refer to the electrical circuit diagrams and check the low pressure fuel system fuel pressure sensor circuit for short circuit to ground, open circuit, high resistance. Repair the wiring harness as necessary
			  ▪ Using the Jaguar Land Rover approved diagnostic equipment, check the powertrain control module for Diagnostic Trouble Code(s) P2451 and/or P2452. if the current DTC is set in conjunction with P2451 or P2452, then install a new fuel low pressure sensor as required
			  ▪ Using the Jaguar Land Rover approved diagnostic equipment, run application - ECU Diagnostics / Powertrain Control Module (PCM) / Clear Diagnostic Trouble Code(s). Retest

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION

P008A-77	Low Pressure Fuel System Pressure - Too Low - Commanded position not reachable	▪ Electrical Cause
		  ▪ Yes
		▪ Mechanical Cause
		  ▪ Yes
		▪ Control Module Cavity
		  ▪ Circuit reference - O_V_5VRAILPS -
		  ▪ Circuit reference - I_A_FLPS -
		  ▪ Circuit reference - G_R_RAILPS -
		▪ Monitor Description
		  ▪ Monitoring of the low fuel pressure control deviation
		▪ Prioritized List of Possible Causes
		  ▪ Low pressure fuel system fuel pressure sensor circuit short circuit to ground, open circuit, high resistance
		  ▪ Low pressure fuel system fuel pressure sensor failure
			▪ Vehicle Conditions to Enable DTC Logging Strategy
			  ▪ Engine running
			▪ Prioritized Checks to Perform
			  ▪ Refer to the electrical circuit diagrams and check the low pressure fuel system fuel pressure sensor circuit for short circuit to ground, open circuit, high resistance. Repair the wiring harness or install a new low pressure fuel system fuel pressure sensor as necessary
			  ▪ Install a new low pressure fuel system fuel pressure sensor
			  ▪ Using the Jaguar Land Rover approved diagnostic equipment, run application - ECU Diagnostics / Powertrain Control Module (PCM) / Clear Diagnostic Trouble Code(s). Retest

P008A-84	Low Pressure Fuel System Pressure - Too Low - Signal below allowable range	▪ Monitor Description
		  ▪ The low pressure fuel signal is below a threshold
		▪ Prioritized List of Possible Causes
		  ▪ Fuel low pressure sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance
		  ▪ Fuel system leakage from the fuel tank and lines
		  ▪ Low pressure fuel pump failure
			▪ Prioritized Checks to Perform
			  ▪ Refer to the electrical circuit diagrams and check fuel low pressure sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance
			  ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion
			  ▪ Check for fuel system leakage from the fuel tank and lines both internal and external of the fuel tank
			  ▪ Using the Jaguar Land Rover approved diagnostic equipment, run application - ECU Diagnostics / Powertrain Control Module (PCM) / Clear Diagnostic Trouble Code(s). Retest
			  ▪ Install a new low pressure fuel pump as required

P008B-07	Low Pressure Fuel System Pressure - Too High - Mechanical failures	▪ Electrical Cause
		  ▪ Yes
		▪ Mechanical Cause
		  ▪ Yes
		▪ Monitor Description
		  ▪ Monitoring of the low fuel pressure control deviation
		▪ Prioritized List of Possible Causes
		  ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion
		  ▪ Low pressure fuel system fuel pressure sensor circuit short circuit to ground, open circuit, high resistance
		  ▪ Low pressure fuel system fuel pressure sensor failure
			▪ Vehicle Conditions to Enable DTC Logging Strategy
			  ▪ Engine running
			▪ Prioritized Checks to Perform
			  ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion
			  ▪ Refer to the electrical circuit diagrams and check the low pressure fuel system fuel pressure sensor circuit for short circuit to ground, open circuit, high resistance. Repair the wiring harness as necessary
			  ▪ Using the Jaguar Land Rover approved diagnostic equipment, check the powertrain control module for Diagnostic Trouble Code(s) P2451 and/or P2452. if the current DTC is set in conjunction with P2451 or P2452, then install a new fuel low pressure sensor as required
			  ▪ Using the Jaguar Land Rover approved diagnostic equipment, run application - ECU Diagnostics / Powertrain Control Module (PCM) / Clear Diagnostic Trouble Code(s). Retest

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION

P008B-77	Low Pressure Fuel System Pressure - Too High - Commanded position not reachable	▪ Electrical Cause
		  ▪ Yes
		▪ Mechanical Cause
		  ▪ Yes
		▪ Control Module Cavity
		  ▪ Circuit reference - O_V_5VRAILPS -
		  ▪ Circuit reference - I_A_FLPS -
		  ▪ Circuit reference - G_R_RAILPS -
		▪ Monitor Description
		  ▪ Monitoring of the low fuel pressure control deviation
		▪ Prioritized List of Possible Causes
		  ▪ Low pressure fuel system fuel pressure sensor circuit short circuit to ground, open circuit, high resistance
		  ▪ Low pressure fuel system fuel pressure sensor failure
			▪ Vehicle Conditions to Enable DTC Logging Strategy
			  ▪ Engine running
			▪ Prioritized Checks to Perform
			  ▪ Refer to the electrical circuit diagrams and check the low pressure fuel system fuel pressure sensor circuit for short circuit to ground, open circuit, high resistance. Repair the wiring harness or install a new low pressure fuel system fuel pressure sensor as necessary
			  ▪ Install a new low pressure fuel system fuel pressure sensor
			  ▪ Using the Jaguar Land Rover approved diagnostic equipment, run application - ECU Diagnostics / Powertrain Control Module (PCM) / Clear Diagnostic Trouble Code(s). Retest

P008B-85	Low Pressure Fuel System Pressure - Too High - Signal above allowable range	▪ Monitor Description
		  ▪ The Low pressure fuel signal is above a threshold
		▪ Prioritized List of Possible Causes
		  ▪ Low pressure fuel system fuel pressure sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance
		  ▪ Fuel system leakage from the fuel tank and lines
		  ▪ Low pressure fuel pump failure
			▪ Prioritized Checks to Perform
			  ▪ Refer to the electrical circuit diagrams and check the low pressure fuel system fuel pressure sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance
			  ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion
			  ▪ Check for fuel system leakage from the fuel tank and lines both internal and external of the fuel tank
			  ▪ Using the Jaguar Land Rover approved diagnostic equipment, run application - ECU Diagnostics / Powertrain Control Module (PCM) / Clear Diagnostic Trouble Code(s). Retest
			  ▪ Install a new low pressure fuel pump as required

P0096-23	Intake Air Temperature Sensor 2 Circuit Range/Performance Bank 1 - Signal stuck low	▪ Electrical Cause
		  ▪ Yes
		▪ Mechanical Cause
		  ▪ No
		▪ Control Module Cavity
		  ▪ Circuit reference - I_A_BTS -
		▪ Monitor Description
		  ▪ Long term average stuck, no movement in sensor value during drive cycle
		▪ Prioritized List of Possible Causes
		  ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion
		  ▪ Charge air pressure and temperature sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance
			▪ Vehicle Conditions to Enable DTC Logging Strategy
			  ▪ Following soak period greater than 8 hours, drive steady state conditions above 72 km/h for 30 seconds (x3) and steady state below 5 km/h (x3) for 30 seconds
			▪ Prioritized Checks to Perform
			  ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion
			  ▪ Refer to the electrical circuit diagrams and check the charge air pressure and temperature sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance. Repair the wiring harness as necessary
			  ▪ Using the Jaguar Land Rover approved diagnostic equipment, run application - ECU Diagnostics / Powertrain Control Module (PCM) / Clear Diagnostic Trouble Code(s). Retest

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION

P0096-84	Intake Air Temperature Sensor 2 Circuit Range/Performance Bank 1 - Signal below allowable range	▪ Electrical Cause
		  ▪ Yes
		▪ Mechanical Cause
		  ▪ No
		▪ Control Module Cavity
		  ▪ Circuit reference - I_A_BTS -
		▪ Monitor Description
		  ▪ Sensor offset low
		▪ Prioritized List of Possible Causes
		  ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion
		  ▪ Charge air pressure and temperature sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance
			▪ Vehicle Conditions to Enable DTC Logging Strategy
			  ▪ Start engine and allow to idle for 10 seconds following an engine off period greater than 8 hours
			▪ Prioritized Checks to Perform
			  ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion
			  ▪ Refer to the electrical circuit diagrams and check the charge air pressure and temperature sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance. Repair the wiring harness as necessary
			  ▪ Using the Jaguar Land Rover approved diagnostic equipment, run application - ECU Diagnostics / Powertrain Control Module (PCM) / Clear Diagnostic Trouble Code(s). Retest

P0096-85	Intake Air Temperature Sensor 2 Circuit Range/Performance Bank 1 - Signal above allowable range	▪ Electrical Cause
		  ▪ Yes
		▪ Mechanical Cause
		  ▪ No
		▪ Control Module Cavity
		  ▪ Circuit reference - I_A_BTS -
		▪ Monitor Description
		  ▪ Sensor offset high
		▪ Prioritized List of Possible Causes
		  ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion
		  ▪ Charge air pressure and temperature sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance
			▪ Vehicle Conditions to Enable DTC Logging Strategy
			  ▪ Start engine and allow to idle for 10 seconds following an engine off period greater than 8 hours
			▪ Prioritized Checks to Perform
			  ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion
			  ▪ Refer to the electrical circuit diagrams and check the charge air pressure and temperature sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance. Repair the wiring harness as necessary
			  ▪ Using the Jaguar Land Rover approved diagnostic equipment, run application - ECU Diagnostics / Powertrain Control Module (PCM) / Clear Diagnostic Trouble Code(s). Retest

P0097-00	Intake Air Temperature Sensor 2 Circuit Low Bank 1 - No sub type information	▪ Electrical Cause
		  ▪ Yes
		▪ Mechanical Cause
		  ▪ No
		▪ Control Module Cavity
		  ▪ Circuit reference - I_A_BTS -
		  ▪ Circuit reference - G_R_BPTS -
		▪ Prioritized List of Possible Causes
		  ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion
		  ▪ Charge air pressure and temperature sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance
			▪ Vehicle Conditions to Enable DTC Logging Strategy
			  ▪ Ignition on
			▪ Prioritized Checks to Perform
			  ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion
			  ▪ Refer to the electrical circuit diagrams and check the charge air pressure and temperature sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance. Repair the wiring harness as necessary
			  ▪ Using the Jaguar Land Rover approved diagnostic equipment, run application - ECU Diagnostics / Powertrain Control Module (PCM) / Clear Diagnostic Trouble Code(s). Retest

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION

P0098-00	Intake Air Temperature Sensor 2 Circuit High Bank 1 - No sub type information	▪ Electrical Cause
		  ▪ Yes
		▪ Mechanical Cause
		  ▪ No
		▪ Control Module Cavity
		  ▪ Circuit reference - I_A_BTS -
		  ▪ Circuit reference - G_R_BPTS -
		▪ Prioritized List of Possible Causes
		  ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion
		  ▪ Charge air pressure and temperature sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance
			▪ Vehicle Conditions to Enable DTC Logging Strategy
			  ▪ Ignition on
			▪ Prioritized Checks to Perform
			  ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion
			  ▪ Refer to the electrical circuit diagrams and check the charge air pressure and temperature sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance. Repair the wiring harness as necessary
			  ▪ Using the Jaguar Land Rover approved diagnostic equipment, run application - ECU Diagnostics / Powertrain Control Module (PCM) / Clear Diagnostic Trouble Code(s). Retest

P0099-00	Intake Air Temperature Sensor 2 Circuit Intermittent/Erratic Bank 1 - No sub type information	▪ Electrical Cause
		  ▪ Yes
		▪ Mechanical Cause
		  ▪ No
		▪ Control Module Cavity
		  ▪ Circuit reference - I_A_BTS -
		  ▪ Circuit reference - G_R_BPTS -
		▪ Prioritized List of Possible Causes
		  ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion
		  ▪ Charge air pressure and temperature sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance
			▪ Vehicle Conditions to Enable DTC Logging Strategy
			  ▪ Ignition on
			▪ Prioritized Checks to Perform
			  ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion
			  ▪ Refer to the electrical circuit diagrams and check the charge air pressure and temperature sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance. Repair the wiring harness as necessary
			  ▪ Using the Jaguar Land Rover approved diagnostic equipment, run application - ECU Diagnostics / Powertrain Control Module (PCM) / Clear Diagnostic Trouble Code(s). Retest

P009A-84	Intake Air Temperature/Ambient Air Temperature Correlation - Signal below allowable range	▪ Electrical Cause
		  ▪ Yes
		▪ Mechanical Cause
		  ▪ No
		▪ Control Module Cavity
		  ▪ Circuit reference - I_A_IATS -
		  ▪ Circuit reference - G_R_AMS -
		▪ Prioritized List of Possible Causes
		  ▪ Intake air temperature sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance
			▪ Vehicle Conditions to Enable DTC Logging Strategy
			  ▪ Drive the vehicle steadily for a minimum of 20 minutes
			▪ Prioritized Checks to Perform
			  ▪ Refer to the electrical circuit diagrams and check the intake air temperature sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance. Repair the wiring harness or install a new mass air flow and temperature sensor as necessary

<table>
<tr><th>DTC</th><th>DESCRIPTION</th><th>POSSIBLE CAUSES</th><th>ACTION</th></tr>
<tr><td>P009A-85</td><td>Intake Air Temperature/Ambient Air Temperature Correlation - Signal above allowable range</td><td>- Electrical Cause
  - Yes
- Mechanical Cause
  - No
- Control Module Cavity
  - Circuit reference - I_A_IATS -
  - Circuit reference - G_R_AMS -
- Prioritized List of Possible Causes
  - Intake air temperature sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance</td><td>- Vehicle Conditions to Enable DTC Logging Strategy
  - Drive the vehicle steadily for a minimum of 20 minutes
- Prioritized Checks to Perform
  - Refer to the electrical circuit diagrams and check the intake air temperature sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance. Repair the wiring harness or install a new mass air flow and temperature sensor as necessary</td></tr>
<tr><td>P00BC-00</td><td>Mass or Volume Air Flow "A" Circuit Range/Performance - Air Flow Too Low - No sub type information</td><td>- Electrical Cause
  - Yes
- Mechanical Cause
  - Yes
- Control Module Cavity
  - Circuit reference - I_F_AMS -
- Monitor Description
  - Air flow signal reading lower than the model based threshold
- Prioritized List of Possible Causes
  - Other related Diagnostic Trouble Code(s)
  - Connector is disconnected, connector pin is backed out, connector pin corrosion
  - Mass air flow sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance
  - Air induction system components installed incorrectly
  - Air induction system hose clips loose
  - Air induction system air leakage
  - Blocked air filter
  - Mass air flow sensor contamination by oil or debris
  - Electric throttle blade contamination by oil or debris
  - Throttle adaption failure</td><td>- Vehicle Conditions to Enable DTC Logging Strategy
  - Continuously monitored during engine running
- Prioritized Checks to Perform
  - Using the Jaguar Land Rover approved diagnostic equipment check the powertrain control module for related Diagnostic Trouble Code(s) and refer to the relevant DTC index
  - Inspect connectors for signs of water ingress, and pins for damage and/or corrosion
  - Refer to the electrical circuit diagrams and check the mass air flow sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance. Repair the wiring harness as necessary
  - Check the air induction system for correct installation. Rectify as necessary
  - Check the integrity of the air induction system hose clips. Rectify as necessary
  - Check the air induction system for air leaks. Rectify as necessary
  - Check the air filter. Rectify as necessary
  - Check mass air flow sensor for contamination by oil or debris clean as required
  - Check electric throttle blade for contamination by oil or debris clean as required
  - Install a new mass air flow sensor only when diagnosed as failed
  - Using the Jaguar Land Rover approved diagnostic equipment complete throttle adaption routine
  - Using the Jaguar Land Rover approved diagnostic equipment, run application - ECU Diagnostics / Powertrain Control Module (PCM) / Clear Diagnostic Trouble Code(s). Retest</td></tr>
</table>

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION

P00BD-00	Mass or Volume Air Flow "A" Circuit Range/Performance - Air Flow Too High - No sub type information	▪ Electrical Cause
		  ▪ Yes
		▪ Mechanical Cause
		  ▪ Yes
		▪ Control Module Cavity
		  ▪ Circuit reference - I_F_AMS -
		▪ Monitor Description
		  ▪ Air flow signal reading lower than the model based threshold
		▪ Prioritized List of Possible Causes
		  ▪ Other related Diagnostic Trouble Code(s)
		  ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion
		  ▪ Mass air flow sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance
		  ▪ Air induction system components installed incorrectly
		  ▪ Air induction system hose clips loose
		  ▪ Air induction system air leakage
		  ▪ Blocked air filter
		  ▪ Mass air flow sensor contamination by oil or debris
		  ▪ Electric throttle blade contamination by oil or debris
		  ▪ Throttle adaption failure
			▪ Vehicle Conditions to Enable DTC Logging Strategy
			  ▪ Continuously monitored during engine running
			▪ Prioritized Checks to Perform
			  ▪ Using the Jaguar Land Rover approved diagnostic equipment check the powertrain control module for related Diagnostic Trouble Code(s) and refer to the relevant DTC index
			  ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion
			  ▪ Refer to the electrical circuit diagrams and check the mass air flow sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance. Repair the wiring harness as necessary
			  ▪ Check the air induction system for correct installation. Rectify as necessary
			  ▪ Check the integrity of the air induction system hose clips. Rectify as necessary
			  ▪ Check the air induction system for air leaks. Rectify as necessary
			  ▪ Check the air filter. Rectify as necessary
			  ▪ Check mass air flow sensor for contamination by oil or debris clean as required
			  ▪ Check electric throttle blade for contamination by oil or debris clean as required
			  ▪ Install a new mass air flow sensor only when diagnosed as failed
			  ▪ Using the Jaguar Land Rover approved diagnostic equipment complete throttle adaption routine
			  ▪ Using the Jaguar Land Rover approved diagnostic equipment, run application - ECU Diagnostics / Powertrain Control Module (PCM) / Clear Diagnostic Trouble Code(s). Retest

P00C6-00	Fuel Rail Pressure Too Low - Engine Cranking Bank 1 - No sub type information	▪ Electrical Cause
		  ▪ No
		▪ Mechanical Cause
		  ▪ No
		▪ Control Module Cavity
		  ▪ Not applicable
		▪ Monitor Description
		  ▪ The fuel rail pressure is checked against a threshold after a number of engine rotations from the start of cranking. A DTC is set if the fuel rail pressure is below the threshold
		▪ Prioritized List of Possible Causes
		  ▪ Fuel rail has been removed
		  ▪ High pressure fuel pump mechanical failure
			▪ Vehicle Conditions to Enable DTC Logging Strategy
			  ▪ Monitor fuel rail pressure during cranking
			▪ Prioritized Checks to Perform
			  ▪ Using the Jaguar Land Rover approved diagnostic equipment check the fuel rail pressure during cranking. Clear the DTC and attempt another start
			  ▪ Install a new high pressure fuel pump only when diagnosed as failed
			  ▪ Using the Jaguar Land Rover approved diagnostic equipment, run application - ECU Diagnostics / Powertrain Control Module (PCM) / Clear Diagnostic Trouble Code(s). Retest

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION

P00CF-23	Barometric Pressure - Turbocharger/Supercharger Boost Sensor "A" Correlation - Signal stuck low	▪ Electrical Cause
		  ▪ Yes
		▪ Mechanical Cause
		  ▪ Yes
		▪ Control Module Cavity
		  ▪ Circuit reference - I_A_BPS -
		▪ Monitor Description
		  ▪ Compare boost sensor value plus its sensor tolerance to ambient pressure minus a failure threshold
		▪ Prioritized List of Possible Causes
		  ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion
		  ▪ Charge air pressure sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance
		  ▪ Charge air pressure and temperature sensor failure
			▪ Vehicle Conditions to Enable DTC Logging Strategy
			  ▪ Continuously operating
			▪ Prioritized Checks to Perform
			  ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion
			  ▪ Refer to the electrical circuit diagrams and check the charge air pressure sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance. Repair the wiring harness as necessary
			  ▪ Install a new charge air pressure and temperature sensor
			  ▪ Using the Jaguar Land Rover approved diagnostic equipment, run application - ECU Diagnostics / Powertrain Control Module (PCM) / Clear Diagnostic Trouble Code(s). Retest

P00CF-24	Barometric Pressure - Turbocharger/Supercharger Boost Sensor "A" Correlation - Signal stuck high	▪ Electrical Cause
		  ▪ Yes
		▪ Mechanical Cause
		  ▪ Yes
		▪ Control Module Cavity
		  ▪ Circuit reference - I_A_BPS -
		▪ Monitor Description
		  ▪ Compare boost sensor value minus its sensor tolerance to ambient pressure plus a failure threshold
		▪ Prioritized List of Possible Causes
		  ▪ Charge air pressure sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance
		  ▪ Charge air pressure and temperature sensor failure
			▪ Vehicle Conditions to Enable DTC Logging Strategy
			  ▪ Continuously operating
			▪ Prioritized Checks to Perform
			  ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion
			  ▪ Refer to the electrical circuit diagrams and check the charge air pressure sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance. Repair the wiring harness as necessary
			  ▪ Install a new charge air pressure and temperature sensor
			  ▪ Using the Jaguar Land Rover approved diagnostic equipment, run application - ECU Diagnostics / Powertrain Control Module (PCM) / Clear Diagnostic Trouble Code(s). Retest

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION

P00E9-23	Intake Air Temperature Sensor 3 Circuit Range/Performance Bank 1 - Signal stuck low	▪ Electrical Cause
		  ▪ Yes
		▪ Mechanical Cause
		  ▪ Yes
		▪ Control Module Cavity
		  ▪ Circuit reference - I_A_IMTS -
		▪ Monitor Description
		  ▪ Sensor long term average stuck, no movement in sensor value during drive cycle
		▪ Prioritized List of Possible Causes
		  ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion
		  ▪ Manifold absolute pressure and temperature sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance
		  ▪ Manifold absolute pressure and temperature sensor failure
			▪ Vehicle Conditions to Enable DTC Logging Strategy
			  ▪ Following soak period greater than 8 hours, drive steady state cruise conditions above 72 km/h for 30 seconds (x3) and steady state cruise below 5 km/h (x3) for 30 seconds
			▪ Prioritized Checks to Perform
			  ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion
			  ▪ Refer to the electrical circuit diagrams and check the manifold absolute pressure and temperature sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance. Repair the wiring harness as necessary
			  ▪ Install a new manifold absolute pressure and temperature sensor only when diagnosed as failed
			  ▪ Using the Jaguar Land Rover approved diagnostic equipment, run application - ECU Diagnostics / Powertrain Control Module (PCM) / Clear Diagnostic Trouble Code(s). Retest

P00E9-84	Intake Air Temperature Sensor 3 Circuit Range/Performance Bank 1 - Signal below allowable range	▪ Electrical Cause
		  ▪ Yes
		▪ Mechanical Cause
		  ▪ Yes
		▪ Control Module Cavity
		  ▪ Circuit reference - I_A_IMTS -
		▪ Monitor Description
		  ▪ After greater than 8hr engine off time, all temperature sensor reading are averaged at start except the outlier, each temperature sensor is compared to the average, and if there is a large difference a DTC is set after the engine has started
		▪ Prioritized List of Possible Causes
		  ▪ Battery disconnection resulting in errors in engine off time (short soaks may look like long soaks)
		  ▪ Electric block heater applied and not detected
		  ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion
		  ▪ Manifold absolute pressure and temperature sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance
		  ▪ Manifold absolute pressure and temperature sensor failure
			▪ Vehicle Conditions to Enable DTC Logging Strategy
			  ▪ Start engine and allow to idle for 10 seconds following an engine off period greater than 8 hours
			▪ Prioritized Checks to Perform
			  ▪ Check for signs of battery disconnection
			  ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion
			  ▪ Refer to the electrical circuit diagrams and check the manifold absolute pressure and temperature sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance. Repair the wiring harness as necessary
			  ▪ Install a new manifold absolute pressure and temperature sensor only when diagnosed as failed
			  ▪ Using the Jaguar Land Rover approved diagnostic equipment, run application - ECU Diagnostics / Powertrain Control Module (PCM) / Clear Diagnostic Trouble Code(s). Retest

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION

P00E9-85	Intake Air Temperature Sensor 3 Circuit Range/Performance Bank 1 - Signal above allowable range	▪ Electrical Cause
		  ▪ Yes
		▪ Mechanical Cause
		  ▪ Yes
		▪ Control Module Cavity
		  ▪ Circuit reference - I_A_IMTS -
		▪ Monitor Description
		  ▪ After greater than 8hr engine off time, all temperature sensor reading are averaged at start except the outlier, each temperature sensor is compared to the average, and if there is a large difference a DTC is set after the engine has started
		▪ Prioritized List of Possible Causes
		  ▪ Battery disconnection resulting in errors in engine off time (short soaks may look like long soaks)
		  ▪ Electric block heater applied and not detected
		  ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion
		  ▪ Manifold absolute pressure and temperature sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance
		  ▪ Manifold absolute pressure and temperature sensor failure
			▪ Vehicle Conditions to Enable DTC Logging Strategy
			  ▪ Start engine and allow to idle for 10 seconds following an engine off period greater than 8 hours
			▪ Prioritized Checks to Perform
			  ▪ Check for signs of battery disconnection
			  ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion
			  ▪ Refer to the electrical circuit diagrams and check the manifold absolute pressure and temperature sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance. Repair the wiring harness as necessary
			  ▪ Install a new manifold absolute pressure and temperature sensor only when diagnosed as failed
			  ▪ Using the Jaguar Land Rover approved diagnostic equipment, run application - ECU Diagnostics / Powertrain Control Module (PCM) / Clear Diagnostic Trouble Code(s). Retest

P00EA-00	Intake Air Temperature Sensor 3 Circuit Low Bank 1 - No sub type information	▪ Electrical Cause
		  ▪ Yes
		▪ Mechanical Cause
		  ▪ Yes
		▪ Control Module Cavity
		  ▪ Circuit reference - I_A_IMTS -
		▪ Monitor Description
		  ▪ Charged air cooler signal, supply reading high, open circuit above threshold
		▪ Prioritized List of Possible Causes
		  ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion
		  ▪ Manifold absolute pressure and temperature sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance
		  ▪ Manifold absolute pressure and temperature sensor failure
			▪ Vehicle Conditions to Enable DTC Logging Strategy
			  ▪ Ignition on, engine running
			▪ Prioritized Checks to Perform
			  ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion
			  ▪ Refer to the electrical circuit diagrams and check the manifold absolute pressure and temperature sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance. Repair the wiring harness as necessary
			  ▪ Install a new manifold absolute pressure and temperature sensor only when diagnosed as failed
			  ▪ Using the Jaguar Land Rover approved diagnostic equipment, run application - ECU Diagnostics / Powertrain Control Module (PCM) / Clear Diagnostic Trouble Code(s). Retest

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION

P00EB-00	Intake Air Temperature Sensor 3 Circuit High Bank 1 - No sub type information	▪ Electrical Cause
		  ▪ Yes
		▪ Mechanical Cause
		  ▪ Yes
		▪ Control Module Cavity
		  ▪ Circuit reference - I_A_IMTS -
		▪ Monitor Description
		  ▪ Charged air cooler signal , supply reading low, short circuit below threshold
		▪ Prioritized List of Possible Causes
		  ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion
		  ▪ Manifold absolute pressure and temperature sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance
		  ▪ Manifold absolute pressure and temperature sensor failure
			▪ Vehicle Conditions to Enable DTC Logging Strategy
			  ▪ Ignition on, engine running
			▪ Prioritized Checks to Perform
			  ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion
			  ▪ Refer to the electrical circuit diagrams and check the manifold absolute pressure and temperature sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance. Repair the wiring harness as necessary
			  ▪ Install a new manifold absolute pressure and temperature sensor only when diagnosed as failed
			  ▪ Using the Jaguar Land Rover approved diagnostic equipment, run application - ECU Diagnostics / Powertrain Control Module (PCM) / Clear Diagnostic Trouble Code(s). Retest

P00EC-00	Intake Air Temperature Sensor 3 Circuit Intermittent/Erratic Bank 1 - No sub type information	▪ Electrical Cause
		  ▪ Yes
		▪ Mechanical Cause
		  ▪ Yes
		▪ Control Module Cavity
		  ▪ Circuit reference - I_A_IMTS -
		  ▪ Circuit reference - G_R_MAP -
		▪ Monitor Description
		  ▪ Charged air cooler signal intermittent erratic sensor reading, circuit making and breaking, short term average voltage greater than threshold
		▪ Prioritized List of Possible Causes
		  ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion
		  ▪ Manifold absolute pressure and temperature sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance
		  ▪ Manifold absolute pressure and temperature sensor failure
			▪ Vehicle Conditions to Enable DTC Logging Strategy
			  ▪ Ignition on, engine running
			▪ Prioritized Checks to Perform
			  ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion
			  ▪ Refer to the electrical circuit diagrams and check the manifold absolute pressure and temperature sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance. Repair the wiring harness as necessary
			  ▪ Install a new manifold absolute pressure and temperature sensor only when diagnosed as failed
			  ▪ Using the Jaguar Land Rover approved diagnostic equipment, run application - ECU Diagnostics / Powertrain Control Module (PCM) / Clear Diagnostic Trouble Code(s). Retest

P00FF-00	Body Control Module Requested MIL Illumination - No sub type information	▪ The gateway module has detected a system failure and transmitted a MIL illumination request to the powertrain control module	▪ Using the Jaguar Land Rover approved diagnostic equipment, check the gateway module for related Diagnostic Trouble Code(s) and refer to relevant DTC index ▪ Once all gateway module Diagnostic Trouble Code(s) are resolved, clear all stored Diagnostic Trouble Code(s) in the powertrain control module and retest

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION
P0100-38	Mass or Volume Air Flow Sensor "A" Circuit - Signal frequency incorrect	- Electrical Cause - Yes - Mechanical Cause - Yes - Control Module Cavity - Circuit reference I_F_AMS - Circuit reference O_V_5VWGPT O_V_5VAMS - Monitor Description - P0100-38 is set if the feedback voltage from the Air Flow Sensor reads 0 Volts - Open circuit on the 5V reference line (Pin 1) - Open circuit on the airflow sensor signal line (Pin 4) - Mass Air Flow Sensor internal failure	- Check status of DTC's and verify they are pending or confirmed and not historical. If DTCs are historical, take no further action - If P0100-38 is set together with P0111-84 and P0113-00, check and confirm the airflow sensor is electrically connected - If only P0100-38 is stored: - Check the air flow sensor is electrically connected - Using the JLR service tool, read the airflow signal with the engine idling in Park or Neutral - If the airflow value reads zero, manipulate the electrical connector at the airflow sensor when checking the airflow reading. If a non-zero airflow value is then observed investigate further the root cause - Continue manipulation of the harness, working towards the ECU. If at any point a non-zero airflow value is observed, investigate further the root cause and rectify as required - If after manipulation the airflow sensor continues to read zero, check and confirm 5 Volts is present at the airflow sensor and check the integrity of the signal line back to the ECU. Verify sufficient pin tension is present at both the airflow connector and the ECU. Rectify any concerns as required - If the airflow sensor continues to read zero, before replacing the sensor, check for the presence of any DTCs related to airflow too lean/rich or fuelling adaptations. If stored, use the Jaguar Land Rover approved high pressure diagnostic leak detector and check for air intake system leaks. Rectify as necessary. If no leaks are identified, replace the airflow sensor - Upon rectification of the fault, using the Jaguar Land Rover approved diagnostic equipment, clear all stored Diagnostic Trouble Code(s) and retest - Test the vehicle over 3 drive cycles (engine on/off) across varied driving conditions to confirm the repair

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION
P0102-16	Mass or Volume Air Flow Sensor "A" Circuit Low - Circuit voltage below threshold	▪ Electrical Cause ▪ Yes ▪ Mechanical Cause ▪ Yes ▪ Control Module Cavity ▪ Circuit reference - I_F_AMS - ▪ Circuit reference - O_V_5VWGPT O_V_5VAMS - ▪ Monitor Description ▪ Electrical fault of A bank mass air flow sensor ▪ Prioritized List of Possible Causes ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Mass air flow sensor fuse failure ▪ Mass air flow sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Mass air flow sensor failure	**NOTE:** The powertrain control module is unlikely to be the root cause for this DTC and must not be replaced without a Technical Assist Request (TA) being raised for further guidance ▪ Vehicle Conditions to Enable DTC Logging Strategy ▪ Ignition on, start engine and idle for 30 seconds ▪ Prioritized Checks to Perform ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion ▪ Check the integrity of the mass air flow sensor fuse. Rectify as necessary ▪ Refer to the electrical circuit diagrams and check the mass air flow sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance. Repair the wiring harness as necessary ▪ Install a new mass air flow sensor only when diagnosed as failed ▪ Using the Jaguar Land Rover approved diagnostic equipment, run application - ECU Diagnostics / Powertrain Control Module (PCM) / Clear Diagnostic Trouble Code(s). Retest
P0103-17	Mass or Volume Air Flow Sensor "A" Circuit High - Circuit voltage above threshold	▪ Electrical Cause ▪ Yes ▪ Mechanical Cause ▪ Yes ▪ Control Module Cavity ▪ Circuit reference - I_F_AMS - ▪ Circuit reference - O_V_5VWGPT / O_V_5VAMS - ▪ Monitor Description ▪ Electrical fault of A bank mass air flow sensor ▪ Prioritized List of Possible Causes ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Mass air flow sensor fuse failure ▪ Mass air flow sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Mass air flow sensor failure	▪ Vehicle Conditions to Enable DTC Logging Strategy ▪ Ignition on, start engine and idle for 30 seconds ▪ Prioritized Checks to Perform ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion ▪ Check the integrity of the mass air flow sensor fuse. Rectify as necessary ▪ Refer to the electrical circuit diagrams and check the mass air flow sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance. Repair the wiring harness as necessary ▪ Install a new mass air flow sensor only when diagnosed as failed ▪ Using the Jaguar Land Rover approved diagnostic equipment, run application - ECU Diagnostics / Powertrain Control Module (PCM) / Clear Diagnostic Trouble Code(s). Retest

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION
P0106-23	Manifold Absolute Pressure / Barometric Pressure Sensor Circuit Range/Performance - Signal stuck low	▪ Electrical Cause ▪ Yes ▪ Mechanical Cause ▪ Yes ▪ Control Module Signal Name ▪ Circuit reference - O_V_5VMAP - ▪ Circuit reference - I_A_MAP - ▪ Circuit reference - G_R_MAP - ▪ Monitor Description ▪ Pressure sensor plausibility check prior to engine start ▪ Prioritized List of Possible Causes ▪ Manifold absolute pressure and temperature sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Manifold absolute pressure and temperature sensor failure	▪ Prioritized Checks to Perform ▪ Refer to the electrical circuit diagrams and check the manifold absolute pressure and temperature sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion ▪ Install a new manifold absolute pressure and temperature sensor as required ▪ Using the Jaguar Land Rover approved diagnostic equipment, run application - ECU Diagnostics / Powertrain Control Module (PCM) / Clear Diagnostic Trouble Code(s). Retest
P0106-24	Manifold Absolute Pressure / Barometric Pressure Sensor Circuit Range/Performance - Signal stuck high	▪ Electrical Cause ▪ Yes ▪ Mechanical Cause ▪ Yes ▪ Control Module Signal Name ▪ Circuit reference - O_V_5VMAP - ▪ Circuit reference - I_A_MAP - ▪ Circuit reference - G_R_MAP - ▪ Monitor Description ▪ Pressure sensor plausibility check prior to engine start ▪ Prioritized List of Possible Causes ▪ Manifold absolute pressure and temperature sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Manifold absolute pressure and temperature sensor failure	▪ Prioritized Checks to Perform ▪ Refer to the electrical circuit diagrams and check the manifold absolute pressure and temperature sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion ▪ Install a new manifold absolute pressure and temperature sensor as required ▪ Using the Jaguar Land Rover approved diagnostic equipment, run application - ECU Diagnostics / Powertrain Control Module (PCM) / Clear Diagnostic Trouble Code(s). Retest

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION

P0106-62	Manifold Absolute Pressure / Barometric Pressure Sensor Circuit Range/Performance - Signal compare failure	▪ Electrical Cause
		  ▪ Yes
		▪ Mechanical Cause
		  ▪ Yes
		▪ Control Module Signal Name
		  ▪ Circuit reference - O_V_5VMAP -
		  ▪ Circuit reference - I_A_MAP -
		  ▪ Circuit reference - G_R_MAP -
		▪ Monitor Description
		  ▪ Pressure sensor plausibility check against maximum modeled manifold pressure
		▪ Prioritized List of Possible Causes
		  ▪ Manifold absolute pressure and temperature sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance
		  ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion
		  ▪ Manifold absolute pressure and temperature sensor failure
		  ▪ Intake manifold air leak (loose or missing component)
		  ▪ Breather system leak (loose or missing component)
		  ▪ Blocked air filter
		  ▪ Carbon build up around throttle
		  ▪ Damaged/restricted exhaust system
			▪ Prioritized Checks to Perform
			  ▪ Refer to the electrical circuit diagrams and check the manifold absolute pressure and temperature sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance
			  ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion
			  ▪ Install a new manifold absolute pressure and temperature sensor as required
			  ▪ Check air filter is serviceable
			  ▪ Check intake manifold for air leak (loose or missing component)
			  ▪ Check breather system for leak (loose or missing component)
			  ▪ Check electric throttle blade for contamination by oil or debris clean as required
			  ▪ Check for damaged/restricted exhaust system
			  ▪ Using the Jaguar Land Rover approved diagnostic equipment, run application - ECU Diagnostics / Powertrain Control Module (PCM) / Clear Diagnostic Trouble Code(s). Retest

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION

P0106-64	Manifold Absolute Pressure / Barometric Pressure Sensor Circuit Range/Performance - Signal plausibility failure	▪ Electrical Cause
		  ▪ Yes
		▪ Mechanical Cause
		  ▪ Yes
		▪ Control Module Signal Name
		  ▪ Circuit reference - O_V_5VMAP -
		  ▪ Circuit reference - I_A_MAP -
		  ▪ Circuit reference - G_R_MAP -
		▪ Monitor Description
		  ▪ Pressure sensor plausibility check against maximum modeled manifold pressure
		▪ Prioritized List of Possible Causes
		  ▪ Manifold absolute pressure and temperature sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance
		  ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion
		  ▪ Manifold absolute pressure and temperature sensor failure
		  ▪ Intake manifold air leak (loose or missing component)
		  ▪ Breather system leak (loose or missing component)
		  ▪ Blocked air filter
		  ▪ Carbon build up around throttle
		  ▪ Damaged/restricted exhaust system
			▪ Prioritized Checks to Perform
			  ▪ Refer to the electrical circuit diagrams and check the manifold absolute pressure and temperature sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance
			  ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion
			  ▪ Install a new manifold absolute pressure and temperature sensor as required
			  ▪ Check air filter is serviceable
			  ▪ Check intake manifold for air leak (loose or missing component)
			  ▪ Check breather system for leak (loose or missing component)
			  ▪ Check electric throttle blade for contamination by oil or debris clean as required
			  ▪ Check for damaged/restricted exhaust system
			  ▪ Using the Jaguar Land Rover approved diagnostic equipment, run application - ECU Diagnostics / Powertrain Control Module (PCM) / Clear Diagnostic Trouble Code(s). Retest

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION
P0107-16	Manifold Absolute Pressure / Barometric Pressure Sensor Circuit Low - Circuit voltage below threshold	▪ Electrical Cause ▪ Yes ▪ Mechanical Cause ▪ Yes ▪ Control Module Signal Name ▪ Circuit reference - O_V_5VMAP - ▪ Circuit reference - I_A_MAP - ▪ Circuit reference - G_R_MAP - ▪ Monitor Description ▪ Circuit check of the manifold pressure sensor ▪ Prioritized List of Possible Causes ▪ Manifold absolute pressure and temperature sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Manifold absolute pressure and temperature sensor failure	**NOTE:** The powertrain control module is unlikely to be the root cause for this DTC and must not be replaced without a Technical Assist Request (TA) being raised for further guidance ▪ Prioritized Checks to Perform ▪ Refer to the electrical circuit diagrams and check the manifold absolute pressure and temperature sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion ▪ Install a new manifold absolute pressure and temperature sensor as required ▪ Using the Jaguar Land Rover approved diagnostic equipment, run application - ECU Diagnostics / Powertrain Control Module (PCM) / Clear Diagnostic Trouble Code(s). Retest
P0108-17	Manifold Absolute Pressure / Barometric Pressure Sensor Circuit High - Circuit voltage above threshold	▪ Electrical Cause ▪ Yes ▪ Mechanical Cause ▪ Yes ▪ Control Module Signal Name ▪ Circuit reference - O_V_5VMAP - ▪ Circuit reference - I_A_MAP - ▪ Circuit reference - G_R_MAP - ▪ Monitor Description ▪ Circuit check of the manifold pressure sensor ▪ Prioritized List of Possible Causes ▪ Manifold absolute pressure and temperature sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Manifold absolute pressure and temperature sensor failure	▪ Prioritized Checks to Perform ▪ Refer to the electrical circuit diagrams and check the manifold absolute pressure and temperature sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion ▪ Install a new manifold absolute pressure and temperature sensor as required ▪ Using the Jaguar Land Rover approved diagnostic equipment, run application - ECU Diagnostics / Powertrain Control Module (PCM) / Clear Diagnostic Trouble Code(s). Retest

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION

P0111-23	Intake Air Temperature Sensor 1 Circuit Range/Performance Bank 1 - Signal stuck low	▪ Electrical Cause
		  ▪ Yes
		▪ Mechanical Cause
		  ▪ No
		▪ Control Module Cavity
		  ▪ Circuit reference - I_A_IATS -
		▪ Prioritized List of Possible Causes
		  ▪ Intake air temperature sensor signal is lower than expected
		  ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion
		  ▪ Intake air temperature sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance
			▪ Vehicle Conditions to Enable DTC Logging Strategy
			  ▪ Test the vehicle over 3 drive cycles (engine on/off) across varied driving conditions to confirm the repair
			▪ Prioritized Checks to Perform
			  ▪ Leave vehicle turned off for a minimum of 8 hours and allow to soak to a stable temperature
			  ▪ Using the Jaguar Land Rover approved diagnostic equipment, check datalogger parameters - Intake air temperature - Charge air cooler temperature - Ambient air temperature - Manifold air temperature sensor. All sensors should be within 20ºC of each other
			  ▪ Check the intake air temperature sensor (part of the mass air flow sensor) and circuit for sensor contamination
			  ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion
			  ▪ Refer to the electrical circuit diagrams and check connections are secure and wiring integrity
			  ▪ Install a new sensor that is biased higher or lower than the other sensors
			  ▪ Using the Jaguar Land Rover approved diagnostic equipment, run application - ECU Diagnostics / Powertrain Control Module (PCM) / Clear Diagnostic Trouble Code(s). Retest

P0111-84	Intake Air Temperature Sensor 1 Circuit Range/Performance Bank 1 - Signal below allowable range	▪ Electrical Cause
		  ▪ Yes
		▪ Mechanical Cause
		  ▪ No
		▪ Control Module Cavity
		  ▪ Circuit reference I_A_IATS
		▪ Monitor Description
		  ▪ P0111-84 is set if the deviation between the reference temperature and the measured temperature from the air temperature sensor exceeds 20°C after an 8 hour soak period. This DTC is the minimum check therefore the temperature is reading ≥20°C lower than reference
		▪ Open circuit in the signal line
		▪ Short to ground in the signal line
		▪ Offset temperature sensor (temperature sensor integrated within airflow sensor)
			▪ Check status of DTC's and verify they are pending or confirmed and not historical. If DTCs are historical, take no further action
			▪ Leave vehicle turned off for a minimum of 8 hours and allow to soak to a stable temperature
			▪ Using the Jaguar Land Rover approved diagnostic equipment check datalogger signals. All sensors should be within 20ºC of each other
			  ▪ Air filter temperature (at airflow sensor
			  ▪ Charge air cooler temperature
			  ▪ Ambient air temperature and manifold air temperature sensors
			▪ If P0111-84 is set with P0113-00 and P0100-38, check and confirm the airflow sensor is electrically connected
			▪ If P0111-84 is set with P0113-00, check the integrity of the ground line from the sensor back to the ECU, inspect the connector for signs of water ingress or damage and verify sufficient pin tension is present at the airflow connector and the ECU. Rectify any concerns as required
			▪ If P0111-84 is set on its own replace the airflow/temperature sensor
			▪ Using the Jaguar Land Rover approved diagnostic equipment, clear all stored Diagnostic Trouble Code(s) and retest
			  ▪ Test the vehicle over 3 drive cycles (engine on/off) across varied driving conditions to confirm the repair

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION
P0111-85	Intake Air Temperature Sensor 1 Circuit Range/Performance Bank 1 - Signal above allowable range	▪ Electrical Cause ▪ Yes ▪ Mechanical Cause ▪ No ▪ Control Module Cavity ▪ Circuit reference - I_A_IATS - ▪ Prioritized List of Possible Causes ▪ Intake air temperature sensor signal is higher than expected ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Intake air temperature sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance	▪ Vehicle Conditions to Enable DTC Logging Strategy ▪ Test the vehicle over 3 drive cycles (engine on/off) across varied driving conditions to confirm the repair ▪ Prioritized Checks to Perform ▪ Leave vehicle turned off for a minimum of 8 hours and allow to soak to a stable temperature ▪ Using the Jaguar Land Rover approved diagnostic equipment, check datalogger parameters - Intake air temperature - Charge air cooler temperature - Ambient air temperature - Manifold air temperature sensor. All sensors should be within 20ºC of each other ▪ Check the intake air temperature sensor (part of the mass air flow sensor) and circuit for sensor contamination ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion ▪ Refer to the electrical circuit diagrams and check connections are secure and wiring integrity ▪ Install a new sensor that is biased higher or lower than the other sensors ▪ Using the Jaguar Land Rover approved diagnostic equipment, run application - ECU Diagnostics / Powertrain Control Module (PCM) / Clear Diagnostic Trouble Code(s). Retest
P0112-00	Intake Air Temperature Sensor 1 Circuit Low Bank 1 - No sub type information	▪ Electrical Cause ▪ Yes ▪ Mechanical Cause ▪ Yes ▪ Control Module Cavity ▪ Circuit reference - I_A_IATS - ▪ Circuit reference - G_R_AMS - ▪ Prioritized List of Possible Causes ▪ Intake air temperature sensor circuit short circuit to ground ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Intake air temperature sensor failure	▪ Prioritized Checks to Perform ▪ Refer to the electrical circuit diagrams and check the intake air temperature sensor circuit for short circuit to ground ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion ▪ Using the Jaguar Land Rover approved diagnostic equipment, run application - ECU Diagnostics / Powertrain Control Module (PCM) / Clear Diagnostic Trouble Code(s). Retest ▪ Install a new intake air temperature sensor as required

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION

P0113-00	Intake Air Temperature Sensor 1 Circuit High Bank 1 - No sub type information	▪ Electrical Cause
		  ▪ Yes
		▪ Mechanical Cause
		  ▪ Yes
		▪ Control Module Cavity
		  ▪ Circuit reference I_A_IATS
		  ▪ Circuit reference G_R_AMS
		▪ Monitor Description
		  ▪ P0113-00 is set if the feedback voltage from the air temperature sensor is higher than the maximum allowable voltage
		▪ Open circuit in the sensor ground path (Pin 2)
		▪ Open circuit in the sensor signal line (Pin 4)
		▪ Short circuit to battery in the signal line
		▪ Temperature sensor internal failure (integrated within airflow sensor)
			▪ Check status of DTC's and verify they are pending or confirmed and not historical. If DTCs are historical, take no further action
			▪ If P0113-00 is set together with P0111-84 and P0100-38, check and confirm the airflow sensor is electrically connected
			▪ If P0113-00 is set with P0103-17, check the integrity of the ground line from the sensor back to the ECU, inspect the connector for signs of water ingress, corrosion or damage and verify sufficient pin tension is present at the airflow connector and the ECU. Rectify any concerns as required
			▪ If only P0113-00 is set, check the integrity of the signal line from the sensor back to the ECU, inspect the connector for signs of water ingress or damage and verify sufficient pin tension is present at the airflow connector and the ECU. Rectify any concerns as required.
			▪ If no open circuit is found in the signal line, check for a short circuit to battery on the signal line between the sensor and the ECU. Rectify any concerns as required
			▪ If no circuit faults identified, replace the airflow/temperature sensor
			▪ Upon rectification of the fault, using the Jaguar Land Rover approved diagnostic equipment, clear all stored Diagnostic Trouble Code(s) and retest
			  ▪ Test the vehicle over 3 drive cycles (engine on/off) across varied driving conditions to confirm the repair

P0114-00	Intake Air Temperature Sensor 1 Circuit Intermittent Bank 1 - No sub type information	▪ Electrical Cause
		  ▪ Yes
		▪ Mechanical Cause
		  ▪ Yes
		▪ Control Module Cavity
		  ▪ Circuit reference - I_A_IATS -
		  ▪ Circuit reference - G_R_AMS -
		▪ Prioritized List of Possible Causes
		  ▪ Intake air temperature sensor circuit open circuit short circuit to power, short circuit to ground
		  ▪ Connector is disconnected or loose, connector pin is backed out, connector pin corrosion
		  ▪ Intake air temperature sensor failure
			▪ Prioritized Checks to Perform
			  ▪ Refer to the electrical circuit diagrams and check the intake air temperature sensor circuit for open circuit, short circuit to power, short circuit to ground
			  ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion
			  ▪ Using the Jaguar Land Rover approved diagnostic equipment, run application - ECU Diagnostics / Powertrain Control Module (PCM) / Clear Diagnostic Trouble Code(s). Retest
			  ▪ Install a new intake air temperature sensor as required

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION

P0116-2A	Engine Coolant Temperature Sensor 1 Circuit Range/Performance - Signal stuck in range	▪ Electrical Cause
		  ▪ Yes
		▪ Mechanical Cause
		  ▪ Yes
		▪ Control Module Cavity
		  ▪ Circuit reference - I_A_CTS -
		  ▪ Circuit reference - G_R_CTS -
		▪ Monitor Description
		  ▪ Coolant temperature sensor stuck since start
		▪ Prioritized List of Possible Causes
		  ▪ Cooling system mechanical and fluid integrity
		  ▪ Cooling fan system mechanical integrity
		  ▪ Thermostat system mechanical integrity
		  ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion
		  ▪ Engine coolant temperature sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance
		  ▪ Engine coolant temperature sensor contamination
		  ▪ Engine coolant temperature sensor failure
			▪ Prioritized Checks to Perform
			  ▪ Check cooling system for mechanical and fluid integrity
			  ▪ Check cooling fan system for mechanical integrity
			  ▪ Check thermostat system for mechanical integrity
			  ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion
			  ▪ Refer to the electrical circuit diagrams and check the engine coolant temperature sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance
			  ▪ Install a new engine coolant temperature sensor as required
			  ▪ Using the Jaguar Land Rover approved diagnostic equipment, run application - ECU Diagnostics / Powertrain Control Module (PCM) / Clear Diagnostic Trouble Code(s). Retest

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION

P0116-62	Engine Coolant Temperature Sensor 1 Circuit Range/Performance - Signal compare failure	▪ Electrical Cause
		  ▪ Yes
		▪ Mechanical Cause
		  ▪ Yes
		▪ Control Module Cavity
		  ▪ Circuit reference - I_A_CTS -
		  ▪ Circuit reference - G_R_CTS -
		▪ Monitor Description
		  ▪ Coolant temperature sensor reading too high or too low at cold start
		▪ Prioritized List of Possible Causes
		  ▪ Cooling system mechanical and fluid integrity
		  ▪ Cooling fan system mechanical integrity
		  ▪ Thermostat system mechanical integrity
		  ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion
		  ▪ Engine coolant temperature sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance
		  ▪ Engine coolant temperature sensor contamination
		  ▪ Engine coolant temperature sensor failure
			▪ Prioritized Checks to Perform
			  ▪ Check cooling system for mechanical and fluid integrity
			  ▪ Check cooling fan system for mechanical integrity
			  ▪ Check thermostat system for mechanical integrity
			  ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion
			  ▪ Refer to the electrical circuit diagrams and check the engine coolant temperature sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance
			  ▪ Install a new engine coolant temperature sensor as required
			  ▪ Using the Jaguar Land Rover approved diagnostic equipment, run application - ECU Diagnostics / Powertrain Control Module (PCM) / Clear Diagnostic Trouble Code(s). Retest

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION
P0116-64	Engine Coolant Temperature Sensor 1 Circuit Range/Performance - Signal plausibility failure	▪ Electrical Cause ▪ Yes ▪ Mechanical Cause ▪ Yes ▪ Control Module Cavity ▪ Circuit reference - I_A_CTS - ▪ Circuit reference - G_R_CTS - ▪ Monitor Description ▪ Coolant temperature sensor reading too high or too low during engine operation ▪ Prioritized List of Possible Causes ▪ Cooling system mechanical and fluid integrity ▪ Cooling fan system mechanical integrity ▪ Thermostat system mechanical integrity ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Engine coolant temperature sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Engine coolant temperature sensor contamination ▪ Engine coolant temperature sensor failure	▪ Prioritized Checks to Perform ▪ Check cooling system for mechanical and fluid integrity ▪ Check cooling fan system for mechanical integrity ▪ Check thermostat system for mechanical integrity ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion ▪ Refer to the electrical circuit diagrams and check the engine coolant temperature sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance ▪ Install a new engine coolant temperature sensor as required ▪ Using the Jaguar Land Rover approved diagnostic equipment, run application - ECU Diagnostics / Powertrain Control Module (PCM) / Clear Diagnostic Trouble Code(s). Retest

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION

P0117-00	Engine Coolant Temperature Sensor 1 Circuit Low - No sub type information	▪ Electrical Cause
		  ▪ Yes
		▪ Mechanical Cause
		  ▪ Yes
		▪ Control Module Cavity
		  ▪ Circuit reference - I_A_CTS -
		  ▪ Circuit reference - G_R_CTS -
		▪ Monitor Description
		  ▪ Engine coolant outlet temperature sensor short circuit to ground
		▪ Prioritized List of Possible Causes
		  ▪ Cooling system mechanical and fluid integrity
		  ▪ Cooling fan system mechanical integrity
		  ▪ Thermostat system mechanical integrity
		  ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion
		  ▪ Engine coolant temperature sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance
		  ▪ Engine coolant temperature sensor contamination
		  ▪ Engine coolant temperature sensor failure
			▪ Prioritized Checks to Perform
			  ▪ Check cooling system for mechanical and fluid integrity
			  ▪ Check cooling fan system for mechanical integrity
			  ▪ Check thermostat system for mechanical integrity
			  ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion
			  ▪ Refer to the electrical circuit diagrams and check the engine coolant temperature sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance
			  ▪ Install a new engine coolant temperature sensor as required
			  ▪ Using the Jaguar Land Rover approved diagnostic equipment, run application - ECU Diagnostics / Powertrain Control Module (PCM) / Clear Diagnostic Trouble Code(s). Retest

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION
P0118-00	Engine Coolant Temperature Sensor 1 Circuit High - No sub type information	▪ Electrical Cause ▪ Yes ▪ Mechanical Cause ▪ Yes ▪ Control Module Cavity ▪ Circuit reference - I_A_CTS - ▪ Circuit reference - G_R_CTS - ▪ Monitor Description ▪ Engine coolant outlet temperature sensor open circuit or short circuit to battery or 5 volts ▪ Prioritized List of Possible Causes ▪ Cooling system mechanical and fluid integrity ▪ Cooling fan system mechanical integrity ▪ Thermostat system mechanical integrity ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Engine coolant temperature sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Engine coolant temperature sensor contamination ▪ Engine coolant temperature sensor failure	▪ Prioritized Checks to Perform ▪ Check cooling system for mechanical and fluid integrity ▪ Check cooling fan system for mechanical integrity ▪ Check thermostat system for mechanical integrity ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion ▪ Refer to the electrical circuit diagrams and check the engine coolant temperature sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance ▪ Install a new engine coolant temperature sensor as required ▪ Using the Jaguar Land Rover approved diagnostic equipment, run application - ECU Diagnostics / Powertrain Control Module (PCM) / Clear Diagnostic Trouble Code(s). Retest
P0121-00	Throttle/Pedal Position Sensor/Switch "A" Circuit Range/Performance - No sub type information	▪ Electrical Cause ▪ Yes ▪ Mechanical Cause ▪ Yes ▪ Control Module Cavity ▪ Circuit reference - I_A_TPS1 - ▪ Monitor Description ▪ Throttle position sensor A does not align to position sensor B and is also not aligned to a theoretical throttle angle calculated from air charge / load. (Sensor A is electrically within valid range but irrational to operating conditions) ▪ Prioritized List of Possible Causes ▪ Accelerator pedal position sensor circuit short circuit to power, short circuit to ground, high resistance ▪ Accelerator pedal position sensor failure	▪ Vehicle Conditions to Enable DTC Logging Strategy ▪ Make sure all throttle position circuit faults have been rectified prior to investigating this fault. Engine running at idle for 5 seconds, also perform a short less than 10 minutes varied drive cycle to make sure fault has been rectified, make sure light and high engine loads are included ▪ Prioritized Checks to Perform ▪ Refer to the electrical circuit diagrams and check accelerator pedal position sensor circuit for short circuit to power, short circuit to ground, high resistance ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion ▪ Using the Jaguar Land Rover approved diagnostic equipment, with Ignition on but engine off, check accelerator pedal position sensor signal A is aligned to accelerator pedal position sensor signal B ▪ Using the Jaguar Land Rover approved diagnostic equipment, run application - ECU Diagnostics / Powertrain Control Module (PCM) / Clear Diagnostic Trouble Code(s). Retest ▪ Install a new accelerator pedal position sensor as required

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION
P0122-00	Throttle/Pedal Position Sensor/Switch "A" Circuit Low - No sub type information	▪ Electrical Cause ▪ Yes ▪ Mechanical Cause ▪ Yes ▪ Control Module Cavity ▪ Circuit reference - I_A_TPS1 - ▪ Monitor Description ▪ Throttle position sensor A is below valid electrical range ▪ Prioritized List of Possible Causes ▪ Accelerator pedal position sensor circuit is below the valid electrical range ▪ Accelerator pedal position sensor circuit open circuit, short circuit to ground ▪ Accelerator pedal position sensor failure	▪ Vehicle Conditions to Enable DTC Logging Strategy ▪ Ignition on, Engine speed greater than 1200rpm for 5 seconds ▪ Prioritized Checks to Perform ▪ Refer to the electrical circuit diagrams and check accelerator pedal position sensor circuit for open circuit, short circuit to ground ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion ▪ Using the Jaguar Land Rover approved diagnostic equipment, run application - ECU Diagnostics / Powertrain Control Module (PCM) / Clear Diagnostic Trouble Code(s). Retest ▪ Install a new accelerator pedal position sensor as required
P0123-85	Throttle/Pedal Position Sensor/Switch "A" Circuit High - Signal above allowable range	▪ Electrical Cause ▪ Yes ▪ Mechanical Cause ▪ Yes ▪ Control Module Cavity ▪ Circuit reference - I_A_TPS1 - ▪ Monitor Description ▪ Throttle position sensor A is above valid electrical range ▪ Prioritized List of Possible Causes ▪ Accelerator pedal position sensor circuit is above the valid electrical range ▪ Accelerator pedal position sensor circuit short circuit to power ▪ Accelerator pedal position sensor failure	▪ Vehicle Conditions to Enable DTC Logging Strategy ▪ Ignition on, Engine speed greater than 1200rpm for 5 seconds ▪ Prioritized Checks to Perform ▪ Refer to the electrical circuit diagrams and check accelerator pedal position sensor circuit for short circuit to power ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion ▪ Using the Jaguar Land Rover approved diagnostic equipment, run application - ECU Diagnostics / Powertrain Control Module (PCM) / Clear Diagnostic Trouble Code(s). Retest ▪ Install a new accelerator pedal position sensor as required

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION

P0128-00	Coolant Thermostat (Coolant Temperature Below Thermostat Regulating Temperature) - No sub type information	▪ Electrical Cause
		  ▪ No
		▪ Mechanical Cause
		  ▪ Yes
		▪ Thermostat stuck open or partially open
		▪ Debris inside thermostat housing
		▪ Debris inside engine coolant inlet tube
		▪ Debris inside thermostatic valve - automatic transmission fluid cooler
		▪ Debris inside automatic transmission fluid warming valve
		▪ Debris inside the coolant separator
			▪ Check for correct coolant mixture strength (particularly if vehicle has been used in sub zero conditions)
			▪ Check for contamination/debris inside the
			  ▪ thermostat housing
			  ▪ engine coolant inlet tube
			  ▪ thermostatic valve - automatic transmission fluid cooler
			  ▪ automatic transmission fluid warming valve
			  ▪ coolant seperator
			▪ Check and install a new thermostat, only when diagnosed as failed
			▪ Check and install a new engine coolant inlet tube, only when diagnosed as failed
			▪ Check and install a new thermostatic valve - automatic transmission fluid cooler, only when diagnosed as failed
			▪ Check and install a new automatic transmission fluid warming valve, only when diagnosed as failed
			▪ Check and install a new coolant separator, only when diagnosed as failed

P0131-00	O2 Sensor Circuit Low Voltage Bank 1 Sensor 1 - No sub type information	▪ Electrical Cause
		  ▪ Yes
		▪ Mechanical Cause
		  ▪ Yes
		▪ Control Module Cavity
		  ▪ Circuit reference - I_A_LSVN Un -
		  ▪ Circuit reference O_R_LSVG Uref
		  ▪ Circuit reference - I_A_LSCP Ip -
		▪ Monitor Description
		  ▪ Detects low voltage short circuit of the sensor circuit lines
		▪ Prioritized List of Possible Causes
		  ▪ Heated oxygen sensor circuit, short circuit to ground
		  ▪ Heated oxygen sensor failure
			▪ Vehicle Conditions to Enable DTC Logging Strategy
			  ▪ Start engine and allow to idle for 5 minutes
			▪ Prioritized Checks to Perform
			  ▪ Refer to electrical circuit diagrams and check the heated oxygen sensor circuit for short circuit to ground
			  ▪ Install a new heated oxygen sensor as required

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION

P0132-00	O2 Sensor Circuit High Voltage Bank 1 Sensor 1 - No sub type information	▪ Electrical Cause
		  ▪ Yes
		▪ Mechanical Cause
		  ▪ Yes
		▪ Control Module Cavity
		  ▪ Circuit reference - I_A_LSVN Un -
		  ▪ Circuit reference - O_R_LSVG Uref -
		  ▪ Circuit reference - I_A_LSCP Ip -
		▪ Monitor Description
		  ▪ Detects high voltage short circuit of the sensor circuit lines
		▪ Prioritized List of Possible Causes
		  ▪ Heated oxygen sensor circuit, short circuit to power
		  ▪ Heated oxygen sensor failure
			▪ Vehicle Conditions to Enable DTC Logging Strategy
			  ▪ Start engine and allow to idle for 5 minutes
			▪ Prioritized Checks to Perform
			  ▪ Refer to electrical circuit diagrams and check the heated oxygen sensor circuit for short circuit to power
			  ▪ Install a new heated oxygen sensor as required

P0133-00	O2 Sensor Circuit Slow Response Bank 1 Sensor 1 - No sub type information	▪ Electrical Cause
		  ▪ Yes
		▪ Mechanical Cause
		  ▪ Yes
		▪ Control Module Cavity
		  ▪ Circuit reference - I_A_LSVN Un -
		  ▪ Circuit reference - O_R_LSVG Uref -
		  ▪ Circuit reference - I_A_LSCP Ip -
		▪ Monitor Description
		  ▪ Monitor detects when the dynamic response of the upstream oxygen sensor has deteriorated to such an extent that the vehicles emissions exceed legal limits
		▪ Prioritized List of Possible Causes
		  ▪ Heated oxygen sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance
		  ▪ Heated oxygen sensor to powertrain control module wiring shield, open circuit
		  ▪ Air leakage at exhaust manifold
		  ▪ Incorrectly installed heated oxygen sensor
		  ▪ Heated oxygen sensor failure
			▪ Vehicle Conditions to Enable DTC Logging Strategy
			  ▪ Fully warmed-up engine, drive off-idle for a minimum of 10 minutes with several periods of steady operation for more than 20 seconds (no gear shifts, steady accelerator position)
			▪ Prioritized Checks to Perform
			  ▪ Refer to electrical circuit diagrams and check the heated oxygen sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance
			  ▪ Check exhaust manifold for leakage
			  ▪ Check heated oxygen sensor is correctly installed
			  ▪ Install a new heated oxygen sensor as required

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION

P0134-00	O2 Sensor Circuit No Activity Detected Bank 1 Sensor 1 - No sub type information	▪ Electrical Cause
		  ▪ Yes
		▪ Mechanical Cause
		  ▪ Yes
		▪ Control Module Cavity
		  ▪ Circuit reference - I_A_LSVN Un -
		  ▪ Circuit reference - O_R_LSVG Uref -
		  ▪ Circuit reference - I_A_LSCP Ip -
		▪ Monitor Description
		  ▪ Upstream oxygen sensor, bank 1, not correctly installed in the exhaust system or fallen out
		▪ Prioritized List of Possible Causes
		  ▪ Heated oxygen sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance
		  ▪ Heated oxygen sensor to powertrain control module wiring shield open circuit
		  ▪ Air leakage at exhaust manifold
		  ▪ Incorrectly installed heated oxygen sensor
		  ▪ Heated oxygen sensor failure
			▪ Vehicle Conditions to Enable DTC Logging Strategy
			  ▪ Start engine and allow to idle. If low fuel level warning is active allow to idle for at least 10 minutes
			▪ Prioritized Checks to Perform
			  ▪ Refer to electrical circuit diagrams and check the heated oxygen sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance
			  ▪ Check exhaust manifold for leakage
			  ▪ Check heated oxygen sensor is correctly installed
			  ▪ Install a new heated oxygen sensor as required

P0135-00	O2 Sensor Heater Circuit Bank 1 Sensor 1 - No sub type information	▪ Electrical Cause
		  ▪ Yes
		▪ Mechanical Cause
		  ▪ Yes
		▪ Control Module Cavity
		  ▪ Circuit reference - O_T_LSH -
		▪ Monitor Description
		  ▪ Monitors the heating element of the upstream oxygen sensor
		▪ Prioritized List of Possible Causes
		  ▪ Heated oxygen sensor heater control circuit short circuit to ground, short circuit to power, open circuit, high resistance
		  ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion
		  ▪ Heated oxygen sensor failure
			▪ Vehicle Conditions to Enable DTC Logging Strategy
			  ▪ Start engine and allow to idle for 5 minutes
			▪ Prioritized Checks to Perform
			  ▪ Refer to the electrical circuit diagrams and check the compressor recirculation valve circuit for short circuit to ground, short circuit to power, open circuit, high resistance
			  ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion
			  ▪ Using the Jaguar Land Rover approved diagnostic equipment, run application - ECU Diagnostics / Powertrain Control Module (PCM) / Clear Diagnostic Trouble Code(s). Retest
			  ▪ Install a new compressor recirculation valve

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION
P0136-00	O2 Sensor Circuit Bank 1 Sensor 2 - No sub type information	▪ Electrical Cause ▪ Yes ▪ Mechanical Cause ▪ Yes ▪ Control Module Cavity ▪ Circuit reference - G_R_LSF1 - ▪ Circuit reference - I_A_LSF1 - ▪ Monitor Description ▪ Sensor voltage stuck in counter voltage band ▪ Prioritized List of Possible Causes ▪ Heated oxygen sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Heated oxygen sensor failure	▪ Vehicle Conditions to Enable DTC Logging Strategy ▪ Fully warmed-up engine, drive off-idle for a minimum of 5 minutes ▪ Prioritized Checks to Perform ▪ Refer to the electrical circuit diagrams and check the heated oxygen sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion ▪ Using the Jaguar Land Rover approved diagnostic equipment, run application - ECU Diagnostics / Powertrain Control Module (PCM) / Clear Diagnostic Trouble Code(s). Retest ▪ Install a new heated oxygen sensor as required
P0137-00	O2 Sensor Circuit Low Voltage Bank 1 Sensor 2 - No sub type information	▪ Electrical Cause ▪ Yes ▪ Mechanical Cause ▪ Yes ▪ Control Module Cavity ▪ Circuit reference - G_R_LSF1 - ▪ Circuit reference - I_A_LSF1 - ▪ Monitor Description ▪ Sensor voltage below the lower limit ▪ Prioritized List of Possible Causes ▪ Heated oxygen sensor circuit short circuit to ground short circuit to power, open circuit, high resistance ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Heated oxygen sensor failure	▪ Vehicle Conditions to Enable DTC Logging Strategy ▪ Fully warmed-up engine, drive off-idle for a minimum of 5 minutes ▪ Prioritized Checks to Perform ▪ Refer to the electrical circuit diagrams and check the heated oxygen sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion ▪ Using the Jaguar Land Rover approved diagnostic equipment, run application - ECU Diagnostics / Powertrain Control Module (PCM) / Clear Diagnostic Trouble Code(s). Retest ▪ Install a new heated oxygen sensor as required

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION

P0138-00	O2 Sensor Circuit High Voltage Bank 1 Sensor 2 - No sub type information	▪ Electrical Cause
		  ▪ Yes
		▪ Mechanical Cause
		  ▪ Yes
		▪ Control Module Cavity
		  ▪ Circuit reference - G_R_LSF1 -
		  ▪ Circuit reference - I_A_LSF1 -
		▪ Monitor Description
		  ▪ Sensor voltage above the upper limit
		▪ Prioritized List of Possible Causes
		  ▪ Heated oxygen sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance
		  ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion
		  ▪ Heated oxygen sensor failure
			▪ Vehicle Conditions to Enable DTC Logging Strategy
			  ▪ Fully warmed-up engine, drive off-idle for a minimum of 5 minutes
			▪ Prioritized Checks to Perform
			  ▪ Refer to the electrical circuit diagrams and check the heated oxygen sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance
			  ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion
			  ▪ Using the Jaguar Land Rover approved diagnostic equipment, run application - ECU Diagnostics / Powertrain Control Module (PCM) / Clear Diagnostic Trouble Code(s). Retest
			  ▪ Install a new heated oxygen sensor as required

P013A-00	O2 Sensor Slow Response - Rich to Lean Bank 1 Sensor 2 - No sub type information	▪ Electrical Cause
		  ▪ Yes
		▪ Mechanical Cause
		  ▪ Yes
		▪ Monitor Description
		  ▪ Rate of change of sensor response to a change in exhaust gas constitution too slow
		▪ Prioritized List of Possible Causes
		  ▪ Exhaust system leakage
		  ▪ Incorrectly installed heated oxygen sensor
		  ▪ Heated oxygen sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance
		  ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion
		  ▪ Heated oxygen sensor failure
			▪ Prioritized Checks to Perform
			  ▪ Check exhaust system for leakage
			  ▪ Check heated oxygen sensors are installed correctly into the exhaust system
			  ▪ Refer to the electrical circuit diagrams and check the heated oxygen sensor heater circuit for short circuit to ground, short circuit to power, open circuit, high resistance
			  ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion
			  ▪ Using the Jaguar Land Rover approved diagnostic equipment, run application - ECU Diagnostics / Powertrain Control Module (PCM) / Clear Diagnostic Trouble Code(s). Retest
			  ▪ Install a new heated oxygen sensor as required

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION

P013E-00	O2 Sensor Delayed Response - Rich to Lean Bank 1 Sensor 2 - No sub type information	▪ Electrical Cause
		  ▪ Yes
		▪ Mechanical Cause
		  ▪ Yes
		▪ Monitor Description
		  ▪ Time for sensor to respond to a change in exhaust gas constitution too great
		▪ Prioritized List of Possible Causes
		  ▪ Exhaust system leakage
		  ▪ Incorrectly installed heated oxygen sensor
		  ▪ Heated oxygen sensor circuit, short circuit to ground, short circuit to power, open circuit, high resistance
		  ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion
		  ▪ Heated oxygen sensor failure
		  ▪ Variable camshaft timing system fault - Intake camshaft
		  ▪ Variable camshaft timing system fault - Exhaust camshaft
			▪ Prioritized Checks to Perform
			  ▪ Check exhaust system for leakage
			  ▪ Check heated oxygen sensors are installed correctly into the exhaust system
			  ▪ Refer to the electrical circuit diagrams and check heated oxygen sensor heater circuit for short circuit to ground, short circuit to power, open circuit, high resistance
			  ▪ Inspect connectors for signs of water ingress and pins for damage and/or corrosion
			  ▪ Using the Jaguar Land Rover approved diagnostic equipment, run application - ECU Diagnostics / Powertrain Control Module (PCM) / Clear Diagnostic Trouble Code(s). Retest
			  ▪ Install new heated oxygen sensor, only when diagnosed as failed
			  ▪ After timing related repairs and timing adjustment, using the Jaguar Land Rover approved diagnostic equipment, run application - ECU Diagnostics / Powertrain Control Module (PCM) / ECU Functions / Variable Camshaft Timing (VCT) Actuator Intake Test. Perform the relevant corrective actions
			  ▪ After timing related repairs and timing adjustment, using the Jaguar Land Rover approved diagnostic equipment, run application - ECU Diagnostics / Powertrain Control Module (PCM) / ECU Functions / Variable Camshaft Timing (VCT) Actuator Exhaust Test. Perform the relevant corrective actions

P0141-00	O2 Sensor Heater Circuit Bank 1 Sensor 2 - No sub type information	▪ Electrical Cause
		  ▪ Yes
		▪ Mechanical Cause
		  ▪ Yes
		▪ Control Module Cavity
		  ▪ Circuit reference - G_R_LSF1 -
		  ▪ Circuit reference - I_A_LSF1 -
		▪ Monitor Description
		  ▪ Sensor voltage coupled to heater circuit
		▪ Prioritized List of Possible Causes
		  ▪ Heated oxygen sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance
		  ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion
		  ▪ Heated oxygen sensor failure
			▪ Vehicle Conditions to Enable DTC Logging Strategy
			  ▪ Fully warmed-up engine, drive off-idle for a minimum of 5 minutes
			▪ Prioritized Checks to Perform
			  ▪ Refer to the electrical circuit diagrams and check the heated oxygen sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance
			  ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion
			  ▪ Using the Jaguar Land Rover approved diagnostic equipment, run application - ECU Diagnostics / Powertrain Control Module (PCM) / Clear Diagnostic Trouble Code(s). Retest
			  ▪ Install a new heated oxygen sensor as required

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION
P0142-00	O2 Sensor Circuit Bank 1 Sensor 3 - No sub type information	- Electrical Cause - Yes - Mechanical Cause - Yes - Control Module Cavity - Circuit reference - G_R_LSF1 - - Circuit reference - I_A_LSF1 - - Monitor Description - Sensor voltage stuck in counter voltage band - Prioritized List of Possible Causes - Heated oxygen sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance - Connector is disconnected, connector pin is backed out, connector pin corrosion - Heated oxygen sensor failure	**WARNING:** **DO NOT** immediately replace the O2 Sensor based on this Diagnostic Trouble Code - This DTC could be set due to a persistent misfire. DTC P1315 may be set in the PCM - You **MUST** rectify the Diagnostic Trouble Code P1315, before continuing with this action - Vehicle Conditions to Enable Diagnostic Trouble Code Setting Strategy - Fully warmed-up engine, drive off-idle for a minimum of 5 minutes - Prioritized Checks to Perform - Using the Jaguar Land Rover approved diagnostic equipment, scan for Diagnostic Trouble Code(s) set in the PCM. If the DTC P1315 is set, it must be rectified first, before proceeding with this DTC - Refer to the electrical circuit diagrams and check the post catalyst heated oxygen sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance. Check the electrical connectors for evidence of water ingress. Check the terminals for damage and corrosion. Repair the wiring harness as necessary - Using the Jaguar Land Rover approved diagnostic equipment, run application - ECU Diagnostics / Powertrain Control Module (PCM) / Clear Diagnostic Trouble Code(s). Retest - Install a new post catalyst heated oxygen sensor
P0143-00	O2 Sensor Circuit Low Voltage Bank 1 Sensor 3 - No sub type information	- Electrical Cause - Yes - Mechanical Cause - Yes - Control Module Cavity - Circuit reference - G_R_LSF2 - - Circuit reference - I_A_LSF2 - - Monitor Description - Sensor voltage below the lower limit - Prioritized List of Possible Causes - Heated oxygen sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance - Connector is disconnected, connector pin is backed out, connector pin corrosion - Heated oxygen sensor failure	- Vehicle Conditions to Enable DTC Logging Strategy - Fully warmed-up engine, drive off-idle for a minimum of 5 minutes - Prioritized Checks to Perform - Refer to the electrical circuit diagrams and check the heated oxygen sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance - Inspect connectors for signs of water ingress, and pins for damage and/or corrosion - Using the Jaguar Land Rover approved diagnostic equipment, run application - ECU Diagnostics / Powertrain Control Module (PCM) / Clear Diagnostic Trouble Code(s). Retest - Install a new heated oxygen sensor as required

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION

P0144-00	O2 Sensor Circuit High Voltage Bank 1 Sensor 3 - No sub type information	▪ Electrical Cause
		  ▪ Yes
		▪ Mechanical Cause
		  ▪ Yes
		▪ Control Module Cavity
		  ▪ Circuit reference - G_R_LSF2 -
		  ▪ Circuit reference - I_A_LSF2 -
		▪ Monitor Description
		  ▪ Sensor voltage above the upper limit
		▪ Prioritized List of Possible Causes
		  ▪ Heated oxygen sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance
		  ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion
		  ▪ Heated oxygen sensor failure
			▪ Vehicle Conditions to Enable DTC Logging Strategy
			  ▪ Fully warmed-up engine, drive off-idle for a minimum of 5 minutes
			▪ Prioritized Checks to Perform
			  ▪ Refer to the electrical circuit diagrams and check the heated oxygen sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance
			  ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion
			  ▪ Using the Jaguar Land Rover approved diagnostic equipment, run application - ECU Diagnostics / Powertrain Control Module (PCM) / Clear Diagnostic Trouble Code(s). Retest
			  ▪ Install a new heated oxygen sensor as required

P0145-00	O2 Sensor Circuit Slow Response Bank 1 Sensor 3 - No sub type information	▪ Electrical Cause
		  ▪ Yes
		▪ Mechanical Cause
		  ▪ Yes
		▪ Control Module Cavity
		  ▪ Circuit reference - G_R_LSF2 -
		  ▪ Circuit reference - I_A_LSF2 -
		▪ Monitor Description
		  ▪ Monitor detects when the dynamic response of the downstream oxygen sensor has deteriorated to such an extent that the vehicles emissions exceed legal limits
		▪ Prioritized List of Possible Causes
		  ▪ Heated oxygen sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance
		  ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion
		  ▪ Exhaust system leakage
		  ▪ Incorrectly installed heated oxygen sensor
		  ▪ Heated oxygen sensor failure
			▪ Vehicle Conditions to Enable DTC Logging Strategy
			  ▪ Fully warmed-up engine, drive off-idle for a minimum of 10 minutes with several periods of steady operation for more than 20 seconds (no gear shifts, steady accelerator position) followed by an abrupt removal of accelerator input and allow to over-run for at least 5 seconds
			▪ Prioritized Checks to Perform
			  ▪ Refer to the electrical circuit diagrams and check the heated oxygen sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance
			  ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion
			  ▪ Check for exhaust system leakage
			  ▪ Check heated oxygen sensor is correctly installed
			  ▪ Using the Jaguar Land Rover approved diagnostic equipment, run application - ECU Diagnostics / Powertrain Control Module (PCM) / Clear Diagnostic Trouble Code(s). Retest
			  ▪ Install a new heated oxygen sensor as required

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION

P0147-00	O2 Sensor Heater Circuit Bank 1 Sensor 3 - No sub type information	▪ Electrical Cause
		  ▪ Yes
		▪ Mechanical Cause
		  ▪ Yes
		▪ Control Module Cavity
		  ▪ Circuit reference - G_R_LSF2 -
		  ▪ Circuit reference - I_A_LSF2 -
		▪ Monitor Description
		  ▪ Sensor voltage coupled to heater circuit
		▪ Prioritized List of Possible Causes
		  ▪ Heated oxygen sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance
		  ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion
		  ▪ Heated oxygen sensor failure
			▪ Vehicle Conditions to Enable DTC Logging Strategy
			  ▪ Fully warmed-up engine, drive off-idle for a minimum of 5 minutes
			▪ Prioritized Checks to Perform
			  ▪ Refer to the electrical circuit diagrams and check the heated oxygen sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance
			  ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion
			  ▪ Using the Jaguar Land Rover approved diagnostic equipment, run application - ECU Diagnostics / Powertrain Control Module (PCM) / Clear Diagnostic Trouble Code(s). Retest
			  ▪ Install a new heated oxygen sensor as required

P0169-00	Incorrect Fuel Composition - No sub type information	▪ Prioritized List of Possible Causes
		  ▪ Desired fueling control limits implausible. Powertrain control module software error
		  ▪ Powertrain control module failure
			▪ Prioritized Checks to Perform
			  ▪ Refer to the electrical circuit diagrams and check the powertrain control module power and ground circuits for open circuit, high resistance. Repair the wiring harness as necessary
			  ▪ Using the Jaguar Land Rover approved diagnostic equipment, run application - ECU Diagnostics / Powertrain Control Module (PCM) / Update ECU
			  ▪ Install a new powertrain control module. Using the Jaguar Land Rover approved diagnostic equipment, run application - ECU Diagnostics / Powertrain Control Module (PCM) / Replace ECU

P0169-42	Incorrect Fuel Composition - General memory failure	▪ Prioritized List of Possible Causes
		  ▪ Desired lambda was not found plausible. Powertrain control module software error
		  ▪ Powertrain control module failure
			▪ Prioritized Checks to Perform
			  ▪ Refer to the electrical circuit diagrams and check the powertrain control module power and ground circuits for open circuit, high resistance. Repair the wiring harness as necessary
			  ▪ Using the Jaguar Land Rover approved diagnostic equipment, run application - ECU Diagnostics / Powertrain Control Module (PCM) / Update ECU
			  ▪ Install a new powertrain control module. Using the Jaguar Land Rover approved diagnostic equipment, run application - ECU Diagnostics / Powertrain Control Module (PCM) / Replace ECU

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION
P0169-43	Incorrect Fuel Composition - Special memory failure	▪ Prioritized List of Possible Causes ▪ Predicted fuel mass was found not plausible. Powertrain control module software error ▪ Powertrain control module failure	▪ Prioritized Checks to Perform ▪ Refer to the electrical circuit diagrams and check the powertrain control module power and ground circuits for open circuit, high resistance. Repair the wiring harness as necessary ▪ Using the Jaguar Land Rover approved diagnostic equipment, run application - ECU Diagnostics / Powertrain Control Module (PCM) / Update ECU ▪ Install a new powertrain control module. Using the Jaguar Land Rover approved diagnostic equipment, run application - ECU Diagnostics / Powertrain Control Module (PCM) / Replace ECU
P0170-00	Fuel Trim Bank 1 - No sub type information	▪ Prioritized List of Possible Causes ▪ Exhaust system leakage ▪ Post catalyst heated oxygen sensor incorrectly installed ▪ Pre catalyst heated oxygen sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Post catalyst heated oxygen sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Pre catalyst heated oxygen sensor contamination or failure ▪ Post catalyst heated oxygen sensor contamination or failure	▪ Prioritized Checks to Perform ▪ Check exhaust system for leakage. Rectify as required ▪ Check post catalyst heated oxygen sensor is correctly installed ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion ▪ Refer to the electrical circuit diagrams and check the pre catalyst heated oxygen sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance ▪ Refer to the electrical circuit diagrams and check the post catalyst heated oxygen sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance ▪ Using the Jaguar Land Rover approved diagnostic equipment, run application - ECU Diagnostics / Powertrain Control Module (PCM) / Clear Diagnostic Trouble Code(s). Retest ▪ Install a new pre catalyst heated oxygen sensor as required ▪ Install a new post catalyst heated oxygen sensor as required
P0177-24	Fuel Composition Sensor Circuit Range/Performance - Signal stuck high	▪ Electrical Cause ▪ Yes ▪ Mechanical Cause ▪ Yes ▪ Control Module Cavity ▪ Circuit reference - I_T_FFS - ▪ Prioritized List of Possible Causes ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Flex fuel sensor circuit, short circuit to ground, short circuit to power, open circuit, high resistance ▪ Water contamination in the fuel system ▪ Loose or damaged fuel filler cap seal	▪ Prioritized Checks to Perform ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion ▪ Refer to electrical circuit diagrams and check the flex fuel sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance ▪ Drain off the contaminated fuel. Install a new flex fuel filter element as necessary and replace the contaminated fuel. Check the fuel filler cap seal for wear or damage. Using the Jaguar Land Rover approved diagnostic equipment, run application - ECU Diagnostics / Powertrain Control Module (PCM) / Clear Diagnostic Trouble Code(s). Retest

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION
P0177-25	Fuel Composition Sensor Circuit Range/Performance - Signal shape/waveform failure	▪ Electrical Cause ▪ Yes ▪ Mechanical Cause ▪ Yes ▪ Control Module Cavity ▪ Circuit reference - I_T_FFS - ▪ Prioritized List of Possible Causes ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Flex fuel sensor circuit, short circuit to ground, short circuit to power, open circuit, high resistance ▪ Water contamination in the fuel system ▪ Loose or damaged fuel filler cap seal	▪ Prioritized Checks to Perform ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion ▪ Refer to electrical circuit diagrams and check the flex fuel sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance ▪ Drain off the contaminated fuel. Install a new flex fuel filter element as necessary and replace the contaminated fuel. Check the fuel filler cap seal for wear or damage. Using the Jaguar Land Rover approved diagnostic equipment, run application - ECU Diagnostics / Powertrain Control Module (PCM) / Clear Diagnostic Trouble Code(s). Retest
P0177-26	Fuel Composition Sensor Circuit Range/Performance - Signal rate of change below threshold	▪ Electrical Cause ▪ Yes ▪ Mechanical Cause ▪ Yes ▪ Control Module Cavity ▪ Circuit reference - I_T_FFS - ▪ Prioritized List of Possible Causes ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Flex fuel sensor circuit, short circuit to ground, short circuit to power, open circuit, high resistance ▪ Flex fuel sensor failure	▪ Prioritized Checks to Perform ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion ▪ Refer to electrical circuit diagrams and check the flex fuel sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance ▪ Install a new flex fuel sensor as required ▪ Using the Jaguar Land Rover approved diagnostic equipment, run application - ECU Diagnostics / Powertrain Control Module (PCM) / Clear Diagnostic Trouble Code(s). Retest
P0177-36	Fuel Composition Sensor Circuit Range/Performance - Signal frequency too low	▪ Electrical Cause ▪ Yes ▪ Mechanical Cause ▪ Yes ▪ Control Module Cavity ▪ Circuit reference - I_T_FFS - ▪ Prioritized List of Possible Causes ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Flex fuel sensor circuit, short circuit to ground, short circuit to power, open circuit, high resistance ▪ Water contamination in the fuel system ▪ Loose or damaged fuel filler cap seal	▪ Prioritized Checks to Perform ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion ▪ Refer to electrical circuit diagrams and check the flex fuel sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance ▪ Drain off the contaminated fuel. Install a new flex fuel filter element as necessary and replace the contaminated fuel. Check the fuel filler cap seal for wear or damage. Using the Jaguar Land Rover approved diagnostic equipment, run application - ECU Diagnostics / Powertrain Control Module (PCM) / Clear Diagnostic Trouble Code(s). Retest

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION

P0177-37	Fuel Composition Sensor Circuit Range/Performance - Signal frequency too high	▪ Electrical Cause
		  ▪ Yes
		▪ Mechanical Cause
		  ▪ Yes
		▪ Control Module Cavity
		  ▪ Circuit reference - I_T_FFS -
		▪ Prioritized List of Possible Causes
		  ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion
		  ▪ Flex fuel sensor circuit, short circuit to ground, short circuit to power, open circuit, high resistance
		  ▪ Water contamination in the fuel system
		  ▪ Loose or damaged fuel filler cap seal
			▪ Prioritized Checks to Perform
			  ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion
			  ▪ Refer to electrical circuit diagrams and check the flex fuel sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance
			  ▪ Drain off the contaminated fuel. Install a new flex fuel filter element as necessary and replace the contaminated fuel. Check the fuel filler cap seal for wear or damage. Using the Jaguar Land Rover approved diagnostic equipment, run application - ECU Diagnostics / Powertrain Control Module (PCM) / Clear Diagnostic Trouble Code(s). Retest

P0177-38	Fuel Composition Sensor Circuit Range/Performance - Signal frequency incorrect	▪ Electrical Cause
		  ▪ Yes
		▪ Mechanical Cause
		  ▪ Yes
		▪ Control Module Cavity
		  ▪ Circuit reference - I_T_FFS -
		▪ Prioritized List of Possible Causes
		  ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion
		  ▪ Flex fuel sensor circuit, short circuit to ground, short circuit to power, open circuit, high resistance
		  ▪ Flex fuel sensor failure
			▪ Prioritized Checks to Perform
			  ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion
			  ▪ Refer to electrical circuit diagrams and check the flex fuel sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance
			  ▪ Install a new flex fuel sensor as required
			  ▪ Using the Jaguar Land Rover approved diagnostic equipment, run application - ECU Diagnostics / Powertrain Control Module (PCM) / Clear Diagnostic Trouble Code(s). Retest

P0177-40	Fuel Composition Sensor Circuit Range/Performance - Reserved by document	▪ Electrical Cause
		  ▪ Yes
		▪ Mechanical Cause
		  ▪ Yes
		▪ Control Module Cavity
		  ▪ Circuit reference - I_T_FFS -
		▪ Prioritized List of Possible Causes
		  ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion
		  ▪ Flex fuel sensor circuit, short circuit to ground, short circuit to power, open circuit, high resistance
		  ▪ Flex fuel sensor failure
			▪ Prioritized Checks to Perform
			  ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion
			  ▪ Refer to electrical circuit diagrams and check the flex fuel sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance
			  ▪ Install a new flex fuel sensor as required
			  ▪ Using the Jaguar Land Rover approved diagnostic equipment, run application - ECU Diagnostics / Powertrain Control Module (PCM) / Clear Diagnostic Trouble Code(s). Retest

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION
P0177-64	Fuel Composition Sensor Circuit Range/Performance - Signal plausibility failure	▪ Electrical Cause ▪ Yes ▪ Mechanical Cause ▪ Yes ▪ Control Module Cavity ▪ Circuit reference - I_T_FFS - ▪ Prioritized List of Possible Causes ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Flex fuel sensor circuit, short circuit to ground, short circuit to power, open circuit, high resistance ▪ Flex fuel sensor failure	▪ Prioritized Checks to Perform ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion ▪ Refer to electrical circuit diagrams and check the flex fuel sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance ▪ Install a new flex fuel sensor as required ▪ Using the Jaguar Land Rover approved diagnostic equipment, run application - ECU Diagnostics / Powertrain Control Module (PCM) / Clear Diagnostic Trouble Code(s). Retest
P0178-00	Fuel Composition Sensor Circuit Low - No sub type information	▪ Electrical Cause ▪ Yes ▪ Mechanical Cause ▪ Yes ▪ Control Module Cavity ▪ Circuit reference - I_T_FFS - ▪ Prioritized List of Possible Causes ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Flex fuel sensor circuit, short circuit to ground, short circuit to power, open circuit, high resistance ▪ Flex fuel sensor failure	▪ Prioritized Checks to Perform ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion ▪ Refer to electrical circuit diagrams and check the flex fuel sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance ▪ Install a new flex fuel sensor as required ▪ Using the Jaguar Land Rover approved diagnostic equipment, run application - ECU Diagnostics / Powertrain Control Module (PCM) / Clear Diagnostic Trouble Code(s). Retest
P0179-00	Fuel Composition Sensor Circuit High - No sub type information	▪ Electrical Cause ▪ Yes ▪ Mechanical Cause ▪ Yes ▪ Control Module Cavity ▪ Circuit reference - I_T_FFS - ▪ Prioritized List of Possible Causes ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Flex fuel sensor circuit, short circuit to ground, short circuit to power, open circuit, high resistance ▪ Flex fuel sensor failure	▪ Prioritized Checks to Perform ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion ▪ Refer to electrical circuit diagrams and check the flex fuel sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance ▪ Install a new flex fuel sensor as required ▪ Using the Jaguar Land Rover approved diagnostic equipment, run application - ECU Diagnostics / Powertrain Control Module (PCM) / Clear Diagnostic Trouble Code(s). Retest

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION

P017B-2A	Cylinder Head Temperature Sensor Circuit Range/Performance - Signal stuck in range	▪ Electrical Cause
		  ▪ Yes
		▪ Mechanical Cause
		  ▪ Yes
		▪ Control Module Cavity
		  ▪ Circuit reference - I_A_CHT -
		  ▪ Circuit reference - G_R_CHT -
		▪ Monitor Description
		  ▪ Cylinder head temperature sensor stuck since start
		▪ Prioritized List of Possible Causes
		  ▪ Cooling system mechanical and fluid integrity
		  ▪ Cooling fan system mechanical integrity
		  ▪ Thermostat system mechanical integrity
		  ▪ Cylinder head temperature sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance
		  ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion
		  ▪ Cylinder head temperature sensor failure
		  ▪ Engine coolant level or flow failure
			▪ Prioritized Checks to Perform
			  ▪ Check cooling system for mechanical and fluid integrity
			  ▪ Check cooling fan system for mechanical integrity
			  ▪ Check thermostat system for mechanical integrity
			  ▪ Refer to the electrical circuit diagrams and check the cylinder head temperature sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance
			  ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion
			  ▪ Install a new cylinder head temperature sensor as required
			  ▪ Refer to the workshop manual and check the engine cooling system to make sure the coolant condition, level and flow is correct
			  ▪ Using the Jaguar Land Rover approved diagnostic equipment, run application - ECU Diagnostics / Powertrain Control Module (PCM) / Clear Diagnostic Trouble Code(s). Retest

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION
P017B-62	Cylinder Head Temperature Sensor Circuit Range/Performance - Signal compare failure	▪ Electrical Cause ▪ Yes ▪ Mechanical Cause ▪ Yes ▪ Control Module Cavity ▪ Circuit reference - I_A_CHT - ▪ Circuit reference - G_R_CHT - ▪ Monitor Description ▪ Cylinder head temperature sensor reading too high or too low at cold star ▪ Prioritized List of Possible Causes ▪ Cooling system mechanical and fluid integrity ▪ Cooling fan system mechanical integrity ▪ Thermostat system mechanical integrity ▪ Cylinder head temperature sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Cylinder head temperature sensor failure ▪ Engine coolant level or flow failure	▪ Prioritized Checks to Perform ▪ Check cooling system for mechanical and fluid integrity ▪ Check cooling fan system for mechanical integrity ▪ Check thermostat system for mechanical integrity ▪ Refer to the electrical circuit diagrams and check the cylinder head temperature sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion ▪ Install a new cylinder head temperature sensor as required ▪ Refer to the workshop manual and check the engine cooling system to make sure the coolant condition, level and flow is correct ▪ Using the Jaguar Land Rover approved diagnostic equipment, run application - ECU Diagnostics / Powertrain Control Module (PCM) / Clear Diagnostic Trouble Code(s). Retest

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION

P017B-64	Cylinder Head Temperature Sensor Circuit Range/Performance - Signal plausibility failure	▪ Electrical Cause
		  ▪ Yes
		▪ Mechanical Cause
		  ▪ Yes
		▪ Control Module Cavity
		  ▪ Circuit reference - I_A_CHT -
		  ▪ Circuit reference - G_R_CHT -
		▪ Monitor Description
		  ▪ Cylinder head temperature sensor reading too high or too low during engine operation
		▪ Prioritized List of Possible Causes
		  ▪ Cooling system mechanical and fluid integrity
		  ▪ Cooling fan system mechanical integrity
		  ▪ Thermostat system mechanical integrity
		  ▪ Cylinder head temperature sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance
		  ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion
		  ▪ Cylinder head temperature sensor failure
		  ▪ Engine coolant level or flow failure
			▪ Prioritized Checks to Perform
			  ▪ Check cooling system for mechanical and fluid integrity
			  ▪ Check cooling fan system for mechanical integrity
			  ▪ Check thermostat system for mechanical integrity
			  ▪ Refer to the electrical circuit diagrams and check the cylinder head temperature sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance
			  ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion
			  ▪ Install a new cylinder head temperature sensor as required
			  ▪ Refer to the workshop manual and check the engine cooling system to make sure the coolant condition, level and flow is correct
			  ▪ Using the Jaguar Land Rover approved diagnostic equipment, run application - ECU Diagnostics / Powertrain Control Module (PCM) / Clear Diagnostic Trouble Code(s). Retest

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION

P017C-16	Cylinder Head Temperature Sensor Circuit Low - Circuit voltage below threshold	▪ Electrical Cause
		  ▪ Yes
		▪ Mechanical Cause
		  ▪ Yes
		▪ Control Module Cavity
		  ▪ Circuit reference - I_A_CHT -
		  ▪ Circuit reference - G_R_CHT -
		▪ Monitor Description
		  ▪ Measured voltage is below the expected signal range
		▪ Prioritized List of Possible Causes
		  ▪ Cooling system mechanical and fluid integrity
		  ▪ Cooling fan system mechanical integrity
		  ▪ Thermostat system mechanical integrity
		  ▪ Cylinder head temperature sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance
		  ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion
		  ▪ Cylinder head temperature sensor failure
		  ▪ Engine coolant level or flow failure
			▪ Prioritized Checks to Perform
			  ▪ Check cooling system for mechanical and fluid integrity
			  ▪ Check cooling fan system for mechanical integrity
			  ▪ Check thermostat system for mechanical integrity
			  ▪ Refer to the electrical circuit diagrams and check the cylinder head temperature sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance
			  ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion
			  ▪ Install a new cylinder head temperature sensor as required
			  ▪ Refer to the workshop manual and check the engine cooling system to make sure the coolant condition, level and flow is correct
			  ▪ Using the Jaguar Land Rover approved diagnostic equipment, run application - ECU Diagnostics / Powertrain Control Module (PCM) / Clear Diagnostic Trouble Code(s). Retest

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION

P017D-17	Cylinder Head Temperature Sensor Circuit High - Circuit voltage above threshold	▪ Electrical Cause
		  ▪ Yes
		▪ Mechanical Cause
		  ▪ Yes
		▪ Control Module Cavity
		  ▪ Circuit reference - I_A_CHT -
		  ▪ Circuit reference - G_R_CHT -
		▪ Monitor Description
		  ▪ Measured voltage is above the expected signal range
		▪ Prioritized List of Possible Causes
		  ▪ Cooling system mechanical and fluid integrity
		  ▪ Cooling fan system mechanical integrity
		  ▪ Thermostat system mechanical integrity
		  ▪ Cylinder head temperature sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance
		  ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion
		  ▪ Cylinder head temperature sensor failure
		  ▪ Engine coolant level or flow failure
			▪ Prioritized Checks to Perform
			  ▪ Check cooling system for mechanical and fluid integrity
			  ▪ Check cooling fan system for mechanical integrity
			  ▪ Check thermostat system for mechanical integrity
			  ▪ Refer to the electrical circuit diagrams and check the cylinder head temperature sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance
			  ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion
			  ▪ Install a new cylinder head temperature sensor as required
			  ▪ Refer to the workshop manual and check the engine cooling system to make sure the coolant condition, level and flow is correct
			  ▪ Using the Jaguar Land Rover approved diagnostic equipment, run application - ECU Diagnostics / Powertrain Control Module (PCM) / Clear Diagnostic Trouble Code(s). Retest

P0181-21	Fuel Temperature Sensor "A" Circuit Range/Performance - Signal amplitude < minimum	▪ Electrical Cause
		  ▪ Yes
		▪ Mechanical Cause
		  ▪ Yes
		▪ Control Module Cavity
		  ▪ Circuit reference - I_A_RAILFTS -
		  ▪ Circuit reference - G_R_RAILFTS -
		▪ Monitor Description
		  ▪ Comparison of fuel rail temperature against modeled, less than modeled value
		▪ Prioritized List of Possible Causes
		  ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion
		  ▪ Fuel temperature sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance
		  ▪ Fuel temperature sensor failure
			▪ Vehicle Conditions to enable DTC Logging strategy
			  ▪ Drive vehicle under steady-state conditions for up to 30 minutes
			▪ Prioritized Checks to Perform
			  ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion
			  ▪ Refer to the electrical circuit diagrams and check the fuel temperature sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance
			  ▪ Using the Jaguar Land Rover approved diagnostic equipment, run application - ECU Diagnostics / Powertrain Control Module (PCM) / Clear Diagnostic Trouble Code(s). Retest
			  ▪ Install a new fuel temperature sensor as required

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION

P0181-22	Fuel Temperature Sensor "A" Circuit Range/Performance - Signal amplitude > maximum	▪ Electrical Cause
		  ▪ Yes
		▪ Mechanical Cause
		  ▪ Yes
		▪ Control Module Cavity
		  ▪ Circuit reference - I_A_RAILFTS -
		  ▪ Circuit reference - G_R_RAILFTS -
		▪ Monitor Description
		  ▪ Comparison of fuel rail temperature against modeled, greater than modeled value
		▪ Prioritized List of Possible Causes
		  ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion
		  ▪ Fuel temperature sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance
		  ▪ Fuel temperature sensor failure
			▪ Vehicle Conditions to enable DTC Logging strategy
			  ▪ Idle vehicle for 30 minutes
			▪ Prioritized Checks to Perform
			  ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion
			  ▪ Refer to the electrical circuit diagrams and check the fuel temperature sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance
			  ▪ Using the Jaguar Land Rover approved diagnostic equipment, run application - ECU Diagnostics / Powertrain Control Module (PCM) / Clear Diagnostic Trouble Code(s). Retest
			  ▪ Install a new fuel temperature sensor as required

P0182-00	Fuel Temperature Sensor "A" Circuit Low - No sub type information	▪ Electrical Cause
		  ▪ Yes
		▪ Mechanical Cause
		  ▪ Yes
		▪ Control Module Cavity
		  ▪ Circuit reference - I_A_RAILFTS -
		  ▪ Circuit reference - G_R_RAILFTS -
		▪ Monitor Description
		  ▪ Fuel rail temperature sensor voltage below threshold
		▪ Prioritized List of Possible Causes
		  ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion
		  ▪ Fuel temperature sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance
		  ▪ Fuel temperature sensor failure
			▪ Vehicle Conditions to enable DTC Logging strategy
			  ▪ Ignition on, start engine and idle for 30 seconds
			▪ Prioritized Checks to Perform
			  ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion
			  ▪ Refer to the electrical circuit diagrams and check the fuel temperature sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance
			  ▪ Using the Jaguar Land Rover approved diagnostic equipment, run application - ECU Diagnostics / Powertrain Control Module (PCM) / Clear Diagnostic Trouble Code(s). Retest
			  ▪ Install a new fuel temperature sensor as required

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION

P0183-00	Fuel Temperature Sensor "A" Circuit High - No sub type information	▪ Electrical Cause
		  ▪ Yes
		▪ Mechanical Cause
		  ▪ Yes
		▪ Control Module Cavity
		  ▪ Circuit reference Signal I_A_RAILFTS
		  ▪ Circuit reference Ground G_R_RAILFTS
		▪ Monitor Description
		  ▪ Fuel rail temperature sensor voltage above threshold
		▪ Prioritized List of Possible Causes
		  ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion
		  ▪ Fuel temperature sensor circuit, short circuit to ground, short circuit to power, open circuit, high resistance
		  ▪ Fuel temperature sensor failure
			▪ Vehicle Conditions to enable DTC Logging strategy
			  ▪ Ignition on, start engine and idle for 30 seconds
			▪ Prioritized Checks to Perform
			  ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion
			  ▪ Refer to the electrical circuit diagrams and check the fuel temperature sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance
			  ▪ Using the Jaguar Land Rover approved diagnostic equipment, run application - ECU Diagnostics / Powertrain Control Module (PCM) / Clear Diagnostic Trouble Code(s). Retest
			  ▪ Install a new fuel temperature sensor as required

P0191-71	Fuel Rail Pressure Sensor Circuit Range/Performance Bank 1 - Actuator stuck	▪ Electrical Cause
		  ▪ Yes
		▪ Mechanical Cause
		  ▪ Yes
		▪ Control Module Cavity
		  ▪ Circuit reference - O_V_5VRAILPS -
		  ▪ Circuit reference - I_A_RAILPS -
		  ▪ Circuit reference - G_R_RAILPS -
		  ▪ Circuit reference - I_A_RAILFTS -
		▪ Monitor Description
		  ▪ The fuel rail pressure is sampled over 4 injections, if no delta is seen then the fault is raised
		▪ Prioritized List of Possible Causes
		  ▪ Fuel rail pressure sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance
		  ▪ Fuel rail pressure sensor failure
			▪ Vehicle Conditions to Enable DTC Logging Strategy
			  ▪ Engine running below 4520 rpm, fuel pressure above 3 mPa, fuel mass threshold about to 5%, sensor voltage threshold less than 4.5 Volts
			▪ Prioritized Checks to Perform
			  ▪ Refer to the electrical circuit diagrams and check the fuel rail pressure sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance
			  ▪ Using the Jaguar Land Rover approved diagnostic equipment, run application - ECU Diagnostics / Powertrain Control Module (PCM) / Clear Diagnostic Trouble Code(s). Retest
			  ▪ Install a new fuel rail pressure sensor as required

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION

P0191-84	Fuel Rail Pressure Sensor Circuit Range/Performance Bank 1 - Signal below allowable range	▪ Electrical Cause
		  ▪ Yes
		▪ Mechanical Cause
		  ▪ Yes
		▪ Control Module Cavity
		  ▪ Circuit reference - O_V_5VRAILPS -
		  ▪ Circuit reference - I_A_RAILPS -
		  ▪ Circuit reference - G_R_RAILPS -
		  ▪ Circuit reference - I_A_RAILFTS -
		▪ Prioritized List of Possible Causes
		  ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion
		  ▪ Fuel rail pressure sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance
		  ▪ Fuel rail pressure sensor failure
			▪ Vehicle Conditions to Enable DTC Logging Strategy
			  ▪ Engine running for greater than 1 minute
			▪ Prioritized Checks to Perform
			  ▪ Check connector is not disconnected, connector pin is not backed out
			  ▪ Refer to the electrical circuit diagrams and check the fuel rail pressure sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance
			  ▪ Using the Jaguar Land Rover approved diagnostic equipment, run application - ECU Diagnostics / Powertrain Control Module (PCM) / Clear Diagnostic Trouble Code(s). Retest
			  ▪ Install a new fuel rail pressure sensor as required

P0191-85	Fuel Rail Pressure Sensor Circuit Range/Performance Bank 1 - Signal above allowable range	▪ Electrical Cause
		  ▪ Yes
		▪ Mechanical Cause
		  ▪ Yes
		▪ Control Module Cavity
		  ▪ Circuit reference - O_V_5VRAILPS -
		  ▪ Circuit reference - I_A_RAILPS -
		  ▪ Circuit reference - G_R_RAILPS -
		  ▪ Circuit reference - I_A_RAILFTS -
		▪ Prioritized List of Possible Causes
		  ▪ Incorrect calibration installed in to the powertrain control module
		  ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion
		  ▪ Fuel rail pressure sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance
		  ▪ Fuel rail pressure sensor failure
			NOTE:
			If P0191-85 is set it is likely to cause the following Diagnostic Trouble Code(s) to set: P144C-00, P2178-00, P2187-00, P2274-00. Complete the repair actions for P0191-85 first, clear the Diagnostic Trouble Code(s) and retest.
			▪ Vehicle Conditions to Enable DTC Logging Strategy
			  ▪ Engine running for greater than 1 minute
			▪ Prioritized Checks to Perform
			  ▪ Using the Jaguar Land Rover approved diagnostic equipment, reconfigure the powertrain control module with the latest level software
			  ▪ Run the engine until coolant indicates fully warm on the instrument cluster, then switch the engine off for a minimum of 6 hours. Restart the vehicle and check the DTC has not returned. If the DTC returns perform the additional checks below
			  ▪ Check connector is not disconnected, connector pin is not backed out
			  ▪ Refer to the electrical circuit diagrams and check the fuel rail pressure sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance
			  ▪ Using the Jaguar Land Rover approved diagnostic equipment, run application - ECU Diagnostics / Powertrain Control Module (PCM) / Clear Diagnostic Trouble Code(s). Retest
			  ▪ Install a new fuel rail pressure sensor as required

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION

P0192-00	Fuel Rail Pressure Sensor Circuit Low Bank 1 - No sub type information	▪ Electrical Cause
		  ▪ Yes
		▪ Mechanical Cause
		  ▪ Yes
		▪ Control Module Cavity
		  ▪ Circuit reference - O_V_5VRAILPS -
		  ▪ Circuit reference - T_A_RAILPS -
		  ▪ Circuit reference - G_R_RAILPS -
		  ▪ Circuit reference - I_A_RAILFTS -
		▪ Monitor Description
		  ▪ High pressure fuel pressure sensor voltage below threshold
		▪ Prioritized List of Possible Causes
		  ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion
		  ▪ Fuel rail pressure sensor circuit, short circuit to ground
		  ▪ Fuel rail pressure sensor 5 volt power supply circuit, open circuit, high resistance
		  ▪ Fuel rail pressure sensor failure
			▪ Vehicle Conditions to enable DTC Logging strategy
			  ▪ Ignition on, start engine and idle for 30 seconds
			▪ Prioritized Checks to Perform
			  ▪ Check connector is not disconnected, connector pin is not backed out
			  ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion
			  ▪ Refer to the electrical circuit diagrams and check the fuel rail pressure sensor circuit for short circuit to ground
			  ▪ Check fuel rail pressure sensor 5 volt power supply circuit for open circuit, high resistance
			  ▪ Using the Jaguar Land Rover approved diagnostic equipment, run application - ECU Diagnostics / Powertrain Control Module (PCM) / Clear Diagnostic Trouble Code(s). Retest
			  ▪ Install a new fuel rail pressure sensor as required

P0193-00	Fuel Rail Pressure Sensor Circuit High Bank 1 - No sub type information	▪ Electrical Cause
		  ▪ Yes
		▪ Mechanical Cause
		  ▪ Yes
		▪ Control Module Cavity
		  ▪ Circuit reference - O_V_5VRAILPS -
		  ▪ Circuit reference - T_A_RAILPS -
		  ▪ Circuit reference - G_R_RAILPS -
		  ▪ Circuit reference - I_A_RAILFTS -
		▪ Monitor Description
		  ▪ High pressure fuel pressure sensor voltage above threshold
		▪ Prioritized List of Possible Causes
		  ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion
		  ▪ Fuel rail pressure sensor circuit, short circuit to power
		  ▪ Fuel rail pressure sensor ground supply circuit, open circuit, high resistance
		  ▪ Fuel rail pressure sensor failure
			▪ Vehicle Conditions to enable DTC Logging strategy
			  ▪ Ignition on, start engine and idle for 30 seconds
			▪ Prioritized Checks to Perform
			  ▪ Check connector is not disconnected, connector pin is not backed out
			  ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion
			  ▪ Refer to the electrical circuit diagrams and check the fuel rail pressure sensor circuit for short circuit to power
			  ▪ Check fuel rail pressure sensor ground supply circuit for open circuit, high resistance
			  ▪ Using the Jaguar Land Rover approved diagnostic equipment, run application - ECU Diagnostics / Powertrain Control Module (PCM) / Clear Diagnostic Trouble Code(s). Retest
			  ▪ Install a new fuel rail pressure sensor as required

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION
P0196-03	Engine Oil Temperature Sensor "A" Range/Performance - FM (Frequency Modulated) / PWM (Pulse Width Modulated) failures	▪ Electrical Cause ▪ Yes ▪ Mechanical Cause ▪ Yes ▪ Control Module Cavity ▪ Circuit reference - I_T_OPTS - ▪ Circuit reference - O_V_5VOPTS - ▪ Circuit reference - G_R_OPTS - ▪ Monitor Description ▪ Oil temperature sensor - PWM out of range ▪ Prioritized List of Possible Causes ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Oil pressure and temperature sensor fuse failure ▪ Oil pressure and temperature sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Oil pressure and temperature sensor failure	▪ Vehicle Conditions to Enable DTC Logging Strategy ▪ Ignition on, engine running ▪ Prioritized Checks to Perform ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion ▪ Check the integrity of the oil pressure and temperature sensor fuse. Rectify as necessary ▪ Refer to the electrical circuit diagrams and check the oil pressure and temperature sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance. Repair the wiring harness as necessary ▪ Install a new oil pressure and temperature sensor only when diagnosed as failed ▪ Using the Jaguar Land Rover approved diagnostic equipment, run application - ECU Diagnostics / Powertrain Control Module (PCM) / Clear Diagnostic Trouble Code(s). Retest
P0196-2A	Engine Oil Temperature Sensor "A" Range/Performance - Signal stuck in range	▪ Electrical Cause ▪ Yes ▪ Mechanical Cause ▪ Yes ▪ Control Module Cavity ▪ Circuit reference - I_T_OPTS - ▪ Circuit reference - O_V_5VOPTS - ▪ Circuit reference - G_R_OPTS - ▪ Monitor Description ▪ Oil temperature sensor stuck since start ▪ Prioritized List of Possible Causes ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Oil pressure and temperature sensor fuse failure ▪ Oil pressure and temperature sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Oil pressure and temperature sensor failure	▪ Vehicle Conditions to Enable DTC Logging Strategy ▪ Start and drive vehicle at steady state, low load condition for 20 minutes ▪ Prioritized Checks to Perform ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion ▪ Check the integrity of the oil pressure and temperature sensor fuse. Rectify as necessary ▪ Refer to the electrical circuit diagrams and check the oil pressure and temperature sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance. Repair the wiring harness as necessary ▪ Install a new oil pressure and temperature sensor only when diagnosed as failed ▪ Using the Jaguar Land Rover approved diagnostic equipment, run application - ECU Diagnostics / Powertrain Control Module (PCM) / Clear Diagnostic Trouble Code(s). Retest

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION
P0196-62	Engine Oil Temperature Sensor "A" Range/Performance - Signal compare failure	▪ Electrical Cause ▪ Yes ▪ Mechanical Cause ▪ Yes ▪ Control Module Cavity ▪ Circuit reference - I_T_OPTS - ▪ Circuit reference - O_V_5VOPTS - ▪ Circuit reference - G_R_OPTS - ▪ Monitor Description ▪ Oil temperature sensor reading too high or too low at cold start ▪ Prioritized List of Possible Causes ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Oil pressure and temperature sensor fuse failure ▪ Oil pressure and temperature sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Oil pressure and temperature sensor failure	▪ Vehicle Conditions to Enable DTC Logging Strategy ▪ Start engine and allow to idle for 10 seconds following an engine off period greater than 8 hours ▪ Prioritized Checks to Perform ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion ▪ Check the integrity of the oil pressure and temperature sensor fuse. Rectify as necessary ▪ Refer to the electrical circuit diagrams and check the oil pressure and temperature sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance. Repair the wiring harness as necessary ▪ Install a new oil pressure and temperature sensor only when diagnosed as failed ▪ Using the Jaguar Land Rover approved diagnostic equipment, run application - ECU Diagnostics / Powertrain Control Module (PCM) / Clear Diagnostic Trouble Code(s). Retest
P0196-64	Engine Oil Temperature Sensor "A" Range/Performance - Signal plausibility failure	▪ Electrical Cause ▪ Yes ▪ Mechanical Cause ▪ Yes ▪ Control Module Cavity ▪ Circuit reference - I_T_OPTS - ▪ Circuit reference - O_V_5VOPTS - ▪ Circuit reference - G_R_OPTS - ▪ Monitor Description ▪ Oil temperature sensor reading too high or too low during engine operation ▪ Prioritized List of Possible Causes ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Oil pressure and temperature sensor fuse failure ▪ Oil pressure and temperature sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Oil pressure and temperature sensor failure	▪ Vehicle Conditions to Enable DTC Logging Strategy ▪ Start and drive vehicle at steady state, low load condition for 20 minutes ▪ Prioritized Checks to Perform ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion ▪ Check the integrity of the oil pressure and temperature sensor fuse. Rectify as necessary ▪ Refer to the electrical circuit diagrams and check the oil pressure and temperature sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance. Repair the wiring harness as necessary ▪ Install a new oil pressure and temperature sensor only when diagnosed as failed ▪ Using the Jaguar Land Rover approved diagnostic equipment, run application - ECU Diagnostics / Powertrain Control Module (PCM) / Clear Diagnostic Trouble Code(s). Retest

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION
P0196-86	Engine Oil Temperature Sensor "A" Range/Performance - Signal invalid	▪ Electrical Cause ▪ Yes ▪ Mechanical Cause ▪ Yes ▪ Control Module Cavity ▪ Circuit reference - I_T_OPTS - ▪ Circuit reference - O_V_5VOPTS - ▪ Circuit reference - G_R_OPTS - ▪ Monitor Description ▪ Oil temperature sensor hardware fault ▪ Prioritized List of Possible Causes ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Oil pressure and temperature sensor fuse failure ▪ Oil pressure and temperature sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Oil pressure and temperature sensor failure	▪ Vehicle Conditions to Enable DTC Logging Strategy ▪ Ignition on, engine running ▪ Prioritized Checks to Perform ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion ▪ Check the integrity of the oil pressure and temperature sensor fuse. Rectify as necessary ▪ Refer to the electrical circuit diagrams and check the oil pressure and temperature sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance. Repair the wiring harness as necessary ▪ Install a new oil pressure and temperature sensor only when diagnosed as failed ▪ Using the Jaguar Land Rover approved diagnostic equipment, run application - ECU Diagnostics / Powertrain Control Module (PCM) / Clear Diagnostic Trouble Code(s). Retest
P01BA-64	Engine Oil Temperature Sensor "B" Range/Performance - Signal plausibility failure	▪ Electrical Cause ▪ Yes ▪ Mechanical Cause ▪ Yes ▪ Control Module Cavity ▪ Circuit reference - I_A_UOTS - ▪ Circuit reference - G_R_UOTS - ▪ Prioritized List of Possible Causes ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Oil temperature sensor fuse failure ▪ Oil temperature sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Oil temperature sensor failure	▪ Prioritized Checks to Perform ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion ▪ Check the integrity of the oil temperature sensor fuse. Rectify as necessary ▪ Refer to the electrical circuit diagrams and check the oil temperature sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance. Repair the wiring harness as necessary ▪ Install a new oil temperature sensor only when diagnosed as failed ▪ Using the Jaguar Land Rover approved diagnostic equipment, run application - ECU Diagnostics / Powertrain Control Module (PCM) / Clear Diagnostic Trouble Code(s). Retest

<table>
<tr><th>DTC</th><th>DESCRIPTION</th><th>POSSIBLE CAUSES</th><th>ACTION</th></tr>
<tr>
<td>P01BA-84</td>
<td>Engine Oil Temperature Sensor "B" Range/Performance - Signal below allowable range</td>
<td>

Electrical Cause
Yes
Mechanical Cause
Yes
Control Module Cavity
Circuit reference I_A_UOTS
Circuit reference G_R_UOTS
Monitor Description
Oil temperature rationality low
Component
Uniair oil temperature sensor
Prioritized List of Possible Causes
Oil temperature sensor failure
Oil temperature sensor - Wiring integrity
Short circuit to ground
Short circuit to power
Open circuit
High resistance
Connector is disconnected, connector pin is backed out, connector pin corrosion

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Vehicle Conditions to enable DTC Logging Strategy
Start and drive vehicle at various part load conditions for 20 minutes
Prioritized Checks to Perform
Refer to the electrical circuit diagrams and check the resistance values at the oil temperature sensor. At an ambient air temperature (23°C), resistance values measured between the two poles of the oil temperature sensor should be greater than 2100 Ohms. Visually inspect the oil temperature sensor connector(s) for signs of water ingress, and pins for damage and/or corrosion. If resistance values are not to specification and/or if the oil temperature sensor connector is damaged, install a new oil temperature sensor
Inspect connectors for signs of water ingress, and pins for damage and/or corrosion
Refer to the electrical circuit diagrams and check the oil temperature sensor circuit for
Short circuit to ground
Short circuit to power
Open circuit
High resistance - resistance should be less than 0.1 Ohm
Signs of water ingress, pins damage and/or corrosion of harness connector(s)
If harness resistance values are not to specification and/or if the harness connector(s) are damaged, repair or replace the harness as required
Visually inspect the oil temperature sensor circuit connector(s) at the powertrain control module for signs of water ingress, and pins for damage and/or corrosion. If the connector is damaged on the powertrain control module, install a new powertrain control module as required
Using the Jaguar Land Rover approved diagnostic equipment, check the powertrain control module for cooling system related Diagnostic Trouble Code(s) and refer to the relevant DTC index
Visually inspect the engine compartment for obstructions to the expected air flow around the engine and/or for the presence of any non-standard or additional insulating materials. Rectify as required
Using the Jaguar Land Rover approved diagnostic equipment, clear all stored Diagnostic Trouble Code(s) and retest

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DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION
P01BA-85	Engine Oil Temperature Sensor "B" Range/Performance - Signal above allowable range	▪ Electrical Cause ▪ Yes ▪ Mechanical Cause ▪ Yes ▪ Control Module Cavity ▪ Circuit reference I_A_UOTS ▪ Circuit reference G_R_UOTS ▪ Monitor Description ▪ Oil temperature rationality high ▪ Component ▪ Uniair oil temperature sensor ▪ Prioritized List of Possible Causes ▪ Oil temperature sensor failure ▪ Oil temperature sensor - Wiring integrity ▪ Short circuit to ground ▪ Short circuit to power ▪ Open circuit ▪ High resistance ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion	▪ Vehicle Conditions to enable DTC Logging Strategy ▪ Start and drive vehicle at various part load conditions for 20 minutes ▪ Prioritized Checks to Perform ▪ Refer to the electrical circuit diagrams and check the resistance values at the oil temperature sensor. At an ambient air temperature (23°C), resistance values measured between the two poles of the oil temperature sensor should be greater than 2100 Ohms. Visually inspect the oil temperature sensor connector(s) for signs of water ingress, and pins for damage and/or corrosion. If resistance values are not to specification and/or if the oil temperature sensor connector is damaged, install a new oil temperature sensor ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion ▪ Refer to the electrical circuit diagrams and check the oil temperature sensor circuit for ▪ Short circuit to ground ▪ Short circuit to power ▪ Open circuit ▪ High resistance - resistance should be less than 0.1 Ohm ▪ Signs of water ingress, pins damage and/or corrosion of harness connector(s) ▪ If harness resistance values are not to specification and/or if the harness connector(s) are damaged, repair or replace the harness as required ▪ Visually inspect the oil temperature sensor circuit connector(s) at the powertrain control module for signs of water ingress, and pins for damage and/or corrosion. If the connector is damaged on the powertrain control module, install a new powertrain control module as required ▪ Using the Jaguar Land Rover approved diagnostic equipment, check the powertrain control module for cooling system related Diagnostic Trouble Code(s) and refer to the relevant DTC index ▪ Visually inspect the engine compartment for obstructions to the expected air flow around the engine and/or for the presence of any non-standard or additional insulating materials. Rectify as required ▪ Using the Jaguar Land Rover approved diagnostic equipment, clear all stored Diagnostic Trouble Code(s) and retest

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION

P01BB-00	Engine Oil Temperature Sensor "B" Circuit Low - No sub type information	▪ Electrical Cause
		  ▪ Yes
		▪ Mechanical Cause
		  ▪ No
		▪ Control Module Cavity
		  ▪ Circuit reference I_A_UOTS
		  ▪ Circuit reference G_R_UOTS
		▪ Monitor Description
		  ▪ Continuously Variable Camshaft Timing lift oil temp sensor short circuit to ground
		▪ Component
		  ▪ Uniair oil temperature sensor
		▪ Prioritized List of Possible Causes
		▪ Oil temperature sensor failure
		▪ Oil temperature sensor - Wiring integrity
		  ▪ Short circuit to ground
		  ▪ Short circuit to power
		  ▪ Open circuit
		  ▪ High resistance
		▪ Connector is disconnected, connector pin is backed out, connector pin corrosion
			▪ Vehicle Conditions to enable DTC Logging Strategy
			  ▪ Ignition on, Engine running
			▪ Prioritized Checks to Perform
			▪ Refer to the electrical circuit diagrams and check the resistance values at the oil temperature sensor. At an ambient air temperature (23°C), resistance values measured between the two poles of the oil temperature sensor should be greater than 2100 Ohms. Visually inspect the oil temperature sensor connector(s) for signs of water ingress, and pins for damage and/or corrosion. If resistance values are not to specification and/or if the oil temperature sensor connector is damaged, install a new oil temperature sensor
			▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion
			▪ Refer to the electrical circuit diagrams and check the oil temperature sensor circuit for
			  ▪ Short circuit to ground
			  ▪ Short circuit to power
			  ▪ Open circuit
			  ▪ High resistance - resistance should be less than 0.1 Ohm
			  ▪ Signs of water ingress, pins damage and/or corrosion of harness connector(s)
			▪ If harness resistance values are not to specification and/or if the harness connector(s) are damaged, repair or replace the harness as required
			▪ Visually inspect the oil temperature sensor circuit connector(s) at the powertrain control module for signs of water ingress, and pins for damage and/or corrosion. If the connector is damaged on the powertrain control module, install a new powertrain control module as required
			▪ Using the Jaguar Land Rover approved diagnostic equipment, clear all stored Diagnostic Trouble Code(s) and retest

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION
P01BC-00	Engine Oil Temperature Sensor "B" Circuit High - No sub type information	- Electrical Cause - Yes - Mechanical Cause - No - Control Module Cavity - Circuit reference I_A_UOTS - Circuit reference G_R_UOTS - Monitor Description - Continuously Variable Camshaft Timing lift oil temp sensor open circuit, short circuit to power - Component - Uniair oil temperature sensor - Prioritized List of Possible Causes - Oil temperature sensor failure - Oil temperature sensor - Wiring integrity - Short circuit to ground - Short circuit to power - Open circuit - High resistance - Connector is disconnected, connector pin is backed out, connector pin corrosion	- Vehicle Conditions to enable DTC Logging Strategy - Ignition on, Engine running - Prioritized Checks to Perform - Refer to the electrical circuit diagrams and check the resistance values at the oil temperature sensor. At an ambient air temperature (23°C), resistance values measured between the two poles of the oil temperature sensor should be greater than 2100 Ohms. Visually inspect the oil temperature sensor connector(s) for signs of water ingress, and pins for damage and/or corrosion. If resistance values are not to specification and/or if the oil temperature sensor connector is damaged, install a new oil temperature sensor - Inspect connectors for signs of water ingress, and pins for damage and/or corrosion - Refer to the electrical circuit diagrams and check the oil temperature sensor circuit for - Short circuit to ground - Short circuit to power - Open circuit - High resistance - resistance should be less than 0.1 Ohm - Signs of water ingress, pins damage and/or corrosion of harness connector(s) - If harness resistance values are not to specification and/or if the harness connector(s) are damaged, repair or replace the harness as required - Visually inspect the oil temperature sensor circuit connector(s) at the powertrain control module for signs of water ingress, and pins for damage and/or corrosion. If the connector is damaged on the powertrain control module, install a new powertrain control module as required - Using the Jaguar Land Rover approved diagnostic equipment, clear all stored Diagnostic Trouble Code(s) and retest

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION

P01BD-86	Engine Oil Temperature Sensor "B" Circuit Intermittent/Erratic - Signal invalid	▪ Electrical Cause
		  ▪ Yes
		▪ Mechanical Cause
		  ▪ Yes
		▪ Control Module Cavity
		  ▪ Circuit reference I_A_UOTS
		  ▪ Circuit reference G_R_UOTS
		▪ Monitor Description
		  ▪ Maximum variation of continuously Variable Camshaft Timing lift oil temperature
		▪ Component
		  ▪ Uniair oil temperature sensor
		▪ Prioritized List of Possible Causes
		▪ Oil temperature sensor failure
		▪ Oil temperature sensor - Wiring integrity
		  ▪ Short circuit to ground
		  ▪ Short circuit to power
		  ▪ Open circuit
		  ▪ High resistance
		▪ Connector is disconnected, connector pin is backed out, connector pin corrosion
			▪ Vehicle Conditions to enable DTC Logging Strategy
			  ▪ Ignition on, Engine running
			▪ Prioritized Checks to Perform
			▪ Refer to the electrical circuit diagrams and check the resistance values at the oil temperature sensor. At an ambient air temperature (23°C), resistance values measured between the two poles of the oil temperature sensor should be greater than 2100 Ohms. Visually inspect the oil temperature sensor connector(s) for signs of water ingress, and pins for damage and/or corrosion. If resistance values are not to specification and/or if the oil temperature sensor connector is damaged, install a new oil temperature sensor
			▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion
			▪ Refer to the electrical circuit diagrams and check the oil temperature sensor circuit for
			  ▪ Short circuit to ground
			  ▪ Short circuit to power
			  ▪ Open circuit
			  ▪ High resistance - resistance should be less than 0.1 Ohm
			  ▪ Signs of water ingress, pins damage and/or corrosion of harness connector(s)
			▪ If harness resistance values are not to specification and/or if the harness connector(s) are damaged, repair or replace the harness as required
			▪ Visually inspect the oil temperature sensor circuit connector(s) at the powertrain control module for signs of water ingress, and pins for damage and/or corrosion. If the connector is damaged on the powertrain control module, install a new powertrain control module as required
			▪ Using the Jaguar Land Rover approved diagnostic equipment, clear all stored Diagnostic Trouble Code(s) and retest

P01E4-2A	Engine Coolant Temperature Sensor 3 Circuit Range/Performance - Signal stuck in range	▪ Electrical Cause
		  ▪ Yes
		▪ Mechanical Cause
		  ▪ Yes
		▪ Control Module Cavity
		  ▪ Circuit reference - I_A_CBT -
		  ▪ Circuit reference - G_R_CBT -
		▪ Prioritized List of Possible Causes
		  ▪ Cylinder block temperature sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance
		  ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion
		  ▪ Cylinder block temperature sensor failure
			▪ Prioritized Checks to Perform
			  ▪ Refer to the electrical circuit diagrams and check the cylinder block temperature sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance
			  ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion
			  ▪ Install a new cylinder block temperature sensor only when diagnosed as failed
			  ▪ Using the Jaguar Land Rover approved diagnostic equipment, run application - ECU Diagnostics / Powertrain Control Module (PCM) / Clear Diagnostic Trouble Code(s). Retest

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION
P01E4-62	Engine Coolant Temperature Sensor 3 Circuit Range/Performance - Signal compare failure	▪ Electrical Cause ▪ Yes ▪ Mechanical Cause ▪ Yes ▪ Control Module Cavity ▪ Circuit reference - I_A_CBT - ▪ Circuit reference - G_R_CBT - ▪ Prioritized List of Possible Causes ▪ Cylinder block temperature sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Cylinder block temperature sensor failure	▪ Prioritized Checks to Perform ▪ Refer to the electrical circuit diagrams and check the cylinder block temperature sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion ▪ Install a new cylinder block temperature sensor only when diagnosed as failed ▪ Using the Jaguar Land Rover approved diagnostic equipment, run application - ECU Diagnostics / Powertrain Control Module (PCM) / Clear Diagnostic Trouble Code(s). Retest
P01E4-64	Engine Coolant Temperature Sensor 3 Circuit Range/Performance - Signal plausibility failure	▪ Electrical Cause ▪ Yes ▪ Mechanical Cause ▪ Yes ▪ Control Module Cavity ▪ Circuit reference - I_A_CBT - ▪ Circuit reference - G_R_CBT - ▪ Prioritized List of Possible Causes ▪ Cylinder block temperature sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Cylinder block temperature sensor failure	▪ Prioritized Checks to Perform ▪ Refer to the electrical circuit diagrams and check the cylinder block temperature sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion ▪ Install a new cylinder block temperature sensor only when diagnosed as failed ▪ Using the Jaguar Land Rover approved diagnostic equipment, run application - ECU Diagnostics / Powertrain Control Module (PCM) / Clear Diagnostic Trouble Code(s). Retest
P01E5-16	Engine Coolant Temperature Sensor 3 Circuit Low - Circuit voltage below threshold	▪ Electrical Cause ▪ Yes ▪ Mechanical Cause ▪ Yes ▪ Control Module Cavity ▪ Circuit reference - I_A_CBT - ▪ Circuit reference - G_R_CBT - ▪ Prioritized List of Possible Causes ▪ Cylinder block temperature sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Cylinder block temperature sensor failure	▪ Prioritized Checks to Perform ▪ Refer to the electrical circuit diagrams and check the cylinder block temperature sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion ▪ Install a new cylinder block temperature sensor only when diagnosed as failed ▪ Using the Jaguar Land Rover approved diagnostic equipment, run application - ECU Diagnostics / Powertrain Control Module (PCM) / Clear Diagnostic Trouble Code(s). Retest

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION

P01E6-17	Engine Coolant Temperature Sensor 3 Circuit High - Circuit voltage above threshold	▪ Electrical Cause
		  ▪ Yes
		▪ Mechanical Cause
		  ▪ Yes
		▪ Control Module Cavity
		  ▪ Circuit reference - I_A_CBT -
		  ▪ Circuit reference - G_R_CBT -
		▪ Prioritized List of Possible Causes
		  ▪ Cylinder block temperature sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance
		  ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion
		  ▪ Cylinder block temperature sensor failure
			▪ Prioritized Checks to Perform
			  ▪ Refer to the electrical circuit diagrams and check the cylinder block temperature sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance
			  ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion
			  ▪ Install a new cylinder block temperature sensor only when diagnosed as failed
			  ▪ Using the Jaguar Land Rover approved diagnostic equipment, run application - ECU Diagnostics / Powertrain Control Module (PCM) / Clear Diagnostic Trouble Code(s). Retest

P0201-13	Cylinder 1 Injector "A" Circuit - Circuit open	▪ Electrical Cause
		  ▪ Yes
		▪ Mechanical Cause
		  ▪ Yes
		▪ Control Module Cavity
		  ▪ Circuit reference - O_P_INJVH1 -
		  ▪ Circuit reference - O_P_INJVL1 -
		▪ Monitor Description
		  ▪ Injector in cylinder 1 open load
		▪ Prioritized List of Possible Causes
		  ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion
		  ▪ Fuel injector circuit open circuit, high resistance
		  ▪ Fuel injector failure
			▪ Vehicle Conditions to Enable DTC Logging Strategy
			  ▪ Engine running
			▪ Prioritized Checks to Perform
			  ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion
			  ▪ Refer to the electrical circuit diagrams and check fuel injector circuit for open circuit, high resistance
			  ▪ Using the Jaguar Land Rover approved diagnostic equipment, run application - ECU Diagnostics / Powertrain Control Module (PCM) / Clear Diagnostic Trouble Code(s). Retest
			  ▪ Check fuel injector resistance
			  ▪ Install a new fuel injector as required

P0202-13	Cylinder 2 Injector "A" Circuit - Circuit open	▪ Electrical Cause
		  ▪ Yes
		▪ Mechanical Cause
		  ▪ Yes
		▪ Control Module Cavity
		  ▪ Circuit reference - O_P_INJVH2 -
		  ▪ Circuit reference - O_P_INJVL2 -
		▪ Monitor Description
		  ▪ Injector in cylinder 2 open load
		▪ Prioritized List of Possible Causes
		  ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion
		  ▪ Fuel injector circuit open circuit, high resistance
		  ▪ Fuel injector failure
			▪ Vehicle Conditions to Enable DTC Logging Strategy
			  ▪ Engine running
			▪ Prioritized Checks to Perform
			  ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion
			  ▪ Refer to the electrical circuit diagrams and check fuel injector circuit for open circuit, high resistance
			  ▪ Using the Jaguar Land Rover approved diagnostic equipment, run application - ECU Diagnostics / Powertrain Control Module (PCM) / Clear Diagnostic Trouble Code(s). Retest
			  ▪ Check fuel injector resistance
			  ▪ Install a new fuel injector as required

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION

P0203-13	Cylinder 3 Injector "A" Circuit - Circuit open	- Electrical Cause
		  - Yes
		- Mechanical Cause
		  - Yes
		- Control Module Cavity
		  - Circuit reference - O_P_INJVH3 -
		  - Circuit reference - O_P_INJVL3 -
		- Monitor Description
		  - Injector in cylinder 3 open load
		- Prioritized List of Possible Causes
		  - Connector is disconnected, connector pin is backed out, connector pin corrosion
		  - Fuel injector circuit open circuit, high resistance
		  - Fuel injector failure
			- Vehicle Conditions to Enable DTC Logging Strategy
			  - Engine running
			- Prioritized Checks to Perform
			  - Inspect connectors for signs of water ingress, and pins for damage and/or corrosion
			  - Refer to the electrical circuit diagrams and check fuel injector circuit for open circuit, high resistance
			  - Using the Jaguar Land Rover approved diagnostic equipment, run application - ECU Diagnostics / Powertrain Control Module (PCM) / Clear Diagnostic Trouble Code(s). Retest
			  - Check fuel injector resistance
			  - Install a new fuel injector as required

P0204-13	Cylinder 4 Injector "A" Circuit - Circuit open	- Electrical Cause
		  - Yes
		- Mechanical Cause
		  - Yes
		- Control Module Cavity
		  - Circuit reference - O_P_INJVH4 -
		  - Circuit reference - O_P_INJVL4 -
		- Monitor Description
		  - Injector in cylinder 4 open load
		- Prioritized List of Possible Causes
		  - Connector is disconnected, connector pin is backed out, connector pin corrosion
		  - Fuel injector circuit open circuit, high resistance
		  - Fuel injector failure
			- Vehicle Conditions to Enable DTC Logging Strategy
			  - Engine running
			- Prioritized Checks to Perform
			  - Inspect connectors for signs of water ingress, and pins for damage and/or corrosion
			  - Refer to the electrical circuit diagrams and check fuel injector circuit for open circuit, high resistance
			  - Using the Jaguar Land Rover approved diagnostic equipment, run application - ECU Diagnostics / Powertrain Control Module (PCM) / Clear Diagnostic Trouble Code(s). Retest
			  - Check fuel injector resistance
			  - Install a new fuel injector as required

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION

P0221-00	Throttle/Pedal Position Sensor/Switch "B" Circuit Range/Performance - No sub type information	▪ Electrical Cause
		  ▪ Yes
		▪ Mechanical Cause
		  ▪ Yes
		▪ Control Module Cavity
		  ▪ Circuit reference - I_A_TPS2 -
		▪ Monitor Description
		  ▪ Throttle position sensor B does not align to position sensor A and is also not aligned to a theoretical throttle angle calculated from air charge / load. (Sensor A is electrically within valid range but irrational to operating conditions)
		▪ Prioritized List of Possible Causes
		  ▪ Accelerator pedal position sensor circuit short circuit to power, short circuit to ground, high resistance
		  ▪ Accelerator pedal position sensor failure
			▪ Vehicle Conditions to Enable DTC Logging Strategy
			  ▪ Make sure all throttle position circuit faults have been rectified prior to investigating this fault. Engine running at idle for 5 seconds, also perform a short less than 10 minutes varied drive cycle to make sure fault has been rectified, make sure light and high engine loads are included
			▪ Prioritized Checks to Perform
			  ▪ Refer to the electrical circuit diagrams and check accelerator pedal position sensor circuit for short circuit to power, short circuit to ground, high resistance
			  ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion
			  ▪ Using the Jaguar Land Rover approved diagnostic equipment, with Ignition on but engine off, check accelerator pedal position sensor signal A is aligned to accelerator pedal position sensor signal B
			  ▪ Using the Jaguar Land Rover approved diagnostic equipment, run application - ECU Diagnostics / Powertrain Control Module (PCM) / Clear Diagnostic Trouble Code(s). Retest
			  ▪ Install a new accelerator pedal position sensor as required

P0222-00	Throttle/Pedal Position Sensor/Switch "B" Circuit Low - No sub type information	▪ Electrical Cause
		  ▪ Yes
		▪ Mechanical Cause
		  ▪ Yes
		▪ Control Module Cavity
		  ▪ Circuit reference - I_A_TPS2 -
		▪ Monitor Description
		  ▪ Throttle position sensor B is below valid electrical range
		▪ Prioritized List of Possible Causes
		  ▪ Accelerator pedal position sensor circuit is below the valid electrical range
		  ▪ Accelerator pedal position sensor circuit open circuit, short circuit to ground
		  ▪ Accelerator pedal position sensor failure
			▪ Vehicle Conditions to Enable DTC Logging Strategy
			  ▪ Ignition on, Engine speed greater than 1200rpm for 5 seconds
			▪ Prioritized Checks to Perform
			  ▪ Refer to the electrical circuit diagrams and check accelerator pedal position sensor circuit for open circuit, short circuit to ground
			  ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion
			  ▪ Using the Jaguar Land Rover approved diagnostic equipment, run application - ECU Diagnostics / Powertrain Control Module (PCM) / Clear Diagnostic Trouble Code(s). Retest
			  ▪ Install a new accelerator pedal position sensor as required

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION

P0223-85	Throttle/Pedal Position Sensor/Switch "B" Circuit High - Signal above allowable range	▪ Electrical Cause
		  ▪ Yes
		▪ Mechanical Cause
		  ▪ Yes
		▪ Control Module Cavity
		  ▪ Circuit reference - I_A_TPS2 -
		▪ Monitor Description
		  ▪ Throttle position sensor B is above valid electrical range
		▪ Prioritized List of Possible Causes
		  ▪ Accelerator pedal position sensor circuit is above the valid electrical range
		  ▪ Accelerator pedal position sensor circuit short circuit to power
		  ▪ Accelerator pedal position sensor failure
			▪ Vehicle Conditions to Enable DTC Logging Strategy
			  ▪ Ignition on, Engine speed greater than 1200rpm for 5 seconds
			▪ Prioritized Checks to Perform
			  ▪ Refer to the electrical circuit diagrams and check accelerator pedal position sensor circuit for short circuit to power
			  ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion
			  ▪ Using the Jaguar Land Rover approved diagnostic equipment, run application - ECU Diagnostics / Powertrain Control Module (PCM) / Clear Diagnostic Trouble Code(s). Retest
			  ▪ Install a new accelerator pedal position sensor as required

P0230-13	Fuel Pump Primary Circuit - Circuit open	▪ Electrical Cause
		  ▪ Yes
		▪ Mechanical Cause
		  ▪ Yes
		▪ Control Module Cavity
		  ▪ Circuit reference - O_T_PSP -
		▪ Monitor Description
		  ▪ The diagnostic detects failures between the fuel pump driver module and the fuel pump but reports them to the powertrain control module through the PWM line
		▪ Prioritized List of Possible Causes
		  ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion
		  ▪ Harness failure - Wiring integrity
			▪ Vehicle Conditions to Enable DTC Logging Strategy
			  ▪ Ignition running
			▪ Prioritized Checks to Perform
			  ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion
			  ▪ Refer to the electrical circuit diagrams and check connections are secure and wiring integrity

P0230-19	Fuel Pump Primary Circuit - Circuit current above threshold	▪ Electrical Cause
		  ▪ Yes
		▪ Mechanical Cause
		  ▪ Yes
		▪ Control Module Cavity
		  ▪ Circuit reference - O_T_PSP -
		▪ Monitor Description
		  ▪ The diagnostic detects failures between the fuel pump driver module and the fuel pump but reports them to the powertrain control module through the PWM line
		▪ Prioritized List of Possible Causes
		  ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion
		  ▪ Harness failure - Wiring integrity
			▪ Vehicle Conditions to Enable DTC Logging Strategy
			  ▪ Ignition running
			▪ Prioritized Checks to Perform
			  ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion
			  ▪ Refer to the electrical circuit diagrams and check connections are secure and wiring integrity

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION

P0234-00	Turbocharger/Supercharger "A" Overboost Condition - No sub type information	▪ Monitor Description
		  ▪ The monitor compares expected boost pressure when boost pressure control is active against target. If actual is greater than expected then fault is reported
		▪ Prioritized List of Possible Causes
		  ▪ Turbocharger wastegate blocked
		  ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion
		  ▪ Charge air pressure sensor circuit, short circuit to ground, short circuit to power, open circuit, high resistance
			NOTE:
			If a fault is identified with the wastegate system or the actuator, install a new turbocharger. The actuator is not a replaceable part. **DO NOT** remove the actuator or the connecting rod
			▪ Vehicle Conditions to Enable DTC Logging Strategy
			  ▪ Ignition On
			▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion
			▪ Check the powertrain control module connectors for water ingress
			▪ Refer to the electrical circuit diagrams and check the turbocharger electronic wastegate circuit for
			  ▪ Short circuit to ground
			  ▪ Short circuit to power
			  ▪ Open circuit
			  ▪ High resistance
			▪ Check turbocharger wastegate rod movement, start the engine and monitor wastegate rod movement **DO NOT** attempt to disconnect, manipulate or put pressure on the waste gate rod, doing so can result in damage and loss of function of waste gate actuation
			▪ Check turbocharger wastegate voltage at closed position
			▪ Check turbocharger wastegate rod/lever for any mechanical damage, bend or heavy corrosion close to pin For additional information, refer to: **Turbocharger** (303-04D Fuel Charging and Controls - Turbocharger - INGENIUM I4 2.0L Petrol, Diagnosis and Testing).
			▪ Using the Jaguar Land Rover approved diagnostic equipment, clear the Diagnostic Trouble Code(s) and retest

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION

P0237-16	Turbocharger/Supercharger Boost Sensor "A" Circuit Low - Circuit voltage below threshold	▪ Electrical Cause
		  ▪ Yes
		▪ Mechanical Cause
		  ▪ No
		▪ Control Module Cavity
		  ▪ Circuit reference - I_A_BTS -
		▪ Prioritized List of Possible Causes
		  ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion
		  ▪ Charge air pressure and temperature sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance
			NOTE:
			If a fault is identified with the wastegate system or the actuator, install a new turbocharger. The actuator is not a replaceable part. **DO NOT** remove the actuator or the connecting rod
			▪ Vehicle Conditions to Enable DTC Logging Strategy
			  ▪ Ignition On
			▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion
			▪ Check the powertrain control module connectors for water ingress
			▪ Refer to the electrical circuit diagrams and check the turbocharger electronic wastegate circuit for
			  ▪ Short circuit to ground
			  ▪ Short circuit to power
			  ▪ Open circuit
			  ▪ High resistance
			▪ Check turbocharger wastegate rod movement, start the engine and monitor wastegate rod movement **DO NOT** attempt to disconnect, manipulate or put pressure on the waste gate rod, doing so can result in damage and loss of function of waste gate actuation
			▪ Check turbocharger wastegate voltage at closed position
			▪ Check turbocharger wastegate rod/lever for any mechanical damage, bend or heavy corrosion close to pin For additional information, refer to: **Turbocharger** (303-04D Fuel Charging and Controls - Turbocharger - INGENIUM I4 2.0L Petrol, Diagnosis and Testing).
			▪ Using the Jaguar Land Rover approved diagnostic equipment, clear the Diagnostic Trouble Code(s) and retest

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION

P0238-17	Turbocharger/Supercharger Boost Sensor "A" Circuit High - Circuit voltage above threshold	▪ Electrical Cause
		  ▪ Yes
		▪ Mechanical Cause
		  ▪ No
		▪ Control Module Cavity
		  ▪ Circuit reference - I_A_BTS -
		▪ Prioritized List of Possible Causes
		  ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion
		  ▪ Charge air pressure and temperature sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance
			NOTE:
			If a fault is identified with the wastegate system or the actuator, install a new turbocharger. The actuator is not a replaceable part. **DO NOT** remove the actuator or the connecting rod
			▪ Vehicle Conditions to Enable DTC Logging Strategy
			  ▪ Ignition On
			▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion
			▪ Check the powertrain control module connectors for water ingress
			▪ Refer to the electrical circuit diagrams and check the turbocharger electronic wastegate circuit for
			  ▪ Short circuit to ground
			  ▪ Short circuit to power
			  ▪ Open circuit
			  ▪ High resistance
			▪ Check turbocharger wastegate rod movement, start the engine and monitor wastegate rod movement **DO NOT** attempt to disconnect, manipulate or put pressure on the waste gate rod, doing so can result in damage and loss of function of waste gate actuation
			▪ Check turbocharger wastegate voltage at closed position
			▪ Check turbocharger wastegate rod/lever for any mechanical damage, bend or heavy corrosion close to pin For additional information, refer to: **Turbocharger** (303-04D Fuel Charging and Controls - Turbocharger - INGENIUM I4 2.0L Petrol, Diagnosis and Testing).
			▪ Using the Jaguar Land Rover approved diagnostic equipment, clear the Diagnostic Trouble Code(s) and retest

P023A-13	Charge Air Cooler Coolant Pump Control Circuit/Open - Circuit open	▪ Electrical Cause
		  ▪ Yes
		▪ Mechanical Cause
		  ▪ Yes
		▪ Control Module Cavity
		  ▪ Circuit reference - O_T_CCP -
		▪ Prioritized List of Possible Causes
		  ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion
		  ▪ Air charge coolant pump circuit short circuit to ground, short circuit to power, open circuit, high resistance
		  ▪ Air charge coolant pump failure
			▪ Vehicle Conditions to Enable DTC Logging Strategy
			  ▪ Ignition on
			▪ Prioritized Checks to Perform
			  ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion
			  ▪ Refer to the electrical circuit diagrams and check the air charge coolant pump circuit for short circuit to ground, short circuit to power, open circuit, high resistance. Repair the wiring harness as necessary
			  ▪ Install a new air charge coolant pump only when diagnosed as failed
			  ▪ Using the Jaguar Land Rover approved diagnostic equipment, run application - ECU Diagnostics / Powertrain Control Module (PCM) / Clear Diagnostic Trouble Code(s). Retest

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION
P023A-4B	Charge Air Cooler Coolant Pump Control Circuit/Open - Over temperature	▪ Electrical Cause ▪ Yes ▪ Mechanical Cause ▪ Yes ▪ Control Module Cavity ▪ Circuit reference - O_T_CCP - ▪ Prioritized List of Possible Causes ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Air charge coolant pump circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Air charge coolant pump failure	▪ Vehicle Conditions to Enable DTC Logging Strategy ▪ Ignition on ▪ Prioritized Checks to Perform ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion ▪ Refer to the electrical circuit diagrams and check the air charge coolant pump circuit for short circuit to ground, short circuit to power, open circuit, high resistance. Repair the wiring harness as necessary ▪ Install a new air charge coolant pump only when diagnosed as failed ▪ Using the Jaguar Land Rover approved diagnostic equipment, run application - ECU Diagnostics / Powertrain Control Module (PCM) / Clear Diagnostic Trouble Code(s). Retest
P023B-11	Charge Air Cooler Coolant Pump Control Circuit Low - Circuit short to ground	▪ Electrical Cause ▪ Yes ▪ Mechanical Cause ▪ Yes ▪ Control Module Cavity ▪ Circuit reference - O_T_CCP - ▪ Prioritized List of Possible Causes ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Air charge coolant pump circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Air charge coolant pump failure	▪ Vehicle Conditions to Enable DTC Logging Strategy ▪ Ignition on ▪ Prioritized Checks to Perform ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion ▪ Refer to the electrical circuit diagrams and check the air charge coolant pump circuit for short circuit to ground, short circuit to power, open circuit, high resistance. Repair the wiring harness as necessary ▪ Install a new air charge coolant pump only when diagnosed as failed ▪ Using the Jaguar Land Rover approved diagnostic equipment, run application - ECU Diagnostics / Powertrain Control Module (PCM) / Clear Diagnostic Trouble Code(s). Retest
P023C-12	Charge Air Cooler Coolant Pump Control Circuit High - Circuit short to battery	▪ Electrical Cause ▪ Yes ▪ Mechanical Cause ▪ Yes ▪ Control Module Cavity ▪ Circuit reference - O_T_CCP - ▪ Prioritized List of Possible Causes ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Air charge coolant pump circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Air charge coolant pump failure	▪ Vehicle Conditions to Enable DTC Logging Strategy ▪ Ignition on ▪ Prioritized Checks to Perform ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion ▪ Refer to the electrical circuit diagrams and check the air charge coolant pump circuit for short circuit to ground, short circuit to power, open circuit, high resistance. Repair the wiring harness as necessary ▪ Install a new air charge coolant pump only when diagnosed as failed ▪ Using the Jaguar Land Rover approved diagnostic equipment, run application - ECU Diagnostics / Powertrain Control Module (PCM) / Clear Diagnostic Trouble Code(s). Retest

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION

P0243-13	Turbocharger/Supercharger Wastegate Actuator "A" - Circuit open	▪ Electrical Cause
		  ▪ Yes
		▪ Mechanical Cause
		  ▪ Yes
		▪ Control Module Cavity
		  ▪ Circuit reference - O_T_SCPOS -
		  ▪ Circuit reference - O_T_SCNEG -
		▪ Prioritized List of Possible Causes
		  ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion
		  ▪ Turbocharger electronic wastegate circuit short circuit to ground, short circuit to power, open circuit, high resistance
		  ▪ Turbocharger electronic wastegate failure
			NOTE:
			If a fault is identified with the wastegate system or the actuator, install a new turbocharger. The actuator is not a replaceable part. **DO NOT** remove the actuator or the connecting rod
			▪ Vehicle Conditions to Enable DTC Logging Strategy
			  ▪ Ignition On
			▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion
			▪ Check the powertrain control module connectors for water ingress
			▪ Refer to the electrical circuit diagrams and check the turbocharger electronic wastegate circuit for
			  ▪ Short circuit to ground
			  ▪ Short circuit to power
			  ▪ Open circuit
			  ▪ High resistance
			▪ Check turbocharger wastegate rod movement, start the engine and monitor wastegate rod movement **DO NOT** attempt to disconnect, manipulate or put pressure on the waste gate rod, doing so can result in damage and loss of function of waste gate actuation
			▪ Check turbocharger wastegate voltage at closed position
			▪ Check turbocharger wastegate rod/lever for any mechanical damage, bend or heavy corrosion close to pin For additional information, refer to: **Turbocharger** (303-04D Fuel Charging and Controls - Turbocharger - INGENIUM I4 2.0L Petrol, Diagnosis and Testing).
			▪ Using the Jaguar Land Rover approved diagnostic equipment, clear the Diagnostic Trouble Code(s) and retest

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION

P0244-84	Turbocharger/Supercharger Wastegate Actuator "A" Range/Performance - Signal below allowable range	▪ Electrical Cause
		  ▪ Yes
		▪ Mechanical Cause
		  ▪ Yes
		▪ Control Module Cavity
		  ▪ Circuit reference - O_T_SCPOS -
		  ▪ Circuit reference - O_T_SCNEG -
		▪ Prioritized List of Possible Causes
		  ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion
		  ▪ Turbocharger electronic wastegate circuit short circuit to ground, short circuit to power, open circuit, high resistance
		  ▪ Turbocharger electronic wastegate failure
			NOTE:
			If a fault is identified with the wastegate system or the actuator, install a new turbocharger. The actuator is not a replaceable part. **DO NOT** remove the actuator or the connecting rod
			▪ Vehicle Conditions to Enable DTC Logging Strategy
			  ▪ Ignition On
			▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion
			▪ Check the powertrain control module connectors for water ingress
			▪ Refer to the electrical circuit diagrams and check the turbocharger electronic wastegate circuit for
			  ▪ Short circuit to ground
			  ▪ Short circuit to power
			  ▪ Open circuit
			  ▪ High resistance
			▪ Check turbocharger wastegate rod movement, start the engine and monitor wastegate rod movement **DO NOT** attempt to disconnect, manipulate or put pressure on the waste gate rod, doing so can result in damage and loss of function of waste gate actuation
			▪ Check turbocharger wastegate voltage at closed position
			▪ Check turbocharger wastegate rod/lever for any mechanical damage, bend or heavy corrosion close to pin For additional information, refer to: **Turbocharger** (303-04D Fuel Charging and Controls - Turbocharger - INGENIUM I4 2.0L Petrol, Diagnosis and Testing).
			▪ Using the Jaguar Land Rover approved diagnostic equipment, clear the Diagnostic Trouble Code(s) and retest

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION

P0244-85	Turbocharger/Supercharger Wastegate Actuator "A" Range/Performance - Signal above allowable range	▪ Electrical Cause
		  ▪ Yes
		▪ Mechanical Cause
		  ▪ Yes
		▪ Control Module Cavity
		  ▪ Circuit reference - O_T_SCPOS -
		  ▪ Circuit reference - O_T_SCNEG -
		▪ Prioritized List of Possible Causes
		  ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion
		  ▪ Turbocharger electronic wastegate circuit short circuit to ground, short circuit to power, open circuit, high resistance
		  ▪ Turbocharger electronic wastegate failure
			NOTE:
			If a fault is identified with the wastegate system or the actuator, install a new turbocharger. The actuator is not a replaceable part. **DO NOT** remove the actuator or the connecting rod
			▪ Vehicle Conditions to Enable DTC Logging Strategy
			  ▪ Ignition On
			▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion
			▪ Check the powertrain control module connectors for water ingress
			▪ Refer to the electrical circuit diagrams and check the turbocharger electronic wastegate circuit for
			  ▪ Short circuit to ground
			  ▪ Short circuit to power
			  ▪ Open circuit
			  ▪ High resistance
			▪ Check turbocharger wastegate rod movement, start the engine and monitor wastegate rod movement **DO NOT** attempt to disconnect, manipulate or put pressure on the waste gate rod, doing so can result in damage and loss of function of waste gate actuation
			▪ Check turbocharger wastegate voltage at closed position
			▪ Check turbocharger wastegate rod/lever for any mechanical damage, bend or heavy corrosion close to pin For additional information, refer to: **Turbocharger** (303-04D Fuel Charging and Controls - Turbocharger - INGENIUM I4 2.0L Petrol, Diagnosis and Testing).
			▪ Using the Jaguar Land Rover approved diagnostic equipment, clear the Diagnostic Trouble Code(s) and retest

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION

P0245-11	Turbocharger/Supercharger Wastegate Actuator "A" Low - Circuit short to ground	▪ Electrical Cause
		  ▪ Yes
		▪ Mechanical Cause
		  ▪ Yes
		▪ Control Module Cavity
		  ▪ Circuit reference - O_T_SCPOS -
		  ▪ Circuit reference - O_T_SCNEG -
		▪ Prioritized List of Possible Causes
		  ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion
		  ▪ Turbocharger electronic wastegate circuit short circuit to ground, short circuit to power, open circuit, high resistance
		  ▪ Turbocharger electronic wastegate failure
			NOTE:
			If a fault is identified with the wastegate system or the actuator, install a new turbocharger. The actuator is not a replaceable part. **DO NOT** remove the actuator or the connecting rod
			▪ Vehicle Conditions to Enable DTC Logging Strategy
			  ▪ Ignition On
			▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion
			▪ Check the powertrain control module connectors for water ingress
			▪ Refer to the electrical circuit diagrams and check the turbocharger electronic wastegate circuit for
			  ▪ Short circuit to ground
			  ▪ Short circuit to power
			  ▪ Open circuit
			  ▪ High resistance
			▪ Check turbocharger wastegate rod movement, start the engine and monitor wastegate rod movement **DO NOT** attempt to disconnect, manipulate or put pressure on the waste gate rod, doing so can result in damage and loss of function of waste gate actuation
			▪ Check turbocharger wastegate voltage at closed position
			▪ Check turbocharger wastegate rod/lever for any mechanical damage, bend or heavy corrosion close to pin For additional information, refer to: **Turbocharger** (303-04D Fuel Charging and Controls - Turbocharger - INGENIUM I4 2.0L Petrol, Diagnosis and Testing).
			▪ Using the Jaguar Land Rover approved diagnostic equipment, clear the Diagnostic Trouble Code(s) and retest

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION

P0245-14	Turbocharger/Supercharger Wastegate Actuator "A" Low - Circuit short to ground or open	• Electrical Cause
		  ▪ Yes
		• Mechanical Cause
		  ▪ Yes
		• Control Module Cavity
		  ▪ Circuit reference - O_T_SCPOS -
		  ▪ Circuit reference - O_T_SCNEG -
		• Prioritized List of Possible Causes
		  ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion
		  ▪ Turbocharger electronic wastegate circuit short circuit to ground, short circuit to power, open circuit, high resistance
		  ▪ Turbocharger electronic wastegate failure
			NOTE:
			If a fault is identified with the wastegate system or the actuator, install a new turbocharger. The actuator is not a replaceable part. **DO NOT** remove the actuator or the connecting rod
			• Vehicle Conditions to Enable DTC Logging Strategy
			  ▪ Ignition On
			• Inspect connectors for signs of water ingress, and pins for damage and/or corrosion
			• Check the powertrain control module connectors for water ingress
			• Refer to the electrical circuit diagrams and check the turbocharger electronic wastegate circuit for
			  ▪ Short circuit to ground
			  ▪ Short circuit to power
			  ▪ Open circuit
			  ▪ High resistance
			• Check turbocharger wastegate rod movement, start the engine and monitor wastegate rod movement **DO NOT** attempt to disconnect, manipulate or put pressure on the waste gate rod, doing so can result in damage and loss of function of waste gate actuation
			• Check turbocharger wastegate voltage at closed position
			• Check turbocharger wastegate rod/lever for any mechanical damage, bend or heavy corrosion close to pin For additional information, refer to: **Turbocharger** (303-04D Fuel Charging and Controls - Turbocharger - INGENIUM I4 2.0L Petrol, Diagnosis and Testing).
			• Using the Jaguar Land Rover approved diagnostic equipment, clear the Diagnostic Trouble Code(s) and retest

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION

P0246-12	Turbocharger/Supercharger Wastegate Actuator "A" High - Circuit short to battery	▪ Electrical Cause
		  ▪ Yes
		▪ Mechanical Cause
		  ▪ Yes
		▪ Control Module Cavity
		  ▪ Circuit reference - O_T_SCPOS -
		  ▪ Circuit reference - O_T_SCNEG -
		▪ Prioritized List of Possible Causes
		  ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion
		  ▪ Turbocharger electronic wastegate circuit short circuit to ground, short circuit to power, open circuit, high resistance
		  ▪ Turbocharger electronic wastegate failure
			NOTE:
			If a fault is identified with the wastegate system or the actuator, install a new turbocharger. The actuator is not a replaceable part. **DO NOT** remove the actuator or the connecting rod
			▪ Vehicle Conditions to Enable DTC Logging Strategy
			  ▪ Ignition On
			▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion
			▪ Check the powertrain control module connectors for water ingress
			▪ Refer to the electrical circuit diagrams and check the turbocharger electronic wastegate circuit for
			  ▪ Short circuit to ground
			  ▪ Short circuit to power
			  ▪ Open circuit
			  ▪ High resistance
			▪ Check turbocharger wastegate rod movement, start the engine and monitor wastegate rod movement **DO NOT** attempt to disconnect, manipulate or put pressure on the waste gate rod, doing so can result in damage and loss of function of waste gate actuation
			▪ Check turbocharger wastegate voltage at closed position
			▪ Check turbocharger wastegate rod/lever for any mechanical damage, bend or heavy corrosion close to pin For additional information, refer to: **Turbocharger** (303-04D Fuel Charging and Controls - Turbocharger - INGENIUM I4 2.0L Petrol, Diagnosis and Testing).
			▪ Using the Jaguar Land Rover approved diagnostic equipment, clear the Diagnostic Trouble Code(s) and retest

<table>
<tr><th>DTC</th><th>DESCRIPTION</th><th>POSSIBLE CAUSES</th><th>ACTION</th></tr>
<tr><td>P0246-15</td><td>Turbocharger/Supercharger Wastegate Actuator "A" High - Circuit short to battery or open</td><td>Electrical Cause
Yes
Mechanical Cause
Yes
Control Module Cavity
Circuit reference - O_T_SCPOS -
Circuit reference - O_T_SCNEG -
Prioritized List of Possible Causes
Connector is disconnected, connector pin is backed out, connector pin corrosion
Turbocharger electronic wastegate circuit short circuit to ground, short circuit to power, open circuit, high resistance
Turbocharger electronic wastegate failure</td><td>NOTE:
If a fault is identified with the wastegate system or the actuator, install a new turbocharger. The actuator is not a replaceable part. DO NOT remove the actuator or the connecting rod
Vehicle Conditions to Enable DTC Logging Strategy
Ignition On
Inspect connectors for signs of water ingress, and pins for damage and/or corrosion
Check the powertrain control module connectors for water ingress
Refer to the electrical circuit diagrams and check the turbocharger electronic wastegate circuit for
Short circuit to ground
Short circuit to power
Open circuit
High resistance
Check turbocharger wastegate rod movement, start the engine and monitor wastegate rod movement DO NOT attempt to disconnect, manipulate or put pressure on the waste gate rod, doing so can result in damage and loss of function of waste gate actuation
Check turbocharger wastegate voltage at closed position
Check turbocharger wastegate rod/lever for any mechanical damage, bend or heavy corrosion close to pin For additional information, refer to: Turbocharger (303-04D Fuel Charging and Controls - Turbocharger - INGENIUM I4 2.0L Petrol, Diagnosis and Testing).
Using the Jaguar Land Rover approved diagnostic equipment, clear the Diagnostic Trouble Code(s) and retest</td></tr>
<tr><td>P0251-13</td><td>Injection Pump Fuel Metering Control "A" (Cam/Rotor/Injector) - Circuit open</td><td>Electrical Cause
Yes
Mechanical Cause
Yes
Control Module Cavity
Circuit reference - O_V_FSCVH -
Circuit reference - O_V_FSCVL -
Monitor Description
Open load detected by the pump control module, then passed to the main CPU
Prioritized List of Possible Causes
High pressure fuel pump connector is disconnected, connector pin is backed out, connector pin corrosion
High pressure fuel pump circuit, open circuit
High pressure fuel pump failure</td><td>Vehicle Conditions to Enable DTC Logging Strategy
Engine running
Prioritized Checks to Perform
Inspect connectors for signs of water ingress, and pins for damage and/or corrosion
Refer to the electrical circuit diagrams and check the high pressure fuel pump circuit for open circuit
Using the Jaguar Land Rover approved diagnostic equipment, run application - ECU Diagnostics / Powertrain Control Module (PCM) / Clear Diagnostic Trouble Code(s). Retest
Install a new high pressure fuel pump as required</td></tr>
</table>

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION
P0253-00	Injection Pump Fuel Metering Control "A" Low (Cam/Rotor/Injector) - No sub type information	▪ Electrical Cause ▪ Yes ▪ Mechanical Cause ▪ Yes ▪ Control Module Cavity ▪ Circuit reference - O_V_FSCVH - ▪ Circuit reference - O_V_FSCVL - ▪ Monitor Description ▪ Short to ground detected by the pump control module, then passed to the main CPU ▪ Prioritized List of Possible Causes ▪ High pressure fuel pump connector is disconnected, connector pin is backed out, connector pin corrosion ▪ High pressure fuel pump circuit, short circuit to ground ▪ High pressure fuel pump failure	▪ Vehicle Conditions to Enable DTC Logging Strategy ▪ Engine running ▪ Prioritized Checks to Perform ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion ▪ Refer to the electrical circuit diagrams and check the high pressure fuel pump circuit for short circuit to ground ▪ Using the Jaguar Land Rover approved diagnostic equipment, run application - ECU Diagnostics / Powertrain Control Module (PCM) / Clear Diagnostic Trouble Code(s). Retest ▪ Install a new high pressure fuel pump as required
P0254-00	Injection Pump Fuel Metering Control "A" High (Cam/Rotor/Injector) - No sub type information	▪ Electrical Cause ▪ Yes ▪ Mechanical Cause ▪ Yes ▪ Control Module Cavity ▪ Circuit reference - O_V_FSCVH - ▪ Circuit reference - O_V_FSCVL - ▪ Monitor Description ▪ Short to battery voltage detected by the pump control module, then passed to the main CPU ▪ Prioritized List of Possible Causes ▪ High pressure fuel pump connector is disconnected, connector pin is backed out, connector pin corrosion ▪ High pressure fuel pump circuit, short circuit to power ▪ High pressure fuel pump failure	▪ Vehicle Conditions to Enable DTC Logging Strategy ▪ Engine running ▪ Prioritized Checks to Perform ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion ▪ Refer to the electrical circuit diagrams and check the high pressure fuel pump circuit for short circuit to power ▪ Using the Jaguar Land Rover approved diagnostic equipment, run application - ECU Diagnostics / Powertrain Control Module (PCM) / Clear Diagnostic Trouble Code(s). Retest ▪ Install a new high pressure fuel pump as required

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION

P025A-13	Fuel Pump Module "A" Control Circuit/Open - Circuit open	• Electrical Cause
		  ◦ Yes
		• Mechanical Cause
		  ◦ Yes
		• Control Module Cavity
		  ◦ Circuit reference - O_T_PSP -
		• Prioritized List of Possible Causes
		  ◦ Connector is disconnected, connector pin is backed out, connector pin corrosion
		  ◦ Fuel pump driver module circuit short circuit to ground, short circuit to power, open circuit, high resistance
			• Vehicle Conditions to Enable DTC Logging Strategy
			  ◦ Engine running
			• Prioritized Checks to Perform
			  ◦ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion
			  ◦ Refer to the electrical circuit diagrams and check the fuel pump driver module circuit for short circuit to ground, short circuit to power, open circuit, high resistance. Repair the wiring harness as necessary
			  ◦ Using the Jaguar Land Rover approved diagnostic equipment, run application - ECU Diagnostics / Powertrain Control Module (PCM) / Clear Diagnostic Trouble Code(s). Retest

P025B-98	Fuel Pump Module "A" Control Circuit Range/Performance - Component or system over temperature	• Electrical Cause
		  ◦ Yes
		• Mechanical Cause
		  ◦ Yes
		• Control Module Cavity
		  ◦ Circuit reference - O_T_PSP -
		• Prioritized List of Possible Causes
		  ◦ Connector is disconnected, connector pin is backed out, connector pin corrosion
		  ◦ Fuel pump driver module circuit short circuit to ground, short circuit to power, open circuit, high resistance
			• Vehicle Conditions to Enable DTC Logging Strategy
			  ◦ Engine running
			• Prioritized Checks to Perform
			  ◦ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion
			  ◦ Refer to the electrical circuit diagrams and check the fuel pump driver module circuit for short circuit to ground, short circuit to power, open circuit, high resistance. Repair the wiring harness as necessary
			  ◦ Using the Jaguar Land Rover approved diagnostic equipment, run application - ECU Diagnostics / Powertrain Control Module (PCM) / Clear Diagnostic Trouble Code(s). Retest

P025C-11	Fuel Pump Module "A" Control Circuit Low - Circuit short to ground	• Electrical Cause
		  ◦ Yes
		• Mechanical Cause
		  ◦ Yes
		• Control Module Cavity
		  ◦ Circuit reference - O_T_PSP -
		• Prioritized List of Possible Causes
		  ◦ Connector is disconnected, connector pin is backed out, connector pin corrosion
		  ◦ Fuel pump driver module circuit short circuit to ground
			NOTE:
			This fault does not indicate an issue with the fuel driver module. The fuel driver module must not be replaced.
			• Vehicle Conditions to Enable DTC Logging Strategy
			  ◦ Engine running
			• Prioritized Checks to Perform
			  ◦ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion
			  ◦ Refer to the electrical circuit diagrams and check the fuel pump driver module circuit for short circuit to ground
			  ◦ Using the Jaguar Land Rover approved diagnostic equipment, run application - ECU Diagnostics / Powertrain Control Module (PCM) / Clear Diagnostic Trouble Code(s). Retest

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION

P025D-12	Fuel Pump Module "A" Control Circuit High - Circuit short to battery	▪ Electrical Cause
		  ▪ Yes
		▪ Mechanical Cause
		  ▪ Yes
		▪ Control Module Cavity
		  ▪ Circuit reference - O_T_PSP -
		▪ Prioritized List of Possible Causes
		  ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion
		  ▪ Fuel pump driver module circuit, short circuit to power
			▪ Vehicle Conditions to Enable DTC Logging Strategy
			  ▪ Engine running
			▪ Prioritized Checks to Perform
			  ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion
			  ▪ Refer to the electrical circuit diagrams and check the fuel pump driver module circuit for short circuit to power
			  ▪ Using the Jaguar Land Rover approved diagnostic equipment, run application - ECU Diagnostics / Powertrain Control Module (PCM) / Clear Diagnostic Trouble Code(s). Retest

P0261-1C	Cylinder 1 Injector "A" Circuit Low - Circuit voltage out of range	▪ Prioritized List of Possible Causes
		  ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion
		  ▪ Fuel injector circuit short circuit to ground, short circuit to power, open circuit, high resistance
		  ▪ Fuel injector failure
			▪ Prioritized Checks to Perform
			  ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion
			  ▪ Refer to the electrical circuit diagrams and check the fuel injector circuit for short circuit to ground, short circuit to power, open circuit, high resistance
			  ▪ Using the Jaguar Land Rover approved diagnostic equipment, run application - ECU Diagnostics / Powertrain Control Module (PCM) / Clear Diagnostic Trouble Code(s). Retest
			  ▪ Check fuel injector resistance
			  ▪ Install a new fuel injector as required

P0262-1C	Cylinder 1 Injector "A" Circuit High - Circuit voltage out of range	▪ Prioritized List of Possible Causes
		  ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion
		  ▪ Fuel injector circuit short circuit to ground, short circuit to power, open circuit, high resistance
		  ▪ Fuel injector failure
			▪ Prioritized Checks to Perform
			  ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion
			  ▪ Refer to the electrical circuit diagrams and check the fuel injector circuit for short circuit to ground, short circuit to power, open circuit, high resistance
			  ▪ Using the Jaguar Land Rover approved diagnostic equipment, run application - ECU Diagnostics / Powertrain Control Module (PCM) / Clear Diagnostic Trouble Code(s). Retest
			  ▪ Check fuel injector resistance
			  ▪ Install a new fuel injector as required

P0264-1C	Cylinder 2 Injector "A" Circuit Low - Circuit voltage out of range	▪ Prioritized List of Possible Causes
		  ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion
		  ▪ Fuel injector circuit short circuit to ground, short circuit to power, open circuit, high resistance
		  ▪ Fuel injector failure
			▪ Prioritized Checks to Perform
			  ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion
			  ▪ Refer to the electrical circuit diagrams and check the fuel injector circuit for short circuit to ground, short circuit to power, open circuit, high resistance
			  ▪ Using the Jaguar Land Rover approved diagnostic equipment, run application - ECU Diagnostics / Powertrain Control Module (PCM) / Clear Diagnostic Trouble Code(s). Retest
			  ▪ Check fuel injector resistance
			  ▪ Install a new fuel injector as required

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION

P0265-1C	Cylinder 2 Injector "A" Circuit High - Circuit voltage out of range	▪ Prioritized List of Possible Causes
		  ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion
		  ▪ Fuel injector circuit short circuit to ground, short circuit to power, open circuit, high resistance
		  ▪ Fuel injector failure
			▪ Prioritized Checks to Perform
			  ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion
			  ▪ Refer to the electrical circuit diagrams and check the fuel injector circuit for short circuit to ground, short circuit to power, open circuit, high resistance
			  ▪ Using the Jaguar Land Rover approved diagnostic equipment, run application - ECU Diagnostics / Powertrain Control Module (PCM) / Clear Diagnostic Trouble Code(s). Retest
			  ▪ Check fuel injector resistance
			  ▪ Install a new fuel injector as required

P0267-1C	Cylinder 3 Injector "A" Circuit Low - Circuit voltage out of range	▪ Prioritized List of Possible Causes
		  ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion
		  ▪ Fuel injector circuit short circuit to ground, short circuit to power, open circuit, high resistance
		  ▪ Fuel injector failure
			▪ Prioritized Checks to Perform
			  ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion
			  ▪ Refer to the electrical circuit diagrams and check the fuel injector circuit for short circuit to ground, short circuit to power, open circuit, high resistance
			  ▪ Using the Jaguar Land Rover approved diagnostic equipment, run application - ECU Diagnostics / Powertrain Control Module (PCM) / Clear Diagnostic Trouble Code(s). Retest
			  ▪ Check fuel injector resistance
			  ▪ Install a new fuel injector as required

P0268-1C	Cylinder 3 Injector "A" Circuit High - Circuit voltage out of range	▪ Prioritized List of Possible Causes
		  ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion
		  ▪ Fuel injector circuit short circuit to ground, short circuit to power, open circuit, high resistance
		  ▪ Fuel injector failure
			▪ Prioritized Checks to Perform
			  ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion
			  ▪ Refer to the electrical circuit diagrams and check the fuel injector circuit for short circuit to ground, short circuit to power, open circuit, high resistance
			  ▪ Using the Jaguar Land Rover approved diagnostic equipment, run application - ECU Diagnostics / Powertrain Control Module (PCM) / Clear Diagnostic Trouble Code(s). Retest
			  ▪ Check fuel injector resistance
			  ▪ Install a new fuel injector as required

P026B-00	Injection Timing Performance - No sub type information	▪ Electrical Cause
		  ▪ Yes
		▪ Mechanical Cause
		  ▪ Yes
		▪ Prioritized List of Possible Causes
		  ▪ Fuel injector powerstage overheated (internal to powertrain control module)
		  ▪ Other fuel injector related Diagnostic Trouble Code(s)
		  ▪ Fuel injector is mechanically degraded
		  ▪ Fuel injector is carbonized
			▪ Vehicle Conditions to Enable DTC Logging Strategy
			  ▪ Engine running
			▪ Prioritized Checks to Perform
			  ▪ Check powertrain control module and surrounding area for excessive / abnormal heat source
			  ▪ Check powertrain control module for fuel injection related Diagnostic Trouble Code(s) and refer to relevant DTC index
			  ▪ Check fuel injector for mechanical integrity

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION
P026B-29	Injection Timing Performance - Signal signal invalid	- Electrical Cause - Yes - Mechanical Cause - Yes - Prioritized List of Possible Causes - Fuel injector powerstage overheated (internal to powertrain control module) - Other fuel injector related Diagnostic Trouble Code(s) - Fuel injector is mechanically degraded - Fuel injector is carbonized	- Vehicle Conditions to Enable DTC Logging Strategy - Engine running - Prioritized Checks to Perform - Check powertrain control module and surrounding area for excessive / abnormal heat source - Check powertrain control module for fuel injection related Diagnostic Trouble Code(s) and refer to relevant DTC index - Check fuel injector for mechanical integrity
P026E-98	Charge Air Cooler Coolant Pump Performance - Component or system over temperature	- Electrical Cause - Yes - Mechanical Cause - Yes - Control Module Cavity - Circuit reference - O_T_CCP - - Prioritized List of Possible Causes - Coolant system contamination / debris - Connector is disconnected, connector pin is backed out, connector pin corrosion - Charge air cooler coolant pump fuse failure - Charge air cooler coolant pump circuit short circuit to ground, short circuit to power, open circuit, high resistance - Charge air cooler coolant pump failure	- Vehicle Conditions to Enable DTC Logging Strategy - Engine running - Prioritized Checks to Perform - Refer to the relevant section of the workshop manual and check the coolant system for contamination / debris - Inspect connectors for signs of water ingress, and pins for damage and/or corrosion - Check the integrity of the charge air cooler coolant pump fuse. Rectify as necessary - Refer to the electrical circuit diagrams and check the charge air cooler coolant pump circuit for short circuit to ground, short circuit to power, open circuit, high resistance. Repair the wiring harness as necessary - Install a new charge air cooler coolant pump only when diagnosed as failed - Using the Jaguar Land Rover approved diagnostic equipment, run application - ECU Diagnostics / Powertrain Control Module (PCM) / Clear Diagnostic Trouble Code(s). Retest
P0270-1C	Cylinder 4 Injector "A" Circuit Low - Circuit voltage out of range	- Prioritized List of Possible Causes - Connector is disconnected, connector pin is backed out, connector pin corrosion - Fuel injector circuit short circuit to ground, short circuit to power, open circuit, high resistance - Fuel injector failure	- Prioritized Checks to Perform - Inspect connectors for signs of water ingress, and pins for damage and/or corrosion - Refer to the electrical circuit diagrams and check the fuel injector circuit for short circuit to ground, short circuit to power, open circuit, high resistance - Using the Jaguar Land Rover approved diagnostic equipment, run application - ECU Diagnostics / Powertrain Control Module (PCM) / Clear Diagnostic Trouble Code(s). Retest - Check fuel injector resistance - Install a new fuel injector as required

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION

P0271-1C	Cylinder 4 Injector "A" Circuit High - Circuit voltage out of range	▪ Prioritized List of Possible Causes
		  ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion
		  ▪ Fuel injector circuit short circuit to ground, short circuit to power, open circuit, high resistance
		  ▪ Fuel injector failure
			▪ Prioritized Checks to Perform
			  ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion
			  ▪ Refer to the electrical circuit diagrams and check the fuel injector circuit for short circuit to ground, short circuit to power, open circuit, high resistance
			  ▪ Using the Jaguar Land Rover approved diagnostic equipment, run application - ECU Diagnostics / Powertrain Control Module (PCM) / Clear Diagnostic Trouble Code(s). Retest
			  ▪ Check fuel injector resistance
			  ▪ Install a new fuel injector as required

P0299-00	Turbocharger/Supercharger "A" Underboost Condition - No sub type information	▪ Electrical Cause
		  ▪ Yes
		▪ Mechanical Cause
		  ▪ Yes
		▪ Monitor Description
		  ▪ The monitor compares expected boost pressure when boost pressure control is active against target. If actual is less than expected then fault reported
		▪ Prioritized List of Possible Causes
		  ▪ Air intake system, boost air system high pressure boost air leak
		  ▪ Turbocharger seized
		  ▪ Compressor bypass valve stuck open
		  ▪ Air intake system, low pressure intake blocked or restricted
		  ▪ Exhaust system blocked or restricted
		  ▪ Charge air pressure sensor blocked
		  ▪ Charge air pressure sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance
		  ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion
			NOTE:
			If any of the following Diagnostic Trouble Code(s) are set in conjunction with P0299-00, complete the checks below first followed by the checks in the relevant DTC index. P0034-11 P0035-12 P0039-13 P0234-00 P0237-16 P0238-17 P0243-13 P0244-84 P0244-85 P0245-11 P0245-14 P0246-12 P0246-15 P056E-84 P056E-85 P2261-73 P25B3-00 P25B4-00 P2AB8-16 P2AB9-17 P2B81-16 P2B81-17
			▪ Vehicle Conditions to Enable DTC Logging Strategy
			  ▪ Drive vehicle to generate boost pressure for approximately 5 seconds
			▪ Check for manifold absolute pressure sensor Diagnostic Trouble Code(s) (P0106,P0107,P0108). If present, follow the actions in the relevant DTC index
			▪ Check air intake system, low pressure intake for blockages, restriction or leakage
			▪ Check exhaust system, for blockages, restriction or leakage
			▪ Check charge air pressure sensor for damage or blockage
			▪ Check compressor bypass valve for mechanical integrity
			▪ Check turbocharger compressor end and turbine end for any mechanical damage, if any mechanical damage is identified, install a new turbocharger
			▪ Check turbocharger wastegate rod/lever for any mechanical damage/bend For additional information, refer to: **Turbocharger** (303-04D Fuel Charging and Controls - Turbocharger - INGENIUM I4 2.0L Petrol, Diagnosis and Testing).
			▪ Check turbocharger wastegate rod movement, start the engine and monitor wastegate rod movement **DO NOT** attempt to disconnect, manipulate or put pressure on the waste gate rod, doing so can result in damage and loss of function of waste gate actuation
			▪ If a fault is identified with the wastegate system or the actuator, install a new turbocharger. The actuator is not a replaceable part. **DO NOT** remove the actuator or the connecting rod
			▪ Using the Jaguar Land Rover approved diagnostic equipment, clear the Diagnostic Trouble Code(s) and retest

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION
P02A9-00	Engine Coolant Flow Control Valve Position Sensor Circuit Low - No sub type information	▪ Electrical Cause ▪ Yes ▪ Mechanical Cause ▪ Yes ▪ Control Module Cavity ▪ Circuit reference - O_5VCLNTP - ▪ Circuit reference - G_R_VCLNTP - ▪ Circuit reference - I_A_CPP - ▪ Prioritized List of Possible Causes ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Variable coolant pump circuit short circuit to ground, short circuit to power, open circuit, high resistance	▪ Vehicle Conditions to Enable DTC Logging Strategy ▪ Ignition on ▪ Prioritized Checks to Perform ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion ▪ Refer to the electrical circuit diagrams and check the variable coolant pump circuit for short circuit to ground, short circuit to power, open circuit, high resistance. Repair the wiring harness as necessary ▪ Using the Jaguar Land Rover approved diagnostic equipment, run application - ECU Diagnostics / Powertrain Control Module (PCM) / Clear Diagnostic Trouble Code(s). Retest
P02CC-84	Cylinder 1 Fuel Injector Offset Learning at Min Limit - Signal below allowable range	▪ Electrical Cause ▪ Yes ▪ Mechanical Cause ▪ Yes ▪ Prioritized List of Possible Causes ▪ Other fuel injector related Diagnostic Trouble Code(s) ▪ Fuel injector is mechanically degraded ▪ Fuel injector is carbonized	▪ Vehicle Conditions to Enable DTC Logging Strategy ▪ Engine running ▪ Prioritized Checks to Perform ▪ Check powertrain control module for fuel injection related Diagnostic Trouble Code(s) and refer to relevant DTC index ▪ Check fuel injector for mechanical integrity ▪ Install a new fuel injector only when diagnosed as failed ▪ Using the Jaguar Land Rover approved diagnostic equipment, reprogram the injector codes
P02CD-85	Cylinder 1 Fuel Injector Offset Learning at Max Limit - Signal above allowable range	▪ Electrical Cause ▪ Yes ▪ Mechanical Cause ▪ Yes ▪ Prioritized List of Possible Causes ▪ Other fuel injector related Diagnostic Trouble Code(s) ▪ Fuel injector is mechanically degraded ▪ Fuel injector is carbonized	▪ Vehicle Conditions to Enable DTC Logging Strategy ▪ Engine running ▪ Prioritized Checks to Perform ▪ Check powertrain control module for fuel injection related Diagnostic Trouble Code(s) and refer to relevant DTC index ▪ Check fuel injector for mechanical integrity ▪ Install a new fuel injector only when diagnosed as failed ▪ Using the Jaguar Land Rover approved diagnostic equipment, reprogram the injector codes
P02CE-84	Cylinder 2 Fuel Injector Offset Learning at Min Limit - Signal below allowable range	▪ Electrical Cause ▪ Yes ▪ Mechanical Cause ▪ Yes ▪ Prioritized List of Possible Causes ▪ Other fuel injector related Diagnostic Trouble Code(s) ▪ Fuel injector is mechanically degraded ▪ Fuel injector is carbonized	▪ Vehicle Conditions to Enable DTC Logging Strategy ▪ Engine running ▪ Prioritized Checks to Perform ▪ Check powertrain control module for fuel injection related Diagnostic Trouble Code(s) and refer to relevant DTC index ▪ Check fuel injector for mechanical integrity ▪ Install a new fuel injector only when diagnosed as failed ▪ Using the Jaguar Land Rover approved diagnostic equipment, reprogram the injector codes

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION

P02CF-85	Cylinder 2 Fuel Injector Offset Learning at Max Limit - Signal above allowable range	▪ Electrical Cause
		  ▪ Yes
		▪ Mechanical Cause
		  ▪ Yes
		▪ Prioritized List of Possible Causes
		  ▪ Other fuel injector related Diagnostic Trouble Code(s)
		  ▪ Fuel injector is mechanically degraded
		  ▪ Fuel injector is carbonized
			▪ Vehicle Conditions to Enable DTC Logging Strategy
			  ▪ Engine running
			▪ Prioritized Checks to Perform
			  ▪ Check powertrain control module for fuel injection related Diagnostic Trouble Code(s) and refer to relevant DTC index
			  ▪ Check fuel injector for mechanical integrity
			  ▪ Install a new fuel injector only when diagnosed as failed
			  ▪ Using the Jaguar Land Rover approved diagnostic equipment, reprogram the injector codes

P02D0-84	Cylinder 3 Fuel Injector Offset Learning at Min Limit - Signal below allowable range	▪ Electrical Cause
		  ▪ Yes
		▪ Mechanical Cause
		  ▪ Yes
		▪ Prioritized List of Possible Causes
		  ▪ Other fuel injector related Diagnostic Trouble Code(s)
		  ▪ Fuel injector is mechanically degraded
		  ▪ Fuel injector is carbonized
			▪ Vehicle Conditions to Enable DTC Logging Strategy
			  ▪ Engine running
			▪ Prioritized Checks to Perform
			  ▪ Check powertrain control module for fuel injection related Diagnostic Trouble Code(s) and refer to relevant DTC index
			  ▪ Check fuel injector for mechanical integrity
			  ▪ Install a new fuel injector only when diagnosed as failed
			  ▪ Using the Jaguar Land Rover approved diagnostic equipment, reprogram the injector codes

P02D1-85	Cylinder 3 Fuel Injector Offset Learning at Max Limit - Signal above allowable range	▪ Electrical Cause
		  ▪ Yes
		▪ Mechanical Cause
		  ▪ Yes
		▪ Prioritized List of Possible Causes
		  ▪ Other fuel injector related Diagnostic Trouble Code(s)
		  ▪ Fuel injector is mechanically degraded
		  ▪ Fuel injector is carbonized
			▪ Vehicle Conditions to Enable DTC Logging Strategy
			  ▪ Engine running
			▪ Prioritized Checks to Perform
			  ▪ Check powertrain control module for fuel injection related Diagnostic Trouble Code(s) and refer to relevant DTC index
			  ▪ Check fuel injector for mechanical integrity
			  ▪ Install a new fuel injector only when diagnosed as failed
			  ▪ Using the Jaguar Land Rover approved diagnostic equipment, reprogram the injector codes

P02D2-84	Cylinder 4 Fuel Injector Offset Learning at Min Limit - Signal below allowable range	▪ Electrical Cause
		  ▪ Yes
		▪ Mechanical Cause
		  ▪ Yes
		▪ Prioritized List of Possible Causes
		  ▪ Other fuel injector related Diagnostic Trouble Code(s)
		  ▪ Fuel injector is mechanically degraded
		  ▪ Fuel injector is carbonized
			▪ Vehicle Conditions to Enable DTC Logging Strategy
			  ▪ Engine running
			▪ Prioritized Checks to Perform
			  ▪ Check powertrain control module for fuel injection related Diagnostic Trouble Code(s) and refer to relevant DTC index
			  ▪ Check fuel injector for mechanical integrity
			  ▪ Install a new fuel injector only when diagnosed as failed
			  ▪ Using the Jaguar Land Rover approved diagnostic equipment, reprogram the injector codes

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION
P02D3-85	Cylinder 4 Fuel Injector Offset Learning at Max Limit - Signal above allowable range	▪ Electrical Cause ▪ Yes ▪ Mechanical Cause ▪ Yes ▪ Prioritized List of Possible Causes ▪ Other fuel injector related Diagnostic Trouble Code(s) ▪ Fuel injector is mechanically degraded ▪ Fuel injector is carbonized	▪ Vehicle Conditions to Enable DTC Logging Strategy ▪ Engine running ▪ Prioritized Checks to Perform ▪ Check powertrain control module for fuel injection related Diagnostic Trouble Code(s) and refer to relevant DTC index ▪ Check fuel injector for mechanical integrity ▪ Install a new fuel injector only when diagnosed as failed ▪ Using the Jaguar Land Rover approved diagnostic equipment, reprogram the injector codes
P02EE-1C	Cylinder 1 Injector Circuit Range/Performance - Circuit voltage out of range	▪ Electrical Cause ▪ Yes ▪ Mechanical Cause ▪ Yes ▪ Control Module Cavity ▪ Circuit reference - O_P_INJVH1 - ▪ Circuit reference - O_P_INJVL1 - ▪ Monitor Description ▪ Injector in cylinder 1 high side and low side is shorted ▪ Prioritized List of Possible Causes ▪ Fuel injector number 1 high side and low side short circuit to each other ▪ Fuel injector circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Fuel injector failure	▪ Vehicle Conditions to Enable DTC Logging Strategy ▪ Engine running ▪ Prioritized Checks to Perform ▪ Refer to the electrical circuit diagrams and check the fuel injector number 1 for high side and low side short circuit to each other ▪ Refer to the electrical circuit diagrams and check the fuel injector circuit for short circuit to ground, short circuit to power, open circuit, high resistance ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion ▪ Check fuel injector resistance ▪ Install a new fuel injector only when diagnosed as failed ▪ Using the Jaguar Land Rover approved diagnostic equipment, run application - ECU Diagnostics / Powertrain Control Module (PCM) / Clear Diagnostic Trouble Code(s). Retest

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION

P02EF-1C	Cylinder 2 Injector Circuit Range/Performance - Circuit voltage out of range	▪ Electrical Cause
		  ▪ Yes
		▪ Mechanical Cause
		  ▪ Yes
		▪ Control Module Cavity
		  ▪ Circuit reference - O_P_INJVH2 -
		  ▪ Circuit reference - O_P_INJVL2 -
		▪ Monitor Description
		  ▪ Injector in cylinder 2 high side and low side is shorted
		▪ Prioritized List of Possible Causes
		  ▪ Fuel injector number 2 high side and low side short circuit to each other
		  ▪ Fuel injector circuit short circuit to ground, short circuit to power, open circuit, high resistance
		  ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion
		  ▪ Fuel injector failure
			▪ Vehicle Conditions to Enable DTC Logging Strategy
			  ▪ Engine running
			▪ Prioritized Checks to Perform
			  ▪ Refer to the electrical circuit diagrams and check the fuel injector number 2 for high side and low side short circuit to each other
			  ▪ Refer to the electrical circuit diagrams and check the fuel injector circuit for short circuit to ground, short circuit to power, open circuit, high resistance
			  ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion
			  ▪ Check fuel injector resistance
			  ▪ Install a new fuel injector only when diagnosed as failed
			  ▪ Using the Jaguar Land Rover approved diagnostic equipment, run application - ECU Diagnostics / Powertrain Control Module (PCM) / Clear Diagnostic Trouble Code(s). Retest

P02F0-1C	Cylinder 3 Injector Circuit Range/Performance - Circuit voltage out of range	▪ Electrical Cause
		  ▪ Yes
		▪ Mechanical Cause
		  ▪ Yes
		▪ Control Module Cavity
		  ▪ Circuit reference - O_P_INJVH3 -
		  ▪ Circuit reference - O_P_INJVL3 -
		▪ Monitor Description
		  ▪ Injector in cylinder 3 high side and low side is shorted
		▪ Prioritized List of Possible Causes
		  ▪ Fuel injector number 3 high side and low side short circuit to each other
		  ▪ Fuel injector circuit short circuit to ground, short circuit to power, open circuit, high resistance
		  ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion
		  ▪ Fuel injector failure
			▪ Vehicle Conditions to Enable DTC Logging Strategy
			  ▪ Engine running
			▪ Prioritized Checks to Perform
			  ▪ Refer to the electrical circuit diagrams and check the fuel injector number 3 for high side and low side short circuit to each other
			  ▪ Refer to the electrical circuit diagrams and check the fuel injector circuit for short circuit to ground, short circuit to power, open circuit, high resistance
			  ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion
			  ▪ Check fuel injector resistance
			  ▪ Install a new fuel injector only when diagnosed as failed
			  ▪ Using the Jaguar Land Rover approved diagnostic equipment, run application - ECU Diagnostics / Powertrain Control Module (PCM) / Clear Diagnostic Trouble Code(s). Retest

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION

P02F1-1C	Cylinder 4 Injector Circuit Range/Performance - Circuit voltage out of range	- Electrical Cause
		  - Yes
		- Mechanical Cause
		  - Yes
		- Control Module Cavity
		  - Circuit reference - O_P_INJVH4 -
		  - Circuit reference - O_P_INJVL4 -
		- Monitor Description
		  - Injector in cylinder 4 high side and low side is shorted
		- Prioritized List of Possible Causes
		  - Fuel injector number 4 high side and low side short circuit to each other
		  - Fuel injector circuit short circuit to ground, short circuit to power, open circuit, high resistance
		  - Connector is disconnected, connector pin is backed out, connector pin corrosion
		  - Fuel injector failure
			- Vehicle Conditions to Enable DTC Logging Strategy
			  - Engine running
			- Prioritized Checks to Perform
			  - Refer to the electrical circuit diagrams and check the fuel injector number 4 for high side and low side short circuit to each other
			  - Refer to the electrical circuit diagrams and check the fuel injector circuit for short circuit to ground, short circuit to power, open circuit, high resistance
			  - Inspect connectors for signs of water ingress, and pins for damage and/or corrosion
			  - Check fuel injector resistance
			  - Install a new fuel injector only when diagnosed as failed
			  - Using the Jaguar Land Rover approved diagnostic equipment, run application - ECU Diagnostics / Powertrain Control Module (PCM) / Clear Diagnostic Trouble Code(s). Retest

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION

P0300-00	Random/Multiple Cylinder Misfire Detected - No sub type information	- Any of the following Diagnostic Trouble Code(s) if set will cause a misfire. These must be repaired before additional investigations are conducted
		  - P0030-00 P0031-00 P0032-00 P00D1-00 P0131-00 P0132-00 P0133-00 P0134-00 P0192-00 P0193-00 P0444-00 P0458-00 P0459-00 P0496-00 P0497-00 P064D-08 P064D-17 P064D-16 P064D-86 P2096-00 P2096-02 P2097-00 P2097-02 P2231-00 P2237-00 P2237-02 P2237-64 P2243-00 P2251-00 P228C-77 P228D-77 P2626-00
		- The above Diagnostic Trouble Code(s) are relating to the following systems or components
		  - Ignition system
		  - Fuel system
		  - Charge air system
		  - Evaporative emissions canister purge valve
		- Prioritized List of Possible Causes
		  - Fuel level low
		  - Spark plug(s) fouled or failed
		  - Reluctor ring to sensor runout and air gap out of tolerance
		  - Crankshaft position sensor incorrectly installed
		  - Poor fuel quality
		  - Incorrect fuel grade or blend
		  - Contaminated fuel or unauthorized additives
		  - Driveline imbalance - Debris around rotating components, driveline components excessive wear, wheel and tire imbalance
		  - Harness failure - Wiring integrity
		  - Low cylinder compressions - Worn damaged engine components, worn piston rings, incorrect valve clearance, worn valves
		  - Catalyst blocked or damaged
		  - Engine mount damaged
		  - Transmission mount damaged
		  - Accessory drive belt components incorrectly installed
		  - Fuel injector contamination or blocked
		  - Engine build excessive wear
			NOTE:
			The powertrain control module is unlikely to be the root cause for this DTC and must not be replaced without a Technical Assist Request (TA) being raised for further guidance
			- Check powertrain control module for any of the shown Diagnostic Trouble Code(s) and relating to the following systems or components. These must be repaired before additional investigations are conducted
			  - Ignition system
			  - Fuel system
			  - Charge air system
			  - Evaporative emissions canister purge valve
			- Prioritized Checks to Perform
			  - Make sure vehicle has sufficient fuel
			  - Check spark plug(s) for fouling or failure, install new as required
			  - Check reluctor ring for correct installation
			  - Check crankshaft position sensor for correct installation
			  - Drain and refuel with new fuel of correct grade and known quality
			  - Check driveline for debris around rotating components
			  - Check driveline for excessive wear or backlash
			  - Check wheels and tires for snow or mud packing and correct balance
			  - Refer to the electrical circuit diagrams and check connections are secure and wiring integrity
			  - Check cylinder compressions both cold and hot
			  - Check for correct valve clearance
			  - Check catalyst for damage or blockage
			  - Check engine mounts for damage
			  - Check transmission mounts for damage
			  - Check accessory drive belt components are correctly installed
			  - Check fuel injector for contamination or blockage
			  - Check engine oil pressure is to specification
			  - For further detailed information, please refer to the relevant section of the workshop manual. Section 303 - Electronic Engine Controls

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION

P0301-00	Cylinder 1 Misfire Detected - No sub type information	▪ Any of the following Diagnostic Trouble Code(s) if set will cause a misfire. These must be repaired before additional investigations are conducted
		  ▪ P0030-00 P0031-00 P0032-00 P00D1-00 P0131-00 P0132-00 P0133-00 P0134-00 P0192-00 P0193-00 P0444-00 P0458-00 P0459-00 P0496-00 P0497-00 P064D-08 P064D-17 P064D-16 P064D-86 P2096-00 P2096-02 P2097-00 P2097-02 P2231-00 P2237-00 P2237-02 P2237-64 P2243-00 P2251-00 P228C-77 P228D-77 P2626-00
		▪ The above Diagnostic Trouble Code(s) are relating to the following systems or components
		  ▪ Ignition system
		  ▪ Fuel system
		  ▪ Charge air system
		  ▪ Evaporative emissions canister purge valve
		▪ Prioritized List of Possible Causes
		  ▪ Fuel level low
		  ▪ Spark plug(s) fouled or failed
		  ▪ Reluctor ring to sensor runout and air gap out of tolerance
		  ▪ Crankshaft position sensor incorrectly installed
		  ▪ Poor fuel quality
		  ▪ Incorrect fuel grade or blend
		  ▪ Contaminated fuel or unauthorized additives
		  ▪ Driveline imbalance - Debris around rotating components, driveline components excessive wear, wheel and tire imbalance
		  ▪ Harness failure - Wiring integrity
		  ▪ Engine misfire caused by a Continuous Variable Valve Lift (CVVL) mechanical failure
		  ▪ Low cylinder compressions - Worn damaged engine components, worn piston rings, incorrect valve clearance, worn valves
		  ▪ Catalyst blocked or damaged
		  ▪ Engine mount damaged
		  ▪ Transmission mount damaged
		  ▪ Accessory drive belt components incorrectly installed
		  ▪ Fuel injector contamination or blocked
		  ▪ Engine build excessive wear
			▪ Check powertrain control module for any of the shown Diagnostic Trouble Code(s) and relating to the following systems or components. These must be repaired before additional investigations are conducted
			  ▪ Ignition system
			  ▪ Fuel system
			  ▪ Charge air system
			  ▪ Evaporative emissions canister purge valve
			▪ **WARNING:** When completing the cylinder compression tests, if the test result is 0%, or no pressure, **DO NOT** immediately replace the engine
			NOTE: The 0% reading on the compression test could be due to the inlet valves not opening, caused by a mechanical failure in the CVVL unit
			When instructed to complete the cylinder compression test for this DTC, refer to the warning and note above this text
			▪ Prioritized Checks to Perform
			  ▪ Make sure vehicle has sufficient fuel
			  ▪ Check spark plug(s) for fouling or failure, install new as required
			  ▪ Check reluctor ring for correct installation
			  ▪ Check crankshaft position sensor for correct installation
			  ▪ Drain and refuel with new fuel of correct grade and known quality
			  ▪ Check driveline for debris around rotating components
			  ▪ Check driveline for excessive wear or backlash
			  ▪ Check wheels and tires for snow or mud packing and correct balance
			  ▪ Refer to the electrical circuit diagrams and check connections are secure and wiring integrity
			  ▪ Continuous Variable Vale Lift (CVVL) system - mechanical failure. If the vehicle has the CVVL system fitted to the engine, remove the CVVL unit and inspect the components for mechanical damage or a broken spring. Replace as necessary
			  ▪ Check cylinder compressions both cold and hot
			  ▪ Check for correct valve clearance
			  ▪ Check catalyst for damage or blockage
			  ▪ Check engine mounts for damage
			  ▪ Check transmission mounts for damage
			  ▪ Check accessory drive belt components are correctly installed
			  ▪ Check fuel injector for contamination or blockage
			  ▪ Check engine oil pressure is to specification

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION
			▪ For further detailed information, please refer to the relevant section of the workshop manual. Section 303 - Electronic Engine Controls

<table>
<tr><th>DTC</th><th>DESCRIPTION</th><th>POSSIBLE CAUSES</th><th>ACTION</th></tr>
<tr>
<td>P0302-00</td>
<td>Cylinder 2 Misfire Detected - No sub type information</td>
<td>

Any of the following Diagnostic Trouble Code(s) if set will cause a misfire. These must be repaired before additional investigations are conducted

P0030-00 P0031-00 P0032-00 P00D1-00 P0131-00 P0132-00 P0133-00 P0134-00 P0192-00 P0193-00 P0444-00 P0458-00 P0459-00 P0496-00 P0497-00 P064D-08 P064D-17 P064D-16 P064D-86 P2096-00 P2096-02 P2097-00 P2097-02 P2231-00 P2237-00 P2237-02 P2237-64 P2243-00 P2251-00 P228C-77 P228D-77 P2626-00

The above Diagnostic Trouble Code(s) are relating to the following systems or components

Ignition system
Fuel system
Charge air system
Evaporative emissions canister purge valve

Prioritized List of Possible Causes

Fuel level low
Spark plug(s) fouled or failed
Reluctor ring to sensor runout and air gap out of tolerance
Crankshaft position sensor incorrectly installed
Poor fuel quality
Incorrect fuel grade or blend
Contaminated fuel or unauthorized additives
Driveline imbalance - Debris around rotating components, driveline components excessive wear, wheel and tire imbalance
Harness failure - Wiring integrity
Engine misfire caused by a Continuous Variable Valve Lift (CVVL) mechanical failure
Low cylinder compressions - Worn damaged engine components, worn piston rings, incorrect valve clearance, worn valves
Catalyst blocked or damaged
Engine mount damaged
Transmission mount damaged
Accessory drive belt components incorrectly installed
Fuel injector contamination or blocked
Engine build excessive wear

</td>
<td>

Check powertrain control module for any of the shown Diagnostic Trouble Code(s) and relating to the following systems or components. These must be repaired before additional investigations are conducted

Ignition system
Fuel system
Charge air system
Evaporative emissions canister purge valve

WARNING:
When completing the cylinder compression tests, if the test result is 0%, or no pressure, DO NOT immediately replace the engine

NOTE:
The 0% reading on the compression test could be due to the inlet valves not opening, caused by a mechanical failure in the CVVL unit

When instructed to complete the cylinder compression test for this DTC, refer to the warning and note above this text
Prioritized Checks to Perform

Make sure vehicle has sufficient fuel
Check spark plug(s) for fouling or failure, install new as required
Check reluctor ring for correct installation
Check crankshaft position sensor for correct installation
Drain and refuel with new fuel of correct grade and known quality
Check driveline for debris around rotating components
Check driveline for excessive wear or backlash
Check wheels and tires for snow or mud packing and correct balance
Refer to the electrical circuit diagrams and check connections are secure and wiring integrity
Continuous Variable Vale Lift (CVVL) system - mechanical failure. If the vehicle has the CVVL system fitted to the engine, remove the CVVL unit and inspect the components for mechanical damage or a broken spring. Replace as necessary
Check cylinder compressions both cold and hot
Check for correct valve clearance
Check catalyst for damage or blockage
Check engine mounts for damage
Check transmission mounts for damage
Check accessory drive belt components are correctly installed
Check fuel injector for contamination or blockage
Check engine oil pressure is to specification

</td>
</tr>
</table>

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION
			▪ For further detailed information, please refer to the relevant section of the workshop manual. Section 303 - Electronic Engine Controls

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION

P0303-00	Cylinder 3 Misfire Detected - No sub type information	- Any of the following Diagnostic Trouble Code(s) if set will cause a misfire. These must be repaired before additional investigations are conducted
		  - P0030-00 P0031-00 P0032-00 P00D1-00 P0131-00 P0132-00 P0133-00 P0134-00 P0192-00 P0193-00 P0444-00 P0458-00 P0459-00 P0496-00 P0497-00 P064D-08 P064D-17 P064D-16 P064D-86 P2096-00 P2096-02 P2097-00 P2097-02 P2231-00 P2237-00 P2237-02 P2237-64 P2243-00 P2251-00 P228C-77 P228D-77 P2626-00
		- The above Diagnostic Trouble Code(s) are relating to the following systems or components
		  - Ignition system
		  - Fuel system
		  - Charge air system
		  - Evaporative emissions canister purge valve
		- Prioritized List of Possible Causes
		  - Fuel level low
		  - Spark plug(s) fouled or failed
		  - Reluctor ring to sensor runout and air gap out of tolerance
		  - Crankshaft position sensor incorrectly installed
		  - Poor fuel quality
		  - Incorrect fuel grade or blend
		  - Contaminated fuel or unauthorized additives
		  - Driveline imbalance - Debris around rotating components, driveline components excessive wear, wheel and tire imbalance
		  - Harness failure - Wiring integrity
		  - Engine misfire caused by a Continuous Variable Valve Lift (CVVL) mechanical failure
		  - Low cylinder compressions - Worn damaged engine components, worn piston rings, incorrect valve clearance, worn valves
		  - Catalyst blocked or damaged
		  - Engine mount damaged
		  - Transmission mount damaged
		  - Accessory drive belt components incorrectly installed
		  - Fuel injector contamination or blocked
		  - Engine build excessive wear
			NOTE:
			The powertrain control module is unlikely to be the root cause for this DTC and must not be replaced without a Technical Assist Request (TA) being raised for further guidance
			- **WARNING:**
			When completing the cylinder compression tests, if the test result is 0%, or no pressure, **DO NOT** immediately replace the engine
			NOTE:
			The 0% reading on the compression test could be due to the inlet valves not opening, caused by a mechanical failure in the CVVL unit
			When instructed to complete the cylinder compression test for this DTC, refer to the warning and note above this text
			- Check powertrain control module for any of the shown Diagnostic Trouble Code(s) and relating to the following systems or components. These must be repaired before additional investigations are conducted
			  - Ignition system
			  - Fuel system
			  - Charge air system
			  - Evaporative emissions canister purge valve
			- Prioritized Checks to Perform
			  - Make sure vehicle has sufficient fuel
			  - Check spark plug(s) for fouling or failure, install new as required
			  - Check reluctor ring for correct installation
			  - Check crankshaft position sensor for correct installation
			  - Drain and refuel with new fuel of correct grade and known quality
			  - Check driveline for debris around rotating components
			  - Check driveline for excessive wear or backlash
			  - Check wheels and tires for snow or mud packing and correct balance
			  - Refer to the electrical circuit diagrams and check connections are secure and wiring integrity
			  - Continuous Variable Vale Lift (CVVL) system - mechanical failure. If the vehicle has the CVVL system fitted to the engine, remove the CVVL unit and inspect the components for mechanical damage or a broken spring. Replace as necessary
			  - Check cylinder compressions both cold and hot
			  - Check for correct valve clearance
			  - Check catalyst for damage or blockage
			  - Check engine mounts for damage

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION
			▪ Check transmission mounts for damage ▪ Check accessory drive belt components are correctly installed ▪ Check fuel injector for contamination or blockage ▪ Check engine oil pressure is to specification ▪ For further detailed information, please refer to the relevant section of the workshop manual. Section 303 - Electronic Engine Controls

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION

P0304-00	Cylinder 4 Misfire Detected - No sub type information	▪ Any of the following Diagnostic Trouble Code(s) if set will cause a misfire. These must be repaired before additional investigations are conducted
		  ▪ P0030-00 P0031-00 P0032-00 P00D1-00 P0131-00 P0132-00 P0133-00 P0134-00 P0192-00 P0193-00 P0444-00 P0458-00 P0459-00 P0496-00 P0497-00 P064D-08 P064D-17 P064D-16 P064D-86 P2096-00 P2096-02 P2097-00 P2097-02 P2231-00 P2237-00 P2237-02 P2237-64 P2243-00 P2251-00 P228C-77 P228D-77 P2626-00
		▪ The above Diagnostic Trouble Code(s) are relating to the following systems or components
		  ▪ Ignition system
		  ▪ Fuel system
		  ▪ Charge air system
		  ▪ Evaporative emissions canister purge valve
		▪ Prioritized List of Possible Causes
		  ▪ Fuel level low
		  ▪ Spark plug(s) fouled or failed
		  ▪ Reluctor ring to sensor runout and air gap out of tolerance
		  ▪ Crankshaft position sensor incorrectly installed
		  ▪ Poor fuel quality
		  ▪ Incorrect fuel grade or blend
		  ▪ Contaminated fuel or unauthorized additives
		  ▪ Driveline imbalance - Debris around rotating components, driveline components excessive wear, wheel and tire imbalance
		  ▪ Harness failure - Wiring integrity
		  ▪ Engine misfire caused by a Continuous Variable Valve Lift (CVVL) mechanical failure
		  ▪ Low cylinder compressions - Worn damaged engine components, worn piston rings, incorrect valve clearance, worn valves
		  ▪ Catalyst blocked or damaged
		  ▪ Engine mount damaged
		  ▪ Transmission mount damaged
		  ▪ Accessory drive belt components incorrectly installed
		  ▪ Fuel injector contamination or blocked
		  ▪ Engine build excessive wear
			NOTE:
			The powertrain control module is unlikely to be the root cause for this DTC and must not be replaced without a Technical Assist Request (TA) being raised for further guidance
			▪ Check powertrain control module for any of the shown Diagnostic Trouble Code(s) and relating to the following systems or components. These must be repaired before additional investigations are conducted
			  ▪ Ignition system
			  ▪ Fuel system
			  ▪ Charge air system
			  ▪ Evaporative emissions canister purge valve
			▪ **WARNING:**
			When completing the cylinder compression tests, if the test result is 0%, or no pressure, **DO NOT** immediately replace the engine
			NOTE:
			The 0% reading on the compression test could be due to the inlet valves not opening, caused by a mechanical failure in the CVVL unit
			When instructed to complete the cylinder compression test for this DTC, refer to the warning and note above this text
			▪ Prioritized Checks to Perform
			  ▪ Make sure vehicle has sufficient fuel
			  ▪ Check spark plug(s) for fouling or failure, install new as required
			  ▪ Check reluctor ring for correct installation
			  ▪ Check crankshaft position sensor for correct installation
			  ▪ Drain and refuel with new fuel of correct grade and known quality
			  ▪ Check driveline for debris around rotating components
			  ▪ Check driveline for excessive wear or backlash
			  ▪ Check wheels and tires for snow or mud packing and correct balance
			  ▪ Refer to the electrical circuit diagrams and check connections are secure and wiring integrity
			  ▪ Continuous Variable Vale Lift (CVVL) system - mechanical failure. If the vehicle has the CVVL system fitted to the engine, remove the CVVL unit and inspect the components for mechanical damage or a broken spring. Replace as necessary
			  ▪ Check cylinder compressions both cold and hot
			  ▪ Check for correct valve clearance
			  ▪ Check catalyst for damage or blockage
			  ▪ Check engine mounts for damage

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION
			▪ Check transmission mounts for damage ▪ Check accessory drive belt components are correctly installed ▪ Check fuel injector for contamination or blockage ▪ Check engine oil pressure is to specification ▪ For further detailed information, please refer to the relevant section of the workshop manual. Section 303 - Electronic Engine Controls

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION

P0316-00	Engine Misfire Detected On Startup (First 1000 Revolutions) - No sub type information	• Any of the following Diagnostic Trouble Code(s) if set will cause a misfire. These must be repaired before additional investigations are conducted
		  ▪ P0030-00 P0031-00 P0032-00 P00D1-00 P0131-00 P0132-00 P0133-00 P0134-00 P0192-00 P0193-00 P0444-00 P0458-00 P0459-00 P0496-00 P0497-00 P064D-08 P064D-17 P064D-16 P064D-86 P2096-00 P2096-02 P2097-00 P2097-02 P2231-00 P2237-00 P2237-02 P2237-64 P2243-00 P2251-00 P228C-77 P228D-77 P2626-00
		• The above Diagnostic Trouble Code(s) are relating to the following systems or components
		  ▪ Ignition system
		  ▪ Fuel system
		  ▪ Charge air system
		  ▪ Evaporative emissions canister purge valve
		• Prioritized List of Possible Causes
		  ▪ Fuel level low
		  ▪ Spark plug(s) fouled or failed
		  ▪ Reluctor ring to sensor runout and air gap out of tolerance
		  ▪ Crankshaft position sensor incorrectly installed
		  ▪ Poor fuel quality
		  ▪ Incorrect fuel grade or blend
		  ▪ Contaminated fuel or unauthorized additives
		  ▪ Driveline imbalance - Debris around rotating components, driveline components excessive wear, wheel and tire imbalance
		  ▪ Harness failure - Wiring integrity
		  ▪ Low cylinder compressions - Worn damaged engine components, worn piston rings, incorrect valve clearance, worn valves
		  ▪ Catalyst blocked or damaged
		  ▪ Engine mount damaged
		  ▪ Transmission mount damaged
		  ▪ Accessory drive belt components incorrectly installed
		  ▪ Fuel injector contamination or blocked
		  ▪ Engine build excessive wear
			NOTE:
			The powertrain control module is unlikely to be the root cause for this DTC and must not be replaced without a Technical Assist Request (TA) being raised for further guidance
			• Check powertrain control module for any of the shown Diagnostic Trouble Code(s) and relating to the following systems or components. These must be repaired before additional investigations are conducted
			  ▪ Ignition system
			  ▪ Fuel system
			  ▪ Charge air system
			  ▪ Evaporative emissions canister purge valve
			• Prioritized Checks to Perform
			  ▪ Make sure vehicle has sufficient fuel
			  ▪ Check spark plug(s) for fouling or failure, install new as required
			  ▪ Check reluctor ring for correct installation
			  ▪ Check crankshaft position sensor for correct installation
			  ▪ Drain and refuel with new fuel of correct grade and known quality
			  ▪ Check driveline for debris around rotating components
			  ▪ Check driveline for excessive wear or backlash
			  ▪ Check wheels and tires for snow or mud packing and correct balance
			  ▪ Refer to the electrical circuit diagrams and check connections are secure and wiring integrity
			  ▪ Check cylinder compressions both cold and hot
			  ▪ Check for correct valve clearance
			  ▪ Check catalyst for damage or blockage
			  ▪ Check engine mounts for damage
			  ▪ Check transmission mounts for damage
			  ▪ Check accessory drive belt components are correctly installed
			  ▪ Check fuel injector for contamination or blockage
			  ▪ Check engine oil pressure is to specification
			  ▪ For further detailed information, please refer to the relevant section of the workshop manual. Section 303 - Electronic Engine Controls

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION

P0324-00	Knock/Combustion Vibration Control System Error - No sub type information	▪ Electrical Cause
		  ▪ Yes
		▪ Mechanical Cause
		  ▪ Yes
		▪ Monitor Description
		  ▪ Plausibility check of the knock window and complex driver circuit
		▪ Prioritized List of Possible Causes
		  ▪ Crankshaft position sensor signal is corrupted
		  ▪ Powertrain control module software failure
		  ▪ Powertrain control module hardware failure
			▪ Vehicle Conditions to Enable DTC Logging Strategy
			  ▪ Normal driving
			▪ Prioritized Checks to Perform
			  ▪ Check powertrain control module for crankshaft position sensor Diagnostic Trouble Code(s) and refer to this DTC index
			  ▪ Using the Jaguar Land Rover approved diagnostic equipment, run application - ECU Diagnostics / Powertrain Control Module (PCM) / Update ECU
			  ▪ Install a new powertrain control module. Using the Jaguar Land Rover approved diagnostic equipment, run application - ECU Diagnostics / Powertrain Control Module (PCM) / Replace ECU

P0327-00	Knock/Combustion Vibration Sensor 1 Circuit Low Bank 1 or Single Sensor - No sub type information	▪ Electrical Cause
		  ▪ Yes
		▪ Mechanical Cause
		  ▪ Yes
		▪ Control Module Cavity
		  ▪ Circuit reference - I_A_KS1B -
		▪ Monitor Description
		  ▪ Short circuit of single knock sensor line
		▪ Prioritized List of Possible Causes
		  ▪ Knock sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance
		  ▪ Knock sensor
		  ▪ Excessive mechanical noise from the engine
			▪ Vehicle Conditions to Enable DTC Logging Strategy
			  ▪ Normal driving
			▪ Prioritized Checks to Perform
			  ▪ Refer to the electrical circuit diagrams and check the knock sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance
			  ▪ Measure knock sensor circuit for correct resistance. Knock sensor resistance should be greater than 4 Mega Ohms
			  ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion
			  ▪ Using the Jaguar Land Rover approved diagnostic equipment, run application - ECU Diagnostics / Powertrain Control Module (PCM) / Clear Diagnostic Trouble Code(s). Retest
			  ▪ Install a new knock sensor as required

P0327-11	Knock/Combustion Vibration Sensor 1 Circuit Low Bank 1 or Single Sensor - Circuit short to ground	▪ Electrical Cause
		  ▪ Yes
		▪ Mechanical Cause
		  ▪ Yes
		▪ Control Module Cavity
		  ▪ Circuit reference - I_A_KS1B -
		▪ Monitor Description
		  ▪ Short circuit to ground of knock sensor line A
		▪ Prioritized List of Possible Causes
		  ▪ Knock sensor circuit short circuit to ground
		  ▪ Knock sensor
		  ▪ Excessive mechanical noise from the engine
			▪ Vehicle Conditions to Enable DTC Logging Strategy
			  ▪ Normal driving
			▪ Prioritized Checks to Perform
			  ▪ Refer to the electrical circuit diagrams and check the knock sensor circuit for short circuit to ground
			  ▪ Measure knock sensor circuit for correct resistance. Knock sensor resistance should be greater than 4 Mega Ohms
			  ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion
			  ▪ Using the Jaguar Land Rover approved diagnostic equipment, run application - ECU Diagnostics / Powertrain Control Module (PCM) / Clear Diagnostic Trouble Code(s). Retest
			  ▪ Install a new knock sensor as required

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION

P0327-21	Knock/Combustion Vibration Sensor 1 Circuit Low Bank 1 or Single Sensor - Signal amplitude < minimum	- Electrical Cause
		  - Yes
		- Mechanical Cause
		  - Yes
		- Control Module Cavity
		  - Circuit reference - I_A_KS1B -
		- Monitor Description
		  - Plausibility of sensor output against engine speed
		- Prioritized List of Possible Causes
		  - Engine misfire
		  - Knock sensor installation
		  - Knock sensor circuit short circuit to ground, short circuit to power
		  - Knock sensor
		  - Excessive mechanical noise from the engine
			- Vehicle Conditions to Enable DTC Logging Strategy
			  - Normal driving
			- Prioritized Checks to Perform
			  - Check powertrain control module for related misfire Diagnostic Trouble Code(s) and refer to this DTC index. Rectify these first
			  - Check knock sensor for correct installation
			  - Refer to the electrical circuit diagrams and check the knock sensor circuit for short circuit to ground, short circuit to power
			  - Measure knock sensor circuit for correct resistance. Knock sensor resistance should be greater than 4 Mega Ohms
			  - Inspect connectors for signs of water ingress, and pins for damage and/or corrosion
			  - Using the Jaguar Land Rover approved diagnostic equipment, run application - ECU Diagnostics / Powertrain Control Module (PCM) / Clear Diagnostic Trouble Code(s). Retest
			  - Install a new knock sensor as required

P0328-00	Knock/Combustion Vibration Sensor 1 Circuit High Bank 1 or Single Sensor - No sub type information	- Electrical Cause
		  - Yes
		- Mechanical Cause
		  - Yes
		- Control Module Cavity
		  - Circuit reference - I_A_KS1B -
		- Monitor Description
		  - Short circuit to battery of knock sensor line
		- Prioritized List of Possible Causes
		  - Knock sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance
		  - Knock sensor
		  - Excessive mechanical noise from the engine
			- Vehicle Conditions to Enable DTC Logging Strategy
			  - Normal driving
			- Prioritized Checks to Perform
			  - Refer to the electrical circuit diagrams and check the knock sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance
			  - Measure knock sensor circuit for correct resistance. Knock sensor resistance should be greater than 4 Mega Ohms
			  - Inspect connectors for signs of water ingress, and pins for damage and/or corrosion
			  - Using the Jaguar Land Rover approved diagnostic equipment, run application - ECU Diagnostics / Powertrain Control Module (PCM) / Clear Diagnostic Trouble Code(s). Retest
			  - Install a new knock sensor as required

P0328-12	Knock/Combustion Vibration Sensor 1 Circuit High Bank 1 or Single Sensor - Circuit short to battery	- Electrical Cause
		  - Yes
		- Mechanical Cause
		  - Yes
		- Control Module Cavity
		  - Circuit reference - I_A_KS1B -
		- Monitor Description
		  - Short circuit to battery of knock sensor 1 line A
		- Prioritized List of Possible Causes
		  - Knock sensor circuit short circuit to power
		  - Knock sensor
		  - Excessive mechanical noise from the engine
			- Vehicle Conditions to Enable DTC Logging Strategy
			  - Normal driving
			- Prioritized Checks to Perform
			  - Refer to the electrical circuit diagrams and check the knock sensor circuit for short circuit to power
			  - Measure knock sensor circuit for correct resistance. Knock sensor resistance should be greater than 4 Mega Ohms
			  - Inspect connectors for signs of water ingress, and pins for damage and/or corrosion
			  - Using the Jaguar Land Rover approved diagnostic equipment, run application - ECU Diagnostics / Powertrain Control Module (PCM) / Clear Diagnostic Trouble Code(s). Retest
			  - Install a new knock sensor as required

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION
P0328-22	Knock/Combustion Vibration Sensor 1 Circuit High Bank 1 or Single Sensor - Signal amplitude > maximum	▪ Electrical Cause ▪ Yes ▪ Mechanical Cause ▪ Yes ▪ Control Module Cavity ▪ Circuit reference - I_A_KS1B - ▪ Monitor Description ▪ Plausibility of sensor output against engine speed ▪ Prioritized List of Possible Causes ▪ Knock sensor circuit short circuit to power ▪ Knock sensor ▪ Excessive mechanical noise from the engine	▪ Vehicle Conditions to Enable DTC Logging Strategy ▪ Normal driving ▪ Prioritized Checks to Perform ▪ Refer to the electrical circuit diagrams and check the knock sensor circuit for short circuit to power ▪ Measure knock sensor circuit for correct resistance. Knock sensor resistance should be greater than 4 Mega Ohms ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion ▪ Check for abnormal engine noise that may corrupt the knock sensor signal ▪ Using the Jaguar Land Rover approved diagnostic equipment, run application - ECU Diagnostics / Powertrain Control Module (PCM) / Clear Diagnostic Trouble Code(s). Retest ▪ Install a new knock sensor as required
P0332-00	Knock/Combustion Vibration Sensor 2 Circuit Low Bank 2 - No sub type information	▪ Electrical Cause ▪ Yes ▪ Mechanical Cause ▪ Yes ▪ Control Module Cavity ▪ Circuit reference - I_A_KS2B - ▪ Monitor Description ▪ Short circuit of single knock sensor line ▪ Prioritized List of Possible Causes ▪ Knock sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Knock sensor	▪ Vehicle Conditions to Enable DTC Logging Strategy ▪ Normal driving ▪ Prioritized Checks to Perform ▪ Refer to the electrical circuit diagrams and check the knock sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance ▪ Measure knock sensor circuit for correct resistance. Knock sensor resistance should be greater than 4 Mega Ohms ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion ▪ Using the Jaguar Land Rover approved diagnostic equipment, run application - ECU Diagnostics / Powertrain Control Module (PCM) / Clear Diagnostic Trouble Code(s). Retest ▪ Install a new knock sensor as required
P0332-11	Knock/Combustion Vibration Sensor 2 Circuit Low Bank 2 - Circuit short to ground	▪ Electrical Cause ▪ Yes ▪ Mechanical Cause ▪ Yes ▪ Control Module Cavity ▪ Circuit reference - I_A_KS2B - ▪ Monitor Description ▪ Short circuit of single knock sensor line ▪ Prioritized List of Possible Causes ▪ Knock sensor circuit short circuit to ground ▪ Knock sensor	▪ Vehicle Conditions to Enable DTC Logging Strategy ▪ Normal driving ▪ Prioritized Checks to Perform ▪ Refer to the electrical circuit diagrams and check the knock sensor circuit for short circuit to ground ▪ Measure knock sensor circuit for correct resistance. Knock sensor resistance should be greater than 4 Mega Ohms ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion ▪ Using the Jaguar Land Rover approved diagnostic equipment, run application - ECU Diagnostics / Powertrain Control Module (PCM) / Clear Diagnostic Trouble Code(s). Retest ▪ Install a new knock sensor as required

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION

P0332-21	Knock/Combustion Vibration Sensor 2 Circuit Low Bank 2 - Signal amplitude < minimum	▪ Electrical Cause
		  ▪ Yes
		▪ Mechanical Cause
		  ▪ Yes
		▪ Control Module Cavity
		  ▪ Circuit reference - I_A_KS2B -
		▪ Monitor Description
		  ▪ Plausibility of sensor output against engine speed
		▪ Prioritized List of Possible Causes
		  ▪ Engine misfire
		  ▪ Knock sensor circuit short circuit to ground, short circuit to power
		  ▪ Knock sensor
			▪ Vehicle Conditions to Enable DTC Logging Strategy
			  ▪ Normal driving
			▪ Prioritized Checks to Perform
			  ▪ Check powertrain control module for related misfire Diagnostic Trouble Code(s) and refer to this DTC index. Rectify these first
			  ▪ Refer to the electrical circuit diagrams and check the knock sensor circuit for short circuit to ground, short circuit to power
			  ▪ Measure knock sensor circuit for correct resistance. Knock sensor resistance should be greater than 4 Mega Ohms
			  ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion
			  ▪ Using the Jaguar Land Rover approved diagnostic equipment, run application - ECU Diagnostics / Powertrain Control Module (PCM) / Clear Diagnostic Trouble Code(s). Retest
			  ▪ Install a new knock sensor as required

P0333-00	Knock/Combustion Vibration Sensor 2 Circuit High Bank 2 - No sub type information	▪ Electrical Cause
		  ▪ Yes
		▪ Mechanical Cause
		  ▪ Yes
		▪ Control Module Cavity
		  ▪ Circuit reference - I_A_KS2B -
		▪ Monitor Description
		  ▪ Short circuit of single knock sensor line
		▪ Prioritized List of Possible Causes
		  ▪ Knock sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance
		  ▪ Knock sensor
			▪ Vehicle Conditions to Enable DTC Logging Strategy
			  ▪ Normal driving
			▪ Prioritized Checks to Perform
			  ▪ Refer to the electrical circuit diagrams and check the knock sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance
			  ▪ Measure knock sensor circuit for correct resistance. Knock sensor resistance should be greater than 4 Mega Ohms
			  ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion
			  ▪ Using the Jaguar Land Rover approved diagnostic equipment, run application - ECU Diagnostics / Powertrain Control Module (PCM) / Clear Diagnostic Trouble Code(s). Retest
			  ▪ Install a new knock sensor as required

P0333-12	Knock/Combustion Vibration Sensor 2 Circuit High Bank 2 - Circuit short to battery	▪ Electrical Cause
		  ▪ Yes
		▪ Mechanical Cause
		  ▪ Yes
		▪ Control Module Cavity
		  ▪ Circuit reference - I_A_KS2B -
		▪ Monitor Description
		  ▪ Short circuit of single knock sensor line
		▪ Prioritized List of Possible Causes
		  ▪ Knock sensor circuit short circuit to power
		  ▪ Knock sensor
			▪ Vehicle Conditions to Enable DTC Logging Strategy
			  ▪ Normal driving
			▪ Prioritized Checks to Perform
			  ▪ Refer to the electrical circuit diagrams and check the knock sensor circuit for short circuit to power
			  ▪ Measure knock sensor circuit for correct resistance. Knock sensor resistance should be greater than 4 Mega Ohms
			  ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion
			  ▪ Using the Jaguar Land Rover approved diagnostic equipment, run application - ECU Diagnostics / Powertrain Control Module (PCM) / Clear Diagnostic Trouble Code(s). Retest
			  ▪ Install a new knock sensor as required

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION
P0333-22	Knock/Combustion Vibration Sensor 2 Circuit High Bank 2 - Signal amplitude > maximum	▪ Electrical Cause ▪ Yes ▪ Mechanical Cause ▪ Yes ▪ Control Module Cavity ▪ Circuit reference I_A_KS2B ▪ Monitor Description ▪ Plausibility of sensor output against engine speed ▪ Prioritized List of Possible Causes ▪ Knock sensor circuit short circuit to power ▪ Knock sensor ▪ Excessive mechanical noise from the engine	▪ Vehicle Conditions to Enable DTC Logging Strategy ▪ Normal driving ▪ Prioritized Checks to Perform ▪ Refer to the electrical circuit diagrams and check the knock sensor circuit for short circuit to power ▪ Measure knock sensor circuit for correct resistance. Knock sensor resistance should be greater than 4 Mega Ohms ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion ▪ Check for abnormal engine noise that may corrupt the knock sensor signal ▪ Using the Jaguar Land Rover approved diagnostic equipment, run application - ECU Diagnostics / Powertrain Control Module (PCM) / Clear Diagnostic Trouble Code(s). Retest ▪ Install a new knock sensor as required
P0335-00	Crankshaft Position Sensor "A" Circuit - No sub type information	▪ Electrical Cause ▪ Yes ▪ Mechanical Cause ▪ Yes ▪ Control Module Cavity ▪ Circuit reference - O_V_5VCRS - ▪ Circuit reference - I_F_CRSPOS - ▪ Circuit reference - G_R_CRS - ▪ Monitor Description ▪ No signal detected from the crankshaft position sensor ▪ Prioritized List of Possible Causes ▪ Crankshaft position sensor circuit, short circuit to ground, short circuit to power, open circuit, high resistance ▪ Crankshaft position sensor air gap to target rotor excessive ▪ Crankshaft position sensor failure	▪ Vehicle Conditions to Enable DTC Logging Strategy ▪ Engine cranking ▪ Engine running ▪ Prioritized Checks to Perform ▪ Refer to electrical circuit diagrams and check the crankshaft position sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion ▪ Check crankshaft position sensor air gap to target rotor is correct ▪ Install a new crankshaft position sensor as required

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION

P0336-00	Crankshaft Position Sensor "A" Circuit Range/Performance - No sub type information	▪ Electrical Cause
		  ▪ Yes
		▪ Mechanical Cause
		  ▪ Yes
		▪ Control Module Cavity
		  ▪ Circuit reference - O_V_5VCRS -
		  ▪ Circuit reference - I_F_CRSPOS -
		  ▪ Circuit reference - G_R_CRS -
		▪ Monitor Description
		  ▪ Disturbance detected on the crankshaft position sensor
		▪ Prioritized List of Possible Causes
		  ▪ Crankshaft position sensor circuit, short circuit to ground, short circuit to power, open circuit, high resistance
		  ▪ Crankshaft position sensor air gap to target rotor excessive
		  ▪ Crankshaft position sensor failure
			▪ Vehicle Conditions to Enable DTC Logging Strategy
			  ▪ Engine cranking
			  ▪ Engine running
			▪ Prioritized Checks to Perform
			  ▪ Refer to electrical circuit diagrams and check the crankshaft position sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance
			  ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion
			  ▪ Check crankshaft position sensor air gap to target rotor is correct
			  ▪ Install a new crankshaft position sensor as required

P0336-03	Crankshaft Position Sensor "A" Circuit Range/Performance - FM (Frequency Modulated) / PWM (Pulse Width Modulated) failures	▪ Electrical Cause
		  ▪ Yes
		▪ Mechanical Cause
		  ▪ Yes
		▪ Control Module Cavity
		  ▪ Circuit reference - O_V_5VCRS -
		  ▪ Circuit reference - I_F_CRSPOS -
		  ▪ Circuit reference - G_R_CRS -
		▪ Monitor Description
		  ▪ Monitors the PWM signal of the sensor. Detects wrong engine stop position, not plausible reverse rotation
		▪ Prioritized List of Possible Causes
		  ▪ Crankshaft position sensor circuit, short circuit to ground, short circuit to power, open circuit, high resistance
		  ▪ Crankshaft position sensor air gap to target rotor excessive
		  ▪ Crankshaft position sensor failure
			▪ Vehicle Conditions to Enable DTC Logging Strategy
			  ▪ Engine cranking
			  ▪ Engine running
			▪ Prioritized Checks to Perform
			  ▪ Refer to electrical circuit diagrams and check the crankshaft position sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance
			  ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion
			  ▪ Check crankshaft position sensor air gap to target rotor is correct
			  ▪ Install a new crankshaft position sensor as required

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION

P0341-00	Camshaft Position Sensor "A" Circuit Range/Performance Bank 1 or Single Sensor - No sub type information	▪ Electrical Cause
		  ▪ Yes
		▪ Mechanical Cause
		  ▪ Yes
		▪ Control Module Cavity
		  ▪ Circuit reference - G_R_CAS -
		  ▪ Circuit reference - I_F_CASI -
		▪ Monitor Description
		  ▪ Disturbance detected on the camshaft signal
		▪ Prioritized List of Possible Causes
		  ▪ Loose camshaft position sensor
		  ▪ Camshaft position sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance
		  ▪ Camshaft position sensor reluctor ring air gap excessive
			▪ Prioritized Checks to Perform
			  ▪ Check camshaft position sensor is installed correctly
			  ▪ Refer to the electrical circuit diagrams and check the camshaft position sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance
			  ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion
			  ▪ Check reluctor ring to sensor run-out and air gap are within specification
			  ▪ After timing related repairs and timing adjustment, using the Jaguar Land Rover approved diagnostic equipment, run application - ECU Diagnostics / Powertrain Control Module (PCM) / ECU Functions / Adaption Clear

P0342-00	Camshaft Position Sensor "A" Circuit Low Bank 1 or Single Sensor - No sub type information	▪ Electrical Cause
		  ▪ Yes
		▪ Mechanical Cause
		  ▪ Yes
		▪ Control Module Cavity
		  ▪ Circuit reference - G_R_CAS -
		  ▪ Circuit reference - I_F_CASI -
		▪ Monitor Description
		  ▪ Detection of a missing camshaft signal, no signal edge present. Level on the input is low
		▪ Prioritized List of Possible Causes
		  ▪ Loose camshaft position sensor
		  ▪ Camshaft position sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance
		  ▪ Camshaft position sensor reluctor ring air gap excessive
			▪ Prioritized Checks to Perform
			  ▪ Check camshaft position sensor is installed correctly
			  ▪ Refer to the electrical circuit diagrams and check the camshaft position sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance
			  ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion
			  ▪ Check reluctor ring to sensor run-out and air gap are within specification
			  ▪ After timing related repairs and timing adjustment, using the Jaguar Land Rover approved diagnostic equipment, run application - ECU Diagnostics / Powertrain Control Module (PCM) / ECU Functions / Adaption Clear

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION
P0343-00	Camshaft Position Sensor "A" Circuit High Bank 1 or Single Sensor - No sub type information	▪ Electrical Cause 　▪ Yes ▪ Mechanical Cause 　▪ Yes ▪ Control Module Cavity 　▪ Circuit reference - G_R_CAS - 　▪ Circuit reference - I_F_CASI - ▪ Monitor Description 　▪ Detection of a missing camshaft signal, no signal edge present. Level on the input is high ▪ Prioritized List of Possible Causes 　▪ Loose camshaft position sensor 　▪ Camshaft position sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance 　▪ Camshaft position sensor reluctor ring air gap excessive	▪ Prioritized Checks to Perform 　▪ Check camshaft position sensor is installed correctly 　▪ Refer to the electrical circuit diagrams and check the camshaft position sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance 　▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion 　▪ Check reluctor ring to sensor run-out and air gap are within specification 　▪ After timing related repairs and timing adjustment, using the Jaguar Land Rover approved diagnostic equipment, run application - ECU Diagnostics / Powertrain Control Module (PCM) / ECU Functions / Adaption Clear
P0350-85	Ignition Coil Primary/Secondary Circuit/Open - Signal above allowable range	▪ Prioritized List of Possible Causes 　▪ The powertrain control module has determined failures where some circuit quantity, reported through serial data, is above a specified range 　▪ Other related Diagnostic Trouble Code(s) 　▪ Power and ground connections to the powertrain control module open circuit 　▪ Powertrain control module failure	▪ Prioritized Checks to Perform 　▪ Check powertrain control module for additional Diagnostic Trouble Code(s) and refer to relevant DTC index. Rectify these first 　▪ Refer to the electrical circuit diagrams and check the power and ground connections to the module 　▪ Install a new powertrain control module as required 　▪ Using the Jaguar Land Rover approved diagnostic equipment, run application - ECU Diagnostics / Powertrain Control Module (PCM) / Clear Diagnostic Trouble Code(s). Retest

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION

P0351-13	Ignition Coil "A" Primary Control Circuit/Open - Circuit open	▪ Electrical Cause
		  ▪ Yes
		▪ Mechanical Cause
		  ▪ Yes
		▪ Control Module Cavity
		  ▪ Circuit reference - O_P_IOS1 -
		▪ Monitor Description
		  ▪ Open circuit detected by the ignition control module, then passed to the main CPU (Internal to the powertrain control module)
		▪ Prioritized List of Possible Causes
		  ▪ The powertrain control module has determined an open circuit through lack of bias voltage, low current flow, no change in the state of an input in response to an output
		  ▪ Ignition coil connector is disconnected, connector pin is backed out, connector pin corrosion
		  ▪ Ignition coil fuse supply failure
		  ▪ Ignition coil circuit open circuit
			▪ Vehicle Conditions to Enable DTC Logging Strategy
			  ▪ Engine running
			▪ Prioritized Checks to Perform
			  ▪ Check ignition coil connector is not disconnected, connector pin is not backed out
			  ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion
			  ▪ Refer to the electrical circuit diagrams and check the ignition coil circuit for open circuit

P0352-13	Ignition Coil "B" Primary Control Circuit/Open - Circuit open	▪ Electrical Cause
		  ▪ Yes
		▪ Mechanical Cause
		  ▪ Yes
		▪ Control Module Cavity
		  ▪ Circuit reference - O_P_IOS2 -
		▪ Monitor Description
		  ▪ Open circuit detected by the ignition control module, then passed to the main CPU (Internal to the powertrain control module)
		▪ Prioritized List of Possible Causes
		  ▪ The powertrain control module has determined an open circuit through lack of bias voltage, low current flow, no change in the state of an input in response to an output
		  ▪ Ignition coil connector is disconnected, connector pin is backed out, connector pin corrosion
		  ▪ Ignition coil fuse supply failure
		  ▪ Ignition coil circuit open circuit
			▪ Vehicle Conditions to Enable DTC Logging Strategy
			  ▪ Engine running
			▪ Prioritized Checks to Perform
			  ▪ Check ignition coil connector is not disconnected, connector pin is not backed out
			  ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion
			  ▪ Refer to the electrical circuit diagrams and check the ignition coil circuit for open circuit

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION

P0353-13	Ignition Coil "C" Primary Control Circuit/Open - Circuit open	▪ Electrical Cause
		  ▪ Yes
		▪ Mechanical Cause
		  ▪ Yes
		▪ Control Module Cavity
		  ▪ Circuit reference - O_P_IOS3 -
		▪ Monitor Description
		  ▪ Open circuit detected by the ignition control module, then passed to the main CPU (Internal to the powertrain control module)
		▪ Prioritized List of Possible Causes
		  ▪ The powertrain control module has determined an open circuit through lack of bias voltage, low current flow, no change in the state of an input in response to an output
		  ▪ Ignition coil connector is disconnected, connector pin is backed out, connector pin corrosion
		  ▪ Ignition coil fuse supply failure
		  ▪ Ignition coil circuit open circuit
			▪ Vehicle Conditions to Enable DTC Logging Strategy
			  ▪ Engine running
			▪ Prioritized Checks to Perform
			  ▪ Check ignition coil connector is not disconnected, connector pin is not backed out
			  ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion
			  ▪ Refer to the electrical circuit diagrams and check the ignition coil circuit for open circuit

P0354-13	Ignition Coil "D" Primary Control Circuit/Open - Circuit open	▪ Electrical Cause
		  ▪ Yes
		▪ Mechanical Cause
		  ▪ Yes
		▪ Control Module Cavity
		  ▪ Circuit reference - O_P_IOS4 -
		▪ Monitor Description
		  ▪ Open circuit detected by the ignition control module, then passed to the main CPU (Internal to the powertrain control module)
		▪ Prioritized List of Possible Causes
		  ▪ The powertrain control module has determined an open circuit through lack of bias voltage, low current flow, no change in the state of an input in response to an output
		  ▪ Ignition coil connector is disconnected, connector pin is backed out, connector pin corrosion
		  ▪ Ignition coil fuse supply failure
		  ▪ Ignition coil circuit open circuit
			▪ Vehicle Conditions to Enable DTC Logging Strategy
			  ▪ Engine running
			▪ Prioritized Checks to Perform
			  ▪ Check ignition coil connector is not disconnected, connector pin is not backed out
			  ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion
			  ▪ Refer to the electrical circuit diagrams and check the ignition coil circuit for open circuit

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION
P0366-00	Camshaft Position Sensor "B" Circuit Range/Performance Bank 1 - No sub type information	- Electrical Cause - Yes - Mechanical Cause - Yes - Control Module Cavity - Circuit reference - G_R_CAS - - Circuit reference - I_F_CASI - - Monitor Description - Disturbance detected on the camshaft signal - Prioritized List of Possible Causes - Loose camshaft position sensor - Camshaft position sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance - Camshaft position sensor reluctor ring air gap excessive	- Prioritized Checks to Perform - Check camshaft position sensor is installed correctly - Refer to the electrical circuit diagrams and check the camshaft position sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance - Inspect connectors for signs of water ingress, and pins for damage and/or corrosion - Check reluctor ring to sensor run-out and air gap are within specification - After timing related repairs and timing adjustment, using the Jaguar Land Rover approved diagnostic equipment, run application - ECU Diagnostics / Powertrain Control Module (PCM) / ECU Functions / Adaption Clear
P0367-00	Camshaft Position Sensor "B" Circuit Low Bank 1 - No sub type information	- Electrical Cause - Yes - Mechanical Cause - Yes - Control Module Cavity - Circuit reference - G_R_CAS - - Circuit reference - I_F_CASI - - Monitor Description - Detection of a missing camshaft signal, no signal edge present. Level on the input is low - Prioritized List of Possible Causes - Loose camshaft position sensor - Camshaft position sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance - Camshaft position sensor reluctor ring air gap excessive	- Prioritized Checks to Perform - Check camshaft position sensor is installed correctly - Refer to the electrical circuit diagrams and check the camshaft position sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance - Inspect connectors for signs of water ingress, and pins for damage and/or corrosion - Check reluctor ring to sensor run-out and air gap are within specification - After timing related repairs and timing adjustment, using the Jaguar Land Rover approved diagnostic equipment, run application - ECU Diagnostics / Powertrain Control Module (PCM) / ECU Functions / Adaption Clear

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION
P0368-00	Camshaft Position Sensor "B" Circuit High Bank 1 - No sub type information	- Electrical Cause - Yes - Mechanical Cause - Yes - Control Module Cavity - Circuit reference - G_R_CAS - - Circuit reference - I_F_CASI - - Monitor Description - Detection of a missing camshaft signal, no signal edge present. Level on the input is high - Prioritized List of Possible Causes - Loose camshaft position sensor - Camshaft position sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance - Camshaft position sensor reluctor ring air gap excessive	- Prioritized Checks to Perform - Check camshaft position sensor is installed correctly - Refer to the electrical circuit diagrams and check the camshaft position sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance - Inspect connectors for signs of water ingress, and pins for damage and/or corrosion - Check reluctor ring to sensor run-out and air gap are within specification - After timing related repairs and timing adjustment, using the Jaguar Land Rover approved diagnostic equipment, run application - ECU Diagnostics / Powertrain Control Module (PCM) / ECU Functions / Adaption Clear
P0420-00	Catalyst System Efficiency Below Threshold Bank 1 - No sub type information	- Prioritized List of Possible Causes - Exhaust system leakage - Catalytic converter failed, caused by lean combustion - Catalytic converter failed, caused by misfire - Catalytic converter failed, caused by excessive oil consumption - Catalytic converter failed, caused by contaminated fuel	- Prioritized Checks to Perform - Using the Jaguar Land Rover approved diagnostic equipment, check powertrain control module, for related Diagnostic Trouble Code(s) and refer to the relevant DTC index - Check exhaust system for leakage - Refer to the relevant sections of the workshop manual and check the induction system for air leaks - Refer to the relevant sections of the workshop manual and check the ignition system for causes of misfire - Refer to the relevant sections of the workshop manual and complete oil consumption checks - Confirm with the customer that the correct fuel has been used

<table>
<thead>
<tr><th>DTC</th><th>DESCRIPTION</th><th>POSSIBLE CAUSES</th><th>ACTION</th></tr>
</thead>
<tbody>
<tr>
<td>P043E-00</td>
<td>Evaporative Emission System Leak Detection Reference Orifice Low Flow - No sub type information</td>
<td>Prioritized List of Possible Causes
The PCM may not be calibrated with the latest level of software. Earlier levels of software may lead to this DTC being set
Diagnostic module tank leakage module internal failure</td>
<td>NOTES:
A guided flow is available for this DTC. To assist with correct diagnosis, please make sure to run the guided flow.
If purge valve related Diagnostic Trouble Code(s) are also set, perform the relevant corrective action(s) first.
If evaporative emissions system leak related Diagnostic Trouble Code(s) are also set, perform the relevant corrective action(s) first.
It is not possible to replicate the failure event as certain entry conditions must be satisfied (for more information, refer to section 303-13: Evaporative Emissions / Description and Operation). To verify the customer concern, perform routine Evaporative System Diagnostic Check and re-read Diagnostic Trouble Code(s).
Prioritized Checks to Perform
Check the PCM is calibrated with the latest level of software. If the latest level of software is NOT installed:
update the powertrain control module to the latest software calibration
Using the Jaguar Land Rover approved diagnostic equipment, clear the DTC and retest
If the fault persists after a software update, continue to the next diagnostic step below
Install a new diagnostic module tank leakage module as necessary</td>
</tr>
<tr>
<td>P043F-00</td>
<td>Evaporative Emission System Leak Detection Reference Orifice High Flow - No sub type information</td>
<td>Prioritized List of Possible Causes
Diagnostic module tank leakage module internal failure</td>
<td>NOTES:
A guided flow is available for this DTC. To assist with correct diagnosis, please make sure to run the guided flow.
If purge valve related Diagnostic Trouble Code(s) are also set, perform the relevant corrective action(s) first.
If evaporative emissions system leak related Diagnostic Trouble Code(s) are also set, perform the relevant corrective action(s) first.
It is not possible to replicate the failure event as certain entry conditions must be satisfied (for more information, refer to section 303-13: Evaporative Emissions / Description and Operation). To verify the customer concern, perform routine Evaporative System Diagnostic Check and re-read Diagnostic Trouble Code(s).
Prioritized Checks to Perform
Install a new diagnostic module tank leakage module as necessary</td>
</tr>
</tbody>
</table>

<table>
<tr><th>DTC</th><th>DESCRIPTION</th><th>POSSIBLE CAUSES</th><th>ACTION</th></tr>
<tr><td>P0440-00</td><td>Evaporative Emission System - No sub type information</td><td>Electrical Cause
No
Mechanical Cause
Yes
Monitor Description
Monitors the purge vent line for a blockage
Prioritized List of Possible Causes
Evaporative system mechanical integrity - Trapped pipe, blockage within pipe, restricted pipe</td><td>Vehicle Conditions to Enable DTC Logging Strategy
Drive vehicle for short period, idle vehicle
Prioritized Checks to Perform
Refer to the relevant section of the workshop manual and check the mechanical integrity of the evaporative system - Trapped pipe, blockage within pipe, restricted pipe
Using the Jaguar Land Rover approved diagnostic equipment, run application - ECU Diagnostics / Powertrain Control Module (PCM) / Clear Diagnostic Trouble Code(s). Retest</td></tr>
<tr><td>P0442-00</td><td>Evaporative Emission System Leak Detected (small leak) - No sub type information</td><td>Electrical Cause
No
Mechanical Cause
Yes
Monitor Description
Engine off, diagnostic module tank leakage module drives the internal diagnostic module tank leakage pump and solenoid valve, can operate for between 200 and 400 seconds
Prioritized List of Possible Causes
Evaporative emissions system leak
Evaporative emission system purge control valve stuck open
Loose fuel tank filler cap
Blocked or frozen diagnostic module tank leakage module air filter (if present during winter ambient conditions)</td><td>NOTES:
A guided flow is available for this DTC. To assist with correct diagnosis, please make sure to run the guided flow.
If purge valve related Diagnostic Trouble Code(s) are also set, perform the relevant corrective action(s) first.
It is not possible to replicate the failure event as certain entry conditions must be satisfied (for more information, refer to section 303-13: Evaporative Emissions / Description and Operation). To verify the customer concern, perform routine Evaporative System Diagnostic Check and re-read Diagnostic Trouble Code(s).
Vehicle Conditions to Enable DTC Logging Strategy
Make sure that the vehicle has a fuel level between 15% and 85%
Ambient temperature is between 32°F and 95°F
Using the Jaguar Land Rover approved diagnostic equipment complete "large leak test"
Prioritized Checks to Perform
Refer to the relevant section of the workshop manual and check the evaporative emissions system for leaks
Inspect connectors for signs of water ingress, and pins for damage and/or corrosion
Check fuel cap installation
Check fuel cap for correct sealing
Check diagnostic module tank leakage air filter</td></tr>
</table>

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION

P0444-00	Evaporative Emission System Purge Control Valve "A" Circuit Open - No sub type information	▪ Electrical Cause
		  ▪ Yes
		▪ Mechanical Cause
		  ▪ No
		▪ Control Module Cavity
		  ▪ Circuit reference - O_T_CPV -
		▪ Monitor Description
		  ▪ The diagnostic function monitors the evaporative emission system purge control valve actuator power stage driver for correct operation
		▪ Prioritized List of Possible Causes
		  ▪ Purge control valve circuit, open circuit, high resistance
		  ▪ Purge control valve failure
			▪ Prioritized Checks to Perform
			  ▪ Refer to electrical circuit diagrams and check the purge control valve circuit for open circuit, high resistance
			  ▪ Install a new purge control valve as required

P0451-2A	EVAP System Pressure Sensor/Switch "A" Circuit Range/Performance - Signal stuck in range	▪ Electrical Cause
		  ▪ Yes
		▪ Mechanical Cause
		  ▪ Yes
		▪ Evaporative emission system integrity
		▪ Incorrect purge valve installed
		▪ Connector is disconnected, connector pin is backed out, connector pin corrosion
		▪ The powertrain control module may not be calibrated with the latest level of software. Earlier levels of software may lead to this DTC being set
		▪ Incorrect powertrain control module installed
		▪ EVAP system differential pressure sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance
		▪ EVAP system differential pressure sensor failure
			▪ Check evaporative emission system integrity
			▪ Check the correct purge valve is installed and rectify as required
			▪ Inspect connectors for secure connection, signs of water ingress, check pins for damage and/or corrosion. Rectify as required
			▪ Using the Jaguar Land Rover approved diagnostic equipment, reconfigure the powertrain control module with the latest level software
			▪ Check the correct powertrain control module is installed and rectify as required
			▪ Refer to the electrical circuit diagrams and check the EVAP system differential pressure sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance
			▪ Check and install a new EVAP system differential pressure sensor as required
			▪ Using the Jaguar Land Rover approved diagnostic equipment, clear the Diagnostic Trouble Code(s) and retest

<table>
<tr><th>DTC</th><th>DESCRIPTION</th><th>POSSIBLE CAUSES</th><th>ACTION</th></tr>
<tr><td>P0451-92</td><td>EVAP System Pressure Sensor/Switch "A" Circuit Range/Performance - Performance or incorrect operation</td><td>Electrical Cause
Yes
Mechanical Cause
Yes
Evaporative emission system integrity
Incorrect purge valve installed
Connector is disconnected, connector pin is backed out, connector pin corrosion
The powertrain control module may not be calibrated with the latest level of software. Earlier levels of software may lead to this DTC being set
Incorrect powertrain control module installed
EVAP system differential pressure sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance
EVAP system differential pressure sensor failure</td><td>Check evaporative emission system integrity
Check the correct purge valve is installed and rectify as required
Inspect connectors for secure connection, signs of water ingress, check pins for damage and/or corrosion. Rectify as required
Using the Jaguar Land Rover approved diagnostic equipment, reconfigure the powertrain control module with the latest level software
Check the correct powertrain control module is installed and rectify as required
Refer to the electrical circuit diagrams and check the EVAP system differential pressure sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance
Check and install a new EVAP system differential pressure sensor as required
Using the Jaguar Land Rover approved diagnostic equipment, clear the Diagnostic Trouble Code(s) and retest</td></tr>
<tr><td>P0452-11</td><td>EVAP System Pressure Sensor/Switch "A" Circuit Low - Circuit short to ground</td><td>Electrical Cause
Yes
Mechanical Cause
Yes
Connector is disconnected, connector pin is backed out, connector pin corrosion
EVAP system differential pressure sensor circuit short circuit to ground
EVAP system differential pressure sensor failure</td><td>Vehicle Conditions to enable DTC Logging strategy
Ignition on
Inspect connectors for signs of water ingress, and pins for damage and/or corrosion
Refer to the electrical circuit diagrams and check the EVAP system differential pressure sensor circuit for short circuit to ground
Using the Jaguar Land Rover approved diagnostic equipment, clear the Diagnostic Trouble Code(s) and retest
Check and install a new EVAP system differential pressure sensor as required</td></tr>
<tr><td>P0453-12</td><td>EVAP System Pressure Sensor/Switch "A" Circuit High - Circuit short to battery</td><td>Electrical Cause
Yes
Mechanical Cause
Yes
Connector is disconnected, connector pin is backed out, connector pin corrosion
EVAP system differential pressure sensor circuit short circuit to power
EVAP system differential pressure sensor failure</td><td>Vehicle Conditions to enable DTC Logging strategy
Ignition on
Inspect connectors for signs of water ingress, and pins for damage and/or corrosion
Refer to the electrical circuit diagrams and check the EVAP system differential pressure sensor circuit for short circuit to power
Using the Jaguar Land Rover approved diagnostic equipment, clear the Diagnostic Trouble Code(s) and retest
Check and install a new EVAP system differential pressure sensor as required</td></tr>
</table>

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION

P0456-00	Evaporative Emission System Leak Detected (very small leak) - No sub type information	• Electrical Cause
		  ▪ No
		• Mechanical Cause
		  ▪ Yes
		• Monitor Description
		  ▪ Engine off, diagnostic module tank leakage module drives the internal diagnostic module tank leakage pump and solenoid valve, can operate for up to 900 seconds
		• Prioritized List of Possible Causes
		  ▪ Leak detection pump system has detected a leak
		  ▪ Leak detection pump circuit, short circuit to ground, short circuit to power, open circuit, high resistance
			NOTE:
			A guided flow is available for this DTC. To assist with correct diagnosis, please make sure to run the guided flow.
			• Prioritized Checks to Perform
			  ▪ Check evaporative emission system for leak using appropriate smoke/leak tester
			  ▪ Refer to electrical circuit diagrams and check the leak detection pump circuit for short circuit to ground, short circuit to power, open circuit, high resistance

P0457-76	Evaporative Emission System Leak Detected (fuel cap loose/off) - Wrong mounting position	• Electrical Cause
		  ▪ No
		• Mechanical Cause
		  ▪ Yes
		• Monitor Description
		  ▪ Operates following a re-fuel event while the vehicle is driving, the diagnostic module tank leakage module drives the internal diagnostic module tank leakage pump and solenoid valve, can run between 200 to 400 seconds
		• Prioritized List of Possible Causes
		  ▪ Evaporative emissions system leak
			• Vehicle Conditions to Enable DTC Logging Strategy
			  ▪ Make sure that the vehicle has a fuel level between 15% and 85%
			  ▪ Ambient temperature is between 32°F and 95°F
			  ▪ Using the Jaguar Land Rover approved diagnostic equipment complete "small leak test"
			• Prioritized Checks to Perform
			  ▪ Refer to the relevant section of the workshop manual and check the evaporative emissions system for leaks
			  ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion
			  ▪ Check fuel cap installation
			  ▪ Check fuel cap for correct sealing
			  ▪ Check diagnostic module tank leakage air filter

P0458-00	Evaporative Emission System Purge Control Valve "A" Circuit Low - No sub type information	• Electrical Cause
		  ▪ Yes
		• Mechanical Cause
		  ▪ No
		• Control Module Cavity
		  ▪ Circuit reference - O_T_CPV -
		• Monitor Description
		  ▪ The diagnostic function monitors the evaporative emission system purge control valve actuator power stage driver for correct operation
		• Prioritized List of Possible Causes
		  ▪ Purge valve circuit short circuit to ground
			• Vehicle Conditions to Enable DTC Logging Strategy
			  ▪ Ignition on. Engine running
			• Prioritized Checks to Perform
			  ▪ Refer to the electrical circuit diagrams and check the purge valve circuit for short circuit to ground

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION

P0459-00	Evaporative Emission System Purge Control Valve "A" Circuit High - No sub type information	▪ Electrical Cause
		  ▪ Yes
		▪ Mechanical Cause
		  ▪ No
		▪ Control Module Cavity
		  ▪ Circuit reference - O_T_CPV -
		▪ Monitor Description
		  ▪ The diagnostic function monitors the evaporative emission system purge control valve actuator power stage driver for correct operation
		▪ Prioritized List of Possible Causes
		  ▪ Purge valve circuit short circuit to power
			▪ Vehicle Conditions to Enable DTC Logging Strategy
			  ▪ Ignition on. Engine running
			▪ Prioritized Checks to Perform
			  ▪ Refer to the electrical circuit diagrams and check the purge valve circuit for short circuit to power

P0480-11	Fan 1 Control Circuit - Circuit short to ground	▪ Electrical Cause
		  ▪ Yes
		▪ Mechanical Cause
		  ▪ Yes
		▪ Control Module Cavity
		  ▪ Circuit reference - O_T_CLG -
		▪ Monitor Description
		  ▪ Circuit check
		▪ Prioritized List of Possible Causes
		  ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion
		  ▪ Engine cooling fan circuit short circuit to ground
		  ▪ Engine cooling fan failure
			▪ Vehicle Conditions to Enable DTC Logging Strategy
			  ▪ Ignition on
			▪ Prioritized Checks to Perform
			  ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion
			  ▪ Refer to the electrical circuit diagrams and check the engine cooling fan circuit for short circuit to ground
			  ▪ Using the Jaguar Land Rover approved diagnostic equipment, run application - ECU Diagnostics / Powertrain Control Module (PCM) / Clear Diagnostic Trouble Code(s). Retest
			  ▪ Install a new engine cooling fan as required

P0480-12	Fan 1 Control Circuit - Circuit short to battery	▪ Electrical Cause
		  ▪ Yes
		▪ Mechanical Cause
		  ▪ Yes
		▪ Control Module Cavity
		  ▪ Circuit reference - O_T_CLG -
		▪ Prioritized List of Possible Causes
		  ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion
		  ▪ Engine cooling fan circuit short circuit to power
		  ▪ Engine cooling fan failure
			▪ Vehicle Conditions to enable DTC Logging strategy
			  ▪ Ignition on
			▪ Prioritized Checks to Perform
			  ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion
			  ▪ Refer to the electrical circuit diagrams and check the engine cooling fan circuit for short circuit to power
			  ▪ Using the Jaguar Land Rover approved diagnostic equipment, run application - ECU Diagnostics / Powertrain Control Module (PCM) / Clear Diagnostic Trouble Code(s). Retest
			  ▪ Install a new engine cooling fan as required

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION

P0480-13	Fan 1 Control Circuit - Circuit open	▪ Electrical Cause
		  ▪ Yes
		▪ Mechanical Cause
		  ▪ Yes
		▪ Control Module Cavity
		  ▪ Circuit reference - O_T_CLG -
		▪ Prioritized List of Possible Causes
		  ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion
		  ▪ Engine cooling fan circuit open circuit, high resistance
		  ▪ Engine cooling fan failure
			▪ Vehicle Conditions to enable DTC Logging strategy
			  ▪ Ignition on
			▪ Prioritized Checks to Perform
			  ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion
			  ▪ Refer to the electrical circuit diagrams and check the engine cooling fan circuit for open circuit, high resistance
			  ▪ Using the Jaguar Land Rover approved diagnostic equipment, run application - ECU Diagnostics / Powertrain Control Module (PCM) / Clear Diagnostic Trouble Code(s). Retest
			  ▪ Install a new engine cooling fan as required

P0480-16	Fan 1 Control Circuit - Circuit voltage below threshold	▪ Electrical Cause
		  ▪ Yes
		▪ Mechanical Cause
		  ▪ Yes
		▪ Control Module Cavity
		  ▪ Circuit reference - O_T_CLG -
		▪ Monitor Description
		  ▪ Battery under voltage has been detected
		▪ Prioritized List of Possible Causes
		  ▪ Vehicle battery under charged
		  ▪ Charging system fault - under charging
		  ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion
		  ▪ Engine cooling fan fuse failure
		  ▪ Engine cooling fan circuit short circuit to ground, short circuit to power, open circuit, high resistance
		  ▪ Engine cooling fan failure
			▪ Vehicle Conditions to Enable DTC Logging Strategy
			  ▪ Engine running
			▪ Prioritized Checks to Perform
			  ▪ Refer to the workshop manual and the battery care manual, inspect the vehicle battery and make sure it is fully charged and serviceable before performing further tests
			  ▪ Check the vehicle charging system performance to make sure the voltage regulation is correct
			  ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion
			  ▪ Check the integrity of the engine cooling fan fuse. Rectify as necessary
			  ▪ Refer to the electrical circuit diagrams and check the engine cooling fan circuit for short circuit to ground, short circuit to power, open circuit, high resistance. Repair the wiring harness as necessary
			  ▪ Install a new engine cooling fan only when diagnosed as failed
			  ▪ Using the Jaguar Land Rover approved diagnostic equipment, run application - ECU Diagnostics / Powertrain Control Module (PCM) / Clear Diagnostic Trouble Code(s). Retest

P0480-17	Fan 1 Control Circuit - Circuit voltage above threshold	▪ Electrical Cause
		  ▪ Yes
		▪ Mechanical Cause
		  ▪ Yes
		▪ Control Module Cavity
		  ▪ Circuit reference - O_T_CLG -
		▪ Monitor Description
		  ▪ Battery over voltage has been detected
		▪ Prioritized List of Possible Causes
		  ▪ Vehicle battery under charged
		  ▪ Charging system fault - over charging
			▪ Vehicle Conditions to Enable DTC Logging Strategy
			  ▪ Engine running
			▪ Prioritized Checks to Perform
			  ▪ Refer to the workshop manual and the battery care manual, inspect the vehicle battery and make sure it is fully charged and serviceable before performing further tests
			  ▪ Check the vehicle charging system performance to make sure the voltage regulation is correct
			  ▪ Using the Jaguar Land Rover approved diagnostic equipment, run application - ECU Diagnostics / Powertrain Control Module (PCM) / Clear Diagnostic Trouble Code(s). Retest

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION

P0480-71	Fan 1 Control Circuit - Actuator stuck	▪ Electrical Cause
		  ▪ Yes
		▪ Mechanical Cause
		  ▪ Yes
		▪ Control Module Cavity
		  ▪ Circuit reference - O_T_CLG -
		▪ Monitor Description
		  ▪ Main fan has stalled
		  ▪ Main fan is blocked
		▪ Prioritized List of Possible Causes
		  ▪ Electric cooling fan unable to move due to obstruction
		  ▪ Engine cooling fan stalling caused by obstruction in fan cowling
		  ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion
		  ▪ Engine cooling fan fuse failure
		  ▪ Engine cooling fan circuit short circuit to ground, short circuit to power, open circuit, high resistance.
		  ▪ Engine cooling fan failure
			▪ Vehicle Conditions to Enable DTC Logging Strategy
			  ▪ Engine running
			▪ Prioritized Checks to Perform
			  ▪ Check there is not any obstruction preventing main electric cooling fan movement
			  ▪ Check for damage to or blockages in fan and fouling of fan cowling. Rectify as required
			  ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion
			  ▪ Check the integrity of the engine cooling fan fuse. Rectify as necessary
			  ▪ Refer to the electrical circuit diagrams and check the engine cooling fan circuit for short circuit to ground, short circuit to power, open circuit, high resistance. Repair the wiring harness as necessary.
			  ▪ Install a new engine cooling fan only when diagnosed as failed
			  ▪ Using the Jaguar Land Rover approved diagnostic equipment, run application - ECU Diagnostics / Powertrain Control Module (PCM) / Clear Diagnostic Trouble Code(s). Retest

P0481-11	Fan 2 Control Circuit - Circuit short to ground	▪ Electrical Cause
		  ▪ Yes
		▪ Mechanical Cause
		  ▪ Yes
		▪ Control Module Cavity
		  ▪ Circuit reference - O_T_CLG2 -
		▪ Monitor Description
		  ▪ Circuit check
		▪ Prioritized List of Possible Causes
		  ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion
		  ▪ Engine cooling fan circuit short circuit to ground
		  ▪ Engine cooling fan failure
			▪ Vehicle Conditions to Enable DTC Logging Strategy
			  ▪ Ignition on
			▪ Prioritized Checks to Perform
			  ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion
			  ▪ Refer to the electrical circuit diagrams and check the engine cooling fan circuit for short circuit to ground
			  ▪ Using the Jaguar Land Rover approved diagnostic equipment, run application - ECU Diagnostics / Powertrain Control Module (PCM) / Clear Diagnostic Trouble Code(s). Retest
			  ▪ Install a new engine cooling fan as required

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION

P0481-12	Fan 2 Control Circuit - Circuit short to battery	▪ Electrical Cause
		  ▪ Yes
		▪ Mechanical Cause
		  ▪ Yes
		▪ Control Module Cavity
		  ▪ Circuit reference - O_T_CLG2 -
		▪ Prioritized List of Possible Causes
		  ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion
		  ▪ Engine cooling fan circuit short circuit to power
		  ▪ Engine cooling fan failure
			▪ Vehicle Conditions to enable DTC Logging strategy
			  ▪ Ignition on
			▪ Prioritized Checks to Perform
			  ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion
			  ▪ Refer to the electrical circuit diagrams and check the engine cooling fan circuit for short circuit to power
			  ▪ Using the Jaguar Land Rover approved diagnostic equipment, run application - ECU Diagnostics / Powertrain Control Module (PCM) / Clear Diagnostic Trouble Code(s). Retest
			  ▪ Install a new engine cooling fan as required

P0481-13	Fan 2 Control Circuit - Circuit open	▪ Electrical Cause
		  ▪ Yes
		▪ Mechanical Cause
		  ▪ Yes
		▪ Control Module Cavity
		  ▪ Circuit reference - O_T_CLG2 -
		▪ Prioritized List of Possible Causes
		  ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion
		  ▪ Engine cooling fan circuit open circuit, high resistance
		  ▪ Engine cooling fan failure
			▪ Vehicle Conditions to enable DTC Logging strategy
			  ▪ Ignition on
			▪ Prioritized Checks to Perform
			  ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion
			  ▪ Refer to the electrical circuit diagrams and check the engine cooling fan circuit for open circuit, high resistance
			  ▪ Using the Jaguar Land Rover approved diagnostic equipment, run application - ECU Diagnostics / Powertrain Control Module (PCM) / Clear Diagnostic Trouble Code(s). Retest
			  ▪ Install a new engine cooling fan as required

P0481-16	Fan 2 Control Circuit - Circuit voltage below threshold	▪ Electrical Cause
		  ▪ Yes
		▪ Mechanical Cause
		  ▪ Yes
		▪ Monitor Description
		  ▪ Battery under voltage has been detected
		▪ Prioritized List of Possible Causes
		  ▪ Vehicle battery under charged
		  ▪ Charging system fault - under charging
		  ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion
		  ▪ Engine cooling fan circuit short circuit to ground, short circuit to power, open circuit, high resistance
		  ▪ Engine cooling fan failure
			▪ Vehicle Conditions to Enable DTC Logging Strategy
			  ▪ Engine running
			▪ Prioritized Checks to Perform
			  ▪ Refer to the workshop manual and the battery care manual, inspect the vehicle battery and make sure it is fully charged and serviceable before performing further tests
			  ▪ Check the vehicle charging system performance to make sure the voltage regulation is correct
			  ▪ Inspect connectors for signs of water ingress and pins for damage and/or corrosion
			  ▪ Refer to the electrical circuit diagrams and check the engine cooling fan circuit for short circuit to ground, short circuit to power, open circuit, high resistance. Repair the wiring harness or install a new engine cooling fan as necessary
			  ▪ Install a new engine cooling fan only when diagnosed as failed

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION

P0481-17	Fan 2 Control Circuit - Circuit voltage above threshold	▪ Electrical Cause
		  ▪ Yes
		▪ Mechanical Cause
		  ▪ Yes
		▪ Monitor Description
		  ▪ Battery over voltage has been detected
		▪ Prioritized List of Possible Causes
		  ▪ Vehicle battery under charged
		  ▪ Charging system fault - over charging
			▪ Vehicle Conditions to Enable DTC Logging Strategy
			  ▪ Engine running
			▪ Prioritized Checks to Perform
			  ▪ Refer to the workshop manual and the battery care manual, inspect the vehicle battery and make sure it is fully charged and serviceable before performing further tests
			  ▪ Check the vehicle charging system performance to make sure the voltage regulation is correct

P0481-71	Fan 2 Control Circuit - Actuator stuck	▪ Electrical Cause
		  ▪ Yes
		▪ Mechanical Cause
		  ▪ Yes
		▪ Monitor Description
		  ▪ Fan has stalled
		  ▪ Fan rotation is blocked
		▪ Prioritized List of Possible Causes
		  ▪ Electric cooling fan unable to move due to obstruction
		  ▪ Engine cooling fan stalling caused by obstruction in fan cowling
		  ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion
		  ▪ Engine cooling fan circuit short circuit to ground, short circuit to power, open circuit, high resistance
		  ▪ Engine cooling fan failure
			▪ Vehicle Conditions to Enable DTC Logging Strategy
			  ▪ Engine running
			▪ Prioritized Checks to Perform
			  ▪ Check there is not any obstruction preventing electric cooling fan movement
			  ▪ Check for damage to or blockages in fan and fouling of fan cowling. Rectify as required
			  ▪ Inspect connectors for signs of water ingress and pins for damage and/or corrosion
			  ▪ Refer to the electrical circuit diagrams and check the engine cooling fan circuit for short circuit to ground, short circuit to power, open circuit, high resistance. Repair the wiring harness or install a new engine cooling fan as necessary
			  ▪ Install a new engine cooling fan only when diagnosed as failed

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION

P0496-00	Evaporative Emission System High Purge Flow - No sub type information	• Electrical Cause
		  ▪ Yes
		• Mechanical Cause
		  ▪ Yes
		• Monitor Description
		  ▪ Federal diagnostic: After 600 seconds of driving, the diagnostic module tank leakage module operates while the purge valve is closed. This pressurizes the evaporative emission system and the DTC is flagged if system is at fault
		  ▪ European diagnostic: After a sufficient time purging with the vehicle driving (or at idle), the monitor will operate an air / fuel ratio mixture test. If the air/fuel ratio test indicates that the purge valve is stuck open then an air flow test is performed at idle in order to confirm a stuck open purge valve fault
		• Prioritized List of Possible Causes
		  ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion
		  ▪ Purge valve hose leaking or disconnected
		  ▪ Purge valve circuit short circuit to power
		  ▪ Purge valve stuck open
			NOTE:
			A guided flow is available for this DTC. To assist with correct diagnosis, please make sure to run the guided flow.
			• Vehicle Conditions to enable DTC Logging strategy
			  ▪ Federal diagnostic: Start and drive vehicle for at least 20 minutes, try to drive in a steady manner below 50mph to maximize the chance of the test operating
			  ▪ European diagnostic: Start and drive vehicle for at least 30 minutes, try to drive in a steady manner below 30mph with periods of engine idling (for 30 second durations) to maximize the chance of the test operating
			• Prioritized Checks to Perform
			  ▪ Using the Jaguar Land Rover approved diagnostic equipment, reconfigure the powertrain control module with the latest level software
			  ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion
			  ▪ Check the integrity of the purge valve hose
			  ▪ Refer to the electrical circuit diagrams and check the purge valve circuit for short circuit to power
			  ▪ Using the Jaguar Land Rover approved diagnostic equipment, perform routine - Purge Valve Self Test. Install a new purge valve as necessary

P0497-00	Evaporative Emission System Low Purge Flow - No sub type information	• Electrical Cause
		  ▪ Yes
		• Mechanical Cause
		  ▪ Yes
		• Monitor Description
		  ▪ Federal diagnostic: Engine running, off idle for DTC / MIL
		  ▪ European diagnostic: Engine running, off idle for mixture test, idle for air test, then DTC / MIL
		• Prioritized List of Possible Causes
		  ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion
		  ▪ Purge valve hose blocked or restricted
		  ▪ Purge valve circuit open circuit, high resistance
		  ▪ Purge valve stuck closed
			• Vehicle Conditions to enable DTC Logging strategy
			  ▪ Federal diagnostic: Start and drive vehicle for at least 20 minutes, try to drive in a steady manner below 50mph to maximize the chance of the test operating
			  ▪ European diagnostic: Start and drive vehicle for at least 30 minutes, try to drive in a steady manner below 30mph with periods of engine idling (for 30 second durations) to maximize the chance of the test operating
			• Prioritized Checks to Perform
			  ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion
			  ▪ Check the integrity of the purge valve hose
			  ▪ Refer to the electrical circuit diagrams and check the purge valve circuit for open circuit, high resistance
			  ▪ Using the Jaguar Land Rover approved diagnostic equipment, perform routine - Purge Valve Self Test. Install a new purge valve as necessary

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION
P0500-81	Vehicle Speed Sensor "A" Circuit - Invalid serial data received	▪ Prioritized List of Possible Causes ▪ Anti-lock brake system control module has detected a failure and reported this to the powertrain control module ▪ Anti-lock brake system fault	▪ Prioritized Checks to Perform ▪ Using the Jaguar Land Rover approved diagnostic equipment, check anti-lock brake system control module for Diagnostic Trouble Code(s) and refer to the relevant DTC index ▪ Check the communication circuit between powertrain control module and anti-lock brake system control module. Repair as necessary
P0501-86	Vehicle Speed Sensor "A" Circuit Range/Performance - Signal invalid	▪ Prioritized List of Possible Causes ▪ Anti-lock brake system fault	▪ Prioritized Checks to Perform ▪ Using the Jaguar Land Rover approved diagnostic equipment, check the anti-lock brake system control module for related Diagnostic Trouble Code(s) and refer to the relevant DTC index
P0502-26	Vehicle Speed Sensor "A" Circuit Low - Signal rate of change below threshold	▪ Prioritized List of Possible Causes ▪ The signal changes more slowly than is allowed ▪ Vehicle speed - range performance	▪ Prioritized Checks to Perform ▪ Using the Jaguar Land Rover approved diagnostic equipment check anti-lock brake system control module for related Diagnostic Trouble Code(s) and refer to the relevant DTC index
P0504-27	Brake Switch "A"/"B" Correlation - Signal rate of change above threshold	▪ Electrical Cause ▪ Yes ▪ Mechanical Cause ▪ Yes ▪ Control Module Cavity ▪ Circuit reference - I_S_BRKMN - ▪ Circuit reference - I_S_BRKRED - ▪ Monitor Description ▪ When the driver is braking and has achieved a high brake line pressure when not pressing the accelerator pedal the brake switches should both = ON/ACTIVE. If both switches are inactive / off in this condition for a prolonged period the switches have failed ▪ Prioritized List of Possible Causes ▪ The signal changes more quickly than is allowed ▪ Brake pedal switch circuit, open circuit, high resistance ▪ Brake diagnostic switch circuit, open circuit, high resistance ▪ Brake pedal switch plunger failure ▪ Brake pedal switch not installed correctly ▪ Brake pedal switch failure	▪ Vehicle Conditions to enable DTC Logging strategy ▪ With engine running press brake pedal to floor and hold for 1 minute, if fault is present DTC will set ▪ Prioritized Checks to Perform ▪ Check for related brake pressure Diagnostic Trouble Code(s) within the anti-lock braking control module ▪ Refer to the electrical circuit diagrams and check the brake pedal switch circuit for open circuit, high resistance ▪ Refer to the electrical circuit diagrams and check brake diagnostic switch circuit for open circuit, high resistance ▪ Check brake pedal switch is installed correctly. Clear DTC, start the engine and press the brake pedal, using maximum travel, for greater than 1 minute taking care not to press the accelerator pedal ▪ Install a new brake pedal switch as required

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION
P0505-22	Idle Control System - Signal amplitude > maximum	▪ Prioritized List of Possible Causes ▪ Intake air system blockage ▪ Front end accessory drive overload, failed or seized component ▪ Power steering pump overload, failed or seized component	▪ Prioritized Checks to Perform ▪ Check air filter is not blocked ▪ Check the intake air system is free from restrictions ▪ Check front end accessory drive belt and driven components for overload failure or seized ▪ Check power steering pump for overload, failed or seized component
P0506-00	Idle Control System - RPM Lower Than Expected - No sub type information	▪ Electrical Cause ▪ No ▪ Mechanical Cause ▪ Yes ▪ Monitor Description ▪ Engine idle speed lower than minimum deviation allowed during normal (warm) operation ▪ Prioritized List of Possible Causes ▪ Intake air system blockage ▪ Front end accessory drive overload, failed or seized component ▪ Power steering pump overload, failed or seized component	▪ Vehicle Conditions to enable DTC Logging strategy ▪ Idle warm engine, operate PAS and AC ▪ Prioritized Checks to Perform ▪ Check air filter is not blocked ▪ Check the intake air system is free from restrictions ▪ Check front end accessory drive belt and driven components for overload failure or seized ▪ Check power steering pump for overload, failed or seized component
P0507-00	Idle Control System - RPM Higher Than Expected - No sub type information	▪ Electrical Cause ▪ No ▪ Mechanical Cause ▪ Yes ▪ Monitor Description ▪ Engine idle speed greater than maximum allowed during normal (warm) operation ▪ Prioritized List of Possible Causes ▪ Intake air system leakage ▪ Intake air system components incorrectly installed, loose or damaged pipework ▪ Engine crankcase breather system leakage ▪ Engine crankcase breather system components incorrectly installed	▪ Vehicle Conditions to enable DTC Logging strategy ▪ Idle warm engine, operate PAS and AC ▪ Prioritized Checks to Perform ▪ Long Term Fueling Trims exceeding +10% ▪ Using the manufacturer approved smoke test system check the intake air system is free from leakage, check that components are correctly and securely installed, check that there is no damage to the pipework that may be allowing leakage. Rectify as required ▪ Check that the engine crankcase breather system is free from external leakage, check that components are correctly and securely installed, check that there is no damage to the pipework that may be allowing external leakage. Rectify as required ▪ Long Term Fueling Trims in normal range (anywhere in the range up to + 10%) ▪ Disconnect the full load breather pipe from the crankcase ventilation unit and check the full load diaphragm for evidence of splitting or being dislodged (not sealing). Rectify as required ▪ Disconnect the part load breather pipe from the crankcase ventilation unit and check the part load diaphragm for evidence of splitting or being dislodged (not sealing). Rectify as required

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION

P050A-22	Cold Start Idle Control System Performance - Signal amplitude > maximum	▪ Prioritized List of Possible Causes
		  ▪ Intake air system blockage
		  ▪ Front end accessory drive overload, failed or seized component
			▪ Prioritized Checks to Perform
			  ▪ Check air filter is not blocked
			  ▪ Check the intake air system is free from restrictions
			  ▪ Check front end accessory drive belt and driven components for overload failure or seized

P050A-23	Cold Start Idle Control System Performance - Signal stuck low	▪ Prioritized List of Possible Causes
		  ▪ Intake air system blockage
		  ▪ Front end accessory drive overload, failed or seized component
			▪ Prioritized Checks to Perform
			  ▪ Check air filter is not blocked
			  ▪ Check the intake air system is free from restrictions
			  ▪ Check front end accessory drive belt and driven components for overload failure or seized

P050A-24	Cold Start Idle Control System Performance - Signal stuck high	▪ Prioritized List of Possible Causes
		  ▪ Intake air system leakage
		  ▪ Intake air system components incorrectly installed, loose or damaged pipework
		  ▪ Engine crankcase breather system leakage
		  ▪ Engine crankcase breather system components incorrectly installed
			▪ Prioritized Checks to Perform
			  ▪ Using the Jaguar Land Rover approved high pressure diagnostic leak detector, check the intake air system is free from leakage, components are correctly installed, not loose or that damaged pipework is allowing leakage
			  ▪ Check engine crankcase breather system is free from leakage, components are correctly installed, not loose or that damaged pipework is allowing leakage

P050B-92	Cold Start Ignition Timing Performance - Performance or incorrect operation	▪ Electrical Cause
		  ▪ No
		▪ Mechanical Cause
		  ▪ Yes
		▪ Monitor Description
		  ▪ Monitors ignition angle during CAT heating in D
		▪ Prioritized List of Possible Causes
		  ▪ The powertrain control module has detected that the actual ignition timing is different to that requested for optimal catalyst heating
		  ▪ Intake air system components incorrectly installed, loose or damaged pipework
		  ▪ Engine crankcase breather system leakage
		  ▪ Engine crankcase breather system components incorrectly installed
		  ▪ Engine vacuum system leak
		  ▪ Brake booster leak
		  ▪ Engine speed error
		  ▪ Incorrect accessory loading
		  ▪ Engine misfire
			▪ Vehicle Conditions to enable DTC Logging strategy
			  ▪ Cold engine, in drive
			▪ Prioritized Checks to Perform
			  ▪ Using the Jaguar Land Rover approved high pressure diagnostic leak detector, check the intake air system is free from leakage, components are correctly installed, pipework is not loose or damaged
			  ▪ Check engine vacuum system is free from leakage, components are correctly installed, valves function correctly, pipework is not loose or damaged
			  ▪ Check engine crankcase breather system is free from leakage, components are correctly installed, valves function correctly, pipework is not loose or damaged
			  ▪ Check air filter(s) are not blocked
			  ▪ Check the intake air system is free from restrictions
			  ▪ Check front end accessory drive belt and driven components for failure

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION
P050B-93	Cold Start Ignition Timing Performance - No operation	▪ Electrical Cause ▪ No ▪ Mechanical Cause ▪ Yes ▪ Monitor Description ▪ Actual ignition timing different to commanded ignition timing during catalyst heating ▪ Prioritized List of Possible Causes ▪ The powertrain control module has detected that the actual ignition timing is different to that requested for optimal catalyst heating ▪ Intake air system leakage ▪ Intake air system components incorrectly installed, loose or damaged pipework ▪ Engine crankcase breather system leakage ▪ Engine crankcase breather system components incorrectly installed ▪ Engine vacuum system leak ▪ Brake booster leak ▪ Engine speed error ▪ Incorrect accessory loading ▪ Engine misfire	▪ Vehicle Conditions to enable DTC Logging strategy ▪ Cold start and idle for 1 minute after an overnight soak above -7°C ▪ Prioritized Checks to Perform ▪ Using the Jaguar Land Rover approved high pressure diagnostic leak detector, check the intake air system is free from leakage, components are correctly installed, pipework is not loose or damaged ▪ Check engine vacuum system is free from leakage, components are correctly installed, valves function correctly, pipework is not loose or damaged ▪ Check engine crankcase breather system is free from leakage, components are correctly installed, valves function correctly, pipework is not loose or damaged ▪ Check air filter(s) are not blocked ▪ Check the intake air system is free from restrictions ▪ Check front end accessory drive belt and driven components for failure
P050F-01	Brake Assist Vacuum Too Low - General electrical failure	▪ Prioritized List of Possible Causes ▪ Brake vacuum sensor open circuit, high resistance ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion	▪ Prioritized Checks to Perform ▪ Refer to the electrical circuit diagrams and check the brake vacuum sensor circuit for open circuit, high resistance ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion ▪ Using the Jaguar Land Rover approved diagnostic equipment, run application - ECU Diagnostics / Powertrain Control Module (PCM) / Clear Diagnostic Trouble Code(s). Retest
P0512-12	Starter Request Circuit - Circuit short to battery	▪ Electrical Cause ▪ Yes ▪ Mechanical Cause ▪ Yes ▪ Control Module Cavity ▪ Circuit reference - I_S_T50 - ▪ Prioritized List of Possible Causes ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Starter request circuit short circuit to power	▪ Prioritized Checks to Perform ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion ▪ Refer to the electrical circuit diagrams and check the starter request circuit for short circuit to power ▪ Using the Jaguar Land Rover approved diagnostic equipment, run application - ECU Diagnostics / Powertrain Control Module (PCM) / Clear Diagnostic Trouble Code(s). Retest

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION
P0512-73	Starter Request Circuit - Actuator stuck closed	▪ Electrical Cause ▪ Yes ▪ Mechanical Cause ▪ Yes ▪ Control Module Cavity ▪ Circuit reference - I_S_T50 - ▪ Prioritized List of Possible Causes ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Starter request circuit open circuit, short circuit to ground	▪ Prioritized Checks to Perform ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion ▪ Refer to the electrical circuit diagrams and check the starter request circuit for open circuit, short circuit to ground ▪ Using the Jaguar Land Rover approved diagnostic equipment, run application - ECU Diagnostics / Powertrain Control Module (PCM) / Clear Diagnostic Trouble Code(s). Retest
P0513-00	Incorrect Immobilizer Key - No sub type information	▪ Prioritized List of Possible Causes ▪ Invalid signal from body control module immobilizer	▪ Prioritized Checks to Perform ▪ Check for CAN network and powertrain control module related Diagnostic Trouble Code(s) and refer to the relevant DTC index ▪ Check the body control module for related Diagnostic Trouble Code(s) and refer to the relevant DTC index ▪ Using the Jaguar Land Rover approved diagnostic equipment, complete the immobilization procedure
P0520-02	Engine Oil Pressure Sensor/Switch "A" Circuit - General signal failure	▪ Electrical Cause ▪ Yes ▪ Mechanical Cause ▪ Yes ▪ Control Module Cavity ▪ Circuit reference - I_T_OPTS - ▪ Circuit reference - O_V_5VOPTS - ▪ Circuit reference - G_R_OPTS - ▪ Monitor Description ▪ Oil temperature sensor hardware fault ▪ Prioritized List of Possible Causes ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Oil pressure and temperature sensor fuse failure ▪ Oil pressure and temperature sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Oil pressure and temperature sensor failure	▪ Vehicle Conditions to Enable DTC Logging Strategy ▪ Ignition on, engine running ▪ Prioritized Checks to Perform ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion ▪ Check the integrity of the oil pressure and temperature sensor fuse. Rectify as necessary ▪ Refer to the electrical circuit diagrams and check the oil pressure and temperature sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance. Repair the wiring harness as necessary ▪ Install a new oil pressure and temperature sensor only when diagnosed as failed ▪ Using the Jaguar Land Rover approved diagnostic equipment, run application - ECU Diagnostics / Powertrain Control Module (PCM) / Clear Diagnostic Trouble Code(s). Retest

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION
P0520-09	Engine Oil Pressure Sensor/Switch "A" Circuit - Component failures	▪ Electrical Cause ▪ Yes ▪ Mechanical Cause ▪ Yes ▪ Control Module Cavity ▪ Circuit reference - I_T_OPTS - ▪ Circuit reference - O_V_5VOPTS - ▪ Circuit reference - G_R_OPTS - ▪ Monitor Description ▪ Oil temperature sensor hardware fault ▪ Prioritized List of Possible Causes ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Oil pressure and temperature sensor fuse failure ▪ Oil pressure and temperature sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Oil pressure and temperature sensor failure	▪ Vehicle Conditions to Enable DTC Logging Strategy ▪ Ignition on, engine running ▪ Prioritized Checks to Perform ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion ▪ Check the integrity of the oil pressure and temperature sensor fuse. Rectify as necessary ▪ Refer to the electrical circuit diagrams and check the oil pressure and temperature sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance. Repair the wiring harness as necessary ▪ Install a new oil pressure and temperature sensor only when diagnosed as failed ▪ Using the Jaguar Land Rover approved diagnostic equipment, run application - ECU Diagnostics / Powertrain Control Module (PCM) / Clear Diagnostic Trouble Code(s). Retest
P0520-86	Engine Oil Pressure Sensor/Switch "A" Circuit - Signal invalid	▪ Electrical Cause ▪ Yes ▪ Mechanical Cause ▪ Yes ▪ Control Module Cavity ▪ Circuit reference - I_T_OPTS - ▪ Circuit reference - O_V_5VOPTS - ▪ Circuit reference - G_R_OPTS - ▪ Monitor Description ▪ Oil temperature sensor hardware fault ▪ Prioritized List of Possible Causes ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Oil pressure and temperature sensor fuse failure ▪ Oil pressure and temperature sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Oil pressure and temperature sensor failure	▪ Vehicle Conditions to Enable DTC Logging Strategy ▪ Ignition on, engine running ▪ Prioritized Checks to Perform ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion ▪ Check the integrity of the oil pressure and temperature sensor fuse. Rectify as necessary ▪ Refer to the electrical circuit diagrams and check the oil pressure and temperature sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance. Repair the wiring harness as necessary ▪ Install a new oil pressure and temperature sensor only when diagnosed as failed ▪ Using the Jaguar Land Rover approved diagnostic equipment, run application - ECU Diagnostics / Powertrain Control Module (PCM) / Clear Diagnostic Trouble Code(s). Retest

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION

P0521-03	Engine Oil Pressure Sensor/Switch "A" Range/Performance - FM (Frequency Modulated) / PWM (Pulse Width Modulated) failures	▪ Electrical Cause
		  ▪ Yes
		▪ Mechanical Cause
		  ▪ Yes
		▪ Control Module Cavity
		  ▪ Circuit reference - I_T_OPTS -
		  ▪ Circuit reference - O_V_5VOPTS -
		  ▪ Circuit reference - G_R_OPTS -
		▪ Monitor Description
		  ▪ Oil pressure sensor - PWM out of range
		▪ Prioritized List of Possible Causes
		  ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion
		  ▪ Oil pressure and temperature sensor fuse failure
		  ▪ Oil pressure and temperature sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance
		  ▪ Oil pressure and temperature sensor failure
			▪ Vehicle Conditions to Enable DTC Logging Strategy
			  ▪ Ignition on, engine running
			▪ Prioritized Checks to Perform
			  ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion
			  ▪ Check the integrity of the oil pressure and temperature sensor fuse. Rectify as necessary
			  ▪ Refer to the electrical circuit diagrams and check the oil pressure and temperature sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance. Repair the wiring harness as necessary
			  ▪ Install a new oil pressure and temperature sensor only when diagnosed as failed
			  ▪ Using the Jaguar Land Rover approved diagnostic equipment, run application - ECU Diagnostics / Powertrain Control Module (PCM) / Clear Diagnostic Trouble Code(s). Retest

P0521-84	Engine Oil Pressure Sensor/Switch "A" Range/Performance - Signal below allowable range	▪ Electrical Cause
		  ▪ Yes
		▪ Mechanical Cause
		  ▪ Yes
		▪ Control Module Cavity
		  ▪ Circuit reference - O_T_OILP -
		▪ Prioritized List of Possible Causes
		  ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion
		  ▪ Variable oil pump solenoid fuse failure
		  ▪ Variable oil pump solenoid circuit short circuit to ground, short circuit to power, open circuit, high resistance
		  ▪ Variable oil pump solenoid failure
			▪ Vehicle Conditions to Enable DTC Logging Strategy
			  ▪ Ignition on, engine running
			▪ Prioritized Checks to Perform
			  ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion
			  ▪ Check the integrity of the variable oil pump solenoid fuse. Rectify as necessary
			  ▪ Refer to the electrical circuit diagrams and check the variable oil pump solenoid circuit for short circuit to ground, short circuit to power, open circuit, high resistance. Repair the wiring harness as necessary
			  ▪ Install a new variable oil pump solenoid only when diagnosed as failed
			  ▪ Using the Jaguar Land Rover approved diagnostic equipment, run application - ECU Diagnostics / Powertrain Control Module (PCM) / Clear Diagnostic Trouble Code(s). Retest

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION

P0521-86	Engine Oil Pressure Sensor/Switch "A" Range/Performance - Signal invalid	▪ Electrical Cause
		  ▪ Yes
		▪ Mechanical Cause
		  ▪ Yes
		▪ Control Module Cavity
		  ▪ Circuit reference - I_T_OPTS -
		  ▪ Circuit reference - O_V_5VOPTS -
		  ▪ Circuit reference - G_R_OPTS -
		▪ Monitor Description
		  ▪ Oil temperature sensor hardware fault
		▪ Prioritized List of Possible Causes
		  ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion
		  ▪ Oil pressure and temperature sensor fuse failure
		  ▪ Oil pressure and temperature sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance
		  ▪ Oil pressure and temperature sensor failure
			▪ Vehicle Conditions to Enable DTC Logging Strategy
			  ▪ Ignition on, engine running
			▪ Prioritized Checks to Perform
			  ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion
			  ▪ Check the integrity of the oil pressure and temperature sensor fuse. Rectify as necessary
			  ▪ Refer to the electrical circuit diagrams and check the oil pressure and temperature sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance. Repair the wiring harness as necessary
			  ▪ Install a new oil pressure and temperature sensor only when diagnosed as failed
			  ▪ Using the Jaguar Land Rover approved diagnostic equipment, run application - ECU Diagnostics / Powertrain Control Module (PCM) / Clear Diagnostic Trouble Code(s). Retest

P0522-11	Engine Oil Pressure Sensor/Switch "A" Low - Circuit short to ground	▪ Electrical Cause
		  ▪ Yes
		▪ Mechanical Cause
		  ▪ Yes
		▪ Control Module Cavity
		  ▪ Circuit reference - I_T_OPTS -
		  ▪ Circuit reference - O_V_5VOPTS -
		  ▪ Circuit reference - G_R_OPTS -
		▪ Monitor Description
		  ▪ Oil pressure and temperature sensor short circuit to ground
		▪ Prioritized List of Possible Causes
		  ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion
		  ▪ Oil pressure and temperature sensor fuse failure
		  ▪ Oil pressure and temperature sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance
		  ▪ Oil pressure and temperature sensor failure
			▪ Vehicle Conditions to Enable DTC Logging Strategy
			  ▪ Ignition on, engine running
			▪ Prioritized Checks to Perform
			  ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion
			  ▪ Check the integrity of the oil pressure and temperature sensor fuse. Rectify as necessary
			  ▪ Refer to the electrical circuit diagrams and check the oil pressure and temperature sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance. Repair the wiring harness as necessary
			  ▪ Install a new oil pressure and temperature sensor only when diagnosed as failed
			  ▪ Using the Jaguar Land Rover approved diagnostic equipment, run application - ECU Diagnostics / Powertrain Control Module (PCM) / Clear Diagnostic Trouble Code(s). Retest

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION

P0523-12	Engine Oil Pressure Sensor/Switch "A" High - Circuit short to battery	▪ Electrical Cause
		  ▪ Yes
		▪ Mechanical Cause
		  ▪ Yes
		▪ Control Module Cavity
		  ▪ Circuit reference - I_T_OPTS -
		  ▪ Circuit reference - O_V_5VOPTS -
		  ▪ Circuit reference - G_R_OPTS -
		▪ Monitor Description
		  ▪ Oil pressure and temperature sensor short circuit to battery
		▪ Prioritized List of Possible Causes
		  ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion
		  ▪ Oil pressure and temperature sensor fuse failure
		  ▪ Oil pressure and temperature sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance
		  ▪ Oil pressure and temperature sensor failure
			▪ Vehicle Conditions to Enable DTC Logging Strategy
			  ▪ Ignition on, engine running
			▪ Prioritized Checks to Perform
			  ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion
			  ▪ Check the integrity of the oil pressure and temperature sensor fuse. Rectify as necessary
			  ▪ Refer to the electrical circuit diagrams and check the oil pressure and temperature sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance. Repair the wiring harness as necessary
			  ▪ Install a new oil pressure and temperature sensor only when diagnosed as failed
			  ▪ Using the Jaguar Land Rover approved diagnostic equipment, run application - ECU Diagnostics / Powertrain Control Module (PCM) / Clear Diagnostic Trouble Code(s). Retest

P052A-00	Cold Start "A" Camshaft Position Timing Over-Advanced Bank 1 - No sub type information	▪ Electrical Cause
		  ▪ No
		▪ Mechanical Cause
		  ▪ Yes
		▪ Monitor Description
		  ▪ Inlet camshaft target versus actual deviation exceeds allowable limit during catalyst heating phase
		▪ Prioritized List of Possible Causes
		  ▪ Oil level is low
		  ▪ Variable camshaft timing solenoid circuit intake and exhaust connectors swapped
		  ▪ Variable camshaft timing solenoid circuit short circuit to ground, short circuit to power, open circuit, high resistance
		  ▪ Oil pressure is low
		  ▪ Variable camshaft timing actuator mechanical failure
		  ▪ Excessive camshaft friction, restricted movement or stuck
			▪ Vehicle Conditions to Enable DTC Logging Strategy
			  ▪ Engine must be cold or soaked for an extended period (greater than 6 hours) in order to enable catalyst heating, Start and idle vehicle for at least 15 seconds
			▪ Prioritized Checks to Perform
			  ▪ Check the oil level is correct
			  ▪ Check variable camshaft timing solenoid circuit intake and exhaust connectors are not swapped
			  ▪ Refer to the electrical circuit diagrams and check the variable camshaft timing solenoid circuit for short circuit to ground, short circuit to power, open circuit, high resistance
			  ▪ Refer to the relevant sections of the workshop manual and check the oil pressure is within specification. Check oil supply to camshaft is correct and that oil gallery is free from debris
			  ▪ Using the Jaguar Land Rover approved diagnostic equipment, run application - ECU Diagnostics / Powertrain Control Module (PCM) / Clear Diagnostic Trouble Code(s). Retest
			  ▪ Install a new variable camshaft timing actuator as required
			  ▪ Refer to the relevant sections of the workshop manual and check the camshaft operation

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION

P053F-21	Cold Start Fuel Pressure Performance Bank 1 - Signal amplitude < minimum	▪ Monitor Description
		  ▪ The high pressure pump signal is below a threshold during cat heating
		▪ Prioritized List of Possible Causes
		  ▪ External fuel leak from high pressure fuel system
		  ▪ Internal fuel leak into low pressure fuel system
			▪ Vehicle Conditions to Enable DTC Logging Strategy
			  ▪ Engine running greater than 1 minute
			▪ Prioritized Checks to Perform
			  ▪ Check for external fuel leaks from the high pressure fuel pump and fuel rail
			  ▪ Check for internal fuel leaks from high pressure fuel pump into the low pressure fuel system

P053F-22	Cold Start Fuel Pressure Performance Bank 1 - Signal amplitude > maximum	▪ Monitor Description
		  ▪ The high pressure pump signal is below a threshold during cat heating
		▪ Prioritized List of Possible Causes
		  ▪ External fuel leak from high pressure fuel system
		  ▪ Internal fuel leak into low pressure fuel system
			▪ Vehicle Conditions to Enable DTC Logging Strategy
			  ▪ Engine running greater than 1 minute
			▪ Prioritized Checks to Perform
			  ▪ Check for external fuel leaks from the high pressure fuel pump and fuel rail
			  ▪ Check for internal fuel leaks from high pressure fuel pump into the low pressure fuel system

P053F-23	Cold Start Fuel Pressure Performance Bank 1 - Signal stuck low	▪ Monitor Description
		  ▪ The high pressure pump signal is below a threshold during cat heating
		▪ Prioritized List of Possible Causes
		  ▪ External fuel leak from high pressure fuel system
		  ▪ Internal fuel leak into low pressure fuel system
			▪ Vehicle Conditions to Enable DTC Logging Strategy
			  ▪ Engine running greater than 1 minute
			▪ Prioritized Checks to Perform
			  ▪ Check for external fuel leaks from the high pressure fuel pump and fuel rail
			  ▪ Check for internal fuel leaks from high pressure fuel pump into the low pressure fuel system

P053F-24	Cold Start Fuel Pressure Performance Bank 1 - Signal stuck high	▪ Monitor Description
		  ▪ The high pressure pump signal is above a threshold during cat heating
		▪ Prioritized List of Possible Causes
		  ▪ External fuel leak from high pressure fuel system
		  ▪ Internal fuel leak into low pressure fuel system
			▪ Vehicle Conditions to Enable DTC Logging Strategy
			  ▪ Engine running greater than 1 minute
			▪ Prioritized Checks to Perform
			  ▪ Check for external fuel leaks from the high pressure fuel pump and fuel rail
			  ▪ Check for internal fuel leaks from high pressure fuel pump into the low pressure fuel system

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION
P054A-00	Cold Start "B" Camshaft Position Timing Over-Advanced Bank 1 - No sub type information	▪ Electrical Cause ▪ No ▪ Mechanical Cause ▪ Yes ▪ Monitor Description ▪ Exhaust camshaft target versus actual deviation exceeds allowable limit during catalyst heating phase ▪ Prioritized List of Possible Causes ▪ Oil level is low ▪ Variable camshaft timing solenoid circuit intake and exhaust connectors swapped ▪ Variable camshaft timing solenoid circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Oil pressure is low ▪ Variable camshaft timing system fault - Exhaust camshaft ▪ Excessive camshaft friction, restricted movement or stuck	▪ Vehicle Conditions to Enable DTC Setting Strategy ▪ Engine must be cold or soaked for an extended period (greater than 6 hours) in order to enable catalyst heating, Start and idle vehicle for at least 15 seconds ▪ Prioritized Checks to Perform ▪ Check the oil level. Rectify as necessary ▪ Check variable camshaft timing solenoid circuit intake and exhaust connectors are not swapped ▪ Refer to the electrical circuit diagrams and check variable camshaft timing solenoid circuit for short circuit to ground, short circuit to power, open circuit, high resistance. Repair the wiring harness as necessary ▪ Refer to the relevant sections of the workshop manual and check the oil pressure is within specification. Check oil supply to camshaft is correct and that oil gallery is free from debris ▪ After timing related repairs and timing adjustment, using the Jaguar Land Rover approved diagnostic equipment, run application - ECU Diagnostics / Powertrain Control Module (PCM) / ECU Functions / Variable Camshaft Timing (VCT) Actuator Exhaust Test. Perform the relevant corrective actions ▪ Using the Jaguar Land Rover approved diagnostic equipment, run application - ECU Diagnostics / Powertrain Control Module (PCM) / Clear Diagnostic Trouble Code(s). Retest ▪ Refer to the relevant sections of the workshop manual and check the camshaft operation
P0555-81	Brake Booster Pressure Sensor Circuit - Invalid serial data received	▪ Prioritized List of Possible Causes ▪ CAN signal error for brake pressure over CAN bus	▪ Prioritized Checks to Perform ▪ Using the Jaguar Land Rover approved diagnostic equipment, check anti-lock brake system control module for Diagnostic Trouble Code(s) and refer to the relevant DTC index
P0562-00	System Voltage Low - No sub type information	▪ Prioritized List of Possible Causes ▪ Vehicle battery under charged ▪ Charging system fault - under charging	▪ Prioritized Checks to Perform ▪ Refer to the workshop manual and the battery care manual, inspect the vehicle battery and make sure it is fully charged and serviceable before performing further tests ▪ Refer to the electrical circuit diagrams and check the power and ground connections to the module ▪ Check the vehicle charging system performance to make sure the voltage regulation is correct
P0563-00	System Voltage High - No sub type information	▪ Prioritized List of Possible Causes ▪ Vehicle battery over charged ▪ Charging system fault - over charging	▪ Prioritized Checks to Perform ▪ Check the vehicle charging system performance to make sure the voltage regulation is correct

<table>
<tr><th>DTC</th><th>DESCRIPTION</th><th>POSSIBLE CAUSES</th><th>ACTION</th></tr>
<tr>
<td>P056E-84</td>
<td>Cold Start Turbocharger/Supercharger Boost Control "A" Performance - Signal below allowable range</td>
<td>
Electrical Cause
Yes
Mechanical Cause
Yes
Control Module Cavity
Circuit reference - O_T_SCPOS -
Circuit reference - O_T_SCNEG -
Prioritized List of Possible Causes
Turbocharger electronic wastegate mechanical integrity
Connector is disconnected, connector pin is backed out, connector pin corrosion
Turbocharger electronic wastegate circuit short circuit to ground, short circuit to power, open circuit, high resistance
Turbocharger electronic wastegate failure
</td>
<td>NOTE:
If a fault is identified with the wastegate system or the actuator, install a new turbocharger. The actuator is not a replaceable part. DO NOT remove the actuator or the connecting rod

Vehicle Conditions to Enable DTC Logging Strategy
Ignition On
Inspect connectors for signs of water ingress, and pins for damage and/or corrosion
Check the powertrain control module connectors for water ingress
Refer to the electrical circuit diagrams and check the turbocharger electronic wastegate circuit for
Short circuit to ground
Short circuit to power
Open circuit
High resistance
Check turbocharger wastegate rod movement, start the engine and monitor wastegate rod movement DO NOT attempt to disconnect, manipulate or put pressure on the waste gate rod, doing so can result in damage and loss of function of waste gate actuation
Check turbocharger wastegate voltage at closed position
Check turbocharger wastegate rod/lever for any mechanical damage, bend or heavy corrosion close to pin For additional information, refer to: Turbocharger (303-04D Fuel Charging and Controls - Turbocharger - INGENIUM I4 2.0L Petrol, Diagnosis and Testing).
Using the Jaguar Land Rover approved diagnostic equipment, clear the Diagnostic Trouble Code(s) and retest
</td>
</tr>
</table>

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION

P056E-85	Cold Start Turbocharger/Supercharger Boost Control "A" Performance - Signal above allowable range	▪ Electrical Cause
		  ▪ Yes
		▪ Mechanical Cause
		  ▪ Yes
		▪ Control Module Cavity
		  ▪ Circuit reference - O_T_SCPOS -
		  ▪ Circuit reference - O_T_SCNEG -
		▪ Prioritized List of Possible Causes
		  ▪ Turbocharger electronic wastegate mechanical integrity
		  ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion
		  ▪ Turbocharger electronic wastegate circuit short circuit to ground, short circuit to power, open circuit, high resistance
		  ▪ Turbocharger electronic wastegate failure
			NOTE:
			If a fault is identified with the wastegate system or the actuator, install a new turbocharger. The actuator is not a replaceable part. **DO NOT** remove the actuator or the connecting rod
			▪ Vehicle Conditions to Enable DTC Logging Strategy
			  ▪ Ignition On
			▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion
			▪ Check the powertrain control module connectors for water ingress
			▪ Refer to the electrical circuit diagrams and check the turbocharger electronic wastegate circuit for
			  ▪ Short circuit to ground
			  ▪ Short circuit to power
			  ▪ Open circuit
			  ▪ High resistance
			▪ Check turbocharger wastegate rod movement, start the engine and monitor wastegate rod movement **DO NOT** attempt to disconnect, manipulate or put pressure on the waste gate rod, doing so can result in damage and loss of function of waste gate actuation
			▪ Check turbocharger wastegate voltage at closed position
			▪ Check turbocharger wastegate rod/lever for any mechanical damage, bend or heavy corrosion close to pin For additional information, refer to: **Turbocharger** (303-04D Fuel Charging and Controls - Turbocharger - INGENIUM I4 2.0L Petrol, Diagnosis and Testing).
			▪ Using the Jaguar Land Rover approved diagnostic equipment, clear the Diagnostic Trouble Code(s) and retest

P0571-62	Brake Switch "A" Circuit - Signal compare failure	▪ Prioritized List of Possible Causes
		  ▪ The powertrain control module detected a failure when comparing two or more input parameters for plausibility
		  ▪ Other vehicle speed related Diagnostic Trouble Code(s)
		  ▪ Brake pedal switch adjustment
		  ▪ Brake pedal switch circuit short circuit to ground, short circuit to power, open circuit, high resistance
		  ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion
		  ▪ Brake pedal switch failure
			▪ Prioritized Checks to Perform
			  ▪ Check anti-lock brake system control module for vehicle speed related Diagnostic Trouble Code(s) and refer to relevant DTC index
			  ▪ Check brake pedal switch for correct adjustment
			  ▪ Refer to the electrical circuit diagrams and check the brake pedal switch circuit for short circuit to ground, short circuit to power, open circuit, high resistance
			  ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion
			  ▪ Install a new brake pedal switch as required
			  ▪ Using the Jaguar Land Rover approved diagnostic equipment, run application - ECU Diagnostics / Powertrain Control Module (PCM) / Clear Diagnostic Trouble Code(s). Retest

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION

P0575-81	Cruise Control Input Circuit - Invalid serial data received	▪ Prioritized List of Possible Causes
		  ▪ The DTC sets if the cancel, set minus, set plus, resume, headway increase and headway decrease speed control buttons have been pressed for longer than a calibrated period of time. The system then assumes a stuck/damaged button and will cancel and/or disable cruise. The failure will be healed in the next driving cycle if the failure is removed
			▪ Prioritized Checks to Perform
			  ▪ Check speed control buttons are not jammed/contaminated/damaged. Check cruise control module for related Diagnostic Trouble Code(s) and refer to the relevant DTC index
			  ▪ Check steering wheel clock spring and connections to button pack for damage
			  ▪ Install a new cruise control button pack as required

P0597-13	Thermostat Heater Control Circuit/Open - Circuit open	▪ Electrical Cause
		  ▪ Yes
		▪ Mechanical Cause
		  ▪ Yes
		▪ Control Module Cavity
		  ▪ Circuit reference - O_T_ETC -
		▪ Prioritized List of Possible Causes
		  ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion
		  ▪ Coolant thermostat heater circuit short circuit to ground, short circuit to power, open circuit, high resistance
		  ▪ Coolant thermostat heater failure
			▪ Vehicle Conditions to Enable DTC Logging Strategy
			  ▪ Ignition on, engine running
			▪ Prioritized Checks to Perform
			  ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion
			  ▪ Refer to the electrical circuit diagrams and check the coolant thermostat heater circuit for short circuit to ground, short circuit to power, open circuit, high resistance. Repair the wiring harness as necessary
			  ▪ Install a new coolant thermostat heater only when diagnosed as failed
			  ▪ Using the Jaguar Land Rover approved diagnostic equipment, run application - ECU Diagnostics / Powertrain Control Module (PCM) / Clear Diagnostic Trouble Code(s). Retest

P0597-4B	Thermostat Heater Control Circuit/Open - Over temperature	▪ Electrical Cause
		  ▪ Yes
		▪ Mechanical Cause
		  ▪ Yes
		▪ Control Module Cavity
		  ▪ Circuit reference - O_T_ETC -
		▪ Prioritized List of Possible Causes
		  ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion
		  ▪ Coolant thermostat heater circuit short circuit to ground, short circuit to power, open circuit, high resistance
		  ▪ Coolant thermostat heater failure
			▪ Vehicle Conditions to Enable DTC Logging Strategy
			  ▪ Ignition on, engine running
			▪ Prioritized Checks to Perform
			  ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion
			  ▪ Refer to the electrical circuit diagrams and check the coolant thermostat heater circuit for short circuit to ground, short circuit to power, open circuit, high resistance. Repair the wiring harness as necessary
			  ▪ Install a new coolant thermostat heater only when diagnosed as failed
			  ▪ Using the Jaguar Land Rover approved diagnostic equipment, run application - ECU Diagnostics / Powertrain Control Module (PCM) / Clear Diagnostic Trouble Code(s). Retest

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION

P0598-11	Thermostat Heater Control Circuit Low - Circuit short to ground	▪ Electrical Cause
		  ▪ Yes
		▪ Mechanical Cause
		  ▪ Yes
		▪ Control Module Cavity
		  ▪ Circuit reference - O_T_ETC -
		▪ Prioritized List of Possible Causes
		  ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion
		  ▪ Coolant thermostat heater circuit short circuit to ground, short circuit to power, open circuit, high resistance
		  ▪ Coolant thermostat heater failure
			▪ Vehicle Conditions to Enable DTC Logging Strategy
			  ▪ Ignition on, engine running
			▪ Prioritized Checks to Perform
			  ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion
			  ▪ Refer to the electrical circuit diagrams and check the coolant thermostat heater circuit for short circuit to ground, short circuit to power, open circuit, high resistance. Repair the wiring harness as necessary
			  ▪ Install a new coolant thermostat heater only when diagnosed as failed
			  ▪ Using the Jaguar Land Rover approved diagnostic equipment, run application - ECU Diagnostics / Powertrain Control Module (PCM) / Clear Diagnostic Trouble Code(s). Retest

P0599-12	Thermostat Heater Control Circuit High - Circuit short to battery	▪ Electrical Cause
		  ▪ Yes
		▪ Mechanical Cause
		  ▪ Yes
		▪ Control Module Cavity
		  ▪ Circuit reference - O_T_ETC -
		▪ Prioritized List of Possible Causes
		  ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion
		  ▪ Coolant thermostat heater circuit short circuit to ground, short circuit to power, open circuit, high resistance
		  ▪ Coolant thermostat heater failure
			▪ Vehicle Conditions to Enable DTC Logging Strategy
			  ▪ Ignition on, engine running
			▪ Prioritized Checks to Perform
			  ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion
			  ▪ Refer to the electrical circuit diagrams and check the coolant thermostat heater circuit for short circuit to ground, short circuit to power, open circuit, high resistance. Repair the wiring harness as necessary
			  ▪ Install a new coolant thermostat heater only when diagnosed as failed
			  ▪ Using the Jaguar Land Rover approved diagnostic equipment, run application - ECU Diagnostics / Powertrain Control Module (PCM) / Clear Diagnostic Trouble Code(s). Retest

P05A0-97	Active Grille Air Shutter "A" Stuck On - Component or system operation obstructed or blocked	▪ Electrical Cause
		  ▪ Yes
		▪ Mechanical Cause
		  ▪ Yes
		▪ Control Module Cavity
		  ▪ Circuit reference - B_D_LIN -
		▪ Prioritized List of Possible Causes
		  ▪ The powertrain control module has detected that the operation of a component is prevented by an obstruction
		  ▪ Active grille air shutter unable to move due to obstruction
		  ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion
		  ▪ Active grille air shutter circuit short circuit to ground, short circuit to power, open circuit, high resistance
			▪ Prioritized Checks to Perform
			  ▪ Check there is not any obstruction preventing active grille air shutter movement
			  ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion
			  ▪ Refer to the electrical circuit diagrams and check the active grille air shutter circuit for short circuit to ground, short circuit to power, open circuit, high resistance. Repair the wiring harness as necessary

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION

P05A2-01	Active Grille Air Shutter "A" Control Circuit/Open - General electrical failure	▪ Electrical Cause
		  ▪ Yes
		▪ Mechanical Cause
		  ▪ Yes
		▪ Control Module Cavity
		  ▪ Circuit reference - B_D_LIN -
		▪ Prioritized List of Possible Causes
		  ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion
		  ▪ Active grille air shutter circuit short circuit to ground, short circuit to power, open circuit, high resistance
		  ▪ Active grille air shutter unable to move due to obstruction
		  ▪ Active grille air shutter actuator failure
			▪ Prioritized Checks to Perform
			  ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion
			  ▪ Refer to the electrical circuit diagrams and check the active grille air shutter circuit for short circuit to ground, short circuit to power, open circuit, high resistance. Repair the wiring harness as necessary
			  ▪ Check there is not any obstruction preventing active grille air shutter movement
			  ▪ Refer to the electrical circuit diagrams and check the power and ground connections to the active grille air shutter actuator

P05A2-97	Active Grille Air Shutter "A" Control Circuit/Open - Component or system operation obstructed or blocked	▪ Electrical Cause
		  ▪ Yes
		▪ Mechanical Cause
		  ▪ Yes
		▪ Control Module Cavity
		  ▪ Circuit reference - B_D_LIN -
		▪ Prioritized List of Possible Causes
		  ▪ The powertrain control module has detected that the operation of a component is prevented by an obstruction
		  ▪ Active grille air shutter unable to move due to obstruction
		  ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion
		  ▪ Active grille air shutter circuit short circuit to ground, short circuit to power, open circuit, high resistance
			▪ Prioritized Checks to Perform
			  ▪ Check there is not any obstruction preventing active grille air shutter movement
			  ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion
			  ▪ Refer to the electrical circuit diagrams and check the active grille air shutter circuit for short circuit to ground, short circuit to power, open circuit, high resistance. Repair the wiring harness as necessary

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION

P05A6-01	Active Grille Air Shutter "A" Supply Voltage Circuit/Open - General electrical failure	▪ Electrical Cause
		  ▪ Yes
		▪ Mechanical Cause
		  ▪ Yes
		▪ Control Module Cavity
		  ▪ Circuit reference - B_D_LIN -
		▪ Prioritized List of Possible Causes
		  ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion
		  ▪ Active grille air shutter circuit short circuit to ground, short circuit to power, open circuit, high resistance
		  ▪ Active grille air shutter unable to move due to obstruction
		  ▪ Active grille air shutter actuator failure
			▪ Prioritized Checks to Perform
			  ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion
			  ▪ Refer to the electrical circuit diagrams and check the active grille air shutter circuit for short circuit to ground, short circuit to power, open circuit, high resistance. Repair the wiring harness as necessary
			  ▪ Check there is not any obstruction preventing active grille air shutter movement
			  ▪ Refer to the electrical circuit diagrams and check the power and ground connections to the active grille air shutter actuator

P05AF-97	Active Grille Air Shutter "B" Stuck On - Component or system operation obstructed or blocked	▪ Electrical Cause
		  ▪ Yes
		▪ Mechanical Cause
		  ▪ Yes
		▪ Control Module Cavity
		  ▪ Circuit reference - B_D_LIN -
		▪ Prioritized List of Possible Causes
		  ▪ The powertrain control module has detected that the operation of a component is prevented by an obstruction
		  ▪ Active grille air shutter unable to move due to obstruction
		  ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion
		  ▪ Active grille air shutter circuit short circuit to ground, short circuit to power, open circuit, high resistance
			▪ Prioritized Checks to Perform
			  ▪ Check there is not any obstruction preventing active grille air shutter movement
			  ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion
			  ▪ Refer to the electrical circuit diagrams and check the active grille air shutter circuit for short circuit to ground, short circuit to power, open circuit, high resistance. Repair the wiring harness as necessary

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION

P05B1-01	Active Grille Air Shutter "B" Control Circuit/Open - General electrical failure	▪ Electrical Cause
		  ▪ Yes
		▪ Mechanical Cause
		  ▪ Yes
		▪ Control Module Cavity
		  ▪ Circuit reference - B_D_LIN -
		▪ Prioritized List of Possible Causes
		  ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion
		  ▪ Active grille air shutter circuit short circuit to ground, short circuit to power, open circuit, high resistance
		  ▪ Active grille air shutter unable to move due to obstruction
		  ▪ Active grille air shutter actuator failure
			▪ Prioritized Checks to Perform
			  ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion
			  ▪ Refer to the electrical circuit diagrams and check the active grille air shutter circuit for short circuit to ground, short circuit to power, open circuit, high resistance. Repair the wiring harness as necessary
			  ▪ Check there is not any obstruction preventing active grille air shutter movement
			  ▪ Refer to the electrical circuit diagrams and check the power and ground connections to the active grille air shutter actuator

P05B1-97	Active Grille Air Shutter "B" Control Circuit/Open - Component or system operation obstructed or blocked	▪ Electrical Cause
		  ▪ Yes
		▪ Mechanical Cause
		  ▪ Yes
		▪ Control Module Cavity
		  ▪ Circuit reference - B_D_LIN -
		▪ Prioritized List of Possible Causes
		  ▪ The powertrain control module has detected that the operation of a component is prevented by an obstruction
		  ▪ Active grille air shutter unable to move due to obstruction
		  ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion
		  ▪ Active grille air shutter circuit short circuit to ground, short circuit to power, open circuit, high resistance
			▪ Prioritized Checks to Perform
			  ▪ Check there is not any obstruction preventing active grille air shutter movement
			  ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion
			  ▪ Refer to the electrical circuit diagrams and check the active grille air shutter circuit for short circuit to ground, short circuit to power, open circuit, high resistance. Repair the wiring harness as necessary

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION

P05B5-01	Active Grille Air Shutter "B" Supply Voltage Circuit/Open - General electrical failure	▪ Electrical Cause
		  ▪ Yes
		▪ Mechanical Cause
		  ▪ Yes
		▪ Control Module Cavity
		  ▪ Circuit reference - B_D_LIN -
		▪ Prioritized List of Possible Causes
		  ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion
		  ▪ Active grille air shutter circuit short circuit to ground, short circuit to power, open circuit, high resistance
		  ▪ Active grille air shutter unable to move due to obstruction
		  ▪ Active grille air shutter actuator failure
			▪ Prioritized Checks to Perform
			  ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion
			  ▪ Refer to the electrical circuit diagrams and check the active grille air shutter circuit for short circuit to ground, short circuit to power, open circuit, high resistance. Repair the wiring harness as necessary
			  ▪ Check there is not any obstruction preventing active grille air shutter movement
			  ▪ Refer to the electrical circuit diagrams and check the power and ground connections to the active grille air shutter actuator

P05C0-4B	Active Grille Air Shutter Module "A" Over Temperature - Over temperature	▪ Electrical Cause
		  ▪ Yes
		▪ Mechanical Cause
		  ▪ Yes
		▪ Control Module Cavity
		  ▪ Circuit reference - B_D_LIN -
		▪ Prioritized List of Possible Causes
		  ▪ The powertrain control module detected an internal temperature above the expected range
		  ▪ Active grille air shutter unable to move due to obstruction
		  ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion
		  ▪ Active grille air shutter circuit short circuit to ground, short circuit to power, open circuit, high resistance
		  ▪ Active grille air shutter actuator failure
			▪ Prioritized Checks to Perform
			  ▪ Check there is not any obstruction preventing active grille air shutter movement
			  ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion
			  ▪ Refer to the electrical circuit diagrams and check the active grille air shutter circuit for short circuit to ground, short circuit to power, open circuit, high resistance. Repair the wiring harness as necessary
			  ▪ Refer to the electrical circuit diagrams and check the power and ground connections to the active grille air shutter actuator
			  ▪ Install a new active grille air shutter only when diagnosed as failed

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION

P05C1-4B	Active Grille Air Shutter Module "B" Over Temperature - Over temperature	▪ Electrical Cause
		  ▪ Yes
		▪ Mechanical Cause
		  ▪ Yes
		▪ Control Module Cavity
		  ▪ Circuit reference - B_D_LIN -
		▪ Prioritized List of Possible Causes
		  ▪ The powertrain control module detected an internal temperature above the expected range
		  ▪ Active grille air shutter unable to move due to obstruction
		  ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion
		  ▪ Active grille air shutter circuit short circuit to ground, short circuit to power, open circuit, high resistance
		  ▪ Active grille air shutter actuator failure
			▪ Prioritized Checks to Perform
			  ▪ Check there is not any obstruction preventing active grille air shutter movement
			  ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion
			  ▪ Refer to the electrical circuit diagrams and check the active grille air shutter circuit for short circuit to ground, short circuit to power, open circuit, high resistance. Repair the wiring harness as necessary
			  ▪ Refer to the electrical circuit diagrams and check the power and ground connections to the active grille air shutter actuator
			  ▪ Install a new active grille air shutter only when diagnosed as failed

P0602-00	Control Module Programing Error - No sub type information	▪ Electrical Cause
		  ▪ No
		▪ Mechanical Cause
		  ▪ No
		▪ Purpose of the DTC
		▪ To determine if the solenoid valve driver has been programed correctly
		▪ Prioritized List of Possible Causes
		▪ Corrupt powertrain control module software download
		▪ Powertrain control module failure
			▪ Vehicle Conditions to enable DTC Logging Strategy
			  ▪ Ignition ON
			▪ Prioritized Checks to Perform
			▪ Using the Jaguar Land Rover approved diagnostic equipment check and install latest relevant level of software to the powertrain control module, clear the DTC and retest the system
			▪ Install new powertrain control module, only when diagnosed as failed
			  ▪ The powertrain control module now has a liquid cooling system incorporated. The following care points must be followed during removal, installation of component and disconnection, connection of electrical harness connectors
			  ▪ Make sure PCM side connectors are always dry
			  ▪ Make sure mating PCM harness connectors are always dry
			  ▪ Make sure PCM is always handled with connector facing upwards to avoid any accidental exposure of the connector to the coolant from the cooling pipe of the PCM to avoid damage to the PCM
			  ▪ Make sure that the connector from the PCM (harness side) is disconnected and connected only in a dry and clean environment

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION

P0602-08	Control Module Programing Error - Bus signal/message failures	▪ Electrical Cause
		  ▪ No
		▪ Mechanical Cause
		  ▪ No
		▪ Purpose of the DTC
		▪ To determine if the solenoid valve driver has been programed correctly
		▪ Prioritized List of Possible Causes
		▪ Corrupt powertrain control module software download
		▪ Powertrain control module failure
			▪ Vehicle Conditions to enable DTC Logging Strategy
			  ▪ Ignition ON
			▪ Prioritized Checks to Perform
			▪ Using the Jaguar Land Rover approved diagnostic equipment check and install latest relevant level of software to the powertrain control module, clear the DTC and retest the system
			▪ Install new powertrain control module, only when diagnosed as failed
			  ▪ The powertrain control module now has a liquid cooling system incorporated. The following care points must be followed during removal, installation of component and disconnection, connection of electrical harness connectors
			  ▪ Make sure PCM side connectors are always dry
			  ▪ Make sure mating PCM harness connectors are always dry
			  ▪ Make sure PCM is always handled with connector facing upwards to avoid any accidental exposure of the connector to the coolant from the cooling pipe of the PCM to avoid damage to the PCM
			  ▪ Make sure that the connector from the PCM (harness side) is disconnected and connected only in a dry and clean environment

P0606-00	Control Module Processor - No sub type information	▪ Electrical Cause
		  ▪ No
		▪ Mechanical Cause
		  ▪ No
		▪ Control Module Cavity
		  ▪ Not applicable
		▪ Prioritized List of Possible Causes
		  ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion
		  ▪ Powertrain control module failure
			▪ Vehicle Conditions to Enable DTC Logging Strategy
			  ▪ Ignition on for greater than 60 seconds
			▪ Prioritized Checks to Perform
			  ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion
			  ▪ Install a new powertrain control module only when diagnosed as failed
			  ▪ Using the Jaguar Land Rover approved diagnostic equipment, run application - ECU Diagnostics / Powertrain Control Module (PCM) / Clear Diagnostic Trouble Code(s). Retest

P0606-02	Control Module Processor - General signal failure	▪ Electrical Cause
		  ▪ Yes
		▪ Mechanical Cause
		  ▪ No
		▪ Control Module Cavity
		  ▪ Not applicable
		▪ Monitor Description
		  ▪ Internal 5 volt regulator is undervoltage
		  ▪ Internal powertrain control module hardware error
		▪ Prioritized List of Possible Causes
		  ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion
		  ▪ Powertrain control module failure
			▪ Vehicle Conditions to Enable DTC Logging Strategy
			  ▪ Ignition on for greater than 60 seconds
			▪ Prioritized Checks to Perform
			  ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion
			  ▪ Install a new powertrain control module only when diagnosed as failed
			  ▪ Using the Jaguar Land Rover approved diagnostic equipment, run application - ECU Diagnostics / Powertrain Control Module (PCM) / Clear Diagnostic Trouble Code(s). Retest

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION

P0606-17	Control Module Processor - Circuit voltage above threshold	▪ Electrical Cause
		  ▪ Yes
		▪ Mechanical Cause
		  ▪ No
		▪ Control Module Cavity
		  ▪ Not applicable
		▪ Monitor Description
		  ▪ Internal 5 volt regulator is over voltage
		  ▪ Internal powertrain control module hardware error
		▪ Prioritized List of Possible Causes
		  ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion
		  ▪ Powertrain control module failure
			▪ Vehicle Conditions to Enable DTC Logging Strategy
			  ▪ Ignition on for greater than 60 seconds
			▪ Prioritized Checks to Perform
			  ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion
			  ▪ Install a new powertrain control module only when diagnosed as failed
			  ▪ Using the Jaguar Land Rover approved diagnostic equipment, run application - ECU Diagnostics / Powertrain Control Module (PCM) / Clear Diagnostic Trouble Code(s). Retest

P0606-48	Control Module Processor - Supervision software failure	▪ Electrical Cause
		  ▪ No
		▪ Mechanical Cause
		  ▪ No
		▪ Control Module Cavity
		  ▪ Not applicable
		▪ Prioritized List of Possible Causes
		  ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion
		  ▪ Powertrain control module failure
			▪ Vehicle Conditions to Enable DTC Logging Strategy
			  ▪ Ignition on for greater than 60 seconds
			▪ Prioritized Checks to Perform
			  ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion
			  ▪ Install a new powertrain control module only when diagnosed as failed
			  ▪ Using the Jaguar Land Rover approved diagnostic equipment, run application - ECU Diagnostics / Powertrain Control Module (PCM) / Clear Diagnostic Trouble Code(s). Retest

P0606-86	Control Module Processor - Signal invalid	▪ Electrical Cause
		  ▪ Yes
		▪ Mechanical Cause
		  ▪ No
		▪ Control Module Cavity
		  ▪ Not applicable
		▪ Monitor Description
		  ▪ Communication error between monitoring module and powertrain control module powerstage shut off paths (hardware fault within powertrain control module prevents required observation of valid powerstage shut off control)
		▪ Prioritized List of Possible Causes
		  ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion
		  ▪ Powertrain control module failure
			▪ Vehicle Conditions to Enable DTC Logging Strategy
			  ▪ Ignition on for greater than 60 seconds
			▪ Prioritized Checks to Perform
			  ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion
			  ▪ Install a new powertrain control module only when diagnosed as failed
			  ▪ Using the Jaguar Land Rover approved diagnostic equipment, run application - ECU Diagnostics / Powertrain Control Module (PCM) / Clear Diagnostic Trouble Code(s). Retest

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION

P0606-92	Control Module Processor - Performance or incorrect operation	▪ Electrical Cause
		  ▪ Yes
		▪ Mechanical Cause
		  ▪ No
		▪ Control Module Cavity
		  ▪ Not applicable
		▪ Monitor Description
		  ▪ To determine the software functional fault
		▪ Prioritized List of Possible Causes
		  ▪ A software reset has occurred for an abnormal reason
		  ▪ Software reset, low battery reset during start attempt
		  ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion
		  ▪ Powertrain control module failure but only after multiple repeat case of this DTC
			▪ Vehicle Conditions to Enable DTC Logging Strategy
			  ▪ Ignition on and engine idling for greater than 30 seconds
			▪ Prioritized Checks to Perform
			  ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion
			  ▪ Using the Jaguar Land Rover approved diagnostic equipment, run application - ECU Diagnostics / Powertrain Control Module (PCM) / Clear Diagnostic Trouble Code(s). Retest
			  ▪ Complete vehicle test drive (including at least five ignition cycles during the drive). Re-check for Diagnostic Trouble Code(s). If P0606-92 DTC is not present, hand vehicle back to customer. If DTC appears again, repeat clear Diagnostic Trouble Code(s) and vehicle test drive. If DTC repeatedly returns, replace powertrain control module
			  ▪ Install a new powertrain control module only when diagnosed as failed
			  ▪ Using the Jaguar Land Rover approved diagnostic equipment, run application - ECU Diagnostics / Powertrain Control Module (PCM) / Clear Diagnostic Trouble Code(s). Retest

P0606-96	Control Module Processor - Component internal failure	▪ Electrical Cause
		  ▪ No
		▪ Mechanical Cause
		  ▪ No
		▪ Control Module Cavity
		  ▪ Not applicable
		▪ Monitor Description
		  ▪ To provide visibility of software traps and resets
		  ▪ The powertrain control module internal reset monitor logs, reset and trap exceptions, which are triggered by defects in the normal software flow. For each software reset type the visibility in the fault event storage can be configured
		▪ Prioritized List of Possible Causes
		  ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion
		  ▪ Powertrain control module failure
			▪ Vehicle Conditions to Enable DTC Logging Strategy
			  ▪ The reset monitor runs continuously
			▪ Prioritized Checks to Perform
			  ▪ Using the Jaguar Land Rover approved diagnostic equipment, run application - ECU Diagnostics / Powertrain Control Module (PCM) / Update ECU
			  ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion
			  ▪ Install a new powertrain control module only when diagnosed as failed
			  ▪ Using the Jaguar Land Rover approved diagnostic equipment, run application - ECU Diagnostics / Powertrain Control Module (PCM) / Clear Diagnostic Trouble Code(s). Retest

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION
P060A-00	Internal Control Module Monitoring Processor Performance - No sub type information	▪ Electrical Cause ▪ No ▪ Mechanical Cause ▪ No ▪ Control Module Cavity ▪ Not applicable ▪ Monitor Description ▪ Internal software fault of the powertrain control module (processing order, internal communication to monitoring modules ▪ Monitors software processing order, verifies valid data processing by monitoring intermediate unused data from the safety system, inserts pseudo errors within intermediate data points and checks for valid fault detection through monitoring module. All failures indicate PCM internal error and force power stage shutdown through internal hardware resulting in engine stall ▪ Prioritized List of Possible Causes ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Powertrain control module failure	▪ Vehicle Conditions to Enable DTC Logging Strategy ▪ Ignition on for greater than 60 seconds ▪ Prioritized Checks to Perform ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion ▪ Install a new powertrain control module only when diagnosed as failed ▪ Using the Jaguar Land Rover approved diagnostic equipment, run application - ECU Diagnostics / Powertrain Control Module (PCM) / Clear Diagnostic Trouble Code(s). Retest
P060B-00	Internal Control Module A/D Processing Performance - No sub type information	▪ Electrical Cause ▪ Yes ▪ Mechanical Cause ▪ No ▪ Control Module Cavity ▪ Not applicable ▪ Monitor Description ▪ The PCM's internal ADC is verified for correct operation through cyclically forcing a fixed voltage test pulse through the ADC, the low idle test pulse that is output from the ADC is checked for correct magnitude, should the value be out of acceptable voltage ▪ Prioritized List of Possible Causes ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Powertrain control module failure	▪ Vehicle Conditions to Enable DTC Logging Strategy ▪ Ignition on for greater than 60 seconds ▪ Prioritized Checks to Perform ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion ▪ Install a new powertrain control module only when diagnosed as failed ▪ Using the Jaguar Land Rover approved diagnostic equipment, run application - ECU Diagnostics / Powertrain Control Module (PCM) / Clear Diagnostic Trouble Code(s). Retest

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION

P060B-02	Internal Control Module A/D Processing Performance - General signal failure	▪ Electrical Cause
		  ▪ Yes
		▪ Mechanical Cause
		  ▪ No
		▪ Control Module Cavity
		  ▪ Not applicable
		▪ Monitor Description
		  ▪ The PCM's internal ADC is verified for correct operation through cyclically forcing a test pulse through the ADC, the low idle test pulse that is output from the ADC is checked for being updated at the ADC output within expected time frame, should the output not be updated the ADC has failed and the PCM must be replaced
		▪ Prioritized List of Possible Causes
		  ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion
		  ▪ Powertrain control module failure
			▪ Vehicle Conditions to Enable DTC Logging Strategy
			  ▪ Ignition on for greater than 60 seconds
			▪ Prioritized Checks to Perform
			  ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion
			  ▪ Install a new powertrain control module only when diagnosed as failed
			  ▪ Using the Jaguar Land Rover approved diagnostic equipment, run application - ECU Diagnostics / Powertrain Control Module (PCM) / Clear Diagnostic Trouble Code(s). Retest

P060C-00	Internal Control Module Main Processor Performance - No sub type information	▪ Electrical Cause
		  ▪ No
		▪ Mechanical Cause
		  ▪ No
		▪ Control Module Cavity
		  ▪ Not applicable
		▪ Monitor Description
		  ▪ Engine speed calculation was found not plausible due to internal software error, (speed calculated synchronously is compared to speed from 30° segment time)
		▪ Prioritized List of Possible Causes
		  ▪ Engine speed calculation not plausible due to internal powertrain control module software error
		  ▪ Powertrain control module failure
			▪ Vehicle Conditions to Enable DTC Logging Strategy
			  ▪ Engine running greater than 60 seconds, full speed range of engine must be exercised, "Rev engine" or short drive cycle
			▪ Prioritized Checks to Perform
			  ▪ Using the Jaguar Land Rover approved diagnostic equipment, run application - ECU Diagnostics / Powertrain Control Module (PCM) / Update ECU
			  ▪ Install a new powertrain control module only when diagnosed as failed
			  ▪ Using the Jaguar Land Rover approved diagnostic equipment, run application - ECU Diagnostics / Powertrain Control Module (PCM) / Clear Diagnostic Trouble Code(s). Retest

<table>
<tr><th>DTC</th><th>DESCRIPTION</th><th>POSSIBLE CAUSES</th><th>ACTION</th></tr>
<tr>
<td>P060D-00</td>
<td>Internal Control Module Accelerator Pedal Position Performance - No sub type information</td>
<td>
Electrical Cause
No
Mechanical Cause
No
Control Module Cavity
Not applicable
Monitor Description
Internal software fault of the powertrain control module. This DTC indicates that the PCM has failed to correctly detect a synchronization difference between the two accelerator pedal input signals
Prioritized List of Possible Causes
Accelerator pedal diagnostics have not captured a synchronization fault of the accelerator pedal due to an PCM software error
Powertrain control module failure
</td>
<td>
Vehicle Conditions to Enable DTC Logging Strategy
Ignition on 5 seconds, slowly press accelerator pedal from upper rest point to the floor and release slowly
Prioritized Checks to Perform
Using the Jaguar Land Rover approved diagnostic equipment, run application - ECU Diagnostics / Powertrain Control Module (PCM) / Update ECU
Install a new powertrain control module only when diagnosed as failed
Using the Jaguar Land Rover approved diagnostic equipment, run application - ECU Diagnostics / Powertrain Control Module (PCM) / Clear Diagnostic Trouble Code(s). Retest
</td>
</tr>
</table>

cmVlX2FtZzsyMDI1LTEyLTE5VDA3OjAzOjA5Ljc3OFo7MjA3LjE4MS4zLjEzMTtTQUpERDFHWDRMQ0s2NzQ5Nw==